# Amazon Sr SCM, NACF — Complete Interview Set

Loop: Oct 29–30 (PDT) · 5 per-interviewer packets + NACF inbound booklet

· Packet 1 · Snigdha Tripathy (Hiring Manager) · Wed Oct 29, 10:00a

· Packet 2 · Ryan Hoerger (Placement Analytics) · Wed Oct 29, 12:00p

· Packet 3 · Rocky Goforth (Infrastructure) · Thu Oct 30, 10:00a

· Packet 4 · Zhaowen Lan (Sr Mgr Supply Chain) · Thu Oct 30, 11:00a

· Packet 5 · Venkata Josyula (Bar Raiser) · Thu Oct 30, 12:00p

· NACF Inbound Booklet · capacity & inventory

INTERVIEW-PREP PACKET 1 OF 5 · PER-INTERVIEWER · BUSINESS-PROCESS LENS

# Amazon NACF Supply Chain loop — Packet 1: Snigdha Tripathy

Amazon — Senior Supply Chain Manager, NACF Supply Chain (L6)

## Snigdha Tripathy

Senior Manager, Supply Chain — NACF Supply Chain · HIRING MANAGER

**Wednesday, October 29, 2026 · 10:00 am PDT · 55-minute video interview**

**Strategy in one line.** The hiring manager judges fit for the actual work — and her public profile (EU inventory-placement strategy across the crossdock network; readiness for 50+ FC launches; teams built from scratch — public-profile inference in "Read me first") says she will test whether the mechanisms transfer to her world. Open with S11 (a boarding process behind published gates, run from a weekly input/output review) and hold S12 (a committed roadmap re-sequenced on cost-to-serve evidence); lead every answer with the mechanism left behind, the priced trade-off, and the team that grew to own it. Systems are enablers; the process redesign and its business results are the story.

**LP focus — my guess, not confirmed:** Deliver Results · Hire & Develop / Earth's Best Employer · Dive Deep. Amazon does not disclose the assignment; the guesses are reasoned from the interviewer's title, the JD and her public profile in "Read me first" (public-profile inference, labeled as such) and should be held loosely — Think Big / Frugality and Ownership / Earn Trust remain live alternates carried by the twist maps.

**Also expect:** "tell me about yourself" (75-second version, once, here) and "why Amazon" (hiring managers usually ask it); one functional question on how I would approach the NACF process space.

Canon: amazon/amazon-l6-scm-master-packet.html (§1.2 roster, §4.1 ledger). Every number below is that packet's value; every dollar carries its basis label (record · user anchor · modeled · priced).

## Read me first

### The interviewer's lens (inference from title, JD and public profile — the profile parts labeled)

**Inference from the title.** The hiring manager owns the team this role joins and the outcomes it is measured on. A Senior Manager of Supply Chain in NACF is accountable for network-level processes — how inventory is planned, placed, moved and recovered across fulfillment centers — and for the operating cadence (weekly business reviews, goals, mechanisms) that keeps those processes honest. She is the one interviewer who will also judge *fit for the actual work*: can this person define and run a process across teams that do not report to them, make value legible to leadership, and change a committed plan on evidence without breaking trust.

**What that means for the answers.** Lead with scale-with-mechanisms (S11) and roadmap judgment under unit economics (S12). Both are Flexport stories, told as process redesigns: a boarding process behind published gates with a weekly input/output review, and a launch roadmap re-sequenced on cost-to-serve. She will probably also ask the openers — "tell me about yourself" and "why Amazon" — and a functional question on how I would approach the NACF process space in the first 90 days. The JD's own phrase, "designing or modeling the supply chain as opposed to working in it," is the lens to speak to: I designed the boarding process and the review; I did not run the ops desk.

### What her public profile adds (public-profile inference, labeled)

**The profile, as pasted (public-profile inference — directional, not verified biography).** 11+ years at Amazon across operations, launch and learning: EU IXD & Placement Operations Manager, Luxembourg, 2017–21 — defined the EU inventory-placement strategy for the crossdock network (distance-to-customer, speed-vs-cost) and built the team ground-up (+$50MM savings in 2020); Senior Manager EU IXD, 2021–22 — ran the EU crossdock network with a team of six plus interns ($80MM savings in 2021; IXD grown to 60% of inbound, +46% YoY); Senior Launch Program Manager, 2014–17 — readiness for 50+ FC launches including Amazon.in; FC Learning & Development Manager — Kaizen / Lean / CI training, learning curves and skill matrices, delivered an S-Team goal; Dell training background; SP Jain PGPM. **These are her numbers from her profile, quoted for framing only — they are not canon and never enter my stories.**

**What that means for the answers.** Three adjustments. First, lead with the *mechanism left behind* and who owns it now — she built teams and mechanisms herself (a placement strategy, a crossdock network, an L&D program), so "what still runs without you?" is her native question. Second, *price the trade-off* the way she priced distance-to-customer against speed and cost — S11's boarding pace vs automation maturity and S12's operators' two further quarters are exactly her shape of decision. Third, expect *transferability probes* — "how would this apply to a placement / network problem?" — and expect her to spot borrowed Leadership-Principle language: answer in the story's own words (gates, denominators, reversal triggers), never in recited LP slogans.

### Three guessed LP focus areas, with rationale

> **Labeled guess.** Amazon assigns 2–3 Leadership Principles per interviewer and does not disclose them. These three are inferred from the title, the JD and the public profile above (public-profile inference); if the first question lands on a different LP — Think Big / Frugality for S12, Ownership / Earn Trust for the critique-and-private-first threads — the twist maps under each story carry the pivot.

| GUESSED LP | WHY THIS INTERVIEWER WOULD CARRY IT | STORY TO OPEN · BACKUP |
|---|---|---|
| **Deliver Results** | The HM carries the outcome LP almost by default — she needs evidence the candidate lands public targets with the organization intact. S11's two public targets two months early with zero rollbacks is the cleanest record in the set. | S11 (open) · backup: S18 in the master packet — only if S11 is already spent |

| GUESSED LP | WHY THIS INTERVIEWER WOULD CARRY IT | STORY TO OPEN · BACKUP |
|---|---|---|
| **Hire & Develop / Earth's Best Employer (one of the two)** | Her profile is teams built and grown: a placement team built ground-up in Luxembourg, a team of six plus interns running the EU crossdock network, and an S-Team goal delivered as an FC L&D manager on Kaizen / Lean / CI training and skill matrices (public-profile inference). She will probe who grew on my programs: S11's nine input owners signing the definitions they would be red against, wave teams grown into owners; S12's delivery teams now running the burn-up, the swap rule and the re-estimation habit. | S11 (owners own their lines) · S12 (mechanisms the teams now run) |
| **Dive Deep** | She ran a crossdock network hands-on and defined placement strategy at node level (public-profile inference) — expect her to test whether my numbers survive decomposition: CTS $0.66 → $0.03 cross-walked to ≈3.7 touches at $0.18 each; S12's cost translated at the work-item grain (339.5 minutes; 2,200 interactions per departed shipment). The Ownership / Earn Trust probes she may still carry are covered in the twist maps — the critique accepted (S11) and the private-first flip (S12). | S11 (per-container cross-walk) · S12 (work-item grain) |

## Answer strategy — business first, systems as enablers

- **Open with the process problem and its cost, not the system.** First sentence names the broken process (Plan, Source, Make, Deliver or Return), who it hurt (customer, Finance, the floor) and the baseline number. The platform appears only as the enabler of the redesign, in one clause.
- **Use the JD's four outcome words as the results frame:** reliability (did the promise hold?), responsiveness (how much earlier did we see it?), cost efficiency ($ with its basis label), asset utilization (capital, labor-hours, capacity released).
- **Action = operating-model redesign in gated phases** — Discovery → Plan & Design → Build & Test → Cutover → Hypercare, each spoken as entry bar · what I delivered · exit bar. Say the gate that fired, not the meeting that happened.
- **Every dollar carries its basis word in the same breath** — record · user anchor · modeled · priced. A record dollar is said flat; a modeled dollar is said with its inputs. Never promote a modeled number to record.
- **Program-lead voice.** I owned definitions, rules, gates and the signature chain; engineers, SIs and vendor teams configured and built to the spec I signed. The only platform personally configured in canon is Manhattan DOM.
- **The pitch ends on the primary metric.** Then stop. Silence after the pitch is the interviewer filling their STAR worksheet; their probes are the second half of the answer.
- **For the HM specifically:** name the mechanism in Amazon's own vocabulary where it is honest — a weekly input/output review with frozen metric definitions, graduation criteria published before the wave, a reversal trigger written before the flip. Do not claim to have invented the WBR; say it was copied from a trusted review, which is the point.
- **Answer her native question — what did you leave behind, and who owns it now?** Every story lands on a mechanism that still runs without me (the Monday review and the graduation gates; the rebuild-vs-reuse rule and the scope tripwire) with a named owner. She built teams and mechanisms ground-up (public-profile inference); that is the evidence she pattern-matches.
- **Price the trade-off in her units.** She priced distance-to-customer against speed and cost for a crossdock network (public-profile inference); my versions are boarding pace vs automation maturity (S11) and partners-

first vs operators' two further quarters, priced and published (S12). Say the trade and its price, not just the win.

- **Two warnings (public-profile inferences).** She tests transferability — "how would this apply to a placement / network problem?" — so practice the bridges in the twist maps before the call. And she spots borrowed LP language: never recite slogans ("I was customer-obsessed"); say the gate, the denominator, the reversal trigger in the story's own words.
- **Have the openers ready** — the 75-second "tell me about yourself" and the 60-second "why Amazon" from the master packet (newest to oldest; Sephora → Flexport → Elementum → Manhattan). Say the long version once, here.

**Loop-consistency rule.** One question, one story; never the same story to two interviewers — interviewers compare notes. Held for other packets (do not spend here): S1/S3 (Lan) · S2/S18 (Goforth) · S6/S8 (Hoerger) · S16/S15/S10/S14 (Josyula). If pushed for "another example" on an LP, use the named backup in the question bank or the backup bench at the end of this packet, never a story allocated to another interviewer.

**How to read the "L6 level evidence" block under each story.** Amazon's L6 functional bar (user-supplied rubric): *Ambiguity* — strategy or network undefined; I select the stakeholders, set the goals, deliver independently. *Scope & influence* — team-level scope; influences one to two levels up. *Execution* — tactical and strategic; simplifies or extends; applies best practices; force-multiplies; mitigates long-term risk; makes the short-versus-long-term trade-off explicit. *Impact* — long-term team goals. *Process improvement* — operational-excellence best practices institutionalized. *Complex* — visible risks, roadblocks and constraints mitigated through expertise, judgment and influence to consensus. Overview themes the block is tuned to: designing and modeling the supply chain rather than working in it; developing new processes for others to own; root-cause mechanisms that resolve trending issues at scale; recommendations to leadership that outline competing metrics. The block sits right after Results so the evidence is on the page when the probing starts.

## The stories — business-first, canon numbers

Two Flexport flagships. Each: 60-second pitch → Situation/Task → Action in gated phases → Results → Probing tuned to a hiring manager → the key math → Twist map.

### S11 — Flexport — Shipment Visibility launch: a boarding process behind published graduation gates, run from a weekly input/output review (cost-to-serve $0.66 → $0.03 across 7 → 68 customers)

**LPs this story can serve here:** Deliver Results (primary) · Ownership (accepted the critique, owned the measurement) · Customer Obsession (graduation criteria per wave, CSAT in the gate) · Earn Trust (the discarded 99.4% week) · Hire & Develop (backup — wave teams grown into owners)

**Process family:** Deliver — customer onboarding / service-delivery process; weekly operating cadence; launch readiness and staged rollout

**Honesty label.** Program owner across nine teams / ~100 people; the WBR format was copied from an existing leadership review (the LSA review), not invented; build-week counts are illustrative; the ~$1.5M pre-production measurement saving is a separate canonical figure, never merged with cost-to-serve. Nine teams built and operated; I owned the boarding process, the measurement system and the pre-prod proof.

**WHAT IS SHIPMENT VISIBILITY? — THE ONE-MINUTE EXPLAINER (FROM THE PROGRAM DOCUMENTS; RECORD-FROM-DOCS)**

**The minute.** Shipment Visibility — "SV" — is Flexport's client-facing visibility product in the Client App: one pane where a client watches container milestones across their whole book, Flexport-carried and non-Flexport-carried shipments alike, so track-and-trace stops being emails to an ops desk and becomes a product the client logs into. It also created a "visibility-only" client segment — companies that bought visibility first were a deliberate upsell path into Flexport freight. The 2021 plan priced the problem it solved: roughly $3.92M a year of manual track-and-trace salary across the ops org against about $263K of vendor spend, with only 18 of 99 milestones automated at the time. RECORD-FROM-DOCS

**The machinery behind the story.** Product standards: coverage 90%+, accuracy 95%, timeliness 90%, a fail-safe switch, dashboards and alarms — under the operating rule "edits are the enemy": every human correction is an automation defect, not a fix. Rollout: a free pilot in graduated waves — wave 1 two named clients (Ta Chen International, FAM Brands) around Q4 2021, wave 2 four of a seven-client pool around Q1 2022, wave 3 twenty-plus around Q2 2022, full roll-out H2 2022 — each client entering on a ~50-container calibration and graduating on six published criteria. Testing: four gates before launch — a 100-container live test, biweekly bug bashes on the client job stories, shadow accounts fed real Crux data, and UAT gated on data quality (80% of shipments minor-to-no issues). And Crux — the acquired visibility product — was transitioned onto SV by end 2022: 79 customers, $185,892 ARR, 294,581 containers. RECORD-FROM-DOCS

Doc-vs-canon footnotes: the pilot schedule phrases the SLA criterion as "90% of verifications within 24–48h"; canon says "reviewed within 24h" — canon wins, say 24h. The 95% accuracy standard is the 2021 plan's bar for milestone data, not this story's ETA-accuracy metric (60% → 90%, canon) — never mix them. The $3.92M / $263K figures are the plan's ops-org business case, a different basis from canon's per-container CTS — never merged. Wave timing and the six criteria match the master packet.

### 60-SECOND BUSINESS-FIRST PITCH — STOP ON THE PRIMARY METRIC

**Situation.** Flexport's shipment-visibility product had public commitments — 68 monthly active customers, 170 MAU, 32K shipments — and was serving seven pilots at a cost-to-serve of $0.66 a container, most of it human touches. Sales wanted the entire pool boarded at once; leadership said the program was not demonstrating value and it was about to drop from #3 to #14 of 16 initiatives. The process problem was that boarding ran faster than the automation could carry, and nobody could see it. **Task.** As program owner across nine teams, I owned the boarding path, the measurement system and the pre-production proof. **Action.** I rebuilt boarding as a gated process: six graduation criteria published to Sales before wave one, a launch gate of 100 real containers across seven checks, and a weekly input/output review — seven frozen metric definitions, each signed by the owner who would be red against it, every input tagged to the output it moves — copied from a review leadership already trusted and run by hand every Monday for over a year. Waves boarded only through the criteria; wave three was held in public on an ETA-accuracy miss and became the cleanest wave. When a flattering 99.4% week turned out to rest on three touches, I printed a data miss instead. **Result.** Primary metric: cost-to-serve per container $0.66 → $0.03, −95.5%, across customers 7 → 68 — 9.7× — with zero rollbacks and no added ops headcount; automation 40% → 90%, accuracy 60% → 90%, both public targets two months early. The ~$1.5M pre-production measurement saving is a separate record figure and I keep it separate.

## SITUATION / TASK — THE BUSINESS PROCESS PROBLEM, SCOPE, SCALE, BASELINES

**S —** Senior leadership told me the visibility program was not demonstrating value at the anticipated rate — it would drop from #3 to #14 of 16 initiatives. I had been reporting activity, not evidence. The critique was about evidence, not effort, and I accepted that first.

The business process underneath: customer boarding (Deliver) ran as a sales push — board the whole pool now — against an operation where most containers were still touched by hand. Cost-to-serve was $0.66 a container (≈3.7 human interactions at the record cross-walk of $0.18 per interaction); automation covered 40% of containers; ETA accuracy was 60%. The public commitments were 68 monthly active customers, 170 MAU and 32K shipments, off seven pilots.

The cost of leaving it: a bad first impression at scale with rollbacks, ops headcount growing with the customer count, and the program losing its funding slot because value was invisible.

**T —** Program owner across nine teams / ~100 people: own the boarding path, the measurement system and the pre-production proof — and make value legible to the leadership that decides rollout pace and funding, with zero rollbacks and no added ops headcount.

**Baselines on the table:** customers 7 (target 68) · cost-to-serve $0.66/container · automation 40% · ETA accuracy 60% · public targets 68 customers / 170 MAU / 32K shipments · no instrument leadership trusted.

### KEY CHALLENGES — WHAT THE PROGRAM RECORD ADDS (FROM THE PROGRAM DOCUMENTS; RECORD-FROM-DOCS)

- **Rollout sequencing under a free pilot.** Pilot clients pay nothing, so every wave spent engineering, data and ops capacity before revenue; the pool had to be sequenced 2 → 4-of-7 → 20+ against a ~50-container calibration per client and six graduation criteria, with GO/NO-GO decisions taken monthly. That is the doc-level version of "board no faster than the automation can carry." RECORD-FROM-DOCS
- **Testing gates before any launch.** Four gates stood between build and client: a 100-container live test watched end-to-end, biweekly bug bashes on the client job stories, shadow accounts (real Crux data in dedicated demo accounts), and UAT that did not start until 80% of shipments showed minor-to-no issues. The gates are why "zero rollbacks" was achievable. RECORD-FROM-DOCS
- **An acquired product's customers in the middle of the rollout.** Crux's 79 customers — $185,892 ARR, 294,581 containers — had to be transitioned onto SV by end 2022, including cancel-vs-transition decisions by segment and

per-customer calls for the largest accounts, while the pilot waves were still boarding. Migration risk ran in parallel with launch risk. RECORD-FROM-DOCS

- **Data quality as a product standard, not an ops task.** The 2021 plan set coverage 90%+ / accuracy 95% / timeliness 90% with a fail-safe switch, and named the enemy: human edits. "Edits are the enemy" is the sentence behind the story's automation 40% → 90%. RECORD-FROM-DOCS

## ACTION — THE PROCESS / OPERATING-MODEL REDESIGN, IN GATED PHASES

- **Discovery** — wrote the honest diagnosis (activity, not evidence); inventoried what leadership actually decides on (rollout pace, funding); chose to copy the LSA review's format rather than invent one. *Exit bar:* leadership agrees the questions the review must answer weekly; ~50-container calibration entry bar per client.
- **Plan & Design** — froze seven metric definitions (coverage auto, coverage with human, tickets, tickets per active container, 4PL SLA %, 3PL SLA %, delta); input→output tagging; the ±5% week-over-week red/green rule; the 15-day active rule as the honest denominator; six graduation criteria published to Sales before wave 1 — DQ ≥ 90%, ETA ≥ 80%, SLA 90% reviewed within 24h, ≥100 containers + 5 features, stability, CSAT ≥ consignee average; launch gate 100 real containers × 7 checks; pre-prod sample size 300 containers for ±1.6% at 95% confidence. *Exit bar:* each input owner signs the definition they will be red against; first Monday deck dry-run.
- **Build & Test** — ran the review by hand every Monday (1.5–2 hrs, later ~4); pre-decomposed every red with an owner and a dated fix; enforced the data-honesty rule — a truncated week is a data miss, not a result; pre-prod ingest with per-hop monitors and kill switches; load proof 500 containers / 20 min / 99.6%. *Exit bar:* two consecutive Mondays where every red arrives with an owner and the number verifies the following week.
- **Cutover & Go-Live (waves)** — W1 (2 clients) → W2 (4 of a 7-pool) → W3 (20+) held in public on an ETA-accuracy miss traced to one bad source → full rollout. The Monday review became the forum where the rollout decision was taken. *Exit bar:* the VP unlocks the 68-client rollout two months early off the review's own numbers.
- **Hypercare / Close** — discarded a flattering 99.4% three-touch week and printed a data miss; handed the manual pull to a pipeline with the definitions in code (S16 — Josyula's packet; mention only as a pointer); left the critic steering by the numbers. *Exit bar:* automation 40 → 90%; CTS $0.66 → $0.03; same ops team at 68 customers.

**Influence and ownership.** Mine: the diagnosis, the frozen definitions, the review built by hand weekly, the graduation criteria, the hold decisions and the pre-prod measurement design. Nine teams built and operated; each input owner signed the definition they would be red against; the VP took the rollout decisions in the review. The Sales push to board everything now was answered with gates published before wave 1 — so when wave 3 was held, it was the rule firing, not a judgment call; the held customers became the cleanest wave.

> **Designed the process; others own it now.** I designed the review, the definitions and the graduation gates; the nine input owners run their lines and the pipeline (S16) carries the definitions in code — the review steers without me.
> **Trade-off put to leadership.** My recommendation to leadership outlined the competing metrics explicitly — boarding pace and customer count vs automation coverage and ETA accuracy — and proposed to board no faster than the automation could carry; wave 3 was the trade made in public.

## RESULTS — $ · # · % · TIME · CUSTOMER · QUALITY (CANON NUMBERS ONLY)

- **$** — cost-to-serve per container $0.66 → $0.03 (record; −95.5%); direct CTS dollars ≈$50K/yr *modeled* on an illustrative container count — the value is scale without headcount, not the direct dollars; the ~$1.5M pre-production measurement saving is a separate record figure, never merged with CTS.
- **#** — customers 7 → 68 (9.7×) with zero rollbacks and no added ops headcount; targets 68 customers / 170 MAU / 32K shipments (record).
- **%** — CTS −95.5%; automation 40% → 90%; ETA accuracy 60% → 90% (record).
- **Time** — both public targets hit two months early; the rollout unlocked two months early off the review (record); pre-prod measurement path cut rollout proof from 6 months to 1.5.
- **Customer** — six graduation criteria published to Sales before wave 1 — including CSAT ≥ consignee average; wave 3 held in public on an ETA-accuracy miss and became the cleanest wave (record).

- **Quality** — a flattering truncated 99.4% week discarded and printed as a data miss; seven metric definitions frozen and later moved into code (record).

**L6 LEVEL EVIDENCE** — what the interviewer pattern-matches against the Amazon L6 functional rubric; every line is this story's canon, no new numbers

| L6 DIMENSION | EVIDENCE IN THIS STORY |
|---|---|
| **Ambiguity** | Leadership said the program was not demonstrating value (#3 → #14 of 16) and no measurement system existed. I wrote the diagnosis (activity, not evidence), chose the nine teams' input owners as the stakeholders, and set the goals the review had to answer weekly. |
| **Scope & influence** | Nine teams built and operated; Sales held to criteria I published; the VP took the rollout decisions inside the review I ran — influence one to two levels up, from a program seat. |
| **Execution trade-off + mechanism others now own** | Boarding speed (Sales) vs automation maturity and ETA accuracy: wave 3 held in public on the miss. Mechanism others own now: the Monday input/output review and the six graduation criteria, with the seven definitions moved into code (S16). |
| **Impact (long-term)** | Customers 7 → 68 (9.7×) at CTS $0.66 → $0.03 with zero rollbacks and no added ops headcount; both public targets two months early; the review became the standing rollout forum. |
| **Process improvement (standard work institutionalized)** | Seven frozen metric definitions, input→output tagging, the ±5% week-over-week red/green rule, the 15-day active denominator, and the data-honesty rule (a truncated week is a data miss, not a result). |
| **Complexity & consensus** | A critical leadership and a Sales push to board everything now — answered by gates published before wave 1, each input owner signing the definition they would be red against; the VP unlocked the 68-client rollout off the review's own numbers. |
| **Root cause at scale** | The ETA-accuracy miss traced to one bad source and fixed there; the flattering 99.4% week traced to three touches and printed as a data miss — fixed at the definition and the source, not customer by customer. |

## PROBING — CHALLENGE Q&A TUNED TO THIS INTERVIEWER

> **HM — Why is this your Deliver Results story and not just a metrics story?**
>
> Because the results landed behind gates, not around them: both public targets two months early, 7 → 68 customers on the same ops team, zero rollbacks. The review and the graduation criteria are the mechanism; the −95.5% and the 9.7× are the outcome.

> **HM — The direct dollars on a 95% cost reduction are small. Why lead with it?**
>
> Because it is honest and because the point was never the direct dollars: at the 32K-shipment target the direct saving is ≈$50K a year, modeled on an illustrative container count. What it bought is boarding 9.7× the customers without 9.7× the ops headcount — roughly two and a half people never hired — and a rollout unlocked two months early. The ~$1.5M pre-prod figure is a separate record and I never add it to this.

> **HM — Sales wanted everyone boarded. How did you hold the line without becoming the blocker?**
>
> By publishing the six criteria to Sales before wave one, so the rule did the refusing. Wave 3 was held in public on an ETA-accuracy miss traced to one bad source; the customers we held became the cleanest wave, and Sales had seen the gate before it fired.

> **HM — You discarded a 99.4% week. Walk me through that decision.**
>
> Three touches is not a week — a 95% confidence interval on N = 3 is wider than the metric. Printing it would have taught the VP a number that could not survive the next Monday. I printed a data miss. A review that flatters once is not trusted twice.

> **HM — What would you do differently, and how does this translate to my team?**
>
> Start the review at kickoff, not after the critique. At Amazon this is a WBR story told to the company that invented the WBR — the respectful version is that I copied a trusted format, froze definitions with their owners, tagged inputs to outputs and let the review make the rollout decision. Graduation criteria are launch gates published before the wave and held in public.

> **HM — What was yours and what was the team's?**
>
> Mine: the diagnosis, the seven definitions, the Monday build by hand for over a year, the criteria, the holds, the pre-prod measurement design. The nine teams built and operated; each input owner signed their own definition; the VP decided in the room. The format came from the LSA review — I did not invent it.

> **HM — Why waves at all? What did the calibration actually buy?**
>
> The pilot was free, so every boarded client cost engineering, data and ops before revenue — boarding the whole pool would have spent the program's credibility on clients we could not yet serve. The ~50-container calibration per client gave the data team a verified baseline on the client's own freight before scale, and the six published criteria converted "ready" from a feeling into a rule: wave 1 two clients, wave 2 four of a seven-client pool, wave 3 twenty-plus RECORD-FROM-DOCS. The canon version: waves boarded only through the published gates, and the held wave became the cleanest.

> **HM — How did you test before the first paying customer touched it?**
>
> Four gates, all from the program record RECORD-FROM-DOCS: a 100-container live test watched end-to-end; biweekly bug bashes on the client job stories; shadow client accounts — real Crux data fed into dedicated demo accounts so the clients' own data exposed the defects; and UAT that did not start until 80% of shipments showed minor-to-no issues. UAT was gated on data quality, not on the calendar — the testing version of the graduation rule.

> **HM — You were rolling out while an acquired product's customers sat on the old platform. How did you handle that risk?**
>
> As a parallel migration with its own decision rules, not an afterthought RECORD-FROM-DOCS: 79 Crux customers, $185,892 ARR and 294,581 containers, segmented — transition where SV fit, cancel honestly where it did not (two competitor SCM-SaaS accounts were canceled on published dates), per-customer calls for the largest accounts. Internal pushback on migration cost and feature parity was priced into the decision document, not overridden. The pilot gates kept quality honest while the migration ran beside it.

## THE KEY MATH — COST-TO-SERVE, THE SCALE RESULT, AND WHY THE DOLLARS STAY SEPARATE

| STEP | QUANTITY | FORMULA → SUBSTITUTION | BASIS | VALUE |
|---|---|---|---|---|
| 1 | Cost-to-serve before | `$0.66 / $0.18 ≈ 3.7 touches per container` | record; $0.18 per interaction is the record cross-walk | $0.66 — ≈3.7 touches |
| 2 | Cost-to-serve after | `($0.03 − $0.66) / $0.66 = −95.5%; $0.03 / $0.18 = 0.17 touches` | record | $0.03 — ≈0.17 touches |
| 3 | Scale | `68 / 7 = 9.7×; zero rollbacks; no added ops headcount` | record | 9.7× (never "10×") |
| 4 | Direct CTS dollars | `32,000 shipments × 1.5–4 containers = 48–128K; × $0.63 = $30–81K` | modeled (containers per shipment illustrative) | ≈$50K/yr modeled |
| 5 | Pre-prod measurement path | `n = 300 → ±1.6% at 95% CI; rollout proof 1.5 vs 6 months` | record — separate canonical | ~$1.5M, never merged with CTS |
| 6 | Week-over-week rule | `WoW = (B − A)/A = (21.9 − 40.0)/40.0 = −45% → red` | ±5% red/green (record design; values illustrative) | root cause: a carrier format change — measurement, not operators |

Automation 40 → 90% and the WBR hours (1.5–2 → ~4 by hand) are this packet's canon; the loop packet carries different values (§1.3 conflict) — do not mix them in the room.

**Where this lands at Amazon.** This is the JD's "define, create, implement and execute processes" line in its purest form: a boarding process with published graduation criteria, a weekly input/output review with frozen definitions, and the discipline to hold a wave in public. Reliability (zero rollbacks), responsiveness (two months early), cost efficiency (−95.5% CTS), asset utilization (the same ops team at 9.7× the customers).

## TWIST MAP — LIKELY LP QUESTIONS FROM THIS INTERVIEWER → THE ANGLE TO TAKE WITH S11

| LIKELY QUESTION (THIS INTERVIEWER) | LP | ANGLE — WHAT TO EMPHASIZE | SKIP / NEVER SAY |
|---|---|---|---|
| **Tell me about a time you delivered results under pressure / against a target.** | Deliver Results | Open on the critique (dropping #3 → #14) and the public targets; close on both targets two months early, 7 → 68, zero rollbacks. The gates are the how. | Skip the four-system data-pull detail; S16 belongs to the bar raiser. |
| **Tell me about a time you had to make value legible to senior leadership.** | Ownership / Earn Trust | Accepting that the critique was about evidence, not effort; copying a trusted format; signing each definition with its owner; the review becoming the decision forum. | Never say "I invented the WBR." |
| **Tell me about a time you said no to a stakeholder (Sales).** | Have Backbone / Highest Standards | Six criteria published before wave 1; wave 3 held in public; held customers became the cleanest wave. | Do not call Sales wrong — they were right to want speed; the gate did the refusing. |
| **Tell me about a time you chose the harder, honest number.** | Earn Trust | The 99.4% three-touch week printed as a data miss; the 15-day active rule as the honest denominator. | Skip the Wilson-interval math unless asked; say "three touches is not a week." |
| **How did you decide the customer was ready to launch? (Customer Obsession)** | Customer Obsession | CSAT ≥ consignee average inside the gate; ~50-container calibration entry bar; 100 real containers × 7 checks before the first paying customer. | Do not claim customer-satisfaction dollars; none exist in canon. |
| **How would you run a WBR for my team?** | Functional — operating cadence | Copy a trusted format, freeze 5–7 definitions with owners and honest denominators, tag inputs to outputs, pre-decompose reds, let the review take the decision. | Never "$1.00 → $0.03"; never "70% baseline"; never "10×". |
| **How would your wave machine apply to my world — placement / inbound? (transferability probe — public-profile inference)** | Functional — transfer | The wave machine is graduated onboarding behind published criteria — the same shape as onboarding inventory into an IXD / crossdock program: a selection bar published before the first node, a calibration volume per node (~50 containers per client ≈ a trial volume per node), and a public hold when a node misses. Scale follows the gate, never the ask. | Do not lecture on her network; bridge in one sentence and hand it back with a question. |

| LIKELY QUESTION (THIS INTERVIEWER) | LP | ANGLE — WHAT TO EMPHASIZE | SKIP / NEVER SAY |
|---|---|---|---|
| **Your CTS is per container — mine is per node. Does the logic survive? (transferability probe)** | Functional — transfer | Identical logic, different denominator: cost-to-serve per node = touches per unit through the node × cost per touch; the fix is still automation coverage and source quality, and the honest denominator is still defined before the metric. $0.66 → $0.03 is the container-grain instance of it. | Never quote her profile's $50MM / $80MM as if it were mine — those are her numbers, quoted for framing only. |

## S12 — Flexport — FF App export launch: re-sequencing a committed 18-month roadmap on cost-to-serve evidence (value in month 6, 66% less effort, ~$5M cost avoidance)

**LPs this story can serve here:** Think Big (primary — the flip) · Frugality (primary — ~$5M avoided by sequencing) · Have Backbone, Disagree & Commit (changing the manager's plan) · Earn Trust (privately first) · Bias for Action (the reversal trigger written before the swing)
**Process family:** Deliver — export documentation / operator and partner workflow; launch sequencing; unit-economics-driven roadmap governance

**Honesty label.** The roadmap was my manager's; I owned the sequencing recommendation. Engineering produced the re-estimates; the ~$5M is the record's cost-avoidance figure, said once; the run-rate decomposition and the engineer-count split are my model. Loop-packet conflict (§1.3): that document tells this as a Sephora roadmap — the record is Flexport FF App; say Flexport.

### WHAT IS THE FF EXPORT APP? — THE ONE-MINUTE EXPLAINER (FROM THE PROGRAM DOCUMENTS; RECORD-FROM-DOCS)

**The minute.** The Freight Forwarding App — "FFA" — is Flexport's operator-facing execution platform: where import and export shipments actually get run. Its core tenet is work-item-driven execution — every unit of work is a work item assigned to exactly one operator ("one operator, one work item"), so nothing ships on tribal knowledge and every action is attributable. For export, the heart of it is schedule management: a "Generate Schedule" work item builds the shipment's milestone schedule automatically, and a "Replan Schedule" work item fires on defined exceptions — an LCL CFS-cutoff miss, a vessel late against truck pickup, a vessel late past the quoted delivery date, a change of vessel — each with a defined rework path instead of ad-hoc heroics. It shipped FCL first (2022), with Air and LCL behind it (2023). RECORD-FROM-DOCS

**The machinery behind the story.** The schedule work sat inside a five-work-stream FFA program (Schedule Management; Cascading Schedules; Auto-Build Initial Schedule; End-to-End Operator Experience; Operator MiMeC), with schedule automation completing end Q3/early Q4 2022 and export-partner UAT aimed at a March 2023 launch. Because the app serves external partner organizations, it also carries a compliance bar: an immutable audit log of who changed what and when — nine event categories at a 100%-captured standard across ~100 partner orgs and thousands of changes a day, phased Alpha → Beta → MVP with CISO sign-off. RECORD-FROM-DOCS

Doc-vs-canon footnotes: the module documents carry forward-plan dates (export-partner UAT March 2023; schedule automation completing end Q3/early Q4 2022) — the build-out clock after MVP. Canon's "value in month 6" is the sequencing decision's clock: different milestones, no conflict — never merge them. The audit-logs "~100 partner orgs" is platform-wide; canon's 53 is the import-partner migration set — never merge those either.

### 60-SECOND BUSINESS-FIRST PITCH — STOP ON THE PRIMARY METRIC

**Situation.** The Freight Forwarding App export roadmap was operator-first — 18 months to first value, already socialized, with my manager's name on it. The process baselines were 339.5 minutes of handling per shipment and 2,200 operator interactions per departed shipment. The question nobody had asked was time-to-value and unit cost — the plan had been built around persona preference. **Task.** With no authority over the roadmap, re-sequence it on cost-to-serve evidence without abandoning the operators. **Action.** I wrote the baseline memo, had the business cut a bare-minimum scope per persona, and got engineering to re-estimate with confidence levels: operator-first reached −25% cost-to-serve in 9 months with operators still spending 20% of their time in the old app; partner-first reached −50% in 6 months and built the chassis operators would later ride. I added a rebuild-vs-reuse rule per module — over eight dev-weeks, reuse — a scope tripwire in the burn-up, and a written reversal trigger, then read it to my manager privately first; engineering had two days to break it, and the Bellevue onsite ratified the flip with his name on it. Operators' two further quarters on old screens were priced and published, not hidden. **Result.** Primary metric: engineering effort to first value — the 18-month plan → 66% less effort, value in month 6 instead of 18. ~$5M of cost avoidance — the record figure, said once; cost-to-serve −50% versus −25% at the work-item grain; 48 of 53 import partners migrated within a month.

## SITUATION / TASK — THE BUSINESS PROCESS PROBLEM, SCOPE, SCALE, BASELINES

**S —** The inherited launch plan for the freight-forwarding app was operator-persona first, 18 months, already socialized, and my manager owned it. Process baselines: 339.5 minutes of handling per shipment and 2,200 operator interactions per departed shipment — the two numbers everything prices from.

The business problem was time-to-value and unit cost, not persona preference. Left as inherited, operators would reach first value in month 18 while still spending 20% of their time in the old app — half the unit saving (−25% vs −50% at the work-item grain), reached a year later. The risks of re-opening it: a committed, socialized plan; optimistic estimates; and operators feeling abandoned while partners went first.

**T —** Own the sequencing recommendation — and change a plan with my manager's name on it, without authority over the roadmap, in a way he could own.

**Baselines on the table:** 339.5 min/shipment · 2,200 operator interactions per departed shipment · first value in month 18 on the inherited path · cost-to-serve −25% (operator-first) vs −50% (partner-first) at the work-item grain.

### KEY CHALLENGES — WHAT THE PROGRAM RECORD ADDS (FROM THE PROGRAM DOCUMENTS; RECORD-FROM-DOCS)

- **Asia latency surfaced right after MVP.** The monthly business review read P95 page load at 21.19s in Asia against 3.3s in the US; static assets ran ~5s in China against ~500ms in the US; the app's domain had no geo-based DNS or reverse proxy, so Asia traffic rode the public internet. The fixes were sequenced — WAF enablement, a double CDN removed, a regional CDN proposal with smoke tests per change, then China and Asia tuning — with 48 of 53 import partners (90.5%) already migrated and expecting the app to hold. RECORD-FROM-DOCS
- **Compliance was a launch feature, not an afterthought.** The Audit Logs PRD demanded an immutable who/what/when trail across ~100 partner orgs and thousands of changes a day — nine categories at a 100%-captured bar, phased Alpha → Beta → MVP — and the CISO's definition of "important events" was deliberately vague, so PM buy-in and CISO sign-off were written into the plan as the mitigation. RECORD-FROM-DOCS
- **Exceptions had to become work items.** Replanning ran on a defined taxonomy — LCL CFS-cutoff miss, vessel late vs truck pickup, vessel late past quoted delivery date, change of vessel — each with a rework path, under the "one operator, one work item" tenet. The exception list, not the happy path, is where operator trust is won. RECORD-FROM-DOCS
- **A hard external deadline.** Export-partner launch UAT was set for March 2023, with schedule automation completing end Q3/early Q4 2022 and FCL ahead of Air/LCL — a date-driven constraint on top of the unit-economics sequencing. RECORD-FROM-DOCS

## ACTION — THE PROCESS / OPERATING-MODEL REDESIGN, IN GATED PHASES

- **Discovery** — baseline memo (339.5 min; 2,200 interactions); bare-minimum scope per persona cut with the business; the question restated as time-to-value and unit cost. *Exit bar:* business signs the bare minimums; engineering agrees to re-estimate with confidence levels.
- **Plan & Design** — re-estimates 9 months (operator) vs 6 months (partner) with confidence buffers (×1.0 / ×1.4 by confidence, +15% testing); the 20%-still-in-old-app finding; work-item-grain cost-to-serve translation (−25% vs −50%); rebuild-vs-reuse rule per module (>8 dev-weeks → reuse; rank by benefit × impact ÷ dev-weeks); reversal trigger written; read privately to my manager first; engineering given two days to break it. *Exit bar:* my manager puts his name on the flip at the Bellevue onsite; sequencing signed.
- **Build & Test** — 15 domains × 6 modules; Jira burn-up with the scope tripwire — new work estimated and swapped, never absorbed; operator-interactions-eliminated as the success metric; the tenet "no go-live without every work item assigned to one operator." *Exit bar:* burn-up 55.8% → 83.2% across two quarters; every work item owned.
- **Cutover & Go-Live (waves)** — MVP with 11 partners in 10 countries; partner migration plan published; operators' two further quarters on old screens priced and published. *Exit bar:* 48 of 53 import partners migrated within a month.
- **Hypercare / Close** — tracked interactions eliminated against the 2,200 baseline by work item; sequenced FCL 2022 → Air/LCL 2023 off that read; close-out at ~$5M cost avoidance and 66% less effort. *Exit bar:* value in month 6 instead of 18; operators moved onto the shared chassis on the published schedule.

**Influence and ownership.** Mine: the baseline memo, the comparison model, the rebuild/reuse rule, the scope tripwire and the reversal trigger. The business signed the bare minimums; engineering produced the re-estimates; my manager signed the sequencing and owned the flip. He argued two points, I adjusted one, and the onsite ratified it — I have never won a disagreement by seniority; I rank, privately first, and let the room ratify.

> **Designed the process; others own it now.** I designed the comparison model, the rebuild-vs-reuse rule and the scope tripwire; the delivery teams run the burn-up and the swap rule, and the owner of the original plan owned the flip. **Trade-off put to leadership.** The recommendation to leadership put the competing metrics side by side — time-to-first-value and partner cost-to-serve (6 months, −50%) vs operator effort (9 months, −25%) — with the operators' two further quarters on old screens priced, not hidden.

## RESULTS — $ · # · % · TIME · CUSTOMER · QUALITY (CANON NUMBERS ONLY)

- **$** — ~$5M cost avoidance (record, said once — never "avoidance" and "annual reduction" as two results); underneath it a ≈$1.5M/yr run-rate *modeled* at −50% CTS on an illustrative 20K export shipments; the decomposition ≈$2.9M engineering avoided + ≈$1.5M pulled forward ≈ $4.4M is my model, within ~12% of the record.
- **#** — 48 of 53 import partners migrated within a month; MVP with 11 partners in 10 countries; baselines 339.5 min/shipment and 2,200 interactions (record).
- **%** — engineering effort to first value −66%; CTS −50% partner-first vs −25% operator-first at the work-item grain; burn-up 55.8% → 83.2% across two quarters (record).
- **Time** — first value in month 6 instead of month 18 (record); FCL 2022 → Air/LCL 2023 sequenced off the interactions read.
- **Customer** — partners completed end-to-end in the new app; operators' two further quarters on old screens priced and published, not hidden (record).
- **Quality** — new scope estimated and swapped, never absorbed; no go-live without every work item assigned to one operator (record).

**L6 LEVEL EVIDENCE** — what the interviewer pattern-matches against the Amazon L6 functional rubric; every line is this story's canon, no new numbers

| L6 DIMENSION | EVIDENCE IN THIS STORY |
|---|---|
| **Ambiguity** | An 18-month committed roadmap with no cost-to-serve read behind it. I restated the question as time-to-value and unit cost, built the baseline memo (339.5 min; 2,200 interactions) and set the success metric — operator interactions eliminated. |
| **Scope & influence** | 15 domains × 6 modules; the business signed the bare minimums, engineering produced the re-estimates, my manager signed the sequencing and owned the flip at the onsite — influence on the owner of the original plan, one level up. |
| **Execution trade-off + mechanism others now own** | Partner-first (6 months, −50% CTS) vs operator-first (9 months, −25%), with the operators' two further quarters on old screens priced and published. Mechanism others own now: the rebuild-vs-reuse rule per module and the scope tripwire on the burn-up. |
| **Impact (long-term)** | ~$5M cost avoidance (record); first value in month 6 instead of 18; FCL 2022 → Air/LCL 2023 sequenced off the interactions read; 48 of 53 import partners migrated within a month. |
| **Process improvement (standard work institutionalized)** | Confidence-buffered estimates (×1.0 / ×1.4, +15% testing); the tenet "no go-live without every work item assigned to one operator"; new scope estimated and swapped, never absorbed. |
| **Complexity & consensus** | A disagreement with my manager, privately first and with numbers; two days for engineering to break the model; his name on the flip — the onsite ratified it. |
| **Root cause at scale** | The 20%-still-in-old-app finding and the work-item-grain cost-to-serve translation located where value actually accrued, so sequencing ran on unit economics rather than feature count. |

## PROBING — CHALLENGE Q&A TUNED TO THIS INTERVIEWER

**HM — You changed your manager's plan. How, and would you do it to me?**

Privately first, with the numbers — 9 versus 6 months, 20% still in the old app, −25% versus −50%. He argued two points, I adjusted one, and the onsite ratified it with his name on it. Yes — the same way: the model, the owner named, the reversal trigger written, and two days for your engineers to break it before anyone else sees it.

**HM — Is the $5M a saving or an avoidance?**

Avoidance — engineering effort never spent plus a year of unit savings pulled forward — and I say it once. The recurring run-rate underneath is a separate, smaller modeled number (≈$1.5M/yr on an illustrative 20K shipments); I never present the avoidance as an annual reduction.

**HM — What stopped the operators from being abandoned?**

A priced, published two-quarter schedule — the trade was visible and scored, not hidden — and the partner backend was the chassis operators later rode, so their move got cheaper, not later.

**HM — Your engineer count and rate are yours, not the program's.**

Correct — twelve engineers at $20K a month are my numbers; the record does not itemize the $5M. At 10–15 engineers and $15–25K the effort piece runs $1.8–4.5M and the record figure sits inside that band. I would firm it with the two scoping envelopes (66.2 and 99.8 engineer-months) and Finance's loaded rate.

**HM — What made the big swing safe?**

Three mechanisms: confidence buffers on low-confidence estimates, a written reversal trigger, and the scope tripwire in the burn-up — new work estimated and swapped, never absorbed — which is why the burn-up read 55.8 → 83.2% honestly.

**HM — Lesson, and how it shows up in your first quarter here?**

Bring the model, name the owner, write the reversal trigger. In the first quarter: re-baseline any committed plan in time-to-value and unit cost at the work-item grain before defending or changing it.

> **HM — What broke after MVP that the plan had not seen?**
>
> Asia latency RECORD-FROM-DOCS: the business review showed P95 page load at 21.19 seconds in Asia against 3.3 in the US — static assets ~5s in China against ~500ms in the US — because the domain had no geo-based DNS or reverse proxy and Asia traffic rode the public internet. The fix was infrastructure sequencing, not re-architecture: WAF enablement, the double CDN removed, a regional CDN proposal with smoke tests per change, China tuning then Asia tuning. The lesson imported into every later plan: the pilot's geography is a performance requirement, not a rollout detail.

> **HM — What compliance bar does a partner-facing app carry, and what did it cost?**
>
> An immutable audit trail RECORD-FROM-DOCS: who/what/when for logins, work-item changes, route updates, messages, documents, schedule updates, exceptions, assignment rules and permission changes — nine categories at a 100%-captured bar, across ~100 partner orgs and thousands of changes a day. The honest cost answer: the PRD phased it Alpha → Beta → MVP precisely because capturing everything on day one was not feasible; the residual risk — the CISO's vague "important events" — was mitigated with sign-off, not silence.

> **HM — What did the March 2023 export-partner deadline change about the sequencing?**
>
> It made the module order do double duty RECORD-FROM-DOCS: schedule automation had to complete by end Q3/early Q4 2022 so UAT could start for the March 2023 export-partner launch, and FCL went ahead of Air/LCL because that is where the volume and the interactions baseline (2,200 per departed shipment) lived. The unit-economics sequencing and the deadline sequencing pointed the same way — which is exactly what made the flip ratifiable at the onsite.

### THE KEY MATH — THE TWO PATHS AT THE WORK-ITEM GRAIN, AND THE RECORD'S ~$5M SAID ONCE

| STEP | QUANTITY | FORMULA → SUBSTITUTION | BASIS | VALUE |
|---|---|---|---|---|
| 1 | CTS per shipment | `339.5 min × $0.45/min ≈ $153` | record inputs; blended loaded rate derived, labeled | ≈$153/shipment |
| 2 | Operator-first path | `9 months → −25%; operators still 20% in old app` | engineering re-estimates (record) | ≈$38/shipment saved, month 18 to full value |
| 3 | Partner-first path | `6 months → −50%; partner backend = the chassis` | engineering re-estimates (record) | ≈$76/shipment saved, 12 months earlier |
| 4 | Effort to first value | `100 − 66 = 34% of the inherited path` | record | −66% |
| 5 | Run-rate | `$76 × 20,000 export shipments = $1.52M` | volume illustrative (modeled) | ≈$1.5M/yr modeled |
| 6 | Decomposition of the avoidance | `12 eng × 18 mo × 66% × $20K ≈ $2.9M + 12 mo × $1.5M = $1.5M → $4.4M` | counts and rate mine; within ~12% of the record | ~$5M record headline, said once |
| 7 | Confidence rule | `(10 × 1.0 + 10 × 1.4) × 1.15 = 27.6 wks vs 20 naive` | two illustrative 10-week modules | buffers inflate only low-confidence items |

Burn-up trajectory 55.8 → 63.3 → 66.0 → 69.3 → 76.0 → 83.2% (record). Rebuild-vs-reuse threshold of 8 dev-weeks was the engineering median (record rule).

**Where this lands at Amazon.** Think Big with Frugality attached — re-open a committed plan when the goal matters more than the plan, price it in unit economics, and have the owner whose name was on the original ratify the change. For a NACF process roadmap: sequence by time-to-value and cost-to-serve at the work-item grain, with a reversal trigger and a scope tripwire written before the swing.

## TWIST MAP — LIKELY LP QUESTIONS FROM THIS INTERVIEWER → THE ANGLE TO TAKE WITH S12

| LIKELY QUESTION (THIS INTERVIEWER) | LP | ANGLE — WHAT TO EMPHASIZE | SKIP / NEVER SAY |
|---|---|---|---|
| **Tell me about a time you thought big / changed the direction of a program.** | Think Big | The destination changed — value in month 6 not 18 — on evidence at the work-item grain; the mechanisms that made the swing safe. | Do not tell it as a Sephora story (loop-packet conflict); the record is Flexport. |
| **Tell me about a time you saved money / did more with less.** | Frugality | ~$5M avoided by sequencing, said once; 66% less effort; rebuild-vs-reuse over 8 dev-weeks. | Never "$5M annual cost reduction" in the same breath as the avoidance. |
| **Tell me about a time you disagreed with your manager.** | Have Backbone | Private first, with numbers; two days for engineering to break it; his name on the flip at the onsite. | Skip S14 (the quiet-fix refusal) — held for the bar raiser. |
| **How do you prioritize when everything is committed?** | Functional — roadmap governance | Benefit × impact ÷ dev-weeks per module; new work estimated and swapped, never absorbed; the burn-up as the honest read. | Do not claim the roadmap was mine — it was my manager's. |
| **Tell me about a trade-off you made that hurt someone.** | Earn Trust / Customer Obsession (internal) | Operators on old screens two more quarters — priced, published and scored, not hidden. | Do not present the operator delay as costless. |
| **What's a decision you'd reverse if the data changed?** | Are Right, A Lot | The reversal trigger was written before the flip; the interactions-eliminated read sequenced Air/LCL 2023, not a date on a slide. | No YoY framing; the avoidance is one-time. |
| **How does re-sequencing a roadmap relate to launch readiness? (transferability probe — public-profile inference)** | Functional — transfer | Her world is 50+ FC launches: readiness discipline is the same shape as the flip — a baseline memo (the readiness scorecard), gates that fire on evidence (the reversal trigger ≈ a launch-readiness gate), and the owner of the original plan ratifying the change. Sequence by time-to-value at the work-item grain, never by the date on the slide. | Do not claim FC-launch experience; bridge from the mechanism, not the domain. |
| **How would you apply cost-to-serve thinking to a placement network? (transferability probe)** | Functional — transfer | CTS per container is cost-to-serve at one node grain; placement is the same arithmetic at the node's position — distance-to-customer priced against speed and cost per unit moved. The work-item grain (339.5 minutes; 2,200 interactions) is how I keep it honest. | Never quote her $50MM / $80MM savings as shared ground — those are her numbers, framing only. |

## Question bank — likely questions for this interviewer, with a one-line answer plan

Behavioral and functional questions a hiring manager for NACF Supply Chain is likely to ask, in rough order of likelihood. Each plan points at a packet story or a named backup in the master packet. Stories allocated to other interviewers are referenced as pointers only.

| # | LIKELY QUESTION | ANSWER PLAN (ONE LINE) | STORY |
|---|---|---|---|
| 1 | **Tell me about yourself.** | The 75-second newest-to-oldest script from the master packet: Sephora (one signed "available"; dual-DC WMS migration) → Flexport (the review; $0.66 → $0.03) → Elementum (measured lanes, $1.5M out) → Manhattan (99% fill). End on "design the supply chain, keep it honest in a weekly review — at network scale." | Master TMAY |
| 2 | **Why Amazon, why now?** | Three reasons — the forecast (Elementum lanes), the weekly review (Flexport), one version of the truth (Sephora) — and "what I went to Sephora to build is live." | Master Why Amazon |
| 3 | **Tell me about your most significant delivery.** | S11 pitch; end on CTS $0.66 → $0.03 across 7 → 68; stop. | S11 |
| 4 | **Tell me about a time you changed a plan that was already committed.** | S12 pitch; private-first; 66% less effort, value month 6; ~$5M once. | S12 |
| 5 | **Tell me about a time leadership doubted your program.** | S11 Discovery: accepted the critique as being about evidence; built the review; left the critic steering by the numbers. | S11 |
| 6 | **How do you make trade-offs between speed and quality?** | S11: board no faster than automation matures; wave 3 held publicly. S12: confidence buffers over a fast optimistic plan. | S11 / S12 |
| 7 | **Describe how you would run your first 90 days in NACF Supply Chain.** | Book of record and primary metric first; stocks-and-flows map of the process; the invariant broken today; 3 key results with denominators; a weekly input/output review by week 6 (master §3.0, seven steps). | Master §3.0 |
| 8 | **How would you decompose a complex network problem (e.g., inbound placement cost rising)?** | Open as the master's §3.99 script: book of record + metric → stocks and flows → broken invariant → minimal design → sequence by Reach × Impact × Confidence ÷ Effort → one metric, one labeled dollar. | Master §3.99 |
| 9 | **Tell me about a time you failed.** | Full failure story here: the master's UAT-crash bench story (§4.3 — numeric entrance criteria ever since). S18's RF miss (Goforth) and S16's security-review miss (Josyula) are held for other interviewers — pointers only; S4 (the in-transit rule I got wrong) is Lan's backup pointer, so keep it clean. | Bench §4.3 |
| 10 | **How do you measure success for a process you designed?** | A KR without a denominator is not a KR: seven frozen metrics with owners, honest denominators (15-day active rule), inputs tagged to outputs. | S11 |
| 11 | **What's a number you refused to report?** | The 99.4% three-touch week — printed as a data miss. | S11 |
| 12 | **What would your current manager say you need to work on?** | Start the review at kickoff rather than after the critique (S11); back-test a rule before publishing it (master S4 lesson, pointer only). | S11 / S4 pointer |

| # | LIKELY QUESTION | ANSWER PLAN (ONE LINE) | STORY |
|---|---|---|---|
| 13 | **How would you think about inventory-placement strategy for a network like mine? (placement was her 2017–22 world — public-profile inference)** | S11's logic translated: publish the selection bar before the first node (graduation criteria ≈ placement criteria), calibrate per node (~50 containers ≈ a trial volume), hold a node in public when it misses. Then master §3.99: book of record → stocks and flows → broken invariant → minimal design. | S11 / §3.99 |
| 14 | **How do you decide connectivity in a crossdock network — which nodes, which flows?** | S12's sequencing logic: rank flows by benefit × impact ÷ effort at the work-item grain (interactions eliminated per flow), FCL-first as "highest-volume flow first," a reversal trigger written per node before the swing. | S12 |
| 15 | **You ran launch readiness for 50+ buildings — what does readiness mean for a process launch? (her 2014–17 world — public-profile inference)** | FC-readiness discipline translated: a written entry bar per wave (the six criteria), a launch gate on real volume (100 containers × 7 checks), a published rollback rule (the reversal trigger), and the hold taken in public when the bar misses. | S11 / S12 |
| 16 | **Tell me about building a team or capability from scratch.** | S11's mechanism answer: nine input owners recruited to sign the definitions they would be red against; the review run by hand until the pipeline (S16, pointer) carried it; wave teams grown into owners — capability built while delivering, not before. | S11 |

## Cheat-numbers box — canon values with basis labels

Both stories' canon numbers with their basis labels, from the master packet's §4.1 ledger. Say record numbers flat; say modeled numbers with their inputs.

| METRIC | VALUE TO SAY | BASIS | NEVER |
|---|---|---|---|
| **S11 cost-to-serve** | $0.66 → $0.03 per container (−95.5%) | record | "$1.00 → $0.03" |
| **S11 customers** | 7 → 68 = 9.7×; zero rollbacks; no added ops headcount | record | "10×" or "50×" |
| **S11 automation / accuracy** | 40% → 90% / 60% → 90% | record (this packet's canon; loop packet differs — do not mix) | "70% baseline" |
| **S11 targets / timing** | 68 customers · 170 MAU · 32K shipments; both targets two months early | record | — |
| **S11 direct CTS dollars** | ≈$50K/yr (range $30–80K) | modeled on illustrative containers/shipment | presenting it as the value — the value is scale |
| **S11 pre-prod measurement** | 300 containers, ±1.6% at 95%; rollout proof 1.5 vs 6 months; ~$1.5M saved | record — separate canonical | merging the $1.5M with CTS |
| **S11 review by hand** | 1.5–2 → ~4 hrs/wk for over a year; seven frozen metrics; ±5% WoW; 15-day active rule | record | "I invented the WBR" |
| **S12 time-to-value / effort** | month 18 → month 6; 66% less engineering effort | record | — |
| **S12 cost avoidance** | ~$5M, said once | record | "$5M annual cost reduction" alongside it |
| **S12 CTS at work-item grain** | −50% partner-first vs −25% operator-first; 339.5 min/shipment; 2,200 interactions | record | — |
| **S12 partners / burn-up** | 48 of 53 within a month; MVP 11 partners / 10 countries; 55.8% → 83.2% | record | — |
| **S12 run-rate / decomposition** | ≈$1.5M/yr run-rate; ≈$2.9M + ≈$1.5M ≈ $4.4M | modeled (12 eng × $20K; 20K shipments illustrative) | presenting the split as the program's figure |
| **S12 employer** | Flexport FF App export launch | record | telling it as a Sephora roadmap (loop-packet conflict) |

**Never-say, one glance before the call.**

- Never "10×" or "50×" — say 7 → 68 (9.7×).
- Never merge the ~$1.5M pre-production saving with cost-to-serve; never "$1.5M to $2M."
- Never "$5M avoidance" and "$5M annual reduction" as two results — the avoidance is said once.

- Never "I invented the WBR" — copied from the LSA review, which was the point.
- Never claim platform configuration; program-lead voice throughout. The only personally configured platform in canon is Manhattan DOM.
- S16 (the pipeline), S14, S15, S10 are the bar raiser's; S1/S3, S2/S18, S6/S8 belong to Lan, Goforth and Hoerger — pointers only here.

## Backup bench — use only if a primary doesn't fit

Three fallbacks for the hiring manager, none of them another interviewer's primary: S9 (Elementum — the 597-lane Pareto and the ≥95% gate, for Backbone / Think Big / Highest Standards if S11 and S12 are spent), S19 (Elementum · Walmart-client — the promise buffer as one governed dial; a design-flavored Frugality / Invent & Simplify alternate, all model, illustrative labels intact) and S13 (Manhattan — Cyber Monday at 4 AM, the Bias for Action / Customer Obsession alternate). Tell S11 and S12 first.

**Bench discipline.** The two (or four) primaries above are the stories to tell first. Reach for a bench story only when the question genuinely does not fit a primary, or when the interviewer asks for "another example" on an LP already spent. Each entry is a 60-second pitch, a compressed STAR, one results line and two probes — enough to answer and stop; no full probing bank or derivation. Where a bench story is another interviewer's primary, the red note says so: use it only if the question demands it, and keep the facts identical to that packet.

### BACKUP — ONLY IF ASKED S9 — Elementum — 1,000 supplier/carrier routes asked, 597 built first: X12 EDI onboarding behind a ≥95% valid-ASN cohort gate

**LPs this backup can serve:** Have Backbone; Disagree & Commit (primary — 597 vs 1,000, two levels up) · Deliver Results · Insist on the Highest Standards (the gate held twice) · Think Big (the staged 1,000-lane vision) · Broad Responsibility (small carriers kept inside the standard by the portal floor)

**Basis.** I owned the scope analysis and the onboarding standard, influencing an executive two levels above my sponsor; engineers built the maps and harness to the standard; Transportation owned the carrier relationships. Cohort sizes and the number of cohorts are not in the record — do not quote them.

#### 60-SECOND PITCH — STOP ON THE PRIMARY METRIC

**Situation.** The Global Head of Supply Chain wanted inbound visibility on 1,000 lanes in two months; shipment-data completeness was 60% and high-value lanes were blind. **Task.** I owned the scope analysis and the onboarding standard, influencing an executive two levels above my sponsor. **Action.** Discovery: a spend Pareto — 597 lanes carried 80% of freight spend — pre-validated with Transportation and Procurement before I took it up. Plan & Design: the onboarding standard (X12 850/855/856/940/945/214/997 with an XML/portal floor — "the standard was the data, not the transport") and a cohort gate of ≥95% valid ASN. Build & Test: a mapping-and-test harness and a scorecard per lane. Cutover: 597 live in month one; one cohort failed the gate twice on a timestamp format and I held it both times. Hypercare: completeness scorecards to 95%; ~200 lanes the next wave behind the same gate. Lesson: automating dirty data manufactures confident wrong answers — the gate is the product. **Result.** Primary metric: inbound shipment-data completeness 60% → 95%. Freight spend −10% ≈ $2M in six months (record); 597 lanes = 80% of spend live in month one; adoption 8×.

#### COMPRESSED STAR

- **S —** 1,000 global lanes asked in two months — scope, schedule and quality all fixed; completeness 60%; lanes differed by orders of magnitude in spend and carriers differed in EDI capability.
- **T —** Own the scope analysis and the onboarding standard; take the re-scope two levels up with the evidence attached.
- **A —** Spend Pareto (597 = 80% of spend) pre-validated so no lane owner could contradict it in the room; standard + ≥95% valid-ASN gate published to carriers before cohort one; harness so later cohorts were mostly configuration; held a failing cohort twice; ~400 lanes carrying ~20% of spend deferred — priced in spend share, not lane count.
- **R —** Completeness 60 → 95%; 100% visibility on high-value lanes; 597 lanes live month one; freight −10% ≈ $2M in six months (record); method written up as the sourcing SOP.

## RESULTS LINE

Completeness 60% → 95% (record) · 597 of 1,000 lanes = 80% of freight spend live in month one (record) · freight spend −10% ≈ $2M in six months (record — kept as measured, never annualized) · adoption 8× (record) · ≈$0.4M/yr document-handling labor (modeled add-on).

## THE MATH — STEP-BY-STEP DERIVATION

### Table 1 — before: the problem in numbers.

| STEP | QUANTITY | FORMULA → SUBSTITUTION | INPUTS (BASIS) | VALUE |
|---|---|---|---|---|
| 1 | Shipment-data completeness | `480 / 800 = 60%` | expected = 200 shipments × 4 milestones = 800; received 480 (illustrative volumes; 60% record) | 60% — 2 of 5 events missing |
| 2 | What blindness costs | `off-guide spend unseen · premium modes by habit · chase calls per late shipment` | record problem | no instrument, no lever |
| 3 | Manual documents | `597 lanes × ~50 shipments × 3 docs ≈ 90,000 docs/yr` | illustrative volumes | ~90K docs/yr keyed by hand |

### Table 2 — the fix: step by step, values substituted.

| STEP | QUANTITY | FORMULA → SUBSTITUTION | INPUTS (BASIS) | VALUE |
|---|---|---|---|---|
| 1 | Spend Pareto | `S(k) = cumulative spend share; S(597) = 80%` | lanes sorted by annual freight spend (record) | 597 of 1,000 lanes carry 80% — build them first |
| 2 | Cohort gate | `valid-ASN rate ≥ 95% in test before activation` | published to carriers before cohort one (record) | one cohort failed twice on a timestamp format — held both times |
| 3 | Long-tail floor | `XML / portal when EDI is absent` | "the standard was the data, not the transport" (record) | small carriers kept inside the standard |
| 4 | Completeness after | `760 / 800 = 95%` | illustrative volumes; 95% record | 95%; high-value lanes 100% |

### Table 3 — after: the metric, before → after.

| STEP | METRIC | BEFORE | AFTER | BASIS |
|---|---|---|---|---|
| 1 | Shipment-data completeness | 60% | 95%; high-value lanes 100% | record |
| 2 | Lanes live in month one | 0 | 597 = 80% of freight spend | record |
| 3 | Freight spend on onboarded lanes | base | −10% ≈ $2M in six months | record — measured, never annualized |
| 4 | Document handling | ~90K docs/yr keyed | X12/XML validated, portal floor for the tail | modeled |

### Table 4 — the dollars, every multiplication shown.

| STEP | QUANTITY | FORMULA → SUBSTITUTION | INPUTS (BASIS) | VALUE |
|---|---|---|---|---|
| 1 | Implied addressable base | `$2,000,000 / 10% = $20,000,000/yr` | derived from the record pair, stated as such | ≈$20M/yr |
| 2 | The record result | `$20M × 10% = $2M in six months` | record | ≈$2M — said as measured |
| 3 | Slice — routing-guide compliance | `~$0.8M of the $2M` | off-guide spend caught by 856/214 (modeled split) | ~$0.8M |

| STEP | QUANTITY | FORMULA → SUBSTITUTION | INPUTS (BASIS) | VALUE |
|---|---|---|---|---|
| 4 | Slice — mode discipline | `~$0.6M of the $2M` | premium modes → ocean/ground when the ETA is trusted (modeled split) | ~$0.6M |
| 5 | Slice — expedites + detention | `~$0.6M of the $2M` | events replace chase calls (modeled split) | ~$0.6M |
| 6 | Split check | `0.8 + 0.6 + 0.6 = $2.0M` | the slices decompose the record — they never add to it | = the record $2M |
| 7 | Document labor | `90,000 × 6 min = 540,000 min; / 60 = 9,000 h; × $45 = $405,000` | illustrative volumes; benchmark rate (modeled add-on) | ≈$0.4M/yr |

## TWO PROBES

> **You defied a Global Head of Supply Chain's number. How?**
>
> With the spend curve and the lane owners' validation attached — 597 lanes carried 80% of the spend, so the phased plan was argued in spend share, not lane count. He accepted the phasing off the curve; I then committed to the remainder behind the same gate and delivered it.

> **Is the $2M annualized?**
>
> No — it is the record as measured: minus ten percent on the onboarded lanes' spend in six months. My three-way split (routing-guide compliance ~$0.8M, mode discipline ~$0.6M, expedites and detention ~$0.6M) decomposes the record; it never adds to it.

**Canon numbers (basis labels):** 60% → 95% completeness (record) · 597 / 1,000 lanes = 80% of spend (record) · −10% ≈ $2M in 6 months (record) · ≥95% valid-ASN gate, held twice (record) · 8× adoption (record) · ≈$0.4M/yr labor (modeled). Never: cohort counts or sizes; never double the $2M into a run-rate.

## BACKUP — ONLY IF ASKED S19 — Elementum (client: Walmart Marketplace) — the delivery-promise buffer rebuilt as priced risk: one governed dial, +32 points of fast-promise share at the same 95% floor (all model, no record dollar)

**LPs this backup can serve:** Dive Deep (alternate when S8/S10 are spent) · Invent and Simplify (alternate) · Frugality (speed bought with zero new spend) — design-flavored probes

**Basis.** Engagement lead for the promise-policy workstream; the client's promise-engineering and data-science teams owned and built the engine and the transit models — I never configured or deployed any platform. Every quantity is from the engagement's illustrative twelve-cohort model — say "the model" in the same breath; there is no production dollar and none is ever promoted. Labels (metro two-day, mountain rural) are the model's illustrative cohort names.

### 60-SECOND PITCH — STOP ON THE PRIMARY METRIC

**Situation.** A marketplace client printing delivery dates on inventory it never touches: promised date = order time + seller handling + predicted transit + a safety buffer — and the buffer was a flat 1.10 days on every lane regardless of variability. In the engagement model — twelve lane cohorts, about a million orders a week, transit σ from 0.22 to 1.60 days — that flat buffer printed 95.8% accuracy but put only 7.6% of orders on a fast promise: dense metro lanes over-insured and fat-tailed rural lanes under-insured at the same time. **Task.** Own the decomposition, the alternatives, the validation gates and the governance recommendation; the client's data science and engineering did not report to me. **Action.** Discovery: the buffer-waste audit — the flat buffer bought $\Phi(5.0)$, near-certainty, on the tightest cohort and only $\Phi(0.69) \approx 75\%$ on the widest. Plan & Design: four policies on one comparison table — uniform, σ-proportional with one governed dial, minimum-days at the floor, profit-optimal on priced breach economics. Build & Test: $b = 1.645\sigma$ holds 95% on every cohort by construction; the equimarginal check proved the optimum; the Q4 drill showed a frozen summer dial breaking the floor. Cutover: four shadow weeks against logged live orders before any displayed promise. Hypercare: the dial published in the weekly review header, re-solved before peak. **Result.** Primary metric: fast-promise share 7.6% → 39.6%, +32 points at the held 95% floor; average promised day 3.09 → 2.75; +$255K a week, about $13.3M a year — all on the illustrative book, and I say "the model" in the same breath.

### COMPRESSED STAR

- **S —** A flat 1.10-day buffer on every lane: 95.8% headline accuracy hiding over-insurance on tight lanes and under-insurance on loose ones; fast-promise share 7.6% (model).
- **T —** Lead the promise-policy workstream: decompose the waste, put priced alternatives on one table, design the gates, leave a governance instrument — not a tuning habit.
- **A —** Promise identity and accuracy factorization; buffer-waste audit; four policies on one table; proofs ($\Phi(1.645) = 0.95$ per cohort, equimarginal check, Q4 drill at ~1.45× σ); shadow mode for four weeks; the dial k owned by a business owner and printed every Monday.
- **R —** Fast-promise share 7.6 → 39.6% (+32 pts) at exactly the 95% floor; average promised day 3.09 → 2.75; +$255K/wk ≈ $13.3M/yr — model, labeled; the widest cohorts got honestly slower promises (1.10 → 2.63 days) on purpose.

### RESULTS LINE

Fast-promise share 7.6% → 39.6% (+32.0 points) at a held 95.0% floor (model) · average promised day 3.09 → 2.75 (model) · +$255K/wk ≈ $13.3M/yr vs uniform (modeled on the illustrative ~1M-order book) · the profit-optimal frontier (97.8% accuracy, ≈ +$20M/yr) handed to Finance as a priced decision — shown, never claimed, never merged with the $13.3M.

### THE MATH — STEP-BY-STEP DERIVATION

**Table 1 — before: the problem in numbers.**

| STEP | QUANTITY | FORMULA → SUBSTITUTION | INPUTS (BASIS) | VALUE |
|---|---|---|---|---|
| 1 | The promise identity | `EDD = t_order + handling + transit_pred + b; kept ⇔ (handling_err + transit_err) ≤ b` | per order; b = buffer in days (record-from-docs identity) | the buffer is the only policy term |
| 2 | Kept-promise probability per lane | `P(kept) = Φ(b / σ)` | ε ~ N(0, σ²) per cohort — whiteboard version; production uses the empirical quantile (record-from-docs) | accuracy is a function of b and σ only |
| 3 | Incumbent buffer | `b = 1.10 days on every lane` | uniform policy A (illustrative model) | 1.10 d everywhere |
| 4 | Tightest cohort, z-score | `z = b / σ = 1.10 / 0.22 = 5.0; Φ(5.0) ≈ 0.9999997` | metro cohort σ = 0.22 d (illustrative model) | near-certainty bought — promise needlessly slow |
| 5 | Widest cohort, z-score | `z = 1.10 / 1.60 = 0.6875; Φ(0.69) ≈ 0.754` | mountain-rural cohort σ = 1.60 d (illustrative model) | ≈75.4% kept — a known-loss book |
| 6 | Portfolio print | `accuracy 95.8% · fast share 7.6% · avg promised day 3.09 · profit $5.69M/wk` | order-weighted across 12 cohorts, ~1M orders/wk (illustrative model) | a green headline hiding both failures |
| 7 | The asymmetry | `padding fails silently (lost conversion files no ticket); cutting fails loudly (breaches do)` | record-from-docs — why the system drifted slow | bias toward slow, unowned |

**Table 2 — the fix: step by step, values substituted.**

| STEP | QUANTITY | FORMULA → SUBSTITUTION | INPUTS (BASIS) | VALUE |
|---|---|---|---|---|
| 1 | One governed dial | `b_lane = k × σ_lane; Φ(k) = floor ⇒ k = 1.645 for 95%` | Φ(1.645) = 0.95 under normal errors (record-from-docs design) | k = 1.645, one number for the book |
| 2 | Tightest cohort re-buffered | `b = 1.645 × 0.22 = 0.36 days` | σ = 0.22 d (illustrative) | buffer 1.10 → 0.36 d; 0.74 d of speed harvested |
| 3 | Widest cohort re-buffered | `b = 1.645 × 1.60 = 2.63 days` | σ = 1.60 d (illustrative) | buffer 1.10 → 2.63 d — slower on purpose, kept honest |
| 4 | Every cohort at the floor | `P(kept) = Φ(1.645 × σ / σ) = Φ(1.645) = 0.95` | by construction (design) | 95.0% everywhere; no cohort subsidizes another |
| 5 | Portfolio after | `accuracy 95.0% · fast share 39.6% · avg day 2.75 · profit $5.94M/wk` | illustrative model output, policy B | +32 pts fast share at the floor |
| 6 | Peak regime rule | `σ_Q4 ≈ 1.45 × σ; frozen k ⇒ z = 1.645 / 1.45 = 1.13; Φ(1.13) ≈ 0.87 per fully-inflated lane; portfolio prints 91.9% < 95%` | Q4 drill (illustrative model) | k re-solved and published in October, never frozen |
| 7 | Shadow gate | `4 consecutive weeks logged ≥ floor, contacts flat → promote; else hold` | promotion bar (design) | rehearse against reality before any displayed date |

**Table 3 — after: the metric, before → after.**

| STEP | METRIC | BEFORE | AFTER | BASIS |
|---|---|---|---|---|
| 1 | Fast-promise share (≤2-day) | 7.6% | 39.6% (+32 pts) | illustrative model |

| STEP | METRIC | BEFORE | AFTER | BASIS |
|---|---|---|---|---|
| 2 | Portfolio promise accuracy | 95.8% (split: ~100% / 75.4%) | 95.0% — the floor, held on every cohort | illustrative model |
| 3 | Average promised day | 3.09 | 2.75 (−0.34) | illustrative model |
| 4 | Weekly modeled profit | $5.69M | $5.94M (+$255K) | illustrative model |
| 5 | Owner of the speed-accuracy trade | implicit, in the last model tune | one governed dial, published weekly, re-solved per regime | engagement design record |

**Table 4 — the dollars, every multiplication shown.**

| STEP | QUANTITY | FORMULA → SUBSTITUTION | INPUTS (BASIS) | VALUE |
|---|---|---|---|---|
| 1 | Weekly profit, policy A | `$5,690,000 / wk, taken from the model` | illustrative model output | $5.69M/wk |
| 2 | Weekly profit, policy B | `$5,940,000 / wk, taken from the model` | illustrative model output | $5.94M/wk |
| 3 | Weekly delta | `$5,940,000 − $5,690,000 = $250,000 (the model's precise print: +$255K)` | rows 1–2 (illustrative) | ≈ +$255K/wk |
| 4 | Annualized | `$255,000 × 52 = $13,260,000` | row 3 × 52 weeks (modeled) | ≈ $13.3M/yr — "the model," said in the same breath |
| 5 | Frontier: policy D delta | `$6,070,000 − $5,690,000 = $380,000/wk; $380,000 × 52 = $19,760,000` | illustrative model output; ×52 modeled | ≈ $20M/yr — handed to Finance, never claimed |
| 6 | Why D holds 97.8% | `breach EV $12 > margin-per-promise-day $6.50 ⇒ optimum above the 95% floor` | $12 and $6.50 are the model's stated assumptions | the floor and the economics disagree — a governance finding |
| 7 | Headline | `≈ $13.3M/yr modeled; record dollars: none` | rows 3–4 | ≈ $13.3M/yr, always labeled |

## TWO PROBES

> **The whole story rests on an illustrative cohort set.**
>
> It does, deliberately — the model exists so the argument happens in numbers instead of adjectives. If the real σ looks nothing like the model's, k changes and the lane ranking changes; what survives any parameter set is the structure: buffers risk-proportional, the last buffer-day buying the same accuracy everywhere, and a floor that disagrees with the breach economics treated as a governance finding.

> **Why one dial instead of per-lane learned buffers?**
>
> Data science wanted per-lane learned buffers — more model, more value, invisible governance. I priced the difference and recommended the one-dial policy first: it captures most of the value, it is auditable, and σ improvements flow into speed automatically and visibly. The no went into the comparison table as a priced row, not a meeting minute.

**Canon numbers (basis labels):** 7.6% → 39.6% fast-promise share (+32 pts, model) · 95.8% → 95.0% accuracy held at the floor (model) · 3.09 → 2.75 promised days (model) · +$255K/wk ≈ $13.3M/yr (modeled) · 12 cohorts, ~1M orders/wk, σ 0.22–1.60 days, k = 1.645 (illustrative model) · breach ≈$12 vs margin ≈$6.50 (model assumptions). Never a production dollar.

## BACKUP — ONLY IF ASKED S13 — Manhattan — Cyber Monday, 4 AM: allocation across 3 DCs and 100 stores when the scheduler died

**LPs this backup can serve:** Bias for Action (primary) · Customer Obsession (primary backup — the promise ranked first) · Deliver Results

**Basis.** OMS consultant on allocation, promising and the OMS–WMS handshake; engineering fixed the scheduler once I isolated it; the post-mortem mechanisms were mine to drive. $8M / $6M / 99% are record; the margin figures are my model at benchmark rates. Older story — if asked for recent, pair with the UDC robotics throttle (S2): same muscle, 2024–25.

### 60-SECOND PITCH — STOP ON THE PRIMARY METRIC

**Situation.** Manhattan OMS consultant on allocation and promising across 3 DCs and 100 stores; 4 AM Cyber Monday the VP called — backorders piling with inventory present, ~$6M of sales at risk against a 95% fill-rate goal. **Task.** Own the response before the root cause was known. **Action.** Discovery: the contradiction (ATP > 0, backorders rising) was the entry bar to declare an incident. Plan & Design: I ranked by promise-at-risk, not FIFO; node choice = the promise — near 0.85 days vs far 4.60. Build & Test: one bridge, every rebalance timestamped; reason-coded alerts I had already built. Cutover: rebalanced so no DC drowned; status every two hours until 9 PM. Hypercare: root cause — the allocation scheduler died silently at the daylight-saving change; the post-mortem shipped a heartbeat, a DST checklist, a 3σ page rule and a rehearsed rebalance playbook. Lesson: decide reversibly, log it, find the cause later. **Result.** Primary metric: peak fill rate — 95% goal → 99% delivered. ~$8M of peak sales protected against ~$6M at risk (record); my model puts that at ≈$3.2M of gross margin at a 40% benchmark margin; structural stockouts −20% over the engagement.

### COMPRESSED STAR

- **S —** 4 AM Cyber Monday: backorder lines piling with inventory present; promises aging; ~$6M at risk against a 95% fill goal; nobody had written a clearing rule for failure.
- **T —** Own the response before the cause was known, without making it worse.
- **A —** Declared the incident on the contradiction; clearing rule — promise-at-risk, not FIFO — written on the bridge log in the first hour so nobody re-argued it at noon; rebalanced across three DCs; two-hourly status; isolated the DST scheduler death through my own reason-coded alerts; blameless post-mortem → heartbeat, DST checklist, 3σ page rule, rehearsed playbook.
- **R —** Fill 99% vs the 95% goal; ~$8M protected vs ~$6M at risk (record pair); structural stockouts −20% over the engagement; the next peak ran boring.

### RESULTS LINE

Peak fill 99% vs the 95% goal (record) · ~$8M of peak sales protected vs ~$6M at risk (record pair — one-time sales that didn't die, never a saving) · ≈$3.2M gross margin at a 40% benchmark (modeled) · structural stockouts −20% over the engagement (record) · detection moved from a VP phone call to one heartbeat interval.

### THE MATH — STEP-BY-STEP DERIVATION

**Table 1 — before: the problem in numbers.**

| STEP | QUANTITY | FORMULA → SUBSTITUTION | INPUTS (BASIS) | VALUE |
|---|---|---|---|---|
| 1 | 4 AM state | `scheduler dead; backlog = 30,000 orders; ~$6M of sales at risk vs the 95% fill goal` | record risk pair; backlog illustrative | the bridge call |
| 2 | FIFO failure | `FIFO clears by arrival — cheap, distant-promise orders first while at-risk promises age` | record rule problem | value-blind clearing |

| STEP | QUANTITY | FORMULA → SUBSTITUTION | INPUTS (BASIS) | VALUE |
|---|---|---|---|---|
| 3 | Naive node loads | `DC1: 22,000 / 10,000 = 2.2 days; DC3: 2,000 / 6,000 = 0.33 days` | capacities 10K / 8K / 6K orders/day (illustrative) | worst node 2.2 days behind while another idles |
| 4 | No liveness detection | `a dead scheduler was found by a VP at 4 AM, not by a page` | record problem | detection by seniority |

**Table 2 — the fix: step by step, values substituted.**

| STEP | QUANTITY | FORMULA → SUBSTITUTION | INPUTS (BASIS) | VALUE |
|---|---|---|---|---|
| 1 | Promise-at-risk ranking | `risk = value × urgency(promise − now); clear descending` | on the bridge log in hour 1 (record rule) | at-risk promises first, not first-in |
| 2 | Node choice = the promise | `T = handling + transit: near 0.85 d vs far 4.60 d; 4.60 − 0.85 = 3.75 d` | record pair | −3.75 days per re-pointed order |
| 3 | Load levelling | `rebalance 30,000 → 12,500 / 10,000 / 7,500: 12,500/10,000 = 1.25; 10,000/8,000 = 1.25; 7,500/6,000 = 1.25 days` | load = queue / capacity (illustrative) | worst slip 2.2 → 1.25 days |
| 4 | Liveness page (post-mortem) | `page if x < μ − 3σ = 1,200 − 3 × 150 = 1,200 − 450 = 750/hr` | μ = 1,200 allocations/hr, σ = 150 (illustrative); a dead scheduler reads 0/hr | caught in one interval, not by a VP |

**Table 3 — after: the metric, before → after.**

| STEP | METRIC | BEFORE | AFTER | BASIS |
|---|---|---|---|---|
| 1 | Peak fill rate | 95% goal | 99% delivered | record |
| 2 | Peak sales | ~$6M at risk | ~$8M protected | record pair — one-time event |
| 3 | Worst-case promise slip | 2.2 days | 1.25 days | illustrative walk |
| 4 | Detection | a VP at 4 AM | one interval via the 3σ page | post-mortem design |
| 5 | Structural stockouts | — | −20% over the engagement | record |

**Table 4 — the dollars, every multiplication shown.**

| STEP | QUANTITY | FORMULA → SUBSTITUTION | INPUTS (BASIS) | VALUE |
|---|---|---|---|---|
| 1 | Margin protected | `$8,000,000 × 40% = $3,200,000` | benchmark margin 35–45% (modeled) | ≈$3.2M one-time |
| 2 | Orders at risk | `$6,000,000 / $150 = 40,000 orders` | AOV benchmark | 40,000 orders |
| 3 | Cancels / CS avoided | `40,000 × $6.50 = $260,000` | CS contact $5–8 (benchmark; modeled) | ≈$0.26M |
| 4 | Expedite premium avoided | `40,000 × 40% = 16,000 re-pointed; 16,000 × $8 = $128,000` | re-pointed share and premium midpoint of the $5–12 benchmark (both illustrative) | ≈$0.13M |
| 5 | Structural-stockout run-rate | `$400M × 4% = $16M lost sales; × 20% = $3.2M recovered; × 40% = $1,280,000` | illustrative retailer × lost-sales benchmark (2–6%) × −20% record × margin | ≈$1.3M/yr modeled |

## TWO PROBES

### Why act before you knew the cause?

Because waiting had an hourly price — the queue was aging while inventory sat idle — and a reversible, logged intervention did not. I checked with engineering first that manual clears could not corrupt allocation, timestamped every rebalance, and hunted the cause in parallel with alerts I had already built.

### Isn't this a hero story?

The event is; the post-mortem is the point. Carrying the first stretch solo is one of my do-differentlys. What I would present as the result is the mechanism set — heartbeat, DST checklist, 3σ page rule, rehearsed playbook — and that the next peak ran boring.

**Canon numbers (basis labels):** 99% fill vs 95% goal (record) · ~$8M protected vs ~$6M at risk (record) · 3 DCs / 100 stores (record) · near 0.85 d vs far 4.60 d (record) · −20% structural stockouts (record) · ≈$3.2M margin at 40% (modeled). Never "$2MM baseline"; never present $8M as a saving.

INTERVIEW-PREP PACKET 2 OF 5 · PER-INTERVIEWER · BUSINESS-PROCESS LENS

# Amazon NACF Supply Chain loop — Packet 2: Ryan Hoerger

Amazon — Senior Supply Chain Manager, NACF Supply Chain (L6)

## Ryan Hoerger

Manager, Placement Analytics — NACF Supply Chain

**Wednesday, October 29, 2026 · 12:00 pm PDT · 55-minute video interview**

**SHADOW INTERVIEWER: SANDY HSIAO (A SECOND NOTE-TAKER — SAME PACE, SAME NUMBERS; EXPECT ONE QUESTION FROM THE SHADOW NEAR THE END)**

> **Strategy in one line.** The most quantitative interviewer on the panel. Tell two Plan-process redesigns with the math visible and every input labeled: S6 (honest supply definition → shortage alert a lead-time early → $2M released under a two-signal rule) and S8 (brochure lead times → measured lane distributions → p99 promise → ~$1.5M released with zero stockouts). The two programs' dollars never meet. In his language (public-profile inference below): S6 is positioning safety stock where measured errors justify it; S8 is positioning the promise at p99 per lane.

**LP focus — my guess, not confirmed:** Dive Deep · Are Right, A Lot · Learn & Be Curious — with a mentorship angle (his public About line names "a culture of mentorship"; expect one people question). Amazon does not disclose the assignment; the guesses are reasoned from the interviewer's title, the JD and his public profile in "Read me first" (public-profile inference, labeled as such) and should be held loosely — Insist on the Highest Standards remains a live alternate via the 99% bar in S8.

**Shadow etiquette:** address both; the shadow is usually a new interviewer being calibrated — same numbers, same pace; if the shadow asks the last question, answer it with the same discipline (phase asked, then stop).

Canon: amazon/amazon-l6-scm-master-packet.html (§1.2 roster, §4.1 ledger). Every number below is that packet's value; every dollar carries its basis label (record · user anchor · modeled · priced).

## Read me first

### The interviewer's lens (inference from title, JD and public profile — the profile parts labeled)

**Inference from the title.** "Placement Analytics" is the analytical arm of inventory placement — deciding where inventory should sit in the network so that the promise holds at the lowest landed cost. A manager in that function lives on forecast error, lead-time variability, safety-stock policy, service levels and the data-quality underneath them. He will be the most quantitative interviewer on the panel and the most likely to ask "how was that measured?", "what was the denominator?", and "what is the softest input in your model?" He is also the one most likely to recognise a textbook formula being passed off as a program artifact — so the honesty labels matter more here than anywhere.

**What that means for the answers.** Two Elementum planning stories, told as Plan-process redesigns with the math visible: the shortage-alert / real-time supply-demand view (S6 — forecast accuracy 70 → 85%, $2M safety stock released under a two-signal rule) and the measured-lane lead-time program (S8 — promise at p99 on 200 lanes, 580K units / ~$1.5M released with zero stockouts). Both end on *capital released on evidence with service held* — asset utilization and reliability in the JD's words. Keep the two programs' dollars apart in every sentence. A shadow (Sandy Hsiao) is on the call: same pace, same numbers, expect one question from the shadow near the end.

### What his public profile adds (public-profile inference, labeled)

**The profile (public-profile inference — directional, not verified biography).** Manager, Transfer/Placement Analytics at Amazon (Jul 2025–), New York; before that Senior Business Intelligence Engineer at Amazon (Dec 2024–Jul 2025); analytics manager at Nike (2021–24); analyst at Cornerstone Research (2016–20) — an economics-consulting shop where the work product is expert-testimony-grade statistical and financial analysis; Chicago Booth MBA (Siebel Scholar); Duke. His own About line: "rigorous statistical and financial analysis… a blend of data analysis and writing skills… a culture of mentorship." **Quoted for framing only — never referenced in the room; none of it enters my stories.**

**What that means for the answers.** Expect *definition-level probing*: MAPE versus bias and which one steered; the grain (SKU-family? per lane?); the window and refresh; what "99% lead-time accuracy" tolerates in days. Three such definition Q&A are drilled into the S6/S8 probing below. Second, *answer style: crisp and memo-like* — claim, evidence, stop; he writes for a living, and a long wind-up reads as evasion. Third, speak *placement*: S6 is positioning safety stock where the measured errors justify it; S8 is positioning the promise at p99 per lane — his daily work is deciding where inventory sits versus where it is promised. Fourth, the mentorship line in his About suggests one people probe — answer it with planners grown into owners (S6's measurement handover; S8's drift flag owned by planning), not with org charts.

### Three guessed LP focus areas, with rationale

> **Labeled guess.** Amazon assigns 2–3 Leadership Principles per interviewer and does not disclose them. These three are inferred from the title, the JD and the public profile above (public-profile inference); if the first question lands on a different LP — Insist on the Highest Standards via the 99% bar, or Frugality via the capital releases — the twist maps under each story carry the pivot.

| GUESSED LP | WHY THIS INTERVIEWER WOULD CARRY IT | STORY TO OPEN · BACKUP |
|---|---|---|
| **Dive Deep** | An analytics manager tests whether the candidate goes to the data grain — measured lane distributions instead of brochure values; the definition of "supply" before any screen. S8 is the Dive Deep primary in the master map. | S8 (open) · S6 (the supply definition, the MAPE/bias measurement) |

| GUESSED LP | WHY THIS INTERVIEWER WOULD CARRY IT | STORY TO OPEN · BACKUP |
|---|---|---|
| **Are Right, A Lot** | Placement decisions are judgment under uncertainty; he will probe how the candidate was right against the room's belief and how the decision was made safe (release rule, watch window, fat-tail exception). | S8 (measured vs brochure; p99 chosen by planning on the asymmetry) · backup: S6 (refusing unfirmed MRP orders as supply) |
| **Learn & Be Curious (with a mentorship angle)** | A Cornerstone-trained analyst with a Booth MBA who writes about rigor and mentorship (public-profile inference) will test whether the candidate learns at the definition level and grows people at the mechanism level: the MIT toolkit acquired to design what I had been implementing — labeled toolkit, never day job — and planners grown into owners of the measurement, the drift flag and the refresh cadence. Insist on the Highest Standards stays live as the 99% bar before any release and the two-signal rule before any buffer came out. | S8 (99% bar; drift flag owned by planning) · S6 (measurement handed to planners) |

## Answer strategy — business first, systems as enablers

- **Open with the process problem and its cost, not the system.** First sentence names the broken process (Plan, Source, Make, Deliver or Return), who it hurt (customer, Finance, the floor) and the baseline number. The platform appears only as the enabler of the redesign, in one clause.
- **Use the JD's four outcome words as the results frame:** reliability (did the promise hold?), responsiveness (how much earlier did we see it?), cost efficiency ($ with its basis label), asset utilization (capital, labor-hours, capacity released).
- **Action = operating-model redesign in gated phases** — Discovery → Plan & Design → Build & Test → Cutover → Hypercare, each spoken as entry bar · what I delivered · exit bar. Say the gate that fired, not the meeting that happened.
- **Every dollar carries its basis word in the same breath** — record · user anchor · modeled · priced. A record dollar is said flat; a modeled dollar is said with its inputs. Never promote a modeled number to record.
- **Program-lead voice.** I owned definitions, rules, gates and the signature chain; engineers, SIs and vendor teams configured and built to the spec I signed. The only platform personally configured in canon is Manhattan DOM.
- **The pitch ends on the primary metric.** Then stop. Silence after the pitch is the interviewer filling their STAR worksheet; their probes are the second half of the answer.
- **For Placement Analytics specifically:** lead with the planning parameter that was wrong (brochure lead time; unfirmed orders counted as supply), the decision it corrupted (buffer size, shortage response), and the rule that released capital safely. Show one formula only when asked, labeled as the MIT toolkit illustration — the data team built the distributions; planners sized the buffers; Finance signed.
- **Answer style for this room: crisp, memo-like.** Claim, evidence, stop. His background (Cornerstone rigor; a writing-heavy About line — public-profile inference) means a long wind-up reads as evasion; the one-breath basis label ("record", or "modeled, on illustrative inputs") does more credibility work here than any extra sentence.
- **Speak placement.** Frame S6 as positioning safety stock — buffer placed only where measured forecast and lead-time errors justify it — and S8 as positioning the promise at p99 per lane, recomputed on a drift flag. His team decides where inventory sits versus where it is promised (public-profile inference); that is the exact shape of both stories.
- **Keep S6 and S8 apart:** $2M (S6, forecast-driven, two-signal rule) and ~$1.5M / 580K units (S8, lead-time-driven, 99% bar) are different programs — never add them, never swap their mechanisms. "Six sites" —

never "six NA sites"; never paste S6's "3 months" onto the six-site ingest (that is S7, not in this packet).

**Loop-consistency rule.** One question, one story; never the same story to two interviewers — interviewers compare notes. Held for other packets (do not spend here): S11/S12 (Tripathy) · S2/S18 (Goforth) · S1/S3 (Lan) · S16/S15/S10/S14 (Josyula). S7 (CM BOM) and S9 (EDI Pareto) are unallocated master-packet backups — pointers only, and no BOM-explosion content.. If pushed for "another example" on an LP, use the named backup in the question bank or the backup bench at the end of this packet, never a story allocated to another interviewer.

**How to read the "L6 level evidence" block under each story.** Amazon's L6 functional bar (user-supplied rubric): *Ambiguity* — strategy or network undefined; I select the stakeholders, set the goals, deliver independently. *Scope & influence* — team-level scope; influences one to two levels up. *Execution* — tactical and strategic; simplifies or extends; applies best practices; force-multiplies; mitigates long-term risk; makes the short-versus-long-term trade-off explicit. *Impact* — long-term team goals. *Process improvement* — operational-excellence best practices institutionalized. *Complex* — visible risks, roadblocks and constraints mitigated through expertise, judgment and influence to consensus. Overview themes the block is tuned to: designing and modeling the supply chain rather than working in it; developing new processes for others to own; root-cause mechanisms that resolve trending issues at scale; recommendations to leadership that outline competing metrics. The block sits right after Results so the evidence is on the page when the probing starts.

## The stories — business-first, canon numbers

Two Elementum planning stories from the same six-site manufacturer. Each: 60-second pitch → Situation/Task → Action in gated phases → Results → Probing tuned to a placement-analytics manager → the key math → Twist map.

### S6 — Elementum — one real-time supply/demand view across eight data domains: a shortage visible a lead-time early, forecast accuracy 70% → 85%, $2M of safety stock released under a two-signal rule

**LPs this story can serve here:** Dive Deep (the definition of supply; MAPE/bias at the decision grain) · Earn Trust (the two-signal release rule signed with Planning and Finance) · Deliver Results (third — alert live in 3 months) · Think Big (one view over eight domains) · Are Right, A Lot (backup — refusing unfirmed orders as supply)
**Process family:** Plan — S&OP / shortage management / safety-stock policy; data-integrity program behind the plan

**Honesty label.** I owned the definitions, the data-integrity program and the delivery; the client's SAP teams and our integration engineers built the feeds; planners owned the forecast — I owned its measurement and the release rule. I have not owned a demand forecast end-to-end and won't claim it. The safety-stock formula is how I explain it now with the MIT toolkit — an illustration, not a program artifact.

**60-SECOND BUSINESS-FIRST PITCH — STOP ON THE PRIMARY METRIC**

**Situation.** A six-site manufacturer — mixed make-to-order and make-to-stock — learned about a component shortage when the line stopped. S&OP ran off private spreadsheets, shipment data was 60% complete, and supplier, material, BOM, inventory, production, transportation, sales-order and planning data lived in SAP and site systems but never met. Leadership's question — which customer orders are at risk this quarter? — had no data path, so the plan responded with expedites instead of re-plans and buffers sized on noise. **Task.** Nobody owned the seam; I took the definitions, the data-integrity program and the delivery. **Action.** I defined supply honestly before any screen — on-hand plus production plan plus released planned orders only, at SKU × location × date — and refused to count unfirmed MRP orders because they change every run. Domains went live in waves sequenced by planner trust, each behind a published data-quality gate, with daily reconciliation to the SAP source and false alarms worked as defects. The shortage alert fires when days-of-cover falls to the lane's measured p90 lead time — the last moment a re-plan is cheaper than a flight. Safety stock came out only under a rule Planning and Finance signed: release where forecast error and lead-time error had both improved. **Result.** Primary metric: forecast accuracy on the scoped class-A SKUs 70% → 85%. $2M of safety stock released under the two-signal rule — record, one-time working capital — with ≈$0.8M a year of modeled recurring value around it; the shortage alert live in three months; planner effort −50%. That $2M is a different program from the lane program's $1.5M; I never add them.

**SITUATION / TASK — THE BUSINESS PROCESS PROBLEM, SCOPE, SCALE, BASELINES**

**S —** A strategic manufacturing customer with six sites: planners learned about a component shortage when the line stopped; S&OP ran off private spreadsheets; shipment data was 60% complete; eight data domains lived in SAP and site systems but never met. Leadership's question — "which customer orders are at risk this quarter?" — had no data path.

The Plan process underneath was running on an unsafe definition of supply (unfirmed MRP orders counted because they looked better), an unmeasured lead-time error, and a shortage signal that arrived at line-down. The cost: expedites instead of re-plans, buffers sized on noise, and planners spending ~40% of their week reconciling rather than root-causing.

**T —** Nobody owned the seam — no spec, no agreed grain, no definition of "supply," no data-quality bar. I chose all four, as owner of the definitions, the data-integrity program and the delivery. Goal: a real-time supply/demand view with honest definitions and a shortage visible a lead-time early — and planner trust, which was the real product.

**Baselines on the table:** forecast accuracy 70% on scoped class-A SKUs · shortage detected at line-down (target: a lead-time early) · shipment-data completeness 60% · planner effort on reconciliation and firefighting (≈40% of the week, illustrative share; the −50% is record).

## ACTION — THE PROCESS / OPERATING-MODEL REDESIGN, IN GATED PHASES

- **Discovery (4–6 wks)** — profiled every domain (completeness, latency, joinability) to my own gap templates; wrote the definitions memo — grain SKU × location × date; supply = on-hand + production plan + released planned orders only — and the metrics tree (OTIF, inventory $, MAPE/bias, lane lead-time accuracy, completeness, latency, adoption). Refused Planning's request to count unfirmed MRP orders as supply. *Exit bar:* signed gap findings per domain; scope frozen in writing; planners agree the supply definition before any screen.
- **Plan & Design (parallel)** — master-data foundation plan; MPN mapping worklist with its coverage gate; domain-wave sequence by trust; DQ scorecard spec; the 10-step onboarding playbook. *Exit bar:* master-data gates green; unmapped-MPN list under threshold with a named owner; wave order signed by domain owners — who could fail their own wave.
- **Build & Test (8–12 wks)** — domain waves behind published DQ gates — inventory and production first (numbers planners already believed), then purchasing/ASN, then transportation; shortage-alert rule specified (days-of-cover vs lane lead-time p90); daily reconciliation to MD04/COOIS/VA05/MB52; weekly root-cause loop with false alarms worked as defects. *Exit bar:* per-domain completeness ≥ gate; reconciliation passing daily; false alarms trending down; pilot planners using the alert.
- **Cutover & Go-Live (4 wks)** — views live for pilot planners with the at-risk list replacing the Monday spreadsheet; nobody mandated off the spreadsheet — rollback stayed free. *Exit bar:* planners answer "where is my product" from the platform in the live S&OP meeting.
- **Hypercare / Close (4 wks)** — exception workbench staffed; weekly root-cause reviews handed to domain owners; the safety-stock release rule signed with Planning and Finance — release only where forecast error AND lead-time error both improved. *Exit bar:* blockers/day under threshold two consecutive weeks; the release survives audit.

**Influence and ownership.** Mine: the definitions memo, the metrics tree, the DQ scorecard spec, the wave sequence, the release rule and the onboarding playbook. The client's SAP teams and our integration engineers built the feeds; planners owned the forecast — I owned its measurement (MAPE/bias per SKU family) and the release rule; domain owners held sign-off over their own gates. The MPN worklist gated longer than planned — escalated to the sponsor as a re-sequenced date with the coverage number attached, not a slip.

> **Designed the process; others own it now.** I designed the definitions, the DQ gates, the release rule and the onboarding playbook; domain owners now run the gates and the weekly root-cause reviews, and planners run the alert without me. **Trade-off put to leadership.** The recommendation to Planning and Finance outlined the competing metrics — working capital released vs service risk — and resolved it as a rule: release only where forecast error and lead-time error both improved.

## RESULTS — $ · # · % · TIME · CUSTOMER · QUALITY (CANON NUMBERS ONLY)

- **$** — $2M safety stock released (record, one-time working capital, under the both-errors-improved rule signed with Planning and Finance) + ≈$0.8M/yr *modeled* recurring (carry ≈$0.2M at 10% · expedites pre-empted ≈$0.3M · planner capacity ≈$0.3M; range $0.5–1.2M) — never added to S8's $1.5M.
- **#** — eight data domains into one view across six sites; shortage-alert product live in 3 months; adoption 8× (record).
- **%** — forecast accuracy 70% → 85% on scoped class-A SKUs (record); planner effort −50% (record); shipment data 60% complete at the start.
- **Time** — the alert fires at days-of-cover ≤ lane lead-time p90 — a lead-time early instead of at line-down (record mechanism); views live in 3 months (the alert product; the six-site ingest is a separate 5-month program — never paste 3 months onto it).

- **Customer** — "which customer orders are at risk this quarter?" answered from the platform in the live S&OP meeting, not a spreadsheet (record exit bar).
- **Quality** — daily reconciliation to MD04/COOIS/VA05/MB52; false alarms classified weekly to root cause and burned down as defects (record).

**L6 LEVEL EVIDENCE** — what the interviewer pattern-matches against the Amazon L6 functional rubric; every line is this story's canon, no new numbers

| L6 DIMENSION | EVIDENCE IN THIS STORY |
|---|---|
| **Ambiguity** | Eight data domains, six sites and no shared definition of supply. I wrote the definitions memo (grain SKU × location × date; supply = on-hand + production plan + released planned orders only) and the metrics tree, and chose the domain owners who would sign their own gates. |
| **Scope & influence** | The client's SAP teams and our integration engineers built the feeds; planners owned the forecast; Planning and Finance signed the release rule; domain owners held sign-off over their gates — none reporting to me. |
| **Execution trade-off + mechanism others now own** | Working-capital release vs service risk: release only where forecast error AND lead-time error both improved. Mechanisms others own now: the DQ scorecard, the 10-step onboarding playbook and the weekly root-cause reviews handed to domain owners. |
| **Impact (long-term)** | $2M of safety stock released (record); forecast accuracy 70% → 85% on class-A; planner effort −50%; adoption 8×; the shortage alert fires a lead-time early instead of at line-down. |
| **Process improvement (standard work institutionalized)** | Domain waves behind published DQ gates; daily reconciliation to MD04/COOIS/VA05/MB52; false alarms classified weekly to root cause and burned down as defects. |
| **Complexity & consensus** | Nobody was mandated off the Monday spreadsheet — rollback stayed free until planners answered "where is my product" from the platform in the live S&OP meeting; the MPN worklist gated longer than planned and the gate held. |
| **Root cause at scale** | Unfirmed orders inflating supply was a definition defect, not a planner habit — fixed once in the supply definition and applied across all six sites. |

## PROBING — CHALLENGE Q&A TUNED TO THIS INTERVIEWER

> **Analytics — How was 70 → 85% measured, and on what?**
>
> Accuracy = 1 − MAPE on the scoped class-A SKUs, with bias published alongside it from month two, at the SKU-family grain the planners decide on. The planners owned the forecast; I owned the measurement and the input integrity — I have not owned a demand forecast end-to-end and won't claim it.

> **Analytics — Why exclude unfirmed MRP planned orders from supply? That understates it.**
>
> Deliberately. Unfirmed orders breathe every MRP run; counting them amplifies noise and inflates supply, which is exactly how the line-down surprises happened. Released planned orders only — accepted by planners before the first screen was built.

> **Analytics — The $2M — how do I know it is not a prettier average?**
>
> Because of the release rule: buffer came out only on SKU families where forecast error and lead-time error had both demonstrably improved, signed by Planning and Finance. That is why the release survived audit. It is one-time working capital; the recurring value around it is my model and labeled.

> **Analytics — Your expedite row is the softest. Defend it.**
>
> Agreed — about sixty expedites a year pre-empted at a $5K premium, both illustrative. The record is the mechanism: the alert fires at days-of-cover ≤ lane p90. To firm it I would pull the expedited-freight ledger for the six sites twelve months before and after the alert went live and count the premium directly.

> **Analytics — Why p90 for the alert threshold and p99 for the promise (S8)?**
>
> Different decisions. The alert is a re-plan trigger — p90 gives the planner a window before the tail bites. The promise is a commitment — it sits at the tail you can defend. Same measured distribution, two policies read off it.

> **Analytics — Define your two error metrics: MAPE versus bias. Which one steered?**
>
> MAPE is mean absolute percentage error — how wrong the forecast is on average, direction ignored; bias is the signed mean error — whether it runs systematically high or low. MAPE without bias hides a chronic over-forecast inside a decent average, and a chronic over-forecast is exactly how buffer quietly accumulates. So accuracy was reported as 1 − MAPE on the scoped class-A SKUs at the SKU-family grain, with bias published alongside from month two — and the release rule required forecast error and lead-time error both to improve, so neither metric could carry the decision alone.

> **Shadow / closing — What would you do differently?**
>
> Start the MPN-mapping worklist on day one — it is never "done" and it gated more than I planned. And define supply honestly before any screen, which held.

### THE KEY MATH — THE NETTING WALK AT FINISHED-GOODS GRAIN, THE ALERT RULE AND THE TWO-SIGNAL RELEASE (ILLUSTRATIVE VALUES; RECORD PERCENTAGES)

| STEP | QUANTITY | FORMULA → SUBSTITUTION | BASIS | VALUE |
|---|---|---|---|---|
| 1 | Accuracy before / after | `1 − 0.30 = 70% → 1 − 0.15 = 85%` | record | 70% → 85% |
| 2 | Honest supply | `supply(t) = on-hand + in-transit ≤ t + production ≤ t + RELEASED planned orders` | record rule | unfirmed MRP orders excluded |
| 3 | Projected balance (FG SKU) | `PAB(3) = 400 − 500 = −100; PAB(7) = 1,000 − 1,200 = −200; PAB(10) = 1,800 − 1,600 = +200` | illustrative FG example — timing shortage, not volume | fix window = day 9 − day 7 = 2 days |
| 4 | Alert rule | `DoC = 1,200 / 100 = 12 days ≤ lane p90 14 → alert; p90 = 9 → no alert` | record design; values illustrative | a lead-time early, no false alarm on the fast lane |
| 5 | Two-signal release | `SS = z·d·σ_L = 2.33 × 100 × 4 = 932 → σ_L 2: 466; release 466 units — only if BOTH errors improved` | toolkit illustration; the rule is record | release survives audit |
| 6 | Recurring value | `$2M × 10% = $0.2M · 60 × $5K = $0.3M · 2.4 FTE × $120K = $0.3M` | modeled (benchmarks labeled) | ≈$0.8M/yr; $2M one-time record |

MRP math at finished-goods grain only — no BOM explosion in this packet. The six-site ingest (S7) is a separate program; never paste this story's "3 months" onto it.

**Where this lands at Amazon.** Placement rests on two error signals — forecast error and lead-time error — measured at the decision grain. The transferable disciplines: define supply before the screen, alert at the last moment a re-plan is cheaper than an expedite, and release buffer only when both error signals improve. Reliability (shortage seen a lead-time early), cost efficiency (expedites pre-empted), asset utilization ($2M of working capital released on evidence).

### TWIST MAP — LIKELY LP QUESTIONS FROM THIS INTERVIEWER → THE ANGLE TO TAKE WITH S6

| LIKELY QUESTION (THIS INTERVIEWER) | LP | ANGLE — WHAT TO EMPHASIZE | SKIP / NEVER SAY |
|---|---|---|---|
| **Tell me about a time you had to dig into the data to find the real problem.** | Dive Deep | The supply definition — unfirmed orders inflating supply — found by profiling eight domains; MAPE and bias at the grain planners decide on. | Skip the SAP table names unless asked; systems are enablers. |

| LIKELY QUESTION (THIS INTERVIEWER) | LP | ANGLE — WHAT TO EMPHASIZE | SKIP / NEVER SAY |
|---|---|---|---|
| **Tell me about a time you released inventory / improved working capital.** | Frugality / Deliver Results | $2M under the two-signal rule — record, one-time; the recurring ≈$0.8M labeled modeled. | Never add the $2M to S8's $1.5M; never claim $2M per year. |
| **Tell me about a time you were right when the business disagreed.** | Are Right, A Lot | Refusing unfirmed MRP orders as supply because it looked better; refusing a blanket safety-stock release — agreed as a rule instead. | Do not make Planning the villain; the rule did the refusing. |
| **How do you get planners to trust a new signal?** | Earn Trust | Waves sequenced by trust (inventory and production first); false alarms worked as defects; nobody mandated off the spreadsheet; domain owners could fail their own wave. | Do not claim adoption dollars; adoption 8× is the record. |
| **How would you design a shortage / at-risk-order signal for the network?** | Functional — Plan process design | Grain, honest supply, days-of-cover vs measured lane p90, DQ gate per feed, two-signal release — the §3.99 opening: book of record and metric first. | No BOM explosion; finished-goods grain only. |
| **What's the softest number in your model?** | Highest Standards / Earn Trust | The expedite row (60 × $5K); how I would firm it from the expedited-freight ledger. | Never present planner capacity as cash — it is time redeployed. |

## S8 — Elementum — replacing brochure lead times with measured lane distributions: promise at p99 on 200 lanes, 99% lead-time accuracy, 580K units / ~$1.5M of safety stock released with zero stockouts

**LPs this story can serve here:** Dive Deep (primary — measured the distribution, not the average) · Are Right, A Lot (backup — measured vs brochure; p99 on the asymmetry) · Frugality (the release) · Insist on the Highest Standards (the 99% bar before any cut) · Earn Trust (Finance signed each lane; the exception written into the rule)

**Process family:** Plan / Source / Deliver seam — inbound lead-time policy, safety-stock sizing, promise policy; parameter governance with a drift flag

**Honesty label.** I owned the requirement (measured lanes, not brochures), the p99 promise policy and the release rule; the data team built the distributions; planners sized the buffers; Finance signed the release. Lane-cohort counts and week markers are not in the record. The safety-stock formula is the MIT inventory-theory toolkit as I explain it now — an illustration, not a program artifact. Depends on the six-site canonical model (S7 — pointer only).

**60-SECOND BUSINESS-FIRST PITCH — STOP ON THE PRIMARY METRIC**

**Situation.** The same six-site manufacturer planned inventory on carrier-brochure lead times — every buffer sized on a variability nobody had measured: too much on reliable lanes, too little on fat-tailed ones. The logistics manager asked for a "lead-time variability report." **Task.** I reframed it as a parameter and policy problem — nobody owned the seam between transportation reality and planning parameters, and I took it — scoped at 200 lanes by site, lane and mode. **Action.** I wrote the requirement to the data team: mean, spread and tail per lane from actual production cycle plus inbound transit; defined the promise at p99 cumulatively — the smallest day d with $P(L \le d) \ge 0.99$, not average-plus-padding — and showed each lane's asymmetry, line-down cost against carrying cost, and let planning choose the percentile; they chose p99. The release rule was written as a signed planning act with the exception inside it: fat-tail lanes keep their buffer until the source process moves; Finance agreed the rule in the same room, before any cut. I would not endorse a release until lead-time accuracy hit 99% on the 200-lane set; the stockout-and-expedite watch was armed first; every parameter got a drift flag and a refresh cadence so the promise never decays into a new brochure. **Result.** Primary metric: lead-time accuracy on 200 lanes — brochure values → 99% at the p99 promise. 580K units, about $1.5M of safety stock, released without a stockout — record, one-time working capital — with ≈$0.6M a year modeled around it. A different program from the $2M forecast release; never added.

### SITUATION / TASK — THE BUSINESS PROCESS PROBLEM, SCOPE, SCALE, BASELINES

**S —** The six-site manufacturer's logistics manager asked for a "lead-time variability report." The real problem: inventory planning ran on carrier-brochure lead times — every buffer sized on a variability nobody had measured; too much buffer on reliable lanes, too little on fat-tailed ones; the promise to the plan was folklore.

The risk cut both ways: keep planning on brochures and you hold too much on reliable lanes and stock out on fat-tailed ones; cut buffer without evidence and the stockout is yours. And an unmaintained promise decays into a new brochure — which is why every parameter needed a drift flag.

**T —** Nobody owned the seam between transportation reality and planning parameters; I took it. Goal: measured lanes replace catalog values; a promise at a defensible tail; buffer released only on evidence, with service held — scoped at 200 lanes by site, lane and mode.

**Baselines on the table:** lead-time parameters = carrier-brochure values, variability unmeasured · buffer capital sized on catalog values · no lead-time-accuracy measurement · target 99% lead-time accuracy on the 200-lane set before any release.

## ACTION — THE PROCESS / OPERATING-MODEL REDESIGN, IN GATED PHASES

- **Discovery** — reframed the report request as a parameter problem; scoped 200 lanes by site/lane/mode; wrote the success criteria — measured lanes replace catalog values; buffer released only on evidence; fat tails keep buffer; showed one right-skewed lane where the average hid the tail. *Exit bar:* planning and logistics agree the problem is the parameter, not the report.
- **Plan & Design** — requirement to the data team (mean, spread, tail per lane; production cycle + inbound transit vs actual); the p99 promise policy (cumulative, not μ + 2.33σ); the release rule as a signed planning act with the fat-tail exception inside it; the stockout-and-expedite watch; drift flag and refresh cadence. On p95 vs p99, showed the asymmetry per lane and left the choice to planning. *Exit bar:* policy signed by planning; Finance agrees the release rule in the room; data team accepts the spec.
- **Build & Test** — data team built the lane distributions and tracked lead-time accuracy per lane; manual reconciliation stayed alive. *Exit bar:* lead-time accuracy 99% on the 200-lane set — the bar I would not endorse a cut below.
- **Cutover & Go-Live** — planners executed the buffer release lane by lane under the signed rule; fat-tail lanes excluded until their source process moved; watch armed first. *Exit bar:* 580K units / ~$1.5M released; no stockout in the watch window.
- **Hypercare / Close** — read the watch; set drift flags per lane; handed the refresh cadence and the release rule to planning as an act they could re-apply. *Exit bar:* watch clean; the promise stays measured.

**Influence and ownership.** I owned the requirement, the promise policy and the release rule; the data team built the distributions and tracked accuracy; planners sized the buffers and executed; Finance signed each lane. When Finance pushed for a broader release, the rule they had signed did the refusing — fat-tail lanes keep buffer until the source process moves. The percentile fight never happened: I showed each lane's line-down cost against carrying cost and planning chose p99 — their choice surviving an audit is worth more than my choice winning a meeting.

> **Designed the process; others own it now.** I designed the promise policy and the release rule as a signed planning act; planning now re-applies it on its own refresh cadence with a drift flag per lane. **Trade-off put to leadership.** The recommendation to Finance and Planning outlined the competing metrics — ~$1.5M of working capital out vs stockout exposure at the tail — and set the bar: p99, 99% lead-time accuracy, fat-tail lanes excluded until their source process moved.

## RESULTS — $ · # · % · TIME · CUSTOMER · QUALITY (CANON NUMBERS ONLY)

- **$** — 580K units / ~$1.5M of safety stock released without a stockout (record, one-time working capital); ≈$0.6M/yr *modeled* recurring around it (carry ≈$0.3M at a benchmark 20% all-in rate + expedites avoided ≈$0.3M; range $0.3–1.4M) — a different program from S6's $2M; never added.
- **#** — 200 lanes measured by site, lane and mode; 580K units released (record); ≈$2.59/unit derived — low-value components.
- **%** — lead-time accuracy 99% on the 200-lane set at the p99 promise — the bar below which I would not endorse a cut (record).
- **Time** — release only after the 99% bar; watch window clean; refresh cadence and drift flag per lane handed to planning (record).
- **Customer** — service held — zero stockouts on released lanes through the watch window (record).
- **Quality** — release only under the signed rule; fat-tail lanes kept buffer until the source process moved; the promise stays measured, never a new brochure (record).

**L6 LEVEL EVIDENCE** — what the interviewer pattern-matches against the Amazon L6 functional rubric; every line is this story's canon, no new numbers

| L6 DIMENSION | EVIDENCE IN THIS STORY |
|---|---|
| **Ambiguity** | A report request with no agreed parameter behind it. I reframed it as a parameter problem and wrote the success criteria myself: measured lanes replace catalog values; buffer released only on evidence; fat tails keep buffer. |
| **Scope & influence** | 200 lanes by site, lane and mode; the data team built the distributions; planners sized and executed; Finance signed each lane — a planning act I designed and others executed. |
| **Execution trade-off + mechanism others now own** | Working capital out vs stockout risk: promise at p99 (cumulative), a 99% lead-time-accuracy bar below which I would not endorse a cut, fat-tail lanes excluded until their source process moved. Mechanism others own now: the release rule as a signed act, the refresh cadence and a drift flag per lane handed to planning. |
| **Impact (long-term)** | 580K units / ~$1.5M of safety stock released without a stockout (record, one-time working capital); the promise stays measured, never a new brochure. |
| **Process improvement (standard work institutionalized)** | Lead-time accuracy tracked per lane; drift flags; the release rule written so planning can re-apply it without me. |
| **Complexity & consensus** | Finance pushed for a broader release — the rule they had signed did the refusing; the fat-tail exception was inside the rule, so it fired as policy, not as a judgment call. |
| **Root cause at scale** | Brochure lead times were the parameter error behind every excess buffer — fixing the parameter fixed all 200 lanes at once rather than one planner's buffer at a time. |

## PROBING — CHALLENGE Q&A TUNED TO THIS INTERVIEWER

**Analytics — Why p99 and not p95? That is a lot of buffer.**

Per lane, the cost of a line-down on a fat-tailed lane dwarfed the carrying cost of the extra day. I showed the asymmetry lane by lane and let planning choose — they chose p99. On reliable lanes the measured σ was so far below the brochure that p99 still released buffer; on fat-tailed lanes the buffer stayed.

**Analytics — How did you compute p99 on skewed lanes?**

Cumulatively from the empirical distribution — the smallest d with $P(L \le d) \ge 0.99$ — not $\mu + 2.33\sigma$. On one right-skewed lane the normal shortcut understated the true p99 by about 3.4 days; that lane is why the definition is cumulative.

**Analytics — Is the $1.5M measured?**

The release is record: 580K units out, zero stockouts in the watch window, Finance signed each lane. The ≈$2.59/unit is the derived average. What is modeled is the recurring value around it — carry at 20% all-in (capital-only would be ≈$150K at 10%) and avoided expedites — each input labeled.

**Analytics — The expedite number assumes what?**

Ten thousand shipments a year on the 200 lanes and one late arrival in ten forcing an expedite before p99 — illustrative. I would pull the expedite ledger by lane for the year before and after and compare directly.

**Analytics — What is lead-time accuracy, exactly?**

$1 - |\text{actual} - \text{promised}| / \text{promised}$, tracked per lane against the p99 promise; the bar was 99% on the whole 200-lane set. Below it, no release — written into the act before anyone asked.

**Analytics — At what grain did you measure lead time, and over what window?**

At the lane grain, from actual ship-to-receipt durations — one empirical distribution per lane, never a network average. The record carries the per-lane distributions, the refresh cadence and the drift flag (p99 recomputed on refresh; the buffer decision re-opens if accuracy falls); it does not carry a quoted window length, and I will not invent one — what mattered design-wise was a window long enough to hold seasonality and recent enough that a dead lane's history could not vote, with the drift flag as the safety net either way.

> **Analytics — What does "99% lead-time accuracy" tolerate, exactly?**
>
> The metric is a ratio — 1 − |actual − promised| / promised — so the tolerance scales with the lane instead of being a flat day-count: on a 5-day lane the debit starts at roughly a quarter-day of miss; on a 30-day lane, about a day and a half. A flat day tolerance would have been strict on short lanes and blind on long ones. The bar was 99% across the whole 200-lane set before any buffer released — per-lane measurement, no pooling to hide a bad lane inside a good average.

> **Shadow / closing — How does the promise stay honest after you leave?**
>
> Drift flag per lane with a refresh cadence: p99 is recomputed on refresh, and the buffer decision re-opens if accuracy falls. The stockout-and-expedite watch became a standing report owned by planning.

### THE KEY MATH — ONE RELIABLE LANE, ONE FAT-TAIL LANE, AND THE RECURRING VALUE (ILLUSTRATIVE UNITS; RECORD PERCENTAGES AND THE RECORD RELEASE PAIR)

| STEP | QUANTITY | FORMULA → SUBSTITUTION | BASIS | VALUE |
|---|---|---|---|---|
| 1 | Brochure-sized buffer | `SS = z·d·σ_L = 2.33 × 500 × 5 = 5,825 units` | illustrative; toolkit form | 5,825 units of folklore |
| 2 | p99 defined cumulatively | `p99 = min{d : P(L ≤ d) ≥ 0.99}` | record definition | normal shortcut understates a skewed lane by 3.4 days |
| 3 | Reliable lane re-sized | `2.33 × 500 × 1.2 = 1,398; release = 5,825 − 1,398 = 4,427` | measured σ = 1.2 vs brochure 5 (illustrative) | 4,427 units off this lane |
| 4 | Fat-tail lane | `2.33 × 500 × 6.5 = 7,573 — buffer KEPT` | measured σ = 6.5 > brochure 5 (illustrative) | kept until the source process moves |
| 5 | Accuracy gate | `LTacc = 1 − \|actual − promised\| / promised; bar 99% on 200 lanes` | record | 99% held before any cut |
| 6 | The release | `$1,500,000 / 580,000 = $2.59/unit` | record pair; average derived | 580K units / ~$1.5M, 0 stockouts |
| 7 | Recurring value | `$1.5M × 20% = $0.3M; 900 fewer late × 10% × $3K ≈ $0.27M` | modeled (benchmarks labeled) | ≈$0.6M/yr; never added to S6's $2M |

Depends on the six-site canonical model (S7) — six definitions cannot produce one σ. S7 is a pointer only in this packet; say "six sites," never "six NA sites."

**Where this lands at Amazon.** Inbound lead-time variability is half of every buffer and every promise in a placement model. Transferable: measure the lane's own distribution, promise at the tail you can defend, release capital only on evidence with the exception written into the rule, and give every parameter a drift flag. Reliability (zero stockouts), cost efficiency (expedites avoided), asset utilization (~$1.5M of working capital out).

### TWIST MAP — LIKELY LP QUESTIONS FROM THIS INTERVIEWER → THE ANGLE TO TAKE WITH S8

| LIKELY QUESTION (THIS INTERVIEWER) | LP | ANGLE — WHAT TO EMPHASIZE | SKIP / NEVER SAY |
|---|---|---|---|
| **Tell me about a time you went deeper than the request.** | Dive Deep | A "variability report" reframed as a promise-and-buffer policy; mean, spread and tail per lane from actuals; p99 defined cumulatively. | Do not claim to have built the distributions — the data team did, to my requirement. |
| **Tell me about a time the data contradicted conventional wisdom.** | Are Right, A Lot | Brochure values vs measured lanes; one right-skewed lane where the average hid the tail; planning chose p99 on the asymmetry. | S10 (Austria — "the noise said carriers") is the bar raiser's; do not use it here. |

| LIKELY QUESTION (THIS INTERVIEWER) | LP | ANGLE — WHAT TO EMPHASIZE | SKIP / NEVER SAY |
|---|---|---|---|
| **Tell me about a time you held a standard under pressure.** | Insist on the Highest Standards | No release below 99% lead-time accuracy; Finance's push for a broader release answered with the rule they signed. | Do not frame Finance as the obstacle; the exception was in their rule. |
| **How do you set safety stock / service levels?** | Functional — placement policy | Two error signals (forecast, lead time), measured at the lane × SKU grain; policy by cost asymmetry; release by rule with a watch window; drift flags. | Never present the formula as a program artifact — MIT toolkit illustration. |
| **What did you refuse to do?** | Have Backbone | Averaging away a tail; endorsing a cut below 99%; a blanket release — all written into the act before anyone asked. | Keep the $2M (S6) out of this answer. |
| **How did you make Finance comfortable releasing capital?** | Earn Trust | Finance agreed the release rule in the same room as planning signed the policy — before any cut; signed each lane; the watch armed first. | No annualized $1.5M — it is a one-time release. |

## Question bank — likely questions for this interviewer, with a one-line answer plan

Behavioral and functional questions a Placement Analytics manager is likely to ask. Each plan points at a packet story or a named backup in the master packet; stories allocated to other interviewers are pointers only.

| # | LIKELY QUESTION | ANSWER PLAN (ONE LINE) | STORY |
|---|---|---|---|
| 1 | **Walk me through how you improved forecast accuracy.** | S6 pitch — honest supply definition first, MAPE/bias at the decision grain, waves by trust; 70 → 85% on scoped class-A SKUs; planners owned the forecast, I owned its measurement. | S6 |
| 2 | **Tell me about a time you used data to change an inventory policy.** | S8 pitch — brochure → measured lanes, p99 promise, 99% bar, 580K units / ~$1.5M out with zero stockouts. | S8 |
| 3 | **How do you decide where / how much safety stock to hold?** | Two error signals measured at the grain; cost asymmetry sets the percentile; release only by signed rule with a watch; fat tails keep buffer. | S8 → S6 |
| 4 | **What is the softest assumption in your biggest result?** | S6: the expedite count (60 × $5K). S8: the late-arrival share. Say how each would be firmed from a ledger. | S6 / S8 |
| 5 | **How do you measure lead-time performance?** | Lead-time accuracy = 1 − \|actual − promised\| / promised against a per-lane p99 promise; drift flag and refresh cadence. | S8 |
| 6 | **Tell me about a time a stakeholder wanted a number that looked better.** | Planning's request to count unfirmed MRP orders as supply — refused; released planned orders only. | S6 |
| 7 | **How did you handle a dependency that slipped (MPN mapping)?** | Escalated to the sponsor as a re-sequenced date with the coverage number attached; made MPN a standing metric rather than a task. | S6 |
| 8 | **How would you decompose a network placement problem (e.g., inbound placement cost vs. speed)?** | Master §3.99 opening: book of record and metric → stocks and flows → broken invariant → design → sequence; then lead-time and forecast error at the lane × node grain as the two levers. | Master §3.99 |
| 9 | **How do you validate a model before the business acts on it?** | Parallel run against the source (daily reconciliation to SAP; manual sample per cohort); a published gate; the old process stays alive until the match holds. | S6 (S10 pointer — Josyula's) |
| 10 | **What is your quantitative background?** | MIT Supply Chain MicroMasters + Supply Chain Network Design certificate — acquired to design what I had been implementing; the formulas are that toolkit, labeled. Never "I build models for a living." | Master §4.1 toolkit line |
| 11 | **Tell me about a time you improved a process end to end (Plan → Deliver).** | S6 as the Plan redesign (definition → gate → alert → release rule) with S8 as the Source/Deliver parameter feeding it; the two programs stay separate in dollars. | S6 + S8 |
| 12 | **Another example of Dive Deep?** | Pointers only: S10 (Austria — bar raiser's) and S7 (phantom math — unallocated; no BOM-explosion content). If pressed, S7 at the level of "six sites, three waves, five months, zero auto-loaded revision overlaps." | S7 pointer |

## Cheat-numbers box — canon values with basis labels

Canon numbers for S6 and S8 with their basis labels, from the master packet's §4.1 ledger.

| METRIC | VALUE TO SAY | BASIS | NEVER |
|---|---|---|---|
| **S6 forecast accuracy** | 70% → 85% on scoped class-A SKUs | record | claiming I owned the forecast |
| **S6 safety stock** | $2M released under the both-errors-improved rule | record, one-time | "$2M per year"; adding to S8 |
| **S6 recurring** | ≈$0.8M/yr (carry 0.2 · expedites 0.3 · planner capacity 0.3); range $0.5–1.2M | modeled | presenting as record |
| **S6 timing / scope** | shortage alert live in 3 months; eight domains; six sites; adoption 8×; planner effort −50%; shipment data 60% complete at start | record | pasting "3 months" onto the six-site ingest (S7) |
| **S6 alert rule** | days-of-cover ≤ lane lead-time p90 | record mechanism | — |
| **S8 lanes / accuracy** | 200 lanes; 99% lead-time accuracy at the p99 promise | record | — |
| **S8 release** | 580K units / ~$1.5M, zero stockouts in the watch window | record, one-time | "$1.5M to $2M"; annualizing it |
| **S8 recurring** | ≈$0.6M/yr (carry 0.3 at 20% · expedites 0.3); range $0.3–1.4M | modeled | presenting as record |
| **S8 unit value** | ≈$2.59/unit | derived from the record pair | — |
| **Site count** | "six sites" | record | "six NA sites" |
| **Toolkit** | MIT Supply Chain MicroMasters + Network Design certificate; formulas are illustrations | labeled | "I build supply chain models for a living" |
| **S7 (pointer only)** | six sites, three waves, 5 months; hardest site 6 weeks vs 3-week average | record | "500+ product families"; 60→95% inside S7; any BOM-explosion walk |

**Never-say, one glance before the call.**

- Never add $2M (S6) and ~$1.5M (S8) into one number; never "$1.5M to $2M."
- Never "six NA sites" — "six sites." Never paste S6's "3 months" onto the six-site ingest.
- Never claim ownership of the demand forecast or the distributions — measurement, definitions, rule and policy were mine.
- Never present the safety-stock formula as a program artifact; MRP math at finished-goods grain only — no BOM explosion.
- Never present planner capacity as cash — time redeployed, not headcount removed.
- S10 (Austria) is the bar raiser's; S11/S12, S2/S18, S1/S3 belong to Tripathy, Goforth and Lan — pointers only here.

## Backup bench — use only if a primary doesn't fit

Four fallbacks for an analytics manager: three master-packet backups plus one doc-derived Flexport micro-story — S4 (Sephora — the in-transit promise rule I got wrong; the failure / Are Right, A Lot question, promise math at lane σ), S9 (Elementum — the 597-lane spend Pareto and the ≥95% valid-ASN gate; Dive Deep on data completeness, Backbone two levels up) and S7 (Elementum — a contract manufacturer's planning data made safe to plan on, finished-goods grain only; Learn & Be Curious by math, not meetings); the doc-derived fourth is the FFA Asia latency investigation from the Flexport program record (Dive Deep on production evidence; record-from-docs labels, facts identical to the same bench in Josyula's packet). Tell S6 and S8 first.

**Bench discipline.** The two (or four) primaries above are the stories to tell first. Reach for a bench story only when the question genuinely does not fit a primary, or when the interviewer asks for "another example" on an LP already spent. Each entry is a 60-second pitch, a compressed STAR, one results line and two probes — enough to answer and stop; no full probing bank or derivation. Where a bench story is another interviewer's primary, the red note says so: use it only if the question demands it, and keep the facts identical to that packet.

### BACKUP — ONLY IF ASKED S4 — Sephora S/4HANA — the in-transit promise rule I got wrong, fixed through the signed policy, not a side configuration

**LPs this backup can serve:** Are Right, A Lot (the failure question) · Customer Obsession (backup) · Earn Trust (owning the miss) · Learn and Be Curious — the designated policy-failure story

**Basis.** The v1 rule was mine; the indictment came from my own program's reports. Lane σ ≈ 0.3 day and the ~0.49-day window are the record's promise math; the record carries no lane counts and no dollar for the conversion lost while I was wrong — I will not invent the loss figure.

**60-SECOND PITCH — STOP ON THE PRIMARY METRIC**

**Situation.** My first ISM ruleset barred every channel from promising units in transit; the Aging In-Transit report showed dotcom starved of fast movers one day from the building. **Task.** Correct my own policy without weakening oversell protection, through the same three signatories. **Action.** Discovery: conversion-loss evidence from the aging and replenishment reports — the entry bar to reopen the policy; I stated that the v1 rule was mine. Plan & Design: promise math — lane σ ≈ 0.3 day, a ~0.49-day window ≈ 95% coverage; Promisable = in-transit × 1[ETA ≤ window] × 1[on-time ≥ floor]; Aging In-Transit as the kill switch; both blanket options rejected in writing. Build & Test: back-test against receiving history; four regression cells added to the node × channel matrix. Cutover: released on the normal monthly cadence as a signed FSD change, lane by lane — no hotfix, no side configuration. Hypercare: kill switch armed; never fired. Lesson: both sides of a rule get a number before it ships. **Result.** Primary metric: sellable units on qualifying in-transit lanes — starved → restored inside a per-lane promise window, oversell increase = zero, kill switch never fired. ≈$0.5M/yr modeled margin recovered — the record has no dollar for the conversion lost while I was wrong, and I will not invent one.

**COMPRESSED STAR**

- **S —** v1 of my own ISM ruleset: no channel could promise in-transit units — safe-feeling, and starving dotcom of fast movers reliably one day from the building.
- **T —** Fix my own signed policy without weakening oversell protection; take the miss back through the same three signatories.
- **A —** Evidence pack from Aging In-Transit and Replenishment; reframed as promise math (σ ≈ 0.3 d → ~0.49-d window ≈ 95%); rule with on-time floor and kill switch; back-tested against receiving history; four regression cells; released lane by lane on the monthly cadence as an FSD change.

- **R —** Starved lanes restored inside the window; oversell increase = 0; kill switch never fired; re-signed by the same three signatories; lesson institutionalized — the Locus 98% gate (S2) and the 0.5% tolerance (S18) were set from data before the event.

## RESULTS LINE

Sellable units on qualifying in-transit lanes starved → restored inside a back-tested ~0.49-day window (record) · oversell increase = 0; kill switch armed, never fired (record) · four regression cells added (record) · released as a signed FSD change on the normal cadence, no hotfix (record) · ≈$0.5M/yr modeled margin recovered (inputs illustrative).

## THE MATH — STEP-BY-STEP DERIVATION

### Table 1 — before: the problem in numbers.

| STEP | QUANTITY | FORMULA → SUBSTITUTION | INPUTS (BASIS) | VALUE |
|---|---|---|---|---|
| 1 | The v1 rule | `Promisable = 0 for ALL in-transit inventory` | equivalent to claiming σ → ∞ or on-time rate → 0 — infinite variability (record) | everything in transit hidden |
| 2 | One concrete lane | `Promisable = 120 × 0 = 0` | lane with InTransit = 120 units, ETA 0.4 day away (illustrative) | 120 sellable units hidden |
| 3 | Across qualifying lanes | `≈300 fast-mover units/day × 0 promisable` | illustrative volume | ≈300 units/day sat unsellable one day from the building |
| 4 | What starving costs (one SKU) | `100 × 60% × $20 = $1,200/day` | 100 units one day out; 60% would convert; $20 margin (illustrative, not program data) | $1,200/day under-sell |
| 5 | The ignored other side | `100 × (1 − 0.97) = 3 late units; 3 × $15 = $45/day` | on-time rate 97%; $15 expedite (illustrative) | $45/day over-sell exposure — vs $1,200: the “safe” rule priced one side only |

### Table 2 — the fix: step by step, values substituted.

| STEP | QUANTITY | FORMULA → SUBSTITUTION | INPUTS (BASIS) | VALUE |
|---|---|---|---|---|
| 1 | Measure lane variability | `σ = 0.3 day` | trailing receiving history (record) | σ = 0.3 day |
| 2 | Choose coverage | `z at 95% = 1.645` | normal approximation on lane transit (toolkit label) | z = 1.645 |
| 3 | Promise window | `w = z × σ = 1.645 × 0.3 = 0.4935` | rows 1–2 | w ≈ 0.49 day |
| 4 | Coverage check | `Φ(w / σ) = Φ(0.49 / 0.3) = Φ(1.63) ≈ 0.95` | share of arrivals inside the window | ≈95% covered |
| 5 | Corrected rule | `Promisable = InTransit × 1[ETA ≤ w] × 1[OTR ≥ floor]` | OTR = on-time rate over trailing receipts; floor = per-lane admission threshold, 95% here (floor illustrative) | lane by lane, not all-or-nothing |
| 6 | Lane that qualifies | `ETA 0.4 ≤ 0.49 → 1; OTR 97% ≥ 95% → 1; Promisable = 120 × 1 × 1 = 120` | the Table 1 lane, re-evaluated (ETA and OTR illustrative) | its 120 units become promisable |
| 7 | Lane that fails | `ETA 1.2 > 0.49 → 0; Promisable = 80 × 0 = 0` | a slower lane with 80 units in transit (illustrative) | stays at 0 — correctly out |
| 8 | Sample-size gate | `n = z²p(1−p)/e² = 3.84 × 0.95 × 0.05 / 0.05² = 3.84 × 0.0475 / 0.0025 ≈ 73` | OTR estimated within ±5% at 95% CI (illustrative) | a lane qualifies after ~73 receipts |

| STEP | QUANTITY | FORMULA → SUBSTITUTION | INPUTS (BASIS) | VALUE |
|---|---|---|---|---|
| 9 | Kill switch, values in | `trigger = 2 × p90 = 2 × 2.0 = 4.0 days; any unit in transit > 4.0 days → lane out` | lane p90 = 2.0 days (illustrative); armed, reviewed weekly by Inventory Control (record) | never fired (record) |
| 10 | Back-test | `rule replayed against receiving history; 4 regression cells` | before release (record) | no oversell in the back-test |

**Table 3 — after: the metric, before → after.**

| STEP | METRIC | BEFORE | AFTER | BASIS |
|---|---|---|---|---|
| 1 | Sellable units on qualifying in-transit lanes | 0 | ≈300 units/day restored inside the window | record mechanism; volume illustrative |
| 2 | Promise window | none — blanket exclusion | w ≈ 0.49 day at ~95% cover | record (promise math) |
| 3 | Oversell increase from the window | — | 0 | record |
| 4 | Kill switch | not armed | armed, never fired | record |
| 5 | Rule changes after | ad hoc | same 3 signatories, back-test + matrix regression | record — both sides priced from then on |

**Table 4 — the dollars, every multiplication shown.**

| STEP | QUANTITY | FORMULA → SUBSTITUTION | INPUTS (BASIS) | VALUE |
|---|---|---|---|---|
| 1 | Would-be sales | `300 × 60% = 180 sales/day` | units restored (illustrative) × conversion (illustrative) | 180 sales/day |
| 2 | Truly lost without the fix | `180 × 40% = 72 sales/day` | 40% lost-sale share — the rest would have waited (illustrative) | 72 lost sales/day |
| 3 | Margin recovered | `72 × $20 = $1,440/day` | $20 unit margin (illustrative) | ≈$1,440/day |
| 4 | Annualized | `$1,440 × 365 = $525,600` | modeled; the mechanism is record | ≈$0.5M/yr |
| 5 | Cost while v1 stood | `no dollar in the record — not claimed` | record | not claimed |

## TWO PROBES

> **Why not just hotfix it — you knew it was wrong?**
>
> Because the error was in the policy, so the fix had to be too — in the governed artifact, with regression, through the same signatures. A side configuration would have fixed the symptom and left the policy lying about what it allowed.

> **What did the miss cost?**
>
> The record carries no dollar for the conversion lost while v1 stood, and I will not invent one. What I can say is the mechanism: fast movers reliably one day from the building were unsellable on dotcom until the window existed; the ≈$0.5M/yr recovered is my model on illustrative inputs, labeled.

**Canon numbers (basis labels):** Lane σ ≈ 0.3 d; ~0.49-d window ≈ 95% coverage (record promise math) · oversell increase = 0 (record) · 4 regression cells (record) · kill switch never fired (record) · same three signatories re-signed (record) · ≈$0.5M/yr (modeled). Never a lane count; never a dollar for the loss while wrong.

## BACKUP — ONLY IF ASKED S9 — Elementum — 1,000 supplier/carrier routes asked, 597 built first: X12 EDI onboarding behind a ≥95% valid-ASN cohort gate

**LPs this backup can serve:** Have Backbone; Disagree & Commit (primary — 597 vs 1,000, two levels up) · Deliver Results · Insist on the Highest Standards (the gate held twice) · Think Big (the staged 1,000-lane vision) · Broad Responsibility (small carriers kept inside the standard by the portal floor)

**Basis.** I owned the scope analysis and the onboarding standard, influencing an executive two levels above my sponsor; engineers built the maps and harness to the standard; Transportation owned the carrier relationships. Cohort sizes and the number of cohorts are not in the record — do not quote them.

### 60-SECOND PITCH — STOP ON THE PRIMARY METRIC

**Situation.** The Global Head of Supply Chain wanted inbound visibility on 1,000 lanes in two months; shipment-data completeness was 60% and high-value lanes were blind. **Task.** I owned the scope analysis and the onboarding standard, influencing an executive two levels above my sponsor. **Action.** Discovery: a spend Pareto — 597 lanes carried 80% of freight spend — pre-validated with Transportation and Procurement before I took it up. Plan & Design: the onboarding standard (X12 850/855/856/940/945/214/997 with an XML/portal floor — "the standard was the data, not the transport") and a cohort gate of ≥95% valid ASN. Build & Test: a mapping-and-test harness and a scorecard per lane. Cutover: 597 live in month one; one cohort failed the gate twice on a timestamp format and I held it both times. Hypercare: completeness scorecards to 95%; ~200 lanes the next wave behind the same gate. Lesson: automating dirty data manufactures confident wrong answers — the gate is the product. **Result.** Primary metric: inbound shipment-data completeness 60% → 95%. Freight spend −10% ≈ $2M in six months (record); 597 lanes = 80% of spend live in month one; adoption 8×.

### COMPRESSED STAR

- **S —** 1,000 global lanes asked in two months — scope, schedule and quality all fixed; completeness 60%; lanes differed by orders of magnitude in spend and carriers differed in EDI capability.
- **T —** Own the scope analysis and the onboarding standard; take the re-scope two levels up with the evidence attached.
- **A —** Spend Pareto (597 = 80% of spend) pre-validated so no lane owner could contradict it in the room; standard + ≥95% valid-ASN gate published to carriers before cohort one; harness so later cohorts were mostly configuration; held a failing cohort twice; ~400 lanes carrying ~20% of spend deferred — priced in spend share, not lane count.
- **R —** Completeness 60 → 95%; 100% visibility on high-value lanes; 597 lanes live month one; freight −10% ≈ $2M in six months (record); method written up as the sourcing SOP.

### RESULTS LINE

Completeness 60% → 95% (record) · 597 of 1,000 lanes = 80% of freight spend live in month one (record) · freight spend −10% ≈ $2M in six months (record — kept as measured, never annualized) · adoption 8× (record) · ≈$0.4M/yr document-handling labor (modeled add-on).

### THE MATH — STEP-BY-STEP DERIVATION

**Table 1 — before: the problem in numbers.**

| STEP | QUANTITY | FORMULA → SUBSTITUTION | INPUTS (BASIS) | VALUE |
|---|---|---|---|---|
| 1 | Completeness | `480 / 800 = 60%` | expected = 200 shipments × 4 milestones = 800; received 480 (illustrative volumes; 60% record) | 60% — 2 of 5 events missing |

| STEP | QUANTITY | FORMULA → SUBSTITUTION | INPUTS (BASIS) | VALUE |
|---|---|---|---|---|
| 2 | What blindness costs | `off-guide spend unseen · premium modes by habit · chase calls per late shipment` | record problem | no instrument, no lever |
| 3 | Manual documents | `597 lanes × ~50 shipments × 3 docs ≈ 90,000 docs/yr keyed by hand` | illustrative volumes | ~90K docs/yr |

**Table 2 — the fix: step by step, values substituted.**

| STEP | QUANTITY | FORMULA → SUBSTITUTION | INPUTS (BASIS) | VALUE |
|---|---|---|---|---|
| 1 | Spend Pareto | `S(k) = cumulative spend share; S(597) = 80%` | lanes sorted by annual freight spend (record) | 597 of 1,000 lanes carry 80% of spend — do them first |
| 2 | Cohort gate | `validASN rate ≥ 95% in test before activation` | published to carriers before cohort one (record) | one cohort failed twice on a timestamp format — held both times |
| 3 | Long-tail floor | `XML / portal when EDI is absent` | “the standard was the data, not the transport” (record) | small carriers kept inside the standard |
| 4 | Completeness after | `760 / 800 = 95%` | illustrative volumes; 95% record | 95%; high-value lanes 100% |

**Table 3 — after: the metric, before → after.**

| STEP | METRIC | BEFORE | AFTER | BASIS |
|---|---|---|---|---|
| 1 | Shipment-data completeness | 60% | 95%; high-value lanes 100% | record |
| 2 | Lanes live month one | 0 | 597 = 80% of spend | record |
| 3 | Freight spend on onboarded lanes | base | −10% ≈ $2M in 6 months | record |
| 4 | Document handling | ~90K docs/yr keyed | X12/XML validated | modeled |

**Table 4 — the dollars, every multiplication shown.**

| STEP | QUANTITY | FORMULA → SUBSTITUTION | INPUTS (BASIS) | VALUE |
|---|---|---|---|---|
| 1 | Implied addressable base | `$2,000,000 / 10% = $20,000,000/yr` | derived from the record pair, stated as such | ≈$20M/yr |
| 2 | The record result | `$20M × 10% = $2M in 6 months` | record | ≈$2M |
| 3 | Slice — routing-guide compliance | `~$0.8M of the $2M` | off-guide spend caught by 856/214 (modeled split) | ~$0.8M |
| 4 | Slice — mode discipline | `~$0.6M of the $2M` | premium modes → ocean/ground when ETA is trusted (modeled split) | ~$0.6M |
| 5 | Slice — expedites + detention | `~$0.6M of the $2M` | events replace chase calls (modeled split) | ~$0.6M |
| 6 | Split check | `0.8 + 0.6 + 0.6 = $2.0M` | the slices decompose the record — they never add to it | = the record $2M |
| 7 | Document labor | `90,000 × 6 min = 540,000 min; / 60 = 9,000 h; × $45 = $405,000` | illustrative volumes (modeled add-on) | ≈$0.4M/yr |

## TWO PROBES

> **You defied a Global Head of Supply Chain's number. How?**
>
> With the spend curve and the lane owners' validation attached — 597 lanes carried 80% of the spend, so the phased plan was argued in spend share, not lane count. He accepted the phasing off the curve; I then committed to the remainder behind the same gate and delivered it.

> **Is the $2M annualized?**
>
> No — it is the record as measured: minus ten percent on the onboarded lanes' spend in six months. My three-way split (routing-guide compliance ~$0.8M, mode discipline ~$0.6M, expedites and detention ~$0.6M) decomposes the record; it never adds to it.

**Canon numbers (basis labels):** 60% → 95% completeness (record) · 597 / 1,000 lanes = 80% of spend (record) · −10% ≈ $2M in 6 months (record) · ≥95% valid-ASN gate, held twice (record) · 8× adoption (record) · ≈$0.4M/yr labor (modeled). Never: cohort counts or sizes; never double the $2M into a run-rate.

## BACKUP — ONLY IF ASKED S7 — Elementum — making a contract manufacturer's planning data safe to plan on: six sites, three waves, five months, zero auto-loaded revision overlaps

**LPs this backup can serve:** Learn and Be Curious (primary — learned the CM's data by math, not meetings) · Dive Deep · Earn Trust (external partner)

**Basis.** I owned the ingest requirements, the reconciliation gate, the wave plan and the weekly Flextronics call; engineers built the normalization layer and quarantine to the spec; the commercial relationship belonged to the customer's procurement — my leverage was evidence. The $1.5M / 580K belong to the p99 program this backbone enabled (S8). Finished-goods grain only here; no tolerance value — the record does not carry one.

### 60-SECOND PITCH — STOP ON THE PRIMARY METRIC

**Situation.** Flextronics supplied the planning data our component-demand visibility depended on, and it carried two traps documentation never confesses: pass-through assemblies that double-time demand if loaded as stocked, and overlapping revisions that rewrite history if you auto-load one. Six site systems carried six definitions of yield, lead time and order status. **Task.** I owned the ingest requirements, the reconciliation gate, the wave plan and the weekly Flextronics call — with no commercial leverage; evidence was my only lever. **Action.** Discovery: sampled finished goods and reconciled the CM's planning data against its own MRP — the trap signature found by math, named in a discovery memo. Plan & Design: an explicit marker for pass-through items; an effective-date quarantine with human disposition — never auto-load; one supply equation (on-hand + production plan + released planned orders) signed by customer planning. Build & Test: mapping tables owned by named site super-users; a per-site reconciliation report as the go-live gate. Cutover: three waves — two cleanest high-volume sites, then three on parameterized maps, then the hostile legacy ERP by snapshot-diffing at six weeks against a three-week average, priced up front. Hypercare: the weekly call left standing — it had become theirs. Lesson: reconcile to the source's math, not its documentation. **Result.** Primary metric: contract-manufacturer sites with planning-grade data — 0 → 6 in three waves and five months, zero auto-loaded revision overlaps. ≈$0.3M/yr modeled — the six sites and zero overlaps are measured; the dollars are my model; the $1.5M belongs to S8.

### COMPRESSED STAR

- **S —** A contract manufacturer's planning feed with two silent traps; six sites, six definitions; one hostile legacy ERP (flat files, no change-data-capture); no commercial leverage.
- **T —** Own ingest requirements, the reconciliation gate, the 2/3/1 wave plan, and chair the weekly Flextronics call.
- **A —** Found the traps by sampling finished goods against the CM's own MRP; designed the two mechanisms that made the traps impossible rather than policed (explicit marker; effective-date quarantine, human disposition); reconciliation report as the per-site gate, not a date on a slide; wave 3's six weeks priced to leadership as the plan, not a slip; made the weekly call the CM's own protection against blame for misses its feed didn't cause.
- **R —** Planning-grade sites 0 → 6 in 3 waves / 5 months; zero auto-loaded overlaps ever; phantom double-timing eliminated at source; backbone handed to the 200-lane program (S8).

### RESULTS LINE

CM sites with planning-grade data 0 → 6 in three waves / five months (record) · auto-loaded revision overlaps = 0, ever (record) · hardest site 6 weeks vs the 3-week average, priced up front and absorbed in plan (record) · ≈$0.3M/yr (modeled — reconciliation labor, expedites fired at false shortages, overlap rework); the record has no dollar for this story; the $1.5M belongs to S8.

### THE MATH — STEP-BY-STEP DERIVATION

**Table 1 — before: the problem in numbers.**

| STEP | QUANTITY | FORMULA → SUBSTITUTION | INPUTS (BASIS) | VALUE |
|---|---|---|---|---|
| 1 | Planning-grade CM sites | `0 of 6; manual BOM-vs-MRP reconciliation at every site` | record | 0 of 6 |
| 2 | Manual recon burn | `6 h/wk × 6 sites = 36 h/wk` | modeled | 36 h/wk of reconciliation |
| 3 | What MRP computes (finished-goods grain) | `NetReq(FG, t) = GrossReq − OnHand − SchedReceipts` | the netting identity; walk below at FG grain only (illustrative) | substituted in rows 4–5 |
| 4 | Clean row | `NetReq = 500 − 150 − 200 = 150` | Gross 500; on-hand 150; receipts 200 (illustrative) | NetReq = 150; need-date = due − LT_FG |
| 5 | Distorted row — phantom FG record | `NetReq = 600 − 150 − 200 = 250; false shortage = 250 − 150 = 100` | a phantom/stale finished-goods record inflates gross requirements by 100 (illustrative) | an expedite fires on 100 units never needed |
| 6 | How often | `~5/site/mo × 6 sites × 12 mo = 360 false shortages/yr` | modeled count | 360/yr |
| 7 | Timing corruption | `the same stale record pulls the need-date a lead-time early` | record mechanism | planners chase demand that was never real |
| 8 | Revision overlaps | `two revisions both partially true in [d0, d1]; ~24/yr` | auto-loading would rewrite demand for orders released under the prior revision (record rule; count modeled) | ~24 history-rewrite risks/yr |

**Table 2 — the fix: step by step, values substituted.**

| STEP | QUANTITY | FORMULA → SUBSTITUTION | INPUTS (BASIS) | VALUE |
|---|---|---|---|---|
| 1 | Explicit marker | `distorted row reconciled: NetReq = 500 − 150 − 200 = 150` | the phantom record carries an explicit marker; it no longer inflates gross (record mechanism) | NetReq = 150 — matches the site's own MRP |
| 2 | Per-site gate | `count_model vs count_MRP on sampled finished goods must match` | the reconciliation report is the go-live gate, not a date on a slide (record) | a site opens on a matched report |
| 3 | Quarantine | `overlaps held for human disposition per order/effectivity` | record rule | auto-loaded overlaps = 0 — ever |
| 4 | Hostile flat-file site | `completeness = rows received / expected; a 40%-rows file would read as 60% deletions → blocked` | snapshot-diffing with a completeness flag (record mechanism) | partial files never consumed |
| 5 | Wave plan | `W1: 2 × 3 = 6 wk; W2: 3 × 3 = 9 wk; W3: 6 wk; 6 + 9 + 6 = 21 weeks` | hostile site at 6 wks vs the 3-wk average, priced up front (record) | ≈5 months for six sites |
| 6 | Recon labor after | `6 → 1.5 h/wk per site — disposition queue only` | modeled | −4.5 h/wk per site |

**Table 3 — after: the metric, before → after.**

| STEP | METRIC | BEFORE | AFTER | BASIS |
|---|---|---|---|---|
| 1 | Planning-grade CM sites | 0 of 6 | 6 of 6, in 3 waves ≈ 5 months | record |
| 2 | Manual reconciliation | 6 h/wk × 6 sites | 1.5 h/wk × 6 | modeled |
| 3 | Expedites on false shortages | ~72/yr acted on | ~0 — the marker removes the distortion | modeled |
| 4 | Auto-loaded revision overlaps | uncontrolled | 0 — ever | record |

**Table 4 — the dollars, every multiplication shown.**

| STEP | QUANTITY | FORMULA → SUBSTITUTION | INPUTS (BASIS) | VALUE |
|---|---|---|---|---|
| 1 | Reconciliation labor | `(6 − 1.5) = 4.5 h/wk; × 6 sites = 27 h/wk; × 48 wk = 1,296 ≈ 1,300 h; × $60 = $77,760` | modeled | ≈$0.08M/yr |
| 2 | False-shortage expedites | `360 × 20% = 72 acted on; 72 × $3,000 = $216,000` | count from Table 1 row 6; premium benchmark $2–5K (modeled) | ≈$0.2M/yr |
| 3 | Overlap rework avoided | `24 × 16 h = 384 h; 384 × $60 = $23,040` | the quarantine makes the error 0 by construction (modeled) | ≈$0.02M/yr |
| 4 | Total | `0.08 + 0.2 + 0.02 = $0.3M` | the record has no dollar for S7; the $1.5M belongs to S8 | ≈$0.3M/yr |

## TWO PROBES

> **How did you get an external supplier with no obligation to care?**
>
> By making the weekly call theirs: it protected Flextronics from blame for misses their feed didn't cause. Procurement held the commercial relationship; my lever was the evidence in the discovery memo, confirmed by their own engineering.

> **Why three waves instead of one?**
>
> So one surprise could not spend six sites of credibility — the two cleanest high-volume sites first, three on parameterized maps second, the hostile ERP last with its six weeks priced up front. Speed was traded, never quality: a site went live off its matched reconciliation report, not a date.

**Canon numbers (basis labels):** 0 → 6 sites, 3 waves, 5 months (record) · zero auto-loaded overlaps (record) · 6 weeks vs 3-week average (record) · ≈$0.3M/yr (modeled). Never "six NA sites"; never "500+ product families"; 60 → 95% belongs to S9; the $1.5M / 580K belong to S8; no tolerance value.

## BACKUP — ONLY IF ASKED Doc bench — Flexport FFA — the Asia latency investigation: a 21-second P95 traced to routing, decomposed to the layer, fixed in sequence (Dive Deep)

**LPs this backup can serve:** Dive Deep (primary — decomposition to the layer) · Deliver Results · Learn & Be Curious · Insist on the Highest Standards (a smoke-test protocol per change)

**Basis.** Doc-derived micro-story from the program record (the FFA Asia latency investigation, post-MVP 2022) — every figure is RECORD-FROM-DOCS: the investigation document, written when the FF App MVP had launched to 11 partners in 10 countries and 48 of 53 import partners (90.5%) had migrated. Program-lead voice: I owned the investigation, the decomposition and the fix sequence; infrastructure teams executed the WAF and CDN changes. Not in the master packet's S-numbering; supplements canon, never overrides it. The same bench appears in Josyula's packet — facts identical.

### 60-SECOND PITCH — STOP ON THE PRIMARY METRIC

**Situation.** Weeks after the FF App MVP launched to 11 partners in 10 countries — with 48 of 53 import partners already migrated — the monthly business review showed Asia page loads at a P95 of 21.19 seconds against 3.3 in the US. **Task.** Find the layer, not the anecdote, and fix it without a re-architecture in the middle of a live partner-facing launch. **Action.** I decomposed the problem before prescribing: static assets ran ~5s P95 in China against ~500ms in the US, and the app's domain had no geo-based DNS or reverse proxy — Asia traffic was riding the public internet across the Pacific; a double CDN sat on the static path. The fix was a sequence, not a project: enable the WAF, remove the double CDN (direct to S3), then a regional CDN proposal with a smoke-test protocol per change, then China tuning, then Asia tuning — each with a date and an owner. **Result.** Root cause isolated to routing — not the app, not the partners; two fixes shipped inside the investigation window and the regional plan scheduled with dates; and a standing rule imported into every later launch: the pilot's geography is a performance requirement, not a rollout detail. RECORD-FROM-DOCS

### COMPRESSED STAR

- **S —** Post-MVP, the MBR surfaced Asia P95 page load at 21.19s vs 3.3s in the US (17-day measurement window); 48 of 53 import partners (90.5%) already migrated — the app had to hold where the partners were.
- **T —** Isolate the layer causing Asia latency and sequence the fix set without destabilizing a live partner-facing launch.
- **A —** Split static from dynamic: static JS ~5s P95 in China vs ~500ms US; found no geo-based DNS / reverse proxy on the domain (public-internet routing) and a double CDN on static assets; sequenced WAF enablement → double-CDN removal (direct to S3) → regional CDN proposal → China tuning → Asia tuning, each gated by a smoke-test protocol with a named date and owner.
- **R —** Root cause isolated to routing; WAF and double-CDN fixes shipped inside the window; CDN / China / Asia work scheduled with owners; the investigation became the template for launch-geography checks.

### RESULTS LINE

P95 page load 21.19s Asia vs 3.3s US (record-from-docs) · static ~5s China vs ~500ms US (record-from-docs) · 48 of 53 partners migrated = 90.5% (record-from-docs; canon's "48 of 53 within a month" — consistent) · two fixes shipped, three sequenced with dates (record-from-docs) · no post-fix latency figure exists in the record — never invent one.

### THE MATH — STEP-BY-STEP DERIVATION

**Table 1 — before: the problem in numbers.**

| STEP | QUANTITY | FORMULA → SUBSTITUTION | INPUTS (BASIS) | VALUE |
|---|---|---|---|---|
| 1 | Page-load P95 by region | `Asia = 21.19s; US = 3.3s; Others = 4.45s` | Datadog @view.loading_time, 04/24–05/10 window (record-from-docs) | Asia is the outlier |

| STEP | QUANTITY | FORMULA → SUBSTITUTION | INPUTS (BASIS) | VALUE |
|---|---|---|---|---|
| 2 | The gap factor | `21.19 ÷ 3.3 = 6.42` | row 1 (record-from-docs) | 6.42× slower than the US |
| 3 | Traffic behind the number | `6.31 ÷ 2.19 = 2.88` | Asia 6.31k vs US 2.19k page views, same window (record-from-docs) | ≈2.9× the US traffic riding the worst latency |
| 4 | Static-asset leg | `JS P95 = 5s China vs 500ms US; 5 ÷ 0.5 = 10` | static.flexport.com CDN behavior (record-from-docs) | 10× gap on the static leg |
| 5 | Dynamic leg | `no geo-DNS / reverse proxy for trade.flexport.com → Asia GraphQL hairpins over the public internet` | infra review (record-from-docs) | routing penalty on every data call |
| 6 | Known defect on the static path | `double CDN in China — Cloudflare stacked in front of Tencent` | infra review (record-from-docs) | one avoidable hop |

**Table 2 — the fix: step by step, values substituted.**

| STEP | QUANTITY | FORMULA → SUBSTITUTION | INPUTS (BASIS) | VALUE |
|---|---|---|---|---|
| 1 | Decompose before touching | `page load = static resources + GraphQL requests + JS rendering` | the metric's own definition (record-from-docs) | three legs, measured and owned separately |
| 2 | Step 1 — security gate first | `enable WAF for trade.flexport.com: China, then global` | incident-207 follow-up; done 2022/06/30 (record-from-docs) | shipped — fix 1 of 2 |
| 3 | Step 2 — remove the double CDN | `Tencent CDN → direct to the S3 bucket; Cloudflare hop removed for China` | done 2022/06/24 (record-from-docs) | shipped — fix 2 of 2 |
| 4 | Step 3 — static proposal, CN/Asia | `alternatives written, reviewed, signed off` | ETA 2022/07/15 (record-from-docs) | sequenced with a date |
| 5 | Step 4 — China tuning | `e.g. replicate static assets in-region` | ETA 2022/07/30; depends on step 4 (record-from-docs) | sequenced with a date |
| 6 | Step 5 — Asia tuning | `the same playbook beyond China` | ETA 2022/08/15 (record-from-docs) | sequenced with a date |
| 7 | Held in reserve | `global CDN for trade.flexport.com — built only if the metrics still demand it` | 2022 Q3 or later, metrics-gated (record-from-docs) | no speculative spend |
| 8 | Verification per change | `2 user types × 2 instances = 4 smoke passes before + 4 after = 8` | login → workview → logout → re-login; internal SSO + partner users; US + China hosts (record-from-docs; the multiplication counts the checklist) | 8 workflow passes around every change |

**Table 3 — after: the metric, before → after.**

| STEP | METRIC | BEFORE | AFTER | BASIS |
|---|---|---|---|---|
| 1 | Root cause | one number — "Asia is slow" | static leg (10×) and dynamic routing isolated, each with an owner and a sequence | record-from-docs |
| 2 | Fixes | none | 2 shipped (WAF; double-CDN removal); 3 sequenced with dates (07/15, 07/30, 08/15) | record-from-docs |
| 3 | Post-fix P95 | 21.19s | no post-fix figure exists in the record — never claimed | record-from-docs |

| STEP | METRIC | BEFORE | AFTER | BASIS |
|---|---|---|---|---|
| 4 | Spend discipline | — | global CDN gated on metrics, not built on speculation | record-from-docs |

**Table 4 — the dollars, every multiplication shown.**

| STEP | QUANTITY | FORMULA → SUBSTITUTION | INPUTS (BASIS) | VALUE |
|---|---|---|---|---|
| 1 | Partner base exposed | `48 ÷ 53 = 0.9057 ≈ 90.5% of import partners already migrated when the gap surfaced` | migration tracker (record-from-docs) | 90.5% of partners on the affected app |
| 2 | Excess wait served per window | `gap vs US = 21.19 − 3.3 = 17.89s; 6,310 × 17.89 = 112,886 partner-seconds; ÷ 3,600 ≈ 31.4 hours` | 6.31k Asia page views in the 17-day window (record-from-docs); gap arithmetic illustrative | ≈31.4 partner-hours of excess wait every 17 days |
| 3 | Dollar value of the wait | `no dollar in the record — not claimed` | record-from-docs | not claimed |
| 4 | Post-fix saving | `no post-fix latency figure exists → no savings chain exists` | record-from-docs | never invent one |

## TWO PROBES

### How did you know it was routing and not the application?

By decomposition before remediation: the static/dynamic split put ~5s of the China time on assets the app server never touched, and the same assets served in ~500ms from the US — the code was constant, the path was not. The domain's missing geo-DNS / reverse proxy and the double CDN explained both halves of the gap without a single application change.

### Why not just stand up a China region?

A region is a re-architecture with a compliance tail; the evidence said the bottleneck was routing and CDN topology, which fixes in weeks. The sequenced plan — WAF, double-CDN removal, regional CDN, tuning — bought most of the win at a fraction of the risk, and the smoke-test protocol meant each step was reversible.

**Canon numbers (basis labels):** 21.19s vs 3.3s P95 (record-from-docs) · ~5s vs ~500ms static (record-from-docs) · 48/53 = 90.5% migrated (record-from-docs) · 11 partners / 10 countries MVP (record — matches S12 canon) · fix sequence WAF → double-CDN → regional CDN → China → Asia with smoke tests (record-from-docs). Never: a post-fix latency number (none in the record); never present scheduled items as delivered.

INTERVIEW-PREP PACKET 3 OF 5 · PER-INTERVIEWER · BUSINESS-PROCESS LENS

# Amazon NACF Supply Chain loop — Packet 3: Rocky Goforth

Amazon — Senior Supply Chain Manager, NACF Supply Chain (L6)

## Rocky Goforth

Senior Manager, Infrastructure — NACF

**Thursday, October 30, 2026 · 10:00 am PDT · 55-minute video interview**

> **Strategy in one line.** An infrastructure owner wants to know what happens when the plan meets the floor. Open with S2 (one comingle pool instead of two warehouses, a robotics ramp behind a published 98% gate that fired at 94, two go-lives with zero customer-facing incidents) and hold S18 (dual-DC WMS migration: 0.5% tolerance published before mock one, 7% → 1.8% → 0.3%, the RF-device miss owned) for the go-live-gone-wrong / Bias for Action question. Gates, invariants, triggers and hypercare numbers are the vocabulary; vendors configured, DC Ops ran the floor.

**LP focus — my guess, not confirmed:** Insist on the Highest Standards · Bias for Action · Deliver Results or Ownership. Amazon does not disclose the assignment; the guesses are reasoned from the interviewer's title and the JD in "Read me first" and should be held loosely. No interviewer biography is asserted anywhere in this packet — only title-based inference, labeled as such.

Canon: amazon/amazon-l6-scm-master-packet.html (§1.2 roster, §4.1 ledger). Every number below is that packet's value; every dollar carries its basis label (record · user anchor · modeled · priced).

## Read me first

### The interviewer's lens (inference from title + JD — not a biography)

**Inference from the title.** "Infrastructure" in a fulfillment network means the physical and systems substrate the process runs on — buildings, automation and robotics, the WMS/WCS layer, capacity and launches, cutovers and peak readiness. A Senior Manager there is accountable for changes that touch the floor: go-lives that cannot drop a shipment, automation ramps that must not erode throughput, and migrations where inventory accuracy cannot wobble. He will care about gates, rollback, freeze windows, hypercare exits, and what happened when the plan met the floor.

**What that means for the answers.** The two Sephora distribution-center programs: S2 (two systemic warehouses in one building consolidated into one comingle pool, with a WMS upgrade and a robotics ramp on the same critical path and zero customer-facing incidents allowed) and S18 (a dual-DC legacy-WMS migration run to a 0.5% reconciliation tolerance published before the first mock, with the RF-device miss owned). Tell them as Make/Deliver process redesigns — one inventory pool instead of two, a cutover discipline instead of a date — with asset utilization (+15% processing efficiency, labor-hours, capital pooled) and reliability (zero incidents, 0.3% against 0.5%) as the results. The vendor teams configured BlueYonder and Locus; DC Ops ran the floor; I owned the model, the staging, the gates and the go/no-go.

### Three guessed LP focus areas, with rationale

> **Labeled guess.** Amazon assigns 2–3 Leadership Principles per interviewer and does not disclose them. These three are inferred from the title and the JD; if the first question lands on a different LP, the twist maps under each story carry the pivot.

| GUESSED LP | WHY THIS INTERVIEWER WOULD CARRY IT | STORY TO OPEN · BACKUP |
|---|---|---|
| **Insist on the Highest Standards** | Infrastructure owners live by gates that can stop the program: the 98% robotics gate that fired at 94 (S2) and the 0.5% tolerance published with Finance before mock one (S18) are the primary and backup for this LP in the master map. | S2 (open) · S18 (backup) |
| **Bias for Action** | When the floor breaks the plan — a quarter of the RF devices failing the conversion script at cutover — he wants to see a rehearsed contingency fire and a decision taken the same day with the cost priced. | S18 (the paper-pick fallback) · S2 (the throttle to ~30%) |
| **Deliver Results / Ownership (one of the two)** | Two go-lives, peak untouched, +15% signed by DC Ops; an eight-month migration on time and ~3% under budget with a $20M pipeline unblocked. Ownership shows in the RF miss carried with my name on the cause. | S2 (the comingle launch) · S18 (8 months, both DCs) |

### Answer strategy — business first, systems as enablers

- **Open with the process problem and its cost, not the system.** First sentence names the broken process (Plan, Source, Make, Deliver or Return), who it hurt (customer, Finance, the floor) and the baseline number. The platform appears only as the enabler of the redesign, in one clause.
- **Use the JD's four outcome words as the results frame:** reliability (did the promise hold?), responsiveness (how much earlier did we see it?), cost efficiency ($ with its basis label), asset utilization (capital, labor-hours, capacity released).

- **Action = operating-model redesign in gated phases** — Discovery → Plan & Design → Build & Test → Cutover → Hypercare, each spoken as entry bar · what I delivered · exit bar. Say the gate that fired, not the meeting that happened.
- **Every dollar carries its basis word in the same breath** — record · user anchor · modeled · priced. A record dollar is said flat; a modeled dollar is said with its inputs. Never promote a modeled number to record.
- **Program-lead voice.** I owned definitions, rules, gates and the signature chain; engineers, SIs and vendor teams configured and built to the spec I signed. The only platform personally configured in canon is Manhattan DOM.
- **The pitch ends on the primary metric.** Then stop. Silence after the pitch is the interviewer filling their STAR worksheet; their probes are the second half of the answer.
- **For Infrastructure specifically:** speak in gates, invariants and triggers — the published ramp gate and the throttle built in advance; the six cutover invariants (Σ AFS conserved; no SKU sellable in two places; negatives cleared first); rollback triggers as numbers (sync < 99% for 4 h; p95 > 2× for 1 h); hypercare exit as a number (< 2 blockers/day sustained). A gate that cannot stop the program is decoration.
- **Asset utilization is the result frame:** +15% units per labor-hour (signed), one pool instead of two (√2 pooling, modeled and labeled), self-transfers inside one roof eliminated, robotics at full volume by week three, $20M of pipeline unblocked on the date.

**Loop-consistency rule.** One question, one story; never the same story to two interviewers — interviewers compare notes. Held for other packets (do not spend here): S11/S12 (Tripathy) · S6/S8 (Hoerger) · S1/S3 (Lan) · S16/S15/S10/S14 (Josyula). S5 (the architecture council) shares the UDC program's dated chain with S2 — Lan's backup pointer, not yours; S2 and S1 share program-level artifacts (issues log, rollback triggers) — if asked, say so.. If pushed for "another example" on an LP, use the named backup in the question bank or the backup bench at the end of this packet, never a story allocated to another interviewer.

**How to read the "L6 level evidence" block under each story.** Amazon's L6 functional bar (user-supplied rubric): *Ambiguity* — strategy or network undefined; I select the stakeholders, set the goals, deliver independently. *Scope & influence* — team-level scope; influences one to two levels up. *Execution* — tactical and strategic; simplifies or extends; applies best practices; force-multiplies; mitigates long-term risk; makes the short-versus-long-term trade-off explicit. *Impact* — long-term team goals. *Process improvement* — operational-excellence best practices institutionalized. *Complex* — visible risks, roadblocks and constraints mitigated through expertise, judgment and influence to consensus. Overview themes the block is tuned to: designing and modeling the supply chain rather than working in it; developing new processes for others to own; root-cause mechanisms that resolve trending issues at scale; recommendations to leadership that outline competing metrics. The block sits right after Results so the evidence is on the page when the probing starts.

## The stories — business-first, canon numbers

Two Sephora distribution-center programs. Each: 60-second pitch → Situation/Task → Action in gated phases → Results → Probing tuned to an infrastructure owner → the key math → Twist map.

### S2 — Sephora Utah DC — two systemic warehouses in one building → one comingle pool, coupled with a WMS upgrade and a robotics ramp behind a published gate (+15% processing efficiency, zero customer-facing incidents)

**LPs this story can serve here:** Insist on the Highest Standards (primary — the 98% gate that fired at 94) · Deliver Results (two go-lives, peak untouched) · Have Backbone (the throttle) · Strive to be Earth's Best Employer (the floor protected during the ramp) · Bias for Action (the throttle decision the week the gate failed)
**Process family:** Make / Deliver — DC operating model (one inventory pool, three planning identities), automation ramp governance, staged go-lives, cutover runbook, hypercare exit

**Honesty label.** I owned discovery through FSD for the location and inventory model, the staging and the gates; BlueYonder, Locus and the platform teams built and configured; DC Ops ran the floor and measured the +15% post-stabilization. The +15% is Sephora's signed number; every dollar is my model on an illustrative 100K-units/week DC. Program-level artifacts (dated gate chain, 836-row issues log, rollback triggers) are shared with S1/S5 — one umbrella program.

#### 60-SECOND BUSINESS-FIRST PITCH — STOP ON THE PRIMARY METRIC

**Situation.** Sephora's Utah DC ran as two systemic warehouses in one building — Dotcom 801 and Retail 901 — transferring stock to itself and paying duplicate handling for zero customer value. The ask was one omni comingle facility, with a BlueYonder WMS upgrade and Locus robotics on the same critical path, an October–December peak freeze, a dozen core systems and 20+ touched, 60+ stakeholders, and zero customer-facing incidents allowed. **Task.** I owned discovery through the signed design for the location and inventory model, the staging and the acceptance gates. **Action.** I redesigned the operating model as one physical pool with three planning identities, so each channel kept its forecasting lens without a second truth; split the program into two go-lives so each carried one defect class — January proved reserve rules against live demand with the floor untouched, May moved warehouse execution; published the robotics gate at 98% pick-accurate throughput with the throttle dial built in advance and owned by DC Ops; ran three mock cutovers against the 0.5% tolerance and a 343-task runbook whose one irreversible instant was zero-then-publish, so nothing was sellable in two places. When week two of the robotics ramp hit 94 against 98, the gate fired: bots throttled to about 30% of volume for a week, zone and slotting logic root-caused, full volume week three. **Result.** Primary metric: DC processing efficiency — baseline index 100 → 115, +15%, Sephora-signed, measured by DC Ops post-stabilization; zero customer-facing incidents through both go-lives; rollback armed twice, never fired. About $1.6M a year modeled — labor at +15%, pooling carry, fewer split shipments, self-transfer handling removed — and about $2.2M of one-time safety stock released by pooling, also modeled; the +15% and the zero are the record.

#### SITUATION / TASK — THE BUSINESS PROCESS PROBLEM, SCOPE, SCALE, BASELINES

**S —** The Utah DC ran as two systemic warehouses — Dotcom 801 (selling location 800) and Retail 901 — transferring stock to itself and paying duplicate handling for zero customer value. The ask: one omni/comingle facility, with a BlueYonder WMS 8.2 → 2023.1 upgrade and Locus robotics on the same critical path, a dozen core systems (20+ touched), an Oct–Dec peak freeze, zero customer-facing incidents allowed.

The challenges sat on the critical path itself: the WMS upgrade and the robotics ramp shared it, the peak freeze bounded it, and the interim would run two live truths. The register's top three risks — two live truths during the interim, a robotics ramp with no stop condition, and a peak collision — each had a trigger, a mitigation and an owner before sign-off.

**T —** Discovery through FSD for the location and inventory model, the staging and the acceptance gates; 60+ stakeholders (~80 names across ~15 functions) on the steering path; none of the systems reported to me.

**Baselines on the table:** processing efficiency index 100 (units per labor-hour) · customer-facing incidents allowed: 0 · robotics gate 98% published in advance from the pre-ramp baseline and the incident budget · self-transfers and duplicate handling — the standing cost one pool exists to remove.

## ACTION — THE PROCESS / OPERATING-MODEL REDESIGN, IN GATED PHASES

- **Discovery (3/29 → 4/30)** — profiled 801/901 (SKU overlap, order profiles); sponsor objectives and success criteria — one omni facility, self-transfers eliminated, zero customer-facing incidents, peak untouched; systems-impacted table. *Exit bar:* every impacted system named to an owner with a written LOE (landed 7/2); criteria signed by steering.
- **Plan & Design (5/15 → 8/14)** — council pack (three decisions + an out-of-scope fence); the 900/901 location/inventory model with three planning identities over one physical location; staged two-go-live timeline; RAID log; FSD 8/14. *Exit bar:* council sign-off 7/31; FSD 8/14; peak-freeze boundary decided; LOE-backed plan across ~30 build tracks.
- **Build & Test (Aug → Jan/Apr)** — node × channel × bucket-transition matrix as the regression asset; dated E2E days — Inbound 11/22, Dotcom 12/20, Retail 1/30 — every function in one room; Locus gate definition (98%) with the throttle spec; three mock cutovers against the 0.5% tolerance (the S18 discipline reused); the Locus RTL slipped three weeks — escalated the week it surfaced, re-slotted openly, UAT not compressed to absorb it. *Exit bar:* E2E green on three dates; Blocker + Critical open = 0 at each go/no-go; mock 3 inside tolerance.
- **Cutover & Go-Live (January; May 1 → Jul 1)** — runbook of 343 tasks / ~815 hours, each with owner, duration and predecessor; six-step May choreography — clear negatives, transfer AFS to Uncommitted, scan-and-reserve, zero-then-publish; rollback triggers per interface (sync < 99% for 4 h; order-flow p95 > 2× budget for 1 h). *Exit bar:* six invariants held (Σ AFS conserved; no SKU sellable on 800 and 900 at once; negatives cleared first); rollback never fired.
- **Hypercare / Close (Jun → Jul)** — daily burn-down 271 → 65 → 21 → 14 → 4; Locus weekly gate read — week 2 at 94 vs 98 → throttle to ~30% → RCA (zone/slotting logic) → full volume week 3; hypercare exit memo; template handed to the next DC. *Exit bar:* < 2 blockers/day sustained → exit week 3 of 4; +15% measured post-stabilization by DC Ops.

**Influence and ownership.** Mine: the location/inventory model, the staging, the gates, the council pack, the runbook and the go/no-go recommendation. BlueYonder, Locus and the platform teams built and configured; DC Ops ran the floor, owned the robotics gate and the dial before the ramp — so when it fired, the rule fired; I did not improvise. Digital worried a virtual Retail DC would blur their forecasting lens — the three-identity design made the concern the design rather than an argument. The comingle-reserve improvements were held out of the release to protect zero incidents, priced as a later enhancement at steering.

> **Designed the process; others own it now.** I designed the location/inventory model, the gates and the throttle dial; DC Ops owned the dial before the ramp and runs the floor on it, and the hypercare template went to the next program.
> **Trade-off put to leadership.** The recommendation to leadership outlined the competing metrics — +15% processing efficiency and the robotics ramp vs zero customer-facing incidents and peak untouched — and made the gate (98%) and the throttle (~30%) the published answer before the robots arrived.

## RESULTS — $ · # · % · TIME · CUSTOMER · QUALITY (CANON NUMBERS ONLY)

- **$** — ≈$1.6M/yr *modeled* (labor at +15% ≈$0.88M · pooling carry ≈$0.22M · fewer split shipments ≈$0.45M · self-transfer handling ≈$0.10M) + ≈$2.2M one-time safety-stock release *modeled* via √2 pooling; the +15% is Sephora's signed number; every dollar is my model on an illustrative 100K-units/week DC (range $1.1–2.4M/yr).

- # — cutover runbook 343 tasks / 48,885 task-minutes ≈ 815 hrs; issues log 836 rows / 85.4% closed; burn-down 271 → 65 → 21 → 14 → 4 (record).
- **%** — processing efficiency +15% (record, Sephora-signed); robotics gate 98%, week 2 at 94, bots throttled to ~30% of volume for a week, full volume week 3 (record).
- **Time** — council 5/15 → sign-off 7/31 → FSD 8/14; E2E 11/22 · 12/20 · 1/30; January and May go-lives; hypercare exited week 3 of 4 (record).
- **Customer** — zero customer-facing incidents through both go-lives (record) — the constraint every other number moved inside.
- **Quality** — six cutover invariants held; rollback armed for both go-lives, never fired; peak untouched (record).

**L6 LEVEL EVIDENCE** — what the interviewer pattern-matches against the Amazon L6 functional rubric; every line is this story's canon, no new numbers

| L6 DIMENSION | EVIDENCE IN THIS STORY |
|---|---|
| **Ambiguity** | Two systemic warehouses in one building and no agreed operating model. I owned the location/inventory model (three planning identities over one physical location), the staging and the gates, and set the success criteria with the sponsor — one omni facility, self-transfers eliminated, zero customer-facing incidents, peak untouched. |
| **Scope & influence** | BlueYonder, Locus and the platform teams built and configured; DC Ops ran the floor and owned the robotics gate and the dial; the go/no-go recommendation was mine to leadership — influence across a dozen core systems and 60+ stakeholders I did not manage. |
| **Execution trade-off + mechanism others now own** | Robotics throughput vs floor stability: the 98% gate fired at 94 in week 2, bots throttled to ~30% for a week, full volume week 3. Mechanism others own now: the dial, owned by DC Ops before the ramp, and the hypercare template handed to the next program. |
| **Impact (long-term)** | Processing efficiency +15% (Sephora-signed); zero customer-facing incidents through both go-lives; ≈$1.6M/yr modeled plus ≈$2.2M one-time modeled; peak untouched. |
| **Process improvement (standard work institutionalized)** | The node × channel × bucket-transition matrix as the regression asset; a runbook of 343 tasks / ~815 hours with owner, duration and predecessor; six cutover invariants; a daily burn-down 271 → 4. |
| **Complexity & consensus** | Staged two go-lives (January planning / May execution) so each carried one defect class; a peak-freeze boundary; dated E2E days (11/22 · 12/20 · 1/30) with every function in one room. |
| **Root cause at scale** | The week-2 robotics miss was root-caused to zone/slotting logic and fixed before full volume — the rule fired, the cause was removed, the ramp was not pushed through. |

## PROBING — CHALLENGE Q&A TUNED TO THIS INTERVIEWER

> **Infra — Why 98%, and what did you actually do at 94?**
>
> 98% was set from the pre-ramp baseline and the incident budget before the event, and DC Ops owned it. At 94 in week two I throttled the bots to about 30% of volume for a week, root-caused the zone and slotting logic, and ramped again — full volume by week three. A gate that cannot stop the program is decoration; mine did.

> **Infra — Why two go-lives instead of one big one?**
>
> So we never hunted two defect classes in the same week. January proved reserve rules against live demand while the floor was untouched; May left warehouse execution as the only remaining class. One go-live would have put a reserve defect and a floor defect in the same room.

> **Infra — What made you say go in May?**
>
> Counted rows, not the room: Blocker + Critical at zero, mock 3 inside the 0.5% tolerance, the 343-task runbook rehearsed, rollback armed per interface. I would have said no-go on one open Blocker; I never had to.

> **Infra — What were the rollback triggers and did any fire?**
>
> Sync below 99% for four consecutive hours, or order-flow p95 above twice budget for one hour — rollback per interface behind them. Armed for both go-lives; never fired. The one irreversible instant was step 4, zero-then-publish, so no SKU was ever sellable in two places.

> **Infra — The labor number is the biggest piece of your $1.6M. How firm?**
>
> The fifteen percent is measured and signed; the 120 FTE and the $28 loaded rate are my assumptions. Pull the DC's direct hours for the two quarters before and after and multiply by payroll — that turns the model into a record in an afternoon. If the split-shipment row didn't move, the total is about $1.1M.

> **Infra — What would you do differently?**
>
> Staff the interim reporting workstream on day one, not month three — for four months every report had to declare which system of record it read. And the Locus slip: escalated the week it surfaced, but I would have put the RTL on the register as a risk, not a task, from the start.

### THE KEY MATH — THE +15% TO HOURS, THE POOLING RELEASE, AND THE THROTTLE ARITHMETIC (RECORD PERCENTAGES; ILLUSTRATIVE VOLUMES LABELED)

| STEP | QUANTITY | FORMULA → SUBSTITUTION | BASIS | VALUE |
|---|---|---|---|---|
| 1 | Efficiency | `index 100 → 100 × 1.15 = 115` | record (Sephora-signed, post-stabilization) | +15% |
| 2 | Hours for the same volume | `240,000 / 1.15 = 208,700; saved 31,300 h = 13.0%` | 120 FTE × 2,000 h illustrative on a 100K-units/week DC | 31,300 h/yr |
| 3 | Labor dollars | `31,300 × $28 = $876,400` | $28 loaded (modeled; benchmark $24–32) | ≈$0.88M/yr modeled |
| 4 | Pooling release | `$7.5M × (1 − 1/√2) = $7.5M × 0.293 = $2.2M; carry $2.2M × 10% = $0.22M` | √2 pooling on duplicated poolable safety stock (illustrative $50M × 25% × 60%) | ≈$2.2M one-time; ≈$0.22M/yr |
| 5 | Split shipments / self-transfers | `45,000 × $10 = $0.45M; 260,000 × $0.80 × 50% = $0.10M` | modeled (softest rows) | ≈$0.55M/yr |
| 6 | Robotics gate | `T = 94% < 98% → throttle: bot share 100% → 30%; manual covers 70,000 of 100,000 units/wk` | gate record; volume illustrative | RCA, full volume week 3 |
| 7 | Total | `0.88 + 0.22 + 0.45 + 0.10 = $1.65M/yr + $2.2M one-time` | modeled | ≈$1.6M/yr (range $1.1–2.4M) |

Steering charter: one omni facility · self-transfers eliminated · zero customer-facing incidents · peak untouched — all four delivered.

**Where this lands at Amazon.** An FC launch/retrofit with automation on the critical path: publish the ramp gate and the throttle before the robots arrive, stage go-lives so each carries one defect class, hold peak sacred, and write the rollback as a number. Zero-then-publish is the same discipline as never letting one unit be promisable from two nodes at once. Asset utilization (+15% units per labor-hour; one pool), reliability (zero incidents, rollback never fired).

### TWIST MAP — LIKELY LP QUESTIONS FROM THIS INTERVIEWER → THE ANGLE TO TAKE WITH S2

| LIKELY QUESTION (THIS INTERVIEWER) | LP | ANGLE — WHAT TO EMPHASIZE | SKIP / NEVER SAY |
|---|---|---|---|
| **Tell me about a time you held a high standard when it was costly.** | Insist on the Highest Standards | The 98% gate fired at 94; a week of robot volume traded for a root-caused ramp; full volume week three. | Do not say "the robots failed" — the gate worked as designed. |

| LIKELY QUESTION (THIS INTERVIEWER) | LP | ANGLE — WHAT TO EMPHASIZE | SKIP / NEVER SAY |
|---|---|---|---|
| **Tell me about a time you had to slow something down.** | Have Backbone / Bias for Action | The throttle to ~30% the week the gate failed — a decision the same week, owned by DC Ops' own rule. | Do not frame it as caution; it was a pre-built dial firing. |
| **Tell me about launching something with zero room for error.** | Deliver Results / Customer Obsession | Zero customer-facing incidents through both go-lives as the constraint; two go-lives so each carried one defect class; the six invariants. | Skip the Kafka/ESB topology unless asked. |
| **How did you protect the floor / the people during the change?** | Earth's Best Employer | The throttle protected the floor's throughput; DC Ops owned the gate and the dial; the interim's reporting burden named as the price. | Do not invent a headcount or an attrition figure; none exists in canon. |
| **How do you plan a cutover?** | Functional — launch readiness | Runbook with owner/duration/predecessor per task; one irreversible instant identified; rollback triggers as numbers; three mocks against a tolerance; hypercare exit as a number. | Never "400 devices failed" (that is S18's RF miss — ~100 of ~400). |
| **What's the ROI of the consolidation?** | Frugality | +15% signed; ≈$1.6M/yr modeled with the four rows and their inputs; ≈$2.2M one-time pooling, labeled textbook. | Never present the $1.6M as a signed number. |

## S18 — Sephora — legacy WMS → JDA WMS dual-DC migration: a 0.5% reconciliation tolerance published before mock one, three mocks read by cause class (7% → 1.8% → 0.3%), and the RF-device miss owned

**LPs this story can serve here:** Insist on the Highest Standards (backup — the published tolerance) · Bias for Action (the rehearsed fallback fired, the date held) · Deliver Results (backup — 8 months, both DCs, gate cleared) · Learn and Be Curious (first WMS migration; learned the floor in month one) · Ownership (the RF miss with my name on the cause)

**Process family:** Make / Deliver — warehouse execution system cutover across two DCs; inventory-accuracy governance; freeze, failback and hypercare discipline

**Honesty label.** Full lifecycle owner, accountable to sponsor and Finance; I had never run a WMS migration and learned the floor in month one. Month markers are illustrative unless stated; the RF miss is mine — I scored the task by effort when impact was what mattered. Loop-packet conflict (§1.3): that document says ~$800K budget and ~$2.5M/week; this packet's record is ~$1.8M, ~3% under, ~$2.5M/month — say these.

### 60-SECOND BUSINESS-FIRST PITCH — STOP ON THE PRIMARY METRIC

**Situation.** Two Sephora DCs ran on a legacy WMS with no documentation, an eight-month hard deadline, a ~$1.8M budget, and a $20M ROI pipeline of four programs gated behind go-live — about $2.5M a month of slip. Inventory accuracy across two DCs could not wobble at cutover, and I had never run a WMS migration. **Task.** Own the full lifecycle — charter, design sign-off, the ERP ↔ WMS ↔ 3PL ↔ carrier interface inventory, UAT, cutover, hypercare — accountable to the sponsor and Finance. **Action.** I wrote the success criteria in week one — on the date, inside a published reconciliation tolerance, zero accuracy incidents, a numeric hypercare exit — and learned the floor in month one until I could walk receive-to-ship unaided. I built the interface inventory from message logs with an owner and a test cell per contract, set the mapping-table standard as data not code, and published the 0.5% tolerance with Finance before mock one so the number belonged to us before the test. Three mocks were read by cause class, not headline: 7% (units of measure, location names, status vocabularies) → 1.8% (in-transit timing) → 0.3%. Cutover ran a 72-hour in-transit freeze with the legacy system read-only for 30 days. When about 100 of 400 RF devices failed the conversion script, inventory was inside tolerance and the failure was device conversion, not data — so I invoked the rehearsed paper-pick fallback rather than roll back, and told leadership the same day that the training criterion slipped 1.5 weeks while the date did not move. **Result.** Primary metric: inventory reconciliation variance at cutover — 7% → 1.8% → 0.3% against the 0.5% tolerance. On time, about 3% under the ~$1.8M budget (~$54K), the $20M pipeline unblocked the next month — slip avoided, not banked. Hypercare exited week three at under two blockers a day. The RF miss is mine.

### SITUATION / TASK — THE BUSINESS PROCESS PROBLEM, SCOPE, SCALE, BASELINES

**S —** Two DCs on a legacy WMS; zero inherited documentation; a hard eight-month deadline with a $20M ROI pipeline of four programs gated behind go-live (~$2.5M/month of slip); budget ~$1.8M; I had never run a WMS migration. Inventory accuracy across two DCs could not wobble at cutover.

The register at design sign-off: undocumented interface surprises, conversion mismatch above tolerance, and 3PL/carrier freeze non-compliance — each with a trigger, a data-first mitigation and an owner. The one I missed: RF device conversion sat on the log as a task, not a risk.

**T —** Own the full lifecycle — discovery charter, design sign-off, the ERP ↔ WMS ↔ 3PL ↔ carrier interface inventory, UAT, cutover, hypercare — accountable to the sponsor and Finance. Team: 14 core + 6 DC Ops.

**Baselines on the table:** conversion mismatch — gated at the 0.5% tolerance signed with Finance before mock 1 (mock 1 opened at 7%) · accuracy incidents allowed: zero · the date — eight months, hard · budget ~$1.8M ·

hypercare exit < 2 blockers/day.

## ACTION — THE PROCESS / OPERATING-MODEL REDESIGN, IN GATED PHASES

- **Discovery (month 1)** — sponsor charter in week one (objectives, assumptions, success criteria, phase entry/exit criteria); interface inventory v0 from message logs; systems-impacted table with LOE per owner; the floor walked until I could follow receive → put-away → pick → ship unaided. *Exit bar:* charter signed by sponsor and Finance; every interface has a named owner.
- **Plan & Design (months 2–3)** — design sign-off pack; interface inventory v1 with a test cell per contract; conversion approach — masters → open orders → on-hand → in-transit as its own batch; mapping-table standard (data, not code); the 0.5% tolerance published with Finance before mock 1. *Exit bar:* council sign-off; tolerance signed; stories slotted by business criticality × effort.
- **Build & Test (months 4–7)** — QA demo schedule to DC Ops; UAT entrance memo (function testing done, conversion build done, test data handed off, pre-prod + perf ready) — QA could fail a build; three mock cutovers with written root causes by class: 7% → 1.8% → 0.3%; after the RF miss, the training plan and device-readiness table. *Exit bar:* mock 3 ≤ 0.5% → written go-live recommendation; training sign-off (slipped 1.5 weeks, date held).
- **Cutover & Go-Live (month 8)** — runbook; 72-hour transfer freeze (≥ the longest lane transit); conversion batches; post-load reconciliation; legacy read-only switch; the go/no-go call. ~100 of ~400 RF devices failed the conversion script → the rehearsed paper-pick fallback, not a rollback; leadership told the same day with the slip priced. *Exit bar:* post-load mismatch inside tolerance; ship confirms posting; zero accuracy incidents.
- **Hypercare / Close (4 weeks planned)** — blockers/day read daily; 30-day failback watch (legacy read-only); budget closed; exit memo with the single-point-dependency lesson. *Exit bar:* < 2 blockers/day sustained → exit week 3; ~3% under budget; pipeline programs started the next month.

**Influence and ownership.** Mine: the charter, the entrance/exit criteria, the interface inventory, the conversion approach, the mapping-table standard, the tolerance, the go/no-go recommendation and the fallback call; the team built, converted, tested and ran the floor; Finance co-owned the tolerance. Because the success criteria were written in week one and signed by sponsor and Finance, nobody negotiated the exit later. The RF miss is mine — I scored the task by effort when impact was what mattered — and it went upstairs the same afternoon with my name on the cause.

> **Designed the process; others own it now.** I designed the charter, the tolerance, the entrance criteria and the mapping-table standard; the team and Finance now run cutovers against them, and the standard outlived the program. **Trade-off put to leadership.** The recommendation to the sponsor outlined the competing metrics — the eight-month date and the $20M pipeline behind it vs training readiness and conversion variance — and held the date while training slipped 1.5 weeks and the 0.5% tolerance did not move.

## RESULTS — $ · # · % · TIME · CUSTOMER · QUALITY (CANON NUMBERS ONLY)

- **$** — ~3% under the ~$1.8M budget (~$54K, record); the $20M ROI pipeline unblocked the next month — ~$2.5M/month of slip avoided, not banked; ≈$45K *modeled* paper-pick fallback cost paid deliberately to hold the date; ≈$0.1M *modeled* less inventory in dispute at 0.3% vs 0.5% on an illustrative $50M on-hand.
- **#** — ~100 of ~400 RF devices failed the conversion script — the rehearsed fallback fired; team 14 core + 6 DC Ops across two DCs (record).
- **%** — mock variance 7% → 1.8% → 0.3% against the 0.5% tolerance published with Finance before mock 1 (record).
- **Time** — eight-month deadline held; 72-hour in-transit freeze; legacy read-only for 30 days; training slipped 1.5 weeks while the date did not move; hypercare exited week 3 of 4 (record).
- **Customer** — zero accuracy incidents at cutover; ship confirms posting; post-load mismatch inside tolerance (record exit bars).
- **Quality** — mocks read by cause class, each removing a named defect category; failback rehearsed, never used; the paper-pick contingency rehearsed and used (record).

**L6 LEVEL EVIDENCE** — what the interviewer pattern-matches against the Amazon L6 functional rubric; every line is this story's canon, no new numbers

| L6 DIMENSION | EVIDENCE IN THIS STORY |
|---|---|
| **Ambiguity** | An eight-month deadline across two DCs with no charter. I wrote the sponsor charter in week one — objectives, assumptions, success criteria, phase entry/exit criteria — and set the tolerance with Finance before anyone rehearsed. |
| **Scope & influence** | 14 core + 6 DC Ops across two DCs; Finance co-owned the tolerance; QA could fail a build on my UAT entrance memo; the go/no-go recommendation and the fallback call were mine to the sponsor. |
| **Execution trade-off + mechanism others now own** | Date vs readiness: training slipped 1.5 weeks while the date held; a 72-hour transfer freeze (sized to the longest lane) bought conversion integrity. Mechanisms others own now: the 0.5% tolerance, the UAT entrance memo, and the mapping-table standard (data, not code). |
| **Impact (long-term)** | Eight-month deadline held; ~3% under the ~$1.8M budget; the $20M ROI pipeline unblocked the next month — ~$2.5M/month of slip avoided, not banked; zero accuracy incidents at cutover. |
| **Process improvement (standard work institutionalized)** | Three mock cutovers read by cause class (variance 7% → 1.8% → 0.3% against 0.5%); a 30-day failback watch with legacy read-only; a numeric hypercare exit (< 2 blockers/day sustained). |
| **Complexity & consensus** | ~100 of ~400 RF devices failed the conversion script on the day — the rehearsed paper-pick fallback fired, leadership was told the same day, the date did not move. |
| **Root cause at scale** | The single-point-dependency lesson went into the exit memo; each mock removed a named defect category rather than fixing individual records, so the fix scaled to the whole conversion. |

## PROBING — CHALLENGE Q&A TUNED TO THIS INTERVIEWER

**Infra — Why publish a tolerance before you knew you could hit it?**

So the number belonged to Finance and me before the test, not after. Mock one's 7% was the price; mock three's 0.3% was worth it — and the tolerance was never renegotiated.

**Infra — How do you read a mock?**

Never by the headline alone. Every delta got a root-cause class — units of measure, location drift, status vocabularies, then in-transit timing. If the total shrank without a named cause, the mock was not evidence. 7 → 1.8 → 0.3 each removed a category, and the fixes were mapping tables as data, not code.

**Infra — A quarter of the RF devices failed at cutover. Why not roll back?**

Because the fallback was rehearsed and the rollback was not needed: inventory was inside tolerance; the failure was device conversion, not data. I paid about $45K of paper-pick productivity — modeled — to protect about $2.5M a month of pipeline, and took the miss upstairs the same afternoon with my name on the cause.

**Infra — What was the freeze and why 72 hours?**

A 72-hour in-transit freeze, set at or above the longest lane transit, so in-transit could be converted as its own batch with Σ legacy = Σ JDA — the class that took mock two from 1.8% to 0.3%. Legacy stayed read-only for 30 days as the failback.

**Infra — The $875K avoided slip — are you banking it?**

No. It is record arithmetic — 10.5 days at about $83K a day — on a slip that did not happen; it is the price that made the fallback the obvious call. What I would measure is the pipeline's first program start against plan, which was the next month.

**Infra — Lesson, and what rule did you leave behind?**

Any single-point dependency in front of an exit criterion gets a what-if plan regardless of estimated effort. Differently: dual-path device conversion on day one, and a device model × OS × script pass/fail table before training week.

## THE KEY MATH — THE THREE MOCKS, THE SLIP PRICE AND THE FALLBACK (RECORD PERCENTAGES AND RECORD PAIRS; ILLUSTRATIVE UNITS LABELED)

| STEP | QUANTITY | FORMULA → SUBSTITUTION | BASIS | VALUE |
|---|---|---|---|---|
| 1 | Mock 1 | `M = Σ\|Δ\| / Σ on-hand = 140,000 / 2,000,000 = 7.0%` | record % on illustrative units; UoM factor 12 on 1% of SKUs dominates | 7.0% — 14× the tolerance |
| 2 | Mock 2 | `36,000 / 2,000,000 = 1.8%` | UoM / location / status mapping tables as data (record %) | 1.8% |
| 3 | Mock 3 | `6,000 / 2,000,000 = 0.3% ≤ 0.5% → go` | 72-h freeze + in-transit as its own batch (record %) | 0.3% — written go recommendation |
| 4 | Budget | `$1,800,000 × 3% = $54,000` | record | ~$54K under |
| 5 | Slip price | `$2,500,000 / 30 = $83,333/day; 1.5 wk × 7 × $83,333 = $875,000` | record math — avoided, not banked | ≈$875K kept off the critical path |
| 6 | Fallback | `~100 / ~400 = 25% of devices; paper-pick ≈$45K; $875K / $45K ≈ 19 ≈ 20× cheaper than the slip` | modeled (15–40% slower, $25–35/h) | ≈$45K paid deliberately |
| 7 | Dispute delta | `(0.5% − 0.3%) × $50,000,000 = $100,000` | illustrative on-hand (modeled) | ≈$0.1M one-time |

Never "400 devices failed" — ~100 of ~400. Budget and slip framing are this packet's canon; the loop packet differs (§1.3) — do not mix.

**Where this lands at Amazon.** Cutover discipline in its most literal form — a tolerance published with Finance before the first rehearsal, mocks read by cause class, a freeze sized to the longest lane, a tested failback, a numeric hypercare exit nobody can negotiate, and a rehearsed contingency that fired on the day. Reliability (0.3% against 0.5%, zero accuracy incidents), responsiveness (the decision the same day), cost efficiency (~3% under), the $20M pipeline on the date.

## TWIST MAP — LIKELY LP QUESTIONS FROM THIS INTERVIEWER → THE ANGLE TO TAKE WITH S18

| LIKELY QUESTION (THIS INTERVIEWER) | LP | ANGLE — WHAT TO EMPHASIZE | SKIP / NEVER SAY |
|---|---|---|---|
| **Tell me about a time something went wrong at go-live.** | Bias for Action / Ownership | The RF miss: ~100 of ~400 devices; fallback fired, not rollback; leadership told the same day; training slipped 1.5 weeks, the date did not. | Never "400 devices failed"; do not blame the vendor script — the scoring miss was mine. |
| **Tell me about a time you set a standard others thought was too strict.** | Insist on the Highest Standards | 0.5% published with Finance before mock one; mock one at 7% was the price; never renegotiated. | S2's 98% gate is the primary for this LP — use it first if both land here, S18 second. |
| **Tell me about learning something new fast.** | Learn and Be Curious | Never run a WMS migration; charter in week one; the floor walked unaided by end of month one; the mocks were where I proved it. | Do not overclaim WMS configuration — the team converted and built. |
| **Tell me about a time you delivered on a hard date.** | Deliver Results | Eight months, both DCs, 0.3% vs 0.5%, ~3% under, $20M pipeline unblocked the next month. | Never bank the $875K or the $2.5M/month — slip avoided. |

| LIKELY QUESTION (THIS INTERVIEWER) | LP | ANGLE — WHAT TO EMPHASIZE | SKIP / NEVER SAY |
|---|---|---|---|
| **How do you design a freeze / failback for a systems cutover?** | Functional — migration design | Freeze ≥ longest lane transit; in-transit as its own batch; legacy read-only 30 days; rollback triggers as numbers (mismatch > 0.5% for 2 days; ship-confirm posting failing > 2 h). | Keep budget at ~$1.8M and slip at ~$2.5M/month (loop-packet conflict). |
| **What did you miss and what did you change?** | Ownership | RF conversion on the log as a task, not a risk; the rule left behind: any single-point dependency in front of an exit criterion gets a what-if plan. | Do not soften it — "the miss is mine." |

## Question bank — likely questions for this interviewer, with a one-line answer plan

Behavioral and functional questions a Senior Manager of Infrastructure is likely to ask. Each plan points at a packet story or a named backup in the master packet; stories allocated to other interviewers are pointers only.

| # | LIKELY QUESTION | ANSWER PLAN (ONE LINE) | STORY |
|---|---|---|---|
| 1 | **Tell me about the most complex physical / systems change you have led in a DC.** | S2 pitch — one pool, three planning identities, two go-lives, the 98% gate; end on +15% signed and zero incidents. | S2 |
| 2 | **Tell me about a migration or cutover you ran.** | S18 pitch — tolerance before mock one; 7 → 1.8 → 0.3; the RF fallback; on time, ~3% under. | S18 |
| 3 | **How do you decide a program is ready to go live?** | Counted rows, not confidence: Blocker + Critical = 0; mock inside tolerance; runbook rehearsed; rollback armed; a written recommendation. | S2 / S18 |
| 4 | **How do you ramp automation without hurting throughput?** | Publish the gate from the pre-ramp baseline and the incident budget; build the throttle in advance; the floor owner owns the dial; RCA before re-ramp. | S2 |
| 5 | **What are your rollback triggers, typically?** | Numbers per interface: sync < 99% for 4 h; p95 > 2× budget for 1 h (S2); mismatch > 0.5% for two days; ship-confirm posting failing > 2 h (S18). | S2 / S18 |
| 6 | **How do you handle a vendor slipping on your critical path (Locus RTL)?** | Escalate the week it surfaces; re-slot openly; refuse to compress UAT to absorb it; retest after the slip. | S2 |
| 7 | **Tell me about a time a rehearsed contingency saved you.** | Paper-pick fallback for ~100 of ~400 RF devices; ≈$45K modeled vs ≈$875K of slip avoided; the date held. | S18 |
| 8 | **How do you exit hypercare?** | A number written before go-live: < 2 blockers/day sustained; S2 exited week 3 of 4 on 271 → 65 → 21 → 14 → 4; S18 exited week 3. A calendar exit declares victory over open defects. | S2 / S18 |
| 9 | **How would you approach consolidating or launching a node in the NACF network?** | Master §3.99 opening — book of record and metric → stocks and flows → the invariant broken today (a unit sellable in two places) → staged go-lives → ramp gate → hypercare number. | Master §3.99 + S2 |
| 10 | **What is the utilization impact of consolidation?** | +15% units per labor-hour signed; one pool (√2 pooling, modeled); self-transfers to zero by construction; robots at full volume week three. | S2 |
| 11 | **Tell me about a failure.** | S18's RF miss in full — the scoring error, the fallback, the same-day escalation, the rule left behind. | S18 |
| 12 | **How do you keep peak safe during a multi-system program?** | The Oct–Dec freeze decided as a council decision; the staged timeline around it; E2E days dated before and after; no release through peak. | S2 |

## Cheat-numbers box — canon values with basis labels

Canon numbers for S2 and S18 with their basis labels, from the master packet's §4.1 ledger.

| METRIC | VALUE TO SAY | BASIS | NEVER |
|---|---|---|---|
| **S2 efficiency** | +15% processing efficiency (index 100 → 115) | record, Sephora-signed, measured by DC Ops post-stabilization | presenting the dollars as signed |
| **S2 incidents** | zero customer-facing incidents through both go-lives | record | — |
| **S2 robotics gate** | 98% published; week 2 at 94; ~30% throttle for a week; full volume week 3 | record | "the robots failed" |
| **S2 dollars** | ≈$1.6M/yr (0.88 labor · 0.22 carry · 0.45 splits · 0.10 self-transfer); ≈$2.2M one-time pooling; range $1.1–2.4M/yr | modeled on an illustrative 100K-units/week DC | — |
| **S2 scale** | a dozen core systems, 20+ touched; 60+ stakeholders (~80 names, ~15 functions); ~30 build tracks | record | "12 systems" alone or "20 systems" alone; a single stakeholder count |
| **S2 artifacts / dates** | 343 tasks / ~815 hrs; 836 rows / 85.4% closed; council 5/15 → 7/31 → FSD 8/14; E2E 11/22 · 12/20 · 1/30; Jan + May go-lives | record (program-level artifacts shared with S1/S5) | — |
| **S2 rollback triggers** | sync < 99% for 4 h; order-flow p95 > 2× budget for 1 h — armed twice, never fired | record | — |
| **S18 mocks** | 7% → 1.8% → 0.3% vs 0.5% tolerance published with Finance before mock 1 | record | — |
| **S18 date / budget** | 8 months, on time; ~$1.8M, ~3% under (~$54K) | record (this packet's canon; loop packet differs) | "~$800K" |
| **S18 pipeline** | $20M ROI pipeline unblocked the next month; ~$2.5M/month of slip | record; slip avoided, not banked | "$2.5M/week"; banking the ~$875K |
| **S18 RF miss** | ~100 of ~400 devices; paper-pick fallback; training +1.5 weeks; date held | record | "400 devices failed" |
| **S18 freeze / failback / exit** | 72-hour in-transit freeze; legacy read-only 30 days; hypercare exit week 3 at < 2 blockers/day | record | — |
| **S18 modeled** | ≈$45K fallback cost; ≈$0.1M less inventory in dispute (illustrative $50M on-hand) | modeled | — |

**Never-say, one glance before the call.**

- Never "400 devices failed" — ~100 of ~400.
- Never bank the $20M pipeline, the ~$2.5M/month or the ~$875K — slip avoided, not banked.

- Never "12 systems" or "20 systems" alone — a dozen core, 20+ touched; never a single stakeholder count.
- Never present the ≈$1.6M/yr or the ≈$2.2M pooling as signed — the +15% and the zero are the record.
- Never claim BlueYonder or Locus configuration — vendor and platform teams built and configured; I owned the model, staging, gates and go/no-go.
- S1/S3/S5 share the UDC program's dated chain — they are Lan's; S11/S12, S6/S8 and the bar raiser's four stay out of this room.

## Backup bench — use only if a primary doesn't fit

Four fallbacks for an infrastructure owner: S13 (Manhattan — Cyber Monday at 4 AM; the incident-response / Bias for Action alternate to S18, and the older pair for S2's throttle), S5 (Sephora — the architecture council without owning a system; governance of a 20+-system change, same program family as S2 — never pair S5 with S2 for this interviewer unless asked for a second UDC example) and S15 (Sephora — the QA cadence conversion; the people / Hire & Develop question — Josyula's primary, so only if the question demands it); plus a doc-derived fourth from the Flexport program record — the Shipment-Visibility pilot roll-out and its four testing gates (launch discipline behind published graduation criteria; record-from-docs labels; same program family as Tripathy's S11, facts kept identical to her packet). Tell S2 and S18 first.

**Bench discipline.** The two (or four) primaries above are the stories to tell first. Reach for a bench story only when the question genuinely does not fit a primary, or when the interviewer asks for "another example" on an LP already spent. Each entry is a 60-second pitch, a compressed STAR, one results line and two probes — enough to answer and stop; no full probing bank or derivation. Where a bench story is another interviewer's primary, the red note says so: use it only if the question demands it, and keep the facts identical to that packet.

### BACKUP — ONLY IF ASKED S13 — Manhattan — Cyber Monday, 4 AM: allocation across 3 DCs and 100 stores when the scheduler died

**LPs this backup can serve:** Bias for Action (primary) · Customer Obsession (primary backup — the promise ranked first) · Deliver Results

**Basis.** OMS consultant on allocation, promising and the OMS–WMS handshake; engineering fixed the scheduler once I isolated it; the post-mortem mechanisms were mine to drive. $8M / $6M / 99% are record; the margin figures are my model at benchmark rates. Older story — if asked for recent, pair with the UDC robotics throttle (S2): same muscle, 2024–25.

#### 60-SECOND PITCH — STOP ON THE PRIMARY METRIC

**Situation.** Manhattan OMS consultant on allocation and promising across 3 DCs and 100 stores; 4 AM Cyber Monday the VP called — backorders piling with inventory present, ~$6M of sales at risk against a 95% fill-rate goal. **Task.** Own the response before the root cause was known. **Action.** Discovery: the contradiction (ATP > 0, backorders rising) was the entry bar to declare an incident. Plan & Design: I ranked by promise-at-risk, not FIFO; node choice = the promise — near 0.85 days vs far 4.60. Build & Test: one bridge, every rebalance timestamped; reason-coded alerts I had already built. Cutover: rebalanced so no DC drowned; status every two hours until 9 PM. Hypercare: root cause — the allocation scheduler died silently at the daylight-saving change; the post-mortem shipped a heartbeat, a DST checklist, a 3σ page rule and a rehearsed rebalance playbook. Lesson: decide reversibly, log it, find the cause later. **Result.** Primary metric: peak fill rate — 95% goal → 99% delivered. ~$8M of peak sales protected against ~$6M at risk (record); my model puts that at ≈$3.2M of gross margin at a 40% benchmark margin; structural stockouts −20% over the engagement.

#### COMPRESSED STAR

- **S —** 4 AM Cyber Monday: backorder lines piling with inventory present; promises aging; ~$6M at risk against a 95% fill goal; nobody had written a clearing rule for failure.
- **T —** Own the response before the cause was known, without making it worse.
- **A —** Declared the incident on the contradiction; clearing rule — promise-at-risk, not FIFO — written on the bridge log in the first hour so nobody re-argued it at noon; rebalanced across three DCs; two-hourly status; isolated the DST

scheduler death through my own reason-coded alerts; blameless post-mortem → heartbeat, DST checklist, 3σ page rule, rehearsed playbook.

- **R —** Fill 99% vs the 95% goal; ~$8M protected vs ~$6M at risk (record pair); structural stockouts −20% over the engagement; the next peak ran boring.

## RESULTS LINE

Peak fill 99% vs the 95% goal (record) · ~$8M of peak sales protected vs ~$6M at risk (record pair — one-time sales that didn't die, never a saving) · ≈$3.2M gross margin at a 40% benchmark (modeled) · structural stockouts −20% over the engagement (record) · detection moved from a VP phone call to one heartbeat interval.

## THE MATH — STEP-BY-STEP DERIVATION

### Table 1 — before: the problem in numbers.

| STEP | QUANTITY | FORMULA → SUBSTITUTION | INPUTS (BASIS) | VALUE |
|---|---|---|---|---|
| 1 | 4 AM state | `scheduler dead; backlog = 30,000 orders; ~$6M of sales at risk vs the 95% fill goal` | record risk pair; backlog illustrative | the bridge call |
| 2 | FIFO failure | `FIFO clears by arrival — cheap, distant-promise orders first while at-risk promises age` | record rule problem | value-blind clearing |
| 3 | Naive node loads | `DC1: 22,000 / 10,000 = 2.2 days; DC3: 2,000 / 6,000 = 0.33 days` | capacities 10K / 8K / 6K orders/day (illustrative) | worst node 2.2 days behind while another idles |
| 4 | No liveness detection | `a dead scheduler was found by a VP at 4 AM, not by a page` | record problem | detection by seniority |

### Table 2 — the fix: step by step, values substituted.

| STEP | QUANTITY | FORMULA → SUBSTITUTION | INPUTS (BASIS) | VALUE |
|---|---|---|---|---|
| 1 | Promise-at-risk ranking | `risk = value × urgency(promise − now); clear descending` | on the bridge log in hour 1 (record rule) | at-risk promises first, not first-in |
| 2 | Node choice = the promise | `T = handling + transit: near 0.85 d vs far 4.60 d; 4.60 − 0.85 = 3.75 d` | record pair | −3.75 days per re-pointed order |
| 3 | Load levelling | `rebalance 30,000 → 12,500 / 10,000 / 7,500: 12,500/10,000 = 1.25; 10,000/8,000 = 1.25; 7,500/6,000 = 1.25 days` | load = queue / capacity (illustrative) | worst slip 2.2 → 1.25 days |
| 4 | Liveness page (post-mortem) | `page if x < μ − 3σ = 1,200 − 3 × 150 = 1,200 − 450 = 750/hr` | μ = 1,200 allocations/hr, σ = 150 (illustrative); a dead scheduler reads 0/hr | caught in one interval, not by a VP |

### Table 3 — after: the metric, before → after.

| STEP | METRIC | BEFORE | AFTER | BASIS |
|---|---|---|---|---|
| 1 | Peak fill rate | 95% goal | 99% delivered | record |
| 2 | Peak sales | ~$6M at risk | ~$8M protected | record pair — one-time event |
| 3 | Worst-case promise slip | 2.2 days | 1.25 days | illustrative walk |
| 4 | Detection | a VP at 4 AM | one interval via the 3σ page | post-mortem design |
| 5 | Structural stockouts | — | −20% over the engagement | record |

### Table 4 — the dollars, every multiplication shown.

| STEP | QUANTITY | FORMULA → SUBSTITUTION | INPUTS (BASIS) | VALUE |
|---|---|---|---|---|
| 1 | Margin protected | `$8,000,000 × 40% = $3,200,000` | benchmark margin 35–45% (modeled) | ≈$3.2M one-time |
| 2 | Orders at risk | `$6,000,000 / $150 = 40,000 orders` | AOV benchmark | 40,000 orders |
| 3 | Cancels / CS avoided | `40,000 × $6.50 = $260,000` | CS contact $5–8 (benchmark; modeled) | ≈$0.26M |
| 4 | Expedite premium avoided | `40,000 × 40% = 16,000 re-pointed; 16,000 × $8 = $128,000` | re-pointed share and premium midpoint of the $5–12 benchmark (both illustrative) | ≈$0.13M |
| 5 | Structural-stockout run-rate | `$400M × 4% = $16M lost sales; × 20% = $3.2M recovered; × 40% = $1,280,000` | illustrative retailer × lost-sales benchmark (2–6%) × −20% record × margin | ≈$1.3M/yr modeled |

## TWO PROBES

> **Why act before you knew the cause?**
>
> Because waiting had an hourly price — the queue was aging while inventory sat idle — and a reversible, logged intervention did not. I checked with engineering first that manual clears could not corrupt allocation, timestamped every rebalance, and hunted the cause in parallel with alerts I had already built.

> **Isn't this a hero story?**
>
> The event is; the post-mortem is the point. Carrying the first stretch solo is one of my do-differentlys. What I would present as the result is the mechanism set — heartbeat, DST checklist, 3σ page rule, rehearsed playbook — and that the next peak ran boring.

**Canon numbers (basis labels):** 99% fill vs 95% goal (record) · ~$8M protected vs ~$6M at risk (record) · 3 DCs / 100 stores (record) · near 0.85 d vs far 4.60 d (record) · −20% structural stockouts (record) · ≈$3.2M margin at 40% (modeled). Never "$2MM baseline"; never present $8M as a saving.

## BACKUP — ONLY IF ASKED S5 — Sephora — the architecture council without owning any system: three decisions and a fence

**LPs this backup can serve:** Have Backbone; Disagree & Commit (third) · Earn Trust (influence without authority) · Think Big (architecture governance). Same program family as S1/S2 — the governance story

**Basis.** I owned none of the systems; the design decision was the deliverable. Zero escalations and the dated gate chain are the record; the dollars are my model of what governance avoided.

### 60-SECOND PITCH — STOP ON THE PRIMARY METRIC

**Situation.** The UDC location and inventory model had to clear architecture council — a challenge gate — with ~80 named participants across ~15 functions, and I owned none of those systems. **Task.** Take the design from discovery through council to sign-off and FSD without a surprise escalation. **Action.** Discovery: both inventory books profiled before any design argument; business sessions closed 4/30. Plan & Design: every impacted row pre-wired with a signed LOE by 7/2; the council asked for three separable decisions plus an out-of-scope fence (reserve math untouched; MCS decommission belongs to Global ERP). Build & Test: the "build" was the pack — architecture, staged timeline, peak-freeze boundary. Cutover: 5/15 challenge → 7/31 sign-off → 8/14 FSD. Hypercare: the fence held in writing for the whole program; a second live warehouse became a monitored shadow copy, not an argument. Lesson: a no that lives in a document survives. **Result.** Primary metric: surprise escalations on a 60+ stakeholder design — the norm on prior programs → zero; the dated gate chain held. ≈$0.5M one-time modeled (avoided rework on ~3 escalations that never happened + six weeks of benefit not deferred + a fenced second reserve engine not built).

### COMPRESSED STAR

- **S —** A dozen core systems (20+ touched), ~80 participants across ~15 functions, a council that re-litigates designs this size — and I owned none of the systems.
- **T —** Discovery (kickoff 3/29, sessions 4/30) → council 5/15 → sign-off 7/31 → FSD 8/14, without a surprise escalation.
- **A —** Profiled both books first — their numbers, not my opinion; systems-impacted table with an owner on every row; one LOE template, one deadline (7/2), one private conversation per owner before the room; three separable decisions plus the fence; the second-truth objection converted into a monitored shadow copy.
- **R —** Surprise escalations → 0 for the program's life; 100% pre-wire coverage by 7/2; gate chain held to the day with zero re-litigation; fence breaches 0; handed clean into the build (S2).

### RESULTS LINE

Surprise escalations → 0 for the program's life (record) · gate chain 3/29 → 4/30 → 5/15 → 7/2 → 7/31 → 8/14 held to the day (record) · pre-wire coverage 100% — every impacted row with a signed LOE by 7/2 (record) · both scope refusals held in writing; fence breaches 0 (record) · ≈$0.5M one-time (modeled — what did not happen).

### THE MATH — STEP-BY-STEP DERIVATION

**Table 1 — before: the problem in numbers.**

| STEP | QUANTITY | FORMULA → SUBSTITUTION | INPUTS (BASIS) | VALUE |
|---|---|---|---|---|
| 1 | Design size | `~80 named participants × ~15 functions; 20+ systems touched` | record | the blast radius |
| 2 | Norm for this size | `~3 surprise escalations per design` | industry benchmark 2–4 | ~3 expected |
| 3 | Price of one escalation | `960 rework hours × $110/h = $105,600` | modeled | ≈$0.1M each |

| STEP | QUANTITY | FORMULA → SUBSTITUTION | INPUTS (BASIS) | VALUE |
|---|---|---|---|---|
| 4 | Bundle decision risk | `P(all pass) = 0.9 × 0.9 × 0.9 = 0.73` | one up/down vote on the whole design; p = 0.9 per decision (illustrative) | 27% chance one objection unwinds everything |

**Table 2 — the fix: step by step, values substituted.**

| STEP | QUANTITY | FORMULA → SUBSTITUTION | INPUTS (BASIS) | VALUE |
|---|---|---|---|---|
| 1 | Three separable decisions | `architecture · timeline · peak boundary → 2³ = 8 outcomes` | any partial approval preserves progress (record design) | a no on one cannot unwind all |
| 2 | Expected approvals | `E = 0.9 + 0.9 + 0.9 = 2.7 of 3` | vs the bundle's 0.73 chance of everything (illustrative p) | 2.7 vs 0.73 |
| 3 | Pre-wire coverage | `rows with signed LOE / impacted rows = 100% by 7/2` | one template, one deadline, one private conversation per owner (record) | 100% before the room |
| 4 | Gate chain held | `3/29 → 4/30 → 5/15 → 7/2 → 7/31 → 8/14 = 77 days council→sign-off` | held to the day (record) | 0 re-litigation |

**Table 3 — after: the metric, before → after.**

| STEP | METRIC | BEFORE | AFTER | BASIS |
|---|---|---|---|---|
| 1 | Surprise escalations | ~3 (the norm) | 0 for the program's life | record; norm is benchmark |
| 2 | Pre-wire coverage before council | partial | 100% by 7/2 | record |
| 3 | Fence breaches in the log | — | 0 | record |
| 4 | Re-litigation after sign-off | — | 0 across 77 days | record |

**Table 4 — the dollars, every multiplication shown.**

| STEP | QUANTITY | FORMULA → SUBSTITUTION | INPUTS (BASIS) | VALUE |
|---|---|---|---|---|
| 1 | Avoided rework | `3 × 960 h = 2,880 h; 2,880 × $110 = $316,800` | ~3 escalations (benchmark) × rework hours × rate (modeled) | ≈$0.3M |
| 2 | Slip weeks avoided | `6 / 52 = 0.115 yr; 0.115 × $900,000 = $103,800` | six weeks of S2's ≈$0.9M/yr labor value not deferred (modeled) | ≈$0.1M |
| 3 | Fenced engine not built | `2 eng × 12 wk × 40 h = 960 h; 960 × $110 = $105,600` | second reserve engine (modeled) | ≈$0.1M |
| 4 | Total one-time | `0.3 + 0.1 + 0.1 = $0.5M` | all modeled — the dollars are what did not happen | ≈$0.5M |

## TWO PROBES

> **How do you get sixty people to sign LOE you don't control?**
>
> A template, a deadline and a private conversation each — so nobody learned their cost in the room. The council was then asked for three separable decisions it could take one at a time; separability was the design.

> **What did you refuse, and did it hold?**
>
> Retuning the reserve math and folding in the MCS decommission — both written, signed and held; later scope requests were answered from the pack in writing. The legitimate concern (a second live warehouse becoming a second truth) was converted into the design as a monitored shadow copy, because converting a concern costs less than winning it.

**Canon numbers (basis labels):** ~80 participants / ~15 functions; a dozen core systems, 20+ touched (record) · 5/15 → 7/31 → 8/14; LOE 7/2 (record dates) · escalations 0; fence breaches 0 (record) · ≈$0.5M one-time (modeled). Never a dollar as record; never pair with S1 or S2 for the same interviewer.

## BACKUP — ONLY IF ASKED S15 — Sephora — converting a 20-person Product/Engineering/QA team to a four-week cadence in six weeks — and the QA lead who now owns the gate

**LPs this backup can serve:** Hire and Develop the Best (primary) · Strive to be Earth's Best Employer (QA elevated, not blamed) · Insist on the Highest Standards (the gate)

**Josyula's primary (Packet 5, Hire & Develop the Best) — use only if the question demands it, and keep every fact identical to that packet.**

**Basis.** VP air cover, no line authority over anyone in the room; the QA lead authored the definition-of-done — I coached the frame, she owned the gate. The −50% and the six weeks are record; the dollars are my model on illustrative defect counts. Never invent a direct-report headcount.

### 60-SECOND PITCH — STOP ON THE PRIMARY METRIC

**Situation.** The VP asked me to move a 20-person Product/Eng/QA team from a six-week hybrid cadence to four-week agile; QA feared compressed test windows — and they were right. **Task.** Convert in weeks without line authority. **Action.** Discovery: one-on-ones; an influencer per function. Plan & Design: the QA lead authored the definition-of-done — I coached the frame, she owned the gate. Build & Test: Jira benefit-ranking and review-status fields; a QA drop date with reason codes. Cutover: the four-week cadence live in six weeks. Hypercare: QA on product-business calls with pre-approved scenarios; she defended the gate without me. Lesson: development that depends on me staying is not development. **Result.** Primary metric: critical defects per release −50% after the 20-person team moved to a four-week cadence in six weeks. ≈$130K/yr modeled — the −50% and the six weeks are the record; the dollars are my model; no headcount invented.

### COMPRESSED STAR

- **S —** Six-week hybrid cadence; QA (a contracting firm) and engineering (in-house) heading for a standoff; QA's fear of compressed test windows was correct arithmetic, not resistance.
- **T —** Convert to four weeks in weeks, with VP air cover and no line authority.
- **A —** Heard every real objection privately; the QA lead wrote the DoD content and entry criteria that can say no; drop date with reason codes; Jira fields; Sev-1 cause classification; rollback to six weeks named as a two-way door at kickoff and never used; facilitated the retro once and stepped back.
- **R —** Converted in six weeks; critical defects per release −50% while shipping 50% more releases (8.7 → 13/yr modeled from the cadence); the gate survived my leaving; neighbouring teams copied the ceremony from the user guide.

### RESULTS LINE

Critical defects per release −50% (record) · six-week → four-week cadence, converted in six weeks (record) · 20-person team (record); releases/yr 8.7 → 13 (modeled from the cadence) · ≈$130K/yr (modeled; range $80–270K; no headcount invented) · the QA lead defending the gate by cycle 2 (record).

### THE MATH — STEP-BY-STEP DERIVATION

**Table 1 — before: the problem in numbers.**

| STEP | QUANTITY | FORMULA → SUBSTITUTION | INPUTS (BASIS) | VALUE |
|---|---|---|---|---|
| 1 | Cadence | `releases/yr = 52 / 6 = 8.67 ≈ 8.7` | six-week hybrid cycle (record) | 8.7 releases/yr |
| 2 | Criticals | `4 per release × 8.7 = 34.8 ≈ 35/yr` | count illustrative; the record is the −50% | ≈35/yr |

| STEP | QUANTITY | FORMULA → SUBSTITUTION | INPUTS (BASIS) | VALUE |
|---|---|---|---|---|
| 3 | The real constraint | `15 eng × 18 d = 270 eng-days → 270/3 = 90 stories; 4 QA × 18 d = 72 QA-days → 72/1 = 72 stories` | story ≈ 3 eng-days + 1 QA-day (illustrative; theory-of-constraints, toolkit label) | QA caps throughput at 72 — QA is the constraint |
| 4 | Late-land rework | `6 person-days × $800 = $4,800 per release` | re-test + hotfix after last-2-day lands (modeled) | ≈$4.8K per release |

**Table 2 — the fix: step by step, values substituted.**

| STEP | QUANTITY | FORMULA → SUBSTITUTION | INPUTS (BASIS) | VALUE |
|---|---|---|---|---|
| 1 | QA-authored DoD | `entry criteria at Ready-for-QA — stories arrive ready` | the skeptic owns the gate (record) | the gate has an owner |
| 2 | Drop date | `not QA-ready by the drop date → dropped with a reason code` | silent late lands become visible re-scopes (record) | no last-2-day lands |
| 3 | Sev-1 attribution | `share caused by wrong stories vs wrong tests, classified by cause not blame` | QA put on product calls with pre-approved scenarios (record) | the trend reverser |
| 4 | Cadence converted | `releases/yr = 52 / 4 = 13; 13 / 8.7 = +50%` | converted in six weeks (record) | 13 releases/yr |
| 5 | Criticals per release | `4 → 2 = −50%` | record | 2 per release |

**Table 3 — after: the metric, before → after.**

| STEP | METRIC | BEFORE | AFTER | BASIS |
|---|---|---|---|---|
| 1 | Cycle / releases per year | 6 wks / 8.7 | 4 wks / 13 (+50%) | record |
| 2 | Critical defects | ≈35/yr (4 × 8.7) | 2 × 13 = 26/yr — while shipping 50% more | record −50%; counts illustrative |
| 3 | Late-land rework | ≈$4.8K per release | ~0 under the drop rule | modeled |
| 4 | The gate itself | mine to defend | QA lead defending it by cycle 2 | record — survives my leaving |

**Table 4 — the dollars, every multiplication shown.**

| STEP | QUANTITY | FORMULA → SUBSTITUTION | INPUTS (BASIS) | VALUE |
|---|---|---|---|---|
| 1 | Escaped criticals avoided | `35 − 26 = 9/yr; 9 × $10,000 = $90,000` | per-critical cost benchmark $8–15K (modeled) | ≈$90K/yr |
| 2 | Late-land rework removed | `$4,800 × 8.7 releases = $41,760` | → ~0 under the drop rule (modeled) | ≈$40K/yr |
| 3 | Total | `90 + 40 = $130K` | range $80–270K; no headcount invented | ≈$130K/yr |

## TWO PROBES

> **Who did you develop, concretely?**
>
> The QA lead — she authored the definition-of-done and its entry criteria, and by cycle two she was defending the gate on product-business calls without me. The test of development is that it survived my leaving; it did.

> **What did engineering give up?**
>
> Low-impact stories they had to drop at the QA drop date — I covered those with the business. Their objection became co-ownership through review, not a veto over the gate.

**Canon numbers (basis labels):** −50% critical defects per release (record) · 6 weeks to convert; 6-week → 4-week cadence (record) · 20 people (record) · 8.7 → 13 releases/yr (modeled) · ≈$130K/yr (modeled). Never a direct-report headcount.

## BACKUP — ONLY IF ASKED Doc bench — Flexport — the Shipment-Visibility pilot roll-out: graduated waves behind six published criteria, four testing gates, monthly GO/NO-GO (launch discipline)

**LPs this backup can serve:** Insist on the Highest Standards (primary — gates that fire) · Deliver Results · Bias for Action (monthly GO/NO-GO cadence) · Customer Obsession (CSAT inside the gate)

**Basis.** Doc-derived micro-story from the program record (the SV pilot roll-out schedule and the SV product testing approach, 2021–22) — every figure is RECORD-FROM-DOCS. Same program family as Tripathy's S11: if she has already taken S11 deep, use this only for a launch-discipline question and keep the canon numbers identical to her packet. Program-lead voice: I owned the rollout design and the gates; data, ops and CS ran calibration, verification and UAT. Not in the master packet's S-numbering; supplements canon, never overrides it.

### 60-SECOND PITCH — STOP ON THE PRIMARY METRIC

**Situation.** A visibility product with public commitments was running a free pilot — every boarded client cost engineering, data and ops capacity before a dollar of revenue, and boarding clients before the data was trustworthy would manufacture churn at scale. **Task.** Design a roll-out that could say no — and make "no" a rule, not a judgment call. **Action.** Graduated waves: wave 1 two named clients (Ta Chen International, FAM Brands), wave 2 four of a seven-client pool, wave 3 twenty-plus, full roll-out H2 2022 — each client entering on a ~50-container calibration and graduating only on six published criteria: verification SLA 90% reviewed within 24h, data quality ≥90% and at least the incumbent Crux feed, milestone estimation ≥80%, performance to standards, ≥100 containers with key features live, CSAT ≥ the consignee average. Four testing gates stood before any launch: a 100-container live test watched end-to-end, biweekly bug bashes on the client job stories, shadow accounts with real Crux data in dedicated demo accounts, and UAT gated on data quality — 80% of shipments minor-to-no issues — not on the calendar. GO/NO-GO decisions ran monthly. **Result.** A roll-out that never had to roll back: the graduation rule, not heroics, set the pace — the launch-discipline version of S11's canon (waves held on the criteria; the held wave became the cleanest one). RECORD-FROM-DOCS

### COMPRESSED STAR

- **S —** A free pilot under public commitments, with an acquired product (Crux) whose customers would transition mid-roll-out — sequencing and quality bars were the whole game.
- **T —** Own the roll-out design: wave structure, graduation criteria, testing gates and the GO/NO-GO cadence.
- **A —** Published six graduation criteria before wave 1; a ~50-container calibration per client as the entry bar; four testing gates including shadow accounts with real Crux data and a UAT data-quality bar (80% minor-to-no-issues) before any client UAT started; monthly GO/NO-GO from November; wave sizes 2 → 4-of-7 → 20+ → full H2 2022.
- **R —** A roll-out with zero rollbacks (S11 canon); every gate that fired was a rule firing, not a favor refused — the held wave became the cleanest wave.

### RESULTS LINE

Waves 2 → 4-of-7 → 20+ → full H2 2022 (record-from-docs; matches S11 canon) · six graduation criteria published before wave 1 (canon) · ~50-container calibration entry bar (canon) · four testing gates with UAT gated at 80% minor-to-no-issues (record-from-docs) · monthly GO/NO-GO cadence (record-from-docs) · zero rollbacks (canon).

### THE MATH — STEP-BY-STEP DERIVATION

**Table 1 — before: the problem in numbers.**

| STEP | QUANTITY | FORMULA → SUBSTITUTION | INPUTS (BASIS) | VALUE |
|---|---|---|---|---|
| 1 | Client base waiting on the replacement | `79 Crux customers to migrate by end 2022` | Crux transition record (record-from-docs) | 79 accounts exposed to a failed launch |
| 2 | Product state | `0 production clients; data quality unmeasured against the incumbent` | pre-pilot (record-from-docs) | unproven at any scale |
| 3 | The naive alternative | `big-bang = 79 clients × first contact with an unproven product` | the rejected option (record-from-docs) | one bad week burns every reference |
| 4 | Entry bar per client | `≥50 containers of client data required for calibration before onboarding` | graduation criterion 2a (record-from-docs) | every onboarding has a data gate |

**Table 2 — the fix: step by step, values substituted.**

| STEP | QUANTITY | FORMULA → SUBSTITUTION | INPUTS (BASIS) | VALUE |
|---|---|---|---|---|
| 1 | Wave 1 cohort | `2 clients = Ta Chen + FAM Brands` | roll-out schedule (record-from-docs) | 2 live, Dec 2021 |
| 2 | Wave 2 cohort | `4 chosen from a named pool of 7` | roll-out schedule (record-from-docs) | 4-of-7, Feb 2022 |
| 3 | Wave 3 cohort | `20+ clients, Crux + non-Crux` | roll-out schedule (record-from-docs) | gradual expansion, Q2 2022 |
| 4 | Full roll-out | `remaining Crux + net-new + existing 3PL clients` | H2 2022 (record-from-docs) | only after Wave 3 clears the gates |
| 5 | Calibration arithmetic — Wave 1 | `2 × 50 = 100 containers before Wave 1 onboarding` | entry bar from Table 1 row 4 (record-from-docs; multiplication illustrative) | 100 calibration containers |
| 6 | Calibration arithmetic — through Wave 2 | `(2 + 4) × 50 = 6 × 50 = 300 containers` | same bar (record-from-docs; multiplication illustrative) | 300 before any scale-up |
| 7 | Graduation gate | `6 of 6 criteria green before a wave advances — verification SLA 90%, data quality 90%+, milestone accuracy 80%+, production stability, ≥100 containers uploaded per client, zero show-stoppers` | the six published criteria (record-from-docs) | no wave skips the gate |
| 8 | Testing stack behind the gate | `4 tests sequenced: 100-container test → bug bashes every other week → shadow client accounts (Marc Fisher, Michelin) → UAT, starting only when 80% of shipments show minor-to-no issues` | testing-approach doc (record-from-docs) | UAT gated at 80% |
| 9 | Decision cadence | `2 GO/NO-GO meetings before Wave 1 — business (John, Eric) + tech (product & eng leadership)` | roll-out schedule, November (record-from-docs) | every wave is a decision, not a date |

**Table 3 — after: the metric, before → after.**

| STEP | METRIC | BEFORE | AFTER | BASIS |
|---|---|---|---|---|
| 1 | Wave progression | plan only | W1 Dec 2021 → W2 Feb 2022 → W3 Q2 2022 → full H2 2022 | record-from-docs |
| 2 | Clients live on the product | 0 | 2 → 6 → 20+ | record-from-docs |
| 3 | Rollbacks | — | 0 — no wave ever reversed | canon |

| STEP | METRIC | BEFORE | AFTER | BASIS |
|---|---|---|---|---|
| 4 | Speed control | fixed launch date | accelerate / slow down on client feedback, per the published plan | record-from-docs |

**Table 4 — the dollars, every multiplication shown.**

| STEP | QUANTITY | FORMULA → SUBSTITUTION | INPUTS (BASIS) | VALUE |
|---|---|---|---|---|
| 1 | Pilot revenue, by design | `clients × $0 × ~6 months = $0 — free for the full duration of the pilot` | roll-out schedule: "no charge for the full duration of the pilot"; ~6-month estimate (record-from-docs) | $0 forgone deliberately — the gate is the product |
| 2 | Big-bang exposure avoided | `79 × 50 = 3,950 containers of one-shot onboarding risk` | client count and entry bar (record-from-docs; multiplication illustrative) | never spent — waves capped the blast radius at 2, then 6, then 20+ |
| 3 | Cost of a failed Wave 1 | `2 × 50 = 100 calibration containers + one feedback cycle — the maximum the plan allowed to fail at once` | record-from-docs; arithmetic illustrative | failure priced small by design |
| 4 | Dollar result claimed | `no revenue or saving dollar in the record — not claimed` | record-from-docs | not claimed |

## TWO PROBES

### Why gate UAT on data quality instead of a date?

Because a client UAT against dirty data manufactures false confidence — the client signs off on a product the data then betrays. The 80% minor-to-no-issues bar meant UAT started when the product could survive a skeptical user's first week; the date followed the data, never the reverse. Shadow accounts with real Crux data let us find the defects before the client did.

### What did the calibration actually buy?

A verified baseline per client before scale: ~50 containers through verification with the SLA clock running, so the data-quality and estimation criteria were measured on the client's own freight, not on our best lanes. It converted "ready" from a feeling into a measurement — and it made the wave-2 selection from the seven-client pool defensible to the clients not picked.

**Canon numbers (basis labels):** waves 2 → 4-of-7 → 20+ → full H2 2022 (record-from-docs, matches S11 canon) · six criteria incl. DQ ≥90% · ETA ≥80% · SLA 90% within 24h · ≥100 containers + features · stability · CSAT ≥ consignee average (canon; the pilot schedule phrases the SLA as 24–48h — canon wins, footnoted in Tripathy's packet) · ~50-container calibration (canon) · 100-container test · bug bashes · shadow accounts · UAT 80% bar (record-from-docs). Never: a client-revenue claim (the pilot was free); never merge with Crux-transition ARR.

INTERVIEW-PREP PACKET 4 OF 5 · PER-INTERVIEWER · BUSINESS-PROCESS LENS

# Amazon NACF Supply Chain loop — Packet 4: Zhaowen Lan

Amazon — Senior Supply Chain Manager, NACF Supply Chain (L6)

## Zhaowen Lan

Senior Manager, Supply Chain — NACF Supply Chain (peer of the hiring manager)

**Thursday, October 30, 2026 · 11:00 am PDT · 55-minute video interview**

> **Strategy in one line.** A peer Senior Manager tests whether you can reach one truth across organizations that don't report to you and build a process that survives its author. Open with S1 (three orgs, three meanings of "available" → one signed definition per node per channel, policy before configuration; oversells recurring → zero) and hold S3 (Finance and Operations on one Financial On-Hand, the failure direction designed before the happy path) for the Finance / controls / simplification question. Program-lead voice; S4 and S5 are pointers only.

**LP focus — my guess, not confirmed:** Earn Trust · Ownership · Invent and Simplify or Have Backbone. Amazon does not disclose the assignment; the guesses are reasoned from the interviewer's title and the JD in "Read me first" and should be held loosely. No interviewer biography is asserted anywhere in this packet — only title-based inference, labeled as such.

Canon: amazon/amazon-l6-scm-master-packet.html (§1.2 roster, §4.1 ledger). Every number below is that packet's value; every dollar carries its basis label (record · user anchor · modeled · priced).

## Read me first

### The interviewer's lens (inference from title + JD — not a biography)

**Inference from the title.** A second Senior Manager of Supply Chain in the same organization is the peer interviewer: someone whose team will share processes, data definitions and escalation paths with this role. Peers test whether the candidate can get to *one truth* across organizations that do not report to them, design a process that survives its author, and disagree without breaking the relationship. They also notice whether the candidate claims more than they owned — a peer has to live with the person afterwards.

**What that means for the answers.** The two Sephora S/4HANA governance stories, told as process redesigns of the Plan/Deliver seam: S1 (three organizations meant three different things by "available"; one signed definition per node per channel, policy before configuration; oversell incidents recurring → zero through two go-lives) and S3 (Finance and Operations reconciling on-hand by argument; one Financial On-Hand with a safe failure direction and a control report owned by Finance). Program-lead voice throughout — no system reported to me; SAP teams and configurators built to the FSD three orgs signed. S4 (the in-transit rule I got wrong) and S5 (the architecture council) are the same program family and appear only as backup pointers in the question bank.

### Three guessed LP focus areas, with rationale

> **Labeled guess.** Amazon assigns 2–3 Leadership Principles per interviewer and does not disclose them. These three are inferred from the title and the JD; if the first question lands on a different LP, the twist maps under each story carry the pivot.

| GUESSED LP | WHY THIS INTERVIEWER WOULD CARRY IT | STORY TO OPEN · BACKUP |
|---|---|---|
| **Earn Trust** | The peer LP: one signed truth across Supply Chain Strategy, Inventory Management and FSC Inventory Accounting (S1); Finance given the real control report to own instead of a bridge spreadsheet (S3). Both are Earn Trust backups in the master map — the primary (S14) is the bar raiser's. | S1 (open) · S3 (Finance and Ops on one number) |
| **Ownership** | Taking the definition nobody owned without owning any system — the bucket inventory, the asymmetry, the signature chain. A peer wants to know the candidate will own the seam, not the org chart. | S1 (the definition nobody owned) · S3 (the failure direction designed before the happy path) |
| **Invent and Simplify / Have Backbone (one of the two)** | De-aggregation with a safe failure direction is the simplification (S3); pricing Merchandising's "merch reserve promisable" ask and putting the no in the FSD as a row is the backbone (S1). | S3 (one FOH, failure direction) · S1 (the no that lives in a document) |

### Answer strategy — business first, systems as enablers

- **Open with the process problem and its cost, not the system.** First sentence names the broken process (Plan, Source, Make, Deliver or Return), who it hurt (customer, Finance, the floor) and the baseline number. The platform appears only as the enabler of the redesign, in one clause.
- **Use the JD's four outcome words as the results frame:** reliability (did the promise hold?), responsiveness (how much earlier did we see it?), cost efficiency ($ with its basis label), asset utilization (capital, labor-hours, capacity released).
- **Action = operating-model redesign in gated phases** — Discovery → Plan & Design → Build & Test → Cutover → Hypercare, each spoken as entry bar · what I delivered · exit bar. Say the gate that fired, not the

meeting that happened.

- **Every dollar carries its basis word in the same breath** — record · user anchor · modeled · priced. A record dollar is said flat; a modeled dollar is said with its inputs. Never promote a modeled number to record.
- **Program-lead voice.** I owned definitions, rules, gates and the signature chain; engineers, SIs and vendor teams configured and built to the spec I signed. The only platform personally configured in canon is Manhattan DOM.
- **The pitch ends on the primary metric.** Then stop. Silence after the pitch is the interviewer filling their STAR worksheet; their probes are the second half of the answer.
- **For a peer specifically:** name the three organizations and what each owned (Strategy — policy; Inventory Management — execution; Inventory Accounting — the book); say what was pre-wired privately before any council; convert every objection into a document row. "A no that lives in a document survives; a no that lives in a meeting gets re-litigated weekly."
- **Say the decomposition only from this packet's model:** S1 is ≈$2.5M/yr = $1M MCS run cost retired (user anchor) + ≈$1.5M modeled (oversell ≈$0.9M · reconciliation labor ≈$0.33M · change premium ≈$0.3M). S3 is ≈$0.6M/yr, entirely modeled; the record has no pre-state drift dollar. If the room wants one cancel-rate number, keep S1's and drop S3's row.

> **Loop-consistency rule.** One question, one story; never the same story to two interviewers — interviewers compare notes. Held for other packets (do not spend here): S11/S12 (Tripathy) · S6/S8 (Hoerger) · S2/S18 (Goforth) · S16/S15/S10/S14 (Josyula). S4 and S5 are bench stories (backup bench at the end), not primaries; S2 shares the UDC program's dated chain with S1/S5 — if asked, say it is one umbrella program.. If pushed for "another example" on an LP, use the named backup in the question bank or the backup bench at the end of this packet, never a story allocated to another interviewer.

> **How to read the "L6 level evidence" block under each story.** Amazon's L6 functional bar (user-supplied rubric): *Ambiguity* — strategy or network undefined; I select the stakeholders, set the goals, deliver independently. *Scope & influence* — team-level scope; influences one to two levels up. *Execution* — tactical and strategic; simplifies or extends; applies best practices; force-multiplies; mitigates long-term risk; makes the short-versus-long-term trade-off explicit. *Impact* — long-term team goals. *Process improvement* — operational-excellence best practices institutionalized. *Complex* — visible risks, roadblocks and constraints mitigated through expertise, judgment and influence to consensus. Overview themes the block is tuned to: designing and modeling the supply chain rather than working in it; developing new processes for others to own; root-cause mechanisms that resolve trending issues at scale; recommendations to leadership that outline competing metrics. The block sits right after Results so the evidence is on the page when the probing starts.

## The stories — business-first, canon numbers

Two Sephora S/4HANA governance stories from one umbrella program. Each: 60-second pitch → Situation/Task → Action in gated phases → Results → Probing tuned to a peer Senior Manager → the key math → Twist map.

### S1 — Sephora S/4HANA ↔ ISM — one signed meaning of "available" across Store, Retail DC and Dotcom DC: policy before configuration (oversell incidents recurring → zero through two go-lives)

**LPs this story can serve here:** Earn Trust (backup — three orgs signed one definition) · Ownership (backup — took the definition nobody owned) · Think Big (third — the book-of-record migration) · Success & Scale Bring Broad Responsibility (oversell is a customer harm, not a ledger error) · Have Backbone (the priced no to Merchandising)
**Process family:** Plan / Deliver seam — inventory availability governance (available-to-promise policy, bucket ownership, real-time decrement contract, daily reconciliation); book-of-record migration

**Honesty label.** Integration program lead — no system reported to me; every configurator sat in another org. I owned the definitions, the asymmetry, the decrement and reconciliation contract, the signature chain and the council ask; SAP teams and configurators built to the FSD; planners executed the business acts. The record has no pre-state oversell dollar — the ≈$1.5M is my model and I say so. Dated gate chain, issues log and rollback triggers are program-level artifacts shared with S2/S5.

**60-SECOND BUSINESS-FIRST PITCH — STOP ON THE PRIMARY METRIC**

**Situation.** Sephora was moving the inventory book of record from the in-house MCS to SAP S/4HANA, with a real-time layer beside it for the digital demand S/4 cannot see in time — a web order placed, a BOPIS order packed, a POS sale rung but not yet posted. Store, Retail DC and Dotcom DC each meant something different by "available": Merchandising wanted merchandise reserve visible, Finance wanted on-hand to tie to the book, dotcom wanted in-transit promisable. Most oversell bugs are vocabulary bugs, and oversells were recurring across all three node types — a customer harm, not a ledger error. **Task.** As integration program lead for inventory supply — no system reported to me — define every bucket, its owning system, and how available-for-sale is computed and decremented in real time, and get Planning, Finance and operations to sign it before any configuration. **Action.** I built the bucket inventory with exactly one owning system per bucket; wrote the lower-of-two basis and the available-for-sale equation, the real-time decrement contract with an idempotency key per event, and a daily reconciliation of the real-time ledger to the book; pre-wired every row of the systems-impacted table with its owner's level of effort before the council so nobody learned their cost in the room, and asked the council for three decisions rather than a review. Merchandising's ask to make merch reserve promisable was priced — unpickable stock sold — and the no went into the specification as a row, not a meeting minute. Three organizations — Supply Chain Strategy, Inventory Management and FSC Inventory Accounting — signed the FSD on 6/5/25. Then a node × channel × bucket-transition regression matrix, dated end-to-end days, a January planning go-live and a May execution go-live, with the order system reading one published availability number and never computing its own. **Result.** Primary metric: customer-facing oversell incidents — recurring across three node types → zero through two go-lives; one signed availability number per node per channel; MCS retired as book of record. About $2.5M a year: $1M of MCS run cost retired, which is the measured anchor, plus about $1.5M modeled — avoided oversell, dual-book reconciliation labor, legacy change premium — and I say which is which.

## SITUATION / TASK — THE BUSINESS PROCESS PROBLEM, SCOPE, SCALE, BASELINES

**S —** Sephora moved the book of record from MCS to S/4HANA, with ISM beside it to see digital demand S/4 cannot see in real time. Three node types, three working definitions of "available." Merchandising wanted merchandise reserve visible, Finance wanted on-hand to tie to the book, dotcom wanted in-transit promisable. Nobody owned the definition; I owned no system.

The process cost: recurring vocabulary-driven oversells — a customer harm, not a ledger error; two books reconciled by hand (≈4 FTE, modeled); the MCS run cost continuing (≈$1M/yr, user anchor); a legacy change premium on an undocumented stack. Vocabulary drift — a bucket without an owner — sat at the top of my risk register.

**T —** Own the integration design and data contract of inventory supply: which buckets exist, which system owns each, how available-for-sale is computed per node, and how it feeds promising, replenishment and Finance — signed by Planning, Finance and operations before configuration. Sponsor memo with the VP Supply Chain: one signed AFS per node per channel, one Financial On-Hand, zero customer-impacting incidents.

**Baselines on the table:** oversell incidents recurring across three node types (target zero) · availability truths: two books reconciled by hand (target one signed AFS per node per channel with daily drift as the standing check) · MCS running as book of record (target retired) · ~15 functions, ~60 people to move without authority.

## ACTION — THE PROCESS / OPERATING-MODEL REDESIGN, IN GATED PHASES

- **Discovery** — profiled both books (MCS and S/4) for SKU overlap and order profiles per node; node-by-node sessions with ~15 functions; the bucket inventory with one proposed owning system per bucket. *Exit bar:* every bucket owned by exactly one system; sessions closed 4/30; open questions carried with owner and date.
- **Plan & Design** — the lower-of-two basis; the AFS/FOH equations; the real-time decrement contract (event → bucket delta; idempotency key per event; a reversal must reference its original); daily ISM-vs-MB52 reconciliation with tolerance zero; the RAID log; the council pack — FSD v1.1 (3/5/25) → v1.2 (5/16/25). *Exit bar:* FSD signed 6/5/25 by Supply Chain Strategy, Inventory Management and FSC Inventory Accounting with zero open "who owns this bucket" rows.
- **Build & Test** — the node × channel × bucket-transition matrix with before/after assertions — no story enters QA without a matrix cell reference; SIT on the interface contracts before UAT; dated E2E days — Inbound 11/22, Dotcom 12/20, Retail 1/30 — business and engineering in one room. *Exit bar:* 0 open contract defects in SIT; E2E green on three dates; Blocker + Critical open = 0 at go/no-go.
- **Cutover & Go-Live** — go/no-go recommendation from counted rows; the publish-switch plan — OMS reads the published AFS, never computes; reserve-copy invariants; control-screen roles assigned; rollback triggers (sync < 99% for 4 h; order-flow p95 > 2× budget for 1 h) armed for both go-lives, never fired. *Exit bar:* January go-live (reserve rules live against demand); May go-live (execution); no SKU with two published AFS values; Negative AFS clean at T+1.
- **Hypercare / Close** — daily drift report (ISM vs MB52); Negative AFS / AFS Correction audit review; control-owner map — Negative AFS to Inventory Control, Internal Audit and Finance; AFS Correction to the ISM app team; FSD as-built note. *Exit bar:* drift = 0 SKUs on consecutive days; blockers/day under threshold; the design enters the monthly release train.

**Influence and ownership.** Sixty people across three orgs, none reporting to me. The method: pre-wire every row of the systems-impacted table with its owner's level of effort before the council; run working sessions to one table instead of arbitrating in meetings; convert objections into FSD rows; ask the council for three decisions rather than a review. Three signatories because each owned a different truth — Strategy the policy, Inventory Management the execution, Inventory Accounting the book; one signature would have left two orgs free to disagree later. The asymmetry held: oversell is a customer defect, undersell a planning variance Inventory Control chases through the control screens.

> **Designed the process; others own it now.** I designed the availability definition, the decrement contract and the control-owner map; Inventory Control, Internal Audit, Finance and the ISM app team now own the controls, and the drift report runs daily without me. **Trade-off put to leadership.** The recommendation to the council outlined the

> competing metrics — availability to sell vs oversell exposure — and resolved them as an asymmetry in the signed FSD: oversell is a customer defect, undersell a planning variance.

## RESULTS — $ · # · % · TIME · CUSTOMER · QUALITY (CANON NUMBERS ONLY)

- **$** — ≈$2.5M/yr — $1M MCS run/support cost retired (*user anchor*) + ≈$1.5M *modeled* (avoided oversell ≈$0.9M · dual-book reconciliation labor ≈$0.33M · legacy change premium ≈$0.3M); range $1.7–5.2M; upside not in the headline: shrink ≈$0.3M, time-to-market ≈$0.2M. The record has no pre-state oversell dollar.
- **#** — oversell incidents recurring across three node types → zero through two go-lives (record); dual-book reconciliation ≈4 FTE by hand → ≈1 FTE reading the drift report (modeled).
- **%** — issues log 85.4% of 836 rows closed (program-level artifact shared with S2/S5); the oversell rate behind the ≈$0.9M row is 0.10% of orders — modeled, benchmark 0.05–0.25%; the record movement is incidents → zero.
- **Time** — FSD v1.1 3/5/25 → v1.2 5/16/25 → signed 6/5/25; E2E Inbound 11/22 · Dotcom 12/20 · Retail 1/30; January planning and May execution go-lives (record dates).
- **Customer** — customer-facing oversell incidents zero through two go-lives; the asymmetry treats oversell as a customer defect and undersell as a planning variance (record).
- **Quality** — one signed AFS per node per channel; no SKU with two published AFS values; daily ISM-vs-MB52 drift = 0 SKUs on consecutive days at hypercare exit; MCS retired as book of record (record).

**L6 LEVEL EVIDENCE** — what the interviewer pattern-matches against the Amazon L6 functional rubric; every line is this story's canon, no new numbers

| L6 DIMENSION | EVIDENCE IN THIS STORY |
|---|---|
| **Ambiguity** | Three organizations with three meanings of "available" and no owner of the definition. I defined the lower-of-two basis, the AFS/FOH equations and the bucket inventory with one owning system per bucket, and chose the ~15 functions for node-by-node sessions. |
| **Scope & influence** | Sixty people across three orgs, none reporting to me; the council took three decisions I framed; the FSD was signed 6/5/25 by three signatories — influence up to the council and across orgs. |
| **Execution trade-off + mechanism others now own** | Oversell protection vs availability, made explicit as an asymmetry — oversell is a customer defect, undersell a planning variance. Mechanisms others own now: the daily ISM-vs-MB52 drift report and the control-owner map (Negative AFS → Inventory Control, Internal Audit and Finance; AFS Correction → the ISM app team). |
| **Impact (long-term)** | Oversell incidents recurring across three node types → zero through two go-lives; MCS retired as book of record; ≈$2.5M/yr ($1M run cost user anchor + ≈$1.5M modeled). |
| **Process improvement (standard work institutionalized)** | The node × channel × bucket-transition matrix — no story enters QA without a matrix cell reference; SIT on interface contracts before UAT; dated E2E days as entry bars. |
| **Complexity & consensus** | Every row of the systems-impacted table pre-wired with its owner's level of effort before the council; objections converted into FSD rows instead of arbitrated in meetings. |
| **Root cause at scale** | Oversell was traced to two books carrying two meanings — fixed once in a signed definition and a real-time decrement contract (idempotency key per event; a reversal references its original), not incident by incident. |

## PROBING — CHALLENGE Q&A TUNED TO THIS INTERVIEWER

> **Peer — You owned no system. How did you get three organizations to sign one definition?**
>
> Pre-wired every row of the systems-impacted table with its owner's level of effort before the council — nobody learned their cost in the room — ran working sessions to one table, converted objections into FSD rows, and asked the council for three decisions rather than a review. Three signatories because each owned a different truth; one signature would have left two orgs free to disagree later.

> **Peer — Why lower-of-two and not an average or the book?**
>
> Because oversell is a customer defect and undersell is a planning variance. A positive out-of-balance means the book is above the floor; promising the difference risks oversell, so the basis is the minimum, and Inventory Control chases the variance through the Negative AFS and AFS Correction screens. The asymmetry is the policy.

> **Peer — Merchandising wanted merch reserve promisable. What happened?**
>
> I priced it — unpickable stock sold — and offered visibility without promisability. The no went into the FSD as a row, not into a meeting minute. A no that lives in a document survives; one that lives in a meeting gets re-litigated weekly.

> **Peer — How do you know it was a tenth of a percent?**
>
> I don't, to the decimal. The record says incidents were recurring and went to zero; the rate is a benchmark set deliberately low — a twentieth of a typical e-commerce inventory-cancel rate. First thing I would measure: the "item unavailable after order" cancel-reason code for the twelve months before January. If the $1M MCS figure already includes enhancement labor, I drop the change-premium row and the headline is about $2.2M.

> **Peer — What did you get wrong in this program?**
>
> The first in-transit rule I wrote barred every channel from promising units in transit — safe-feeling, and it starved dotcom of fast movers a day from the building. I corrected it through the same three signatories with a per-lane promise window and a kill switch, released on the normal cadence. Lesson: both sides of a rule get a number before it ships, and back-test a ruleset against history before publishing it. (S4 — pointer; full story on request.)

> **Peer — How would this work between your team and mine here?**
>
> Every promise rests on one published availability number per node that downstream systems read and never recompute. Between two teams that means: one definition with one owner per bucket, the policy signed before the code, a daily drift report with tolerance zero and a named control owner — and objections turned into rows we both sign.

### THE KEY MATH — TWO BOOKS, ONE BASIS, ONE AFS — AND THE HEADLINE'S DECOMPOSITION (ILLUSTRATIVE UNITS; RECORD DESIGN; DOLLARS LABELED)

| STEP | QUANTITY | FORMULA → SUBSTITUTION | BASIS | VALUE |
|---|---|---|---|---|
| 1 | Two books disagree | `OOB = S4_Avail − WMS_OnHand = 1,000 − 980 = +20` | illustrative units | +20 — book above floor |
| 2 | Committed buckets | `PickRes 200 + EcomPick 120 + EcomPack 60 + SNB 40 = 420` | illustrative | 420 spoken for |
| 3 | Definition A / B over-promise | `1,000 − 560 = 440; 980 − 560 = 420` | record problem — committed buckets invisible | 440 / 420 units over-promised |
| 4 | One basis, one AFS | `Basis = min(1,000, 980) = 980; AFS = 980 − 420 = 560` | lower-of-two (record design) | AFS = 560; the +20 never promised |
| 5 | Real-time decrement + idempotency | `store rings 3 → AFS 557, SNB 43; replayed key → no-op` | record design; 60-s publish budget | AFS 557 |
| 6 | Daily drift | `ISM.S4_Avail − MB52 = 957 − 957 = 0; tolerance 0` | record design — non-zero drift is a Sev | 0 |
| 7 | Headline | `$1.0M (anchor) + 30,000 × $30 = $0.9M + 3 × $110K = $0.33M + 40 × 60 h × $125 = $0.3M = $2.53M` | anchor + modeled rows (orders illustrative; rate 0.10%) | ≈$2.5M/yr (range $1.7–5.2M) |

Say the decomposition only from this packet's model (the loop packet says ≈$1M/yr support cost — the $1M is the shared core).

**Where this lands at Amazon.** Every fulfillment promise rests on one published availability number per node that downstream systems read and never recompute — at Amazon scale the ATP/promise seam between planning and the

order pipeline. One definition, one owner, policy signed before code. Reliability (oversells to zero), cost efficiency ($1M of run cost retired), responsiveness (a real-time ledger beside the book).

### TWIST MAP — LIKELY LP QUESTIONS FROM THIS INTERVIEWER → THE ANGLE TO TAKE WITH S1

| LIKELY QUESTION (THIS INTERVIEWER) | LP | ANGLE — WHAT TO EMPHASIZE | SKIP / NEVER SAY |
|---|---|---|---|
| **Tell me about a time you built trust across teams that didn't report to you.** | Earn Trust | Three orgs, three truths, one signed FSD on 6/5/25; pre-wiring; objections as rows. | S14 (the $2.5M discrepancy) is the Earn Trust primary — it is the bar raiser's; do not use it here. |
| **Tell me about a time you took ownership of something outside your remit.** | Ownership | Nobody owned the definition of "available"; I owned no system; built the bucket inventory and the signature chain. | Never "I configured S/4" — the SAP teams configured to the signed spec. |
| **Tell me about a decision where you had to say no to a senior stakeholder.** | Have Backbone | Merch reserve promisable — priced, refused, written as an FSD row with visibility offered instead. | Do not caricature Merchandising; their visibility need was met. |
| **Tell me about a large-scale change you led.** | Think Big | The book-of-record migration — MCS retired, one real-time ledger, two go-lives, peak untouched. | Keep S2's floor story out — that is Goforth's; say "one umbrella program" if asked. |
| **How do you define and govern a shared metric / definition between two teams?** | Functional — process governance | Bucket inventory with one owner each; the equation written; signatures before configuration; daily drift with tolerance zero; control owners named. | Keep the oversell $ labeled modeled; the movement is incidents → zero. |
| **Why does this matter beyond the ledger?** | Broad Responsibility / Customer Obsession | Oversell is a customer harm, not a ledger error; the asymmetry accepts visible undersell to make oversell zero by construction. | No YoY revenue claim exists. |

## S3 — Sephora S/4HANA — Shipped-Not-Billed and de-aggregation: Finance and Operations on one Financial On-Hand number, with the failure direction designed before the happy path

**LPs this story can serve here:** Earn Trust (backup — Finance and Ops on one number) · Dive Deep (the grain problem — one posting per aggregate vs transaction lines) · Invent and Simplify (one ledger, one settlement authority, a safe failure direction) · Insist on the Highest Standards (Σ per-line COGS = posting COGS to the cent)

**Process family:** Deliver / Return-to-book seam — sale-to-posting reconciliation, financial on-hand governance, month-end close control

**Honesty label.** The flow, controls and ownership map were mine; S/4 and OMS engineers implemented; Inventory Accounting confirmed in UAT and took over the aging report. The record has no canonical dollar for the pre-state drift; the ≈$0.6M/yr is my model and I say so. Rode the ISM program's dated chain (FSD 6/5/25; May execution go-live).

### 60-SECOND BUSINESS-FIRST PITCH — STOP ON THE PRIMARY METRIC

**Situation.** S/4 posts sales as aggregates — one posting per aggregation number with total quantity and cost — while the point-of-sale sends transaction lines. Between the sale and the posting, units were gone but the book was not decremented: a phantom-availability window in which sold units still showed as available, and Finance and Operations read different on-hand numbers and reconciled by argument, with a bridge spreadsheet at every month-end close. **Task.** Define how the real-time layer tracks that gap and reconciles it so promising, replenishment and Finance agree — under the program's zero-incident constraint, which applied to the book as much as to the customer. **Action.** I named the window and inventoried the events that move units before posting — POS flash sales, omni pending, ship-to-home confirmations — and got Finance and Operations to agree on the description of the gap before any design. The Shipped-Not-Billed bucket increments on those events and decrements only on a matched S/4 aggregate posting; de-aggregation pro-rates cost per line so the sum ties to the cent; and the failure direction was designed before the happy path — an unmatched posting keeps the bucket high, so availability under-states and Replenishment over-orders visibly instead of the customer being oversold invisibly. Finance objected at design review and wanted its own reconciliation; I showed the same number arriving from the physical side first and gave them the unmatched-posting aging report to own — a parallel spreadsheet dies when its owner owns the real report. FSC Inventory Accounting signed inside the FSD on 6/5/25; it rode the May go-live. **Result.** Primary metric: Financial On-Hand definitions — two, Finance's book versus the operations count → one shared number, availability decremented at sale or ship rather than at posting; over-sell from unmatched postings zero by construction; zero incidents. About $0.6M a year modeled — phantom-availability cancels avoided, Finance reconciliation labor, month-end close time; the record has no pre-state drift dollar and I am not claiming one.

### SITUATION / TASK — THE BUSINESS PROCESS PROBLEM, SCOPE, SCALE, BASELINES

**S —** S/4 posts sales as aggregates — one posting per Aggregation# with total quantity and COGS; the Aptos POS sends transaction lines with an Order#. Between the sale and the posting, units are gone but the book is not decremented — the phantom-availability window — and Finance and Operations read different on-hand numbers and reconcile by argument.

The process cost: sold units shown as available for up to a day on the overnight batch; two books reconciled by argument; a bridge spreadsheet at every month-end close; and the gap hiding as oversell if an estimate ever decremented the book.

**T —** Define how ISM tracks the gap in real time and reconciles it so promising, replenishment and Finance agree, with the program's zero-incident constraint applying to the book as much as to the customer. Goal: one Financial

On-Hand, AFS decremented at sale rather than posting, a safe failure direction.

**Baselines on the table:** Financial On-Hand truths: two (target one) · AFS decrement point: at the S/4 posting (target the sale/ship event) · over-sell from unmatched postings: a live risk (target zero by construction; aging report empty or explained at T+1).

## ACTION — THE PROCESS / OPERATING-MODEL REDESIGN, IN GATED PHASES

- **Discovery** — problem statement (the phantom-availability window); event inventory (POS flash sale, omni pending, ship-to-home confirm, S/4 Aggregation# posting); who-reads-which-number map. *Exit bar:* Finance and Ops agree on the description of the gap and which events move units before posting; Internal Audit named as reviewer.
- **Plan & Design** — SNB bucket definition (increment on events; decrement only on matched posting); join spec Aggregation# ↔ POS header ↔ lines; per-line COGS pro-rata de-aggregation with the last line absorbing rounding; the failure-direction statement (unmatched ⇒ SNB stays high); control-owner map. *Exit bar:* signed inside the FSD 6/5/25; Finance accepts one FOH; cost ownership confirmed S/4-owned, ISM visibility-only.
- **Build & Test** — de-aggregation test family (n lines; missing line; duplicate header); the SNB-never-negative assertion; the unmatched-posting aging report; sequenced by event source — POS flash sales, then omni pending, then ship-to-home — each added to SIT only when its increments and matched decrements reconciled. *Exit bar:* every posting matches or ages visibly; Σ per-line COGS = posting COGS to the cent in every cell; UAT = confirmation by Inventory Accounting.
- **Cutover & Go-Live** — riding the May go-live: SNB opening balance reconciled to unposted sales at the snapshot; SNB publish back to S/4 enabled; rollback cheap because S/4 stayed settlement authority. *Exit bar:* aging report empty or explained at T+1; no negative SNB; Finance reads FOH from ISM.
- **Hypercare / Close** — daily aging review with Inventory Accounting; control-owner handoff. *Exit bar:* aging report owned by Finance without me; month-end FOH ties to MB5L through the ISM number, not a bridge spreadsheet (mechanism canonical; days-to-tie not in record).

**Influence and ownership.** Mine: the SNB definition, the join spec, the de-aggregation rule, the failure-direction statement, the control-owner map and the test families; S/4 and OMS engineers implemented; Inventory Accounting confirmed in UAT and took over the aging report. Two refusals: real-time per-line posting to S/4 (throughput and posting cadence are why S/4 posts aggregates — ISM is the real-time ledger, S/4 stays settlement authority) and a Finance-derived SNB (that is two numbers; the whole point was one).

> **Designed the process; others own it now.** I designed the SNB definition, the join spec and the de-aggregation rule; Inventory Accounting owns the aging report and the daily review, and S/4 remained settlement authority. **Trade-off put to leadership.** The recommendation to Finance and Operations outlined the competing metrics — availability accuracy at the sale event vs the integrity of the financial posting — and chose the failure direction before the happy path: under-sell, never over-sell.

## RESULTS — $ · # · % · TIME · CUSTOMER · QUALITY (CANON NUMBERS ONLY)

- **$** — ≈$0.6M/yr *modeled* (phantom-availability cancels avoided ≈$0.45M · Finance reconciliation labor ≈$0.14M · month-end close time ≈$0.05M; range $0.25–1.2M); the record has no pre-state drift dollar and I am not claiming one.
- **#** — Financial On-Hand truths 2 → 1 (record, FSD signed 6/5/25); over-sell from unmatched postings = 0 by construction.
- **%** — design KPI: match rate ≥ 99.5% within 24 h; the modeled phantom-cancel rate is ~0.10 pt of orders → ~0 (modeled, labeled).
- **Time** — AFS decrement point moved from the S/4 posting to the sale/ship event (record); rode the May execution go-live; aging report empty or explained at T+1 (record exit bar).
- **Customer** — the failure direction protects the customer: an unmatched posting under-states availability — under-sell, never over-sell (record mechanism).

- **Quality** — Σ per-line COGS = posting COGS to the cent; SNB never negative; month-end FOH ties to MB5L through the ISM number, not a bridge spreadsheet (record mechanism; days-to-tie not in record).

**L6 LEVEL EVIDENCE** — what the interviewer pattern-matches against the Amazon L6 functional rubric; every line is this story's canon, no new numbers

| L6 DIMENSION | EVIDENCE IN THIS STORY |
|---|---|
| **Ambiguity** | A phantom-availability window between the sale event and the S/4 aggregation posting, with Finance and Operations reading different numbers and nobody owning the gap. I wrote the problem statement, the event inventory and the who-reads-which-number map. |
| **Scope & influence** | Finance (Inventory Accounting), Operations, S/4 and OMS engineers; rode the May go-live; Inventory Accounting confirmed in UAT and took the aging report — influence across a Finance organization I did not sit in. |
| **Execution trade-off + mechanism others now own** | Real-time per-line posting to S/4 (refused — S/4 stays settlement authority) vs an SNB bucket in ISM; the failure direction chosen before the happy path — under-sell, never over-sell. Mechanism others own now: the unmatched-posting aging report, owned by Finance without me. |
| **Impact (long-term)** | Financial On-Hand truths 2 → 1 (FSD signed 6/5/25); month-end FOH ties to MB5L through the ISM number, not a bridge spreadsheet; ≈$0.6M/yr modeled. |
| **Process improvement (standard work institutionalized)** | The de-aggregation test family (n lines; missing line; duplicate header); the SNB-never-negative assertion; the match-rate design KPI (≥ 99.5% within 24 h); a daily aging review handed to Inventory Accounting. |
| **Complexity & consensus** | Two refusals held in writing; sequencing by event source (POS flash sales, then omni pending, then ship-to-home) so each source carried its own defect class. |
| **Root cause at scale** | The structural cause — the AFS decrement point sitting at the S/4 posting instead of the sale/ship event — moved once, for every channel, rather than reconciled by hand each month. |

## PROBING — CHALLENGE Q&A TUNED TO THIS INTERVIEWER

> **Peer — Finance wanted their own reconciliation. How did you turn that around?**
>
> I showed the same number arriving from the physical side first and gave them the real report to own — the unmatched-posting aging report, reviewed daily with Inventory Accounting. A parallel spreadsheet dies when its owner owns the real report. They signed inside the FSD on 6/5/25.

> **Peer — Why decrement only on a matched posting? Isn't that just moving cost to Replenishment?**
>
> Yes, and I priced it that way on purpose: an unmatched posting under-states availability and shows up as an aged row Finance clears the same day, so the over-order is small and visible. Decrementing on an estimate would hide the gap as oversell. The measure is the aging report's p95 in hours and the match rate inside 24 hours; above 99.5% the Replenishment side rounds to zero.

> **Peer — Your cancel row looks borrowed from S1. Double-counted?**
>
> Different window: S1's rate is the vocabulary problem across node types; this is the posting-lag window inside one node. If the room wants one number, I keep S1's and drop this row — the mechanism still stands on the record.

> **Peer — Why not post every line to S/4?**
>
> Throughput and posting cadence — S/4 posts aggregates for a reason. ISM is the real-time ledger; S/4 stays the settlement authority. The price is a matching cycle to maintain; the benefit is one book.

> **Peer — 32 hours a week of Finance time — source?**
>
> Benchmark, not record — four people at eight hours a week is what Inventory Accounting described, not what a timesheet says. I would count manual journal entries and bridge lines per month before and after; that is in the posting log and it is not opinion.

> **Peer — Lesson?**
> Design the failure direction before the happy path. Differently: Internal Audit in discovery, not review — the day Inventory Accounting asked me why a posting had aged before I noticed it, the control had an owner.

## THE KEY MATH — THE WINDOW, THE DE-AGGREGATION TO THE CENT, THE FAILURE DIRECTION, AND THE MODELED DOLLARS

| STEP | QUANTITY | FORMULA → SUBSTITUTION | BASIS | VALUE |
|---|---|---|---|---|
| 1 | Phantom window | `W = t_post − t_sale ≈ 1 day on the overnight batch; 12 units/day × 1 day = 12 phantom units per SKU-day` | record design problem; rate illustrative | sold units shown available up to a day |
| 2 | SNB rule | `SNB += each sale/ship event; SNB −= matched posting only` | record design | window = 12 × 0 = 0 |
| 3 | De-aggregation | `COGS_i = $300 × q_i / 12 → $75 · $50 · $100 · $25 · $50; Σ = $300.00` | illustrative five lines; last line absorbs rounding | ties to the cent |
| 4 | Late line | `one line (q = 2) late → SNB = 12 − 10 = 2 → aged row at 24 h` | illustrative | visible, not phantom |
| 5 | Failure direction | `unmatched ⇒ SNB high ⇒ AFS_reported ≤ AFS_true` | record design | under-sell, never over-sell |
| 6 | Match-rate KPI | `matched / received ≥ 99.5% within 24 h` | design target (record) | ≥ 99.5% |
| 7 | Dollars | `30M × 0.10% × $15 = $0.45M; 32 h × 52 × $85 = $0.14M; 6 × 8 h × $85 × 12 = $0.05M` | all modeled; no pre-state drift dollar in record | ≈$0.6M/yr (range $0.25–1.2M) |

Finance reads FOH from ISM; S/4 remains settlement authority; the aging report with a named Finance owner is the real product.

**Where this lands at Amazon.** Every fulfillment event moves units before Finance posts them — the same window at a vastly larger grain. The transferable design: one real-time ledger, the financial system as settlement authority, a safe failure direction chosen before the happy path, and a control report with a named owner instead of a bridge spreadsheet. Reliability (over-sell zero by construction), cost efficiency (bridge labor and close time), one number for two organizations.

## TWIST MAP — LIKELY LP QUESTIONS FROM THIS INTERVIEWER → THE ANGLE TO TAKE WITH S3

| LIKELY QUESTION (THIS INTERVIEWER) | LP | ANGLE — WHAT TO EMPHASIZE | SKIP / NEVER SAY |
|---|---|---|---|
| **Tell me about a time you got Finance and Operations to agree.** | Earn Trust | FOH 2 → 1; the aging report given to Finance to own; signed inside the FSD on 6/5/25. | Never claim a pre-state drift dollar — the ≈$0.6M is modeled. |
| **Tell me about a problem that needed you to understand the data at its grain.** | Dive Deep | One posting per aggregate vs transaction lines; the Aggregation# ↔ header ↔ lines join; COGS pro-rated to the cent. | Skip the table names unless asked. |
| **Tell me about simplifying something complex.** | Invent and Simplify | One ledger, one settlement authority, one failure direction, one control report — instead of two books and a bridge. | Do not say "I built the matcher" — engineers implemented the design. |

| LIKELY QUESTION (THIS INTERVIEWER) | LP | ANGLE — WHAT TO EMPHASIZE | SKIP / NEVER SAY |
|---|---|---|---|
| **How do you design controls for a process?** | Functional — control design | Choose the failure direction first; a never-negative assertion; a KPI with a denominator (match rate ≥ 99.5% in 24 h); an aging report with a named owner; rollback that is cheap because the authority never moved. | Keep S14's standing reconciliation report out — bar raiser's. |
| **What did you refuse and why?** | Have Backbone | Per-line posting to S/4; a Finance-derived SNB — both refused on mechanism, not preference. | — |
| **What would you do differently?** | Learn and Be Curious | Internal Audit in discovery, not review. | No days-to-tie figure exists — do not invent one. |

## Question bank — likely questions for this interviewer, with a one-line answer plan

Behavioral and functional questions a peer Senior Manager of Supply Chain is likely to ask. Each plan points at a packet story or a bench story at the end of this packet (S4, S5, S9); stories allocated to other interviewers are pointers only.

| # | LIKELY QUESTION | ANSWER PLAN (ONE LINE) | STORY |
|---|---|---|---|
| 1 | **Tell me about a time you aligned multiple organizations on one definition.** | S1 pitch — three truths, one signed FSD 6/5/25; pre-wiring; objections as rows; oversells recurring → zero. | S1 |
| 2 | **Tell me about getting Finance to trust an operational number.** | S3 pitch — one FOH; failure direction; the aging report given to Finance to own. | S3 |
| 3 | **How do you influence without authority?** | Pre-wire every row with its owner's LOE; working sessions to one table; three decisions not a review; a no that lives in a document. (S5, the council pack — same program; pointer only.) | S1 (S5 pointer) |
| 4 | **Tell me about a time you were wrong.** | S4 (bench) — the in-transit rule I wrote too conservatively; starved dotcom of fast movers; fixed through the same three signatories with a per-lane promise window and a kill switch; lesson: both sides of a rule get a number. Keep it to the pointer unless the room asks for the full walk. | S4 pointer |
| 5 | **How would our two teams share a process / a metric here?** | One definition with one owner per bucket; policy signed before code; a daily drift report with tolerance zero and a named control owner; a monthly release train for rule changes. | S1 / S3 |
| 6 | **Tell me about designing for failure.** | S3 — the failure direction before the happy path; unmatched ⇒ under-sell; aging report at T+1. | S3 |
| 7 | **How do you handle a stakeholder who wants a number that looks better (or a capability that sells more)?** | S1 — merch reserve promisable, priced as unpickable stock sold; visibility offered, promisability refused, written as a row. | S1 |
| 8 | **How do you decompose a cross-functional process problem?** | Master §3.99 — book of record and metric first; stocks (every bucket a unit can sit in); flows with their documents; the broken invariant (a unit in two stocks; a number with two owners); minimal design; sequence. | Master §3.99 |
| 9 | **What is the business value of the book-of-record migration?** | ≈$2.5M/yr: $1M measured MCS run cost (anchor) + ≈$1.5M modeled with its three rows; incidents to zero is the record movement. | S1 |
| 10 | **How do you keep a design from being re-litigated after sign-off?** | Three signatories, each owning a different truth; the out-of-scope fence (S5, bench); every change rides the monthly release train as an FSD change, not a side configuration (S4 pointer). | S1 (S5/S4 pointers) |
| 11 | **Tell me about a release / change-management discipline you set up.** | Rule changes ride the monthly cadence as signed FSD changes; the node × channel × bucket matrix is the regression gate; no story enters QA without a matrix cell reference. | S1 |
| 12 | **What's one thing a peer would say is hard about working with you?** | I convert verbal agreements into rows people have to sign — slower in the week, faster over the program. Say it plainly, with the S1 pre-wiring as the example. | S1 |

## Cheat-numbers box — canon values with basis labels

Canon numbers for S1 and S3 (and the S4/S5 bench stories) with their basis labels, from the master packet's §4.1 ledger.

| METRIC | VALUE TO SAY | BASIS | NEVER |
|---|---|---|---|
| **S1 primary metric** | customer-facing oversell incidents recurring across three node types → zero through two go-lives | record | an oversell count or rate as record |
| **S1 headline $** | ≈$2.5M/yr = $1M MCS run cost retired + ≈$1.5M modeled (oversell ≈$0.9M · recon ≈$0.33M · change premium ≈$0.3M); range $1.7–5.2M | user anchor + modeled | presenting the $1.5M as record; adding the shrink/time-to-market upside into the headline |
| **S1 signatures / dates** | FSD v1.1 (3/5/25) → v1.2 (5/16/25) → signed 6/5/25 by Supply Chain Strategy, Inventory Management, FSC Inventory Accounting | record | one signatory |
| **S1 gate chain** | kickoff 3/29 → council 5/15 → LOE 7/2 → sign-off 7/31 → FSD 8/14; E2E 11/22 · 12/20 · 1/30; January planning + May execution go-lives; 836 rows / 85.4% closed | record (shared program artifacts with S2/S5) | — |
| **S1 design** | lower-of-two basis; AFS = Basis − PickRes − EcomPick − EcomPack − SNB; idempotency key per event; daily ISM-vs-MB52 drift, tolerance 0; OMS reads, never computes | record design | "I configured S/4" / "I wrote ABAP" |
| **S1 reconciliation FTE** | ≈4 FTE by hand → ≈1 FTE reading the drift report | modeled | stating it as record |
| **S3 primary metric** | Financial On-Hand truths 2 → 1; AFS decremented at sale/ship, not posting; over-sell from unmatched postings = 0 by construction | record (FSD 6/5/25) | — |
| **S3 dollars** | ≈$0.6M/yr (cancels 0.45 · recon labor 0.14 · close 0.05); range $0.25–1.2M | modeled — no pre-state drift dollar in record | any record drift dollar; double-counting the cancel rate with S1 |
| **S3 design KPI** | match rate ≥ 99.5% within 24 h; aging report empty or explained at T+1; Σ per-line COGS = posting COGS to the cent | record design | a days-to-tie figure |
| **S4 (bench)** | starved lanes → restored; oversell +0; kill switch never fired; ≈$0.5M/yr modeled margin | record / modeled | a dollar for the conversion lost while I was wrong |
| **S5 (bench)** | surprise escalations → 0; ~80 names across ~15 functions; 5/15 → 7/31 → 8/14; ≈$0.5M one-time modeled | record / modeled | — |
| **SAP tenure** | S/4HANA integration program lead 3+ years (Apr 2023 – present); client-SAP data integration at Elementum 2017–19; 15 years full-lifecycle SCM systems | record | "10 years of SAP configuration"; "SAP functional owner" |

**Never-say, one glance before the call.**

- Never "I configured S/4," "I deploy SAP," or "I wrote ABAP" — integration program lead; SAP teams and configurators built to the signed FSD. Manhattan DOM is the only platform personally configured in canon.
- Never present the ≈$1.5M (S1) or the ≈$0.6M (S3) as record; the $1M MCS figure is the user anchor and the only measured dollar here.
- Never give the S1 and S3 cancel rates as two results in one room — keep S1's, drop S3's row.
- Never invent a days-to-tie figure, a direct-report headcount, or a pre-state oversell dollar.
- S4 and S5 are bench stories (full entries at the end of this packet) — reach for them only when the question does not fit S1/S3; one or two sentences in the question bank otherwise, then back to S1/S3.
- S14 (Earn Trust primary) is the bar raiser's; S2 (same umbrella program) is Goforth's — "one umbrella program" if asked, no floor story here.

## Backup bench — use only if a primary doesn't fit

Four fallbacks for a peer Senior Manager — S4 and S5 are the question bank's pointers, here in full bench form: S4 (Sephora — the in-transit promise rule I got wrong, fixed through the same three signatories; the failure / Are Right, A Lot question), S5 (Sephora — the architecture council without owning a system; Backbone and influence without authority — same program family as S1, so use only when the question is about governance rather than the definition) and S9 (Elementum — the 597-lane Pareto two levels up; Backbone / Think Big outside the Sephora program); plus a doc-derived fourth from the Flexport program record — the FFA Audit Logs PRD (controls and process integrity for a partner-facing platform; record-from-docs labels; same platform as Tripathy's S12, facts kept identical to her packet). Tell S1 and S3 first.

**Bench discipline.** The two (or four) primaries above are the stories to tell first. Reach for a bench story only when the question genuinely does not fit a primary, or when the interviewer asks for "another example" on an LP already spent. Each entry is a 60-second pitch, a compressed STAR, one results line and two probes — enough to answer and stop; no full probing bank or derivation. Where a bench story is another interviewer's primary, the red note says so: use it only if the question demands it, and keep the facts identical to that packet.

### BACKUP — ONLY IF ASKED S4 — Sephora S/4HANA — the in-transit promise rule I got wrong, fixed through the signed policy, not a side configuration

**LPs this backup can serve:** Are Right, A Lot (the failure question) · Customer Obsession (backup) · Earn Trust (owning the miss) · Learn and Be Curious — the designated policy-failure story

**Basis.** The v1 rule was mine; the indictment came from my own program's reports. Lane σ ≈ 0.3 day and the ~0.49-day window are the record's promise math; the record carries no lane counts and no dollar for the conversion lost while I was wrong — I will not invent the loss figure.

#### 60-SECOND PITCH — STOP ON THE PRIMARY METRIC

**Situation.** My first ISM ruleset barred every channel from promising units in transit; the Aging In-Transit report showed dotcom starved of fast movers one day from the building. **Task.** Correct my own policy without weakening oversell protection, through the same three signatories. **Action.** Discovery: conversion-loss evidence from the aging and replenishment reports — the entry bar to reopen the policy; I stated that the v1 rule was mine. Plan & Design: promise math — lane σ ≈ 0.3 day, a ~0.49-day window ≈ 95% coverage; Promisable = in-transit × 1[ETA ≤ window] × 1[on-time ≥ floor]; Aging In-Transit as the kill switch; both blanket options rejected in writing. Build & Test: back-test against receiving history; four regression cells added to the node × channel matrix. Cutover: released on the normal monthly cadence as a signed FSD change, lane by lane — no hotfix, no side configuration. Hypercare: kill switch armed; never fired. Lesson: both sides of a rule get a number before it ships. **Result.** Primary metric: sellable units on qualifying in-transit lanes — starved → restored inside a per-lane promise window, oversell increase = zero, kill switch never fired. ≈$0.5M/yr modeled margin recovered — the record has no dollar for the conversion lost while I was wrong, and I will not invent one.

#### COMPRESSED STAR

- **S —** v1 of my own ISM ruleset: no channel could promise in-transit units — safe-feeling, and starving dotcom of fast movers reliably one day from the building.
- **T —** Fix my own signed policy without weakening oversell protection; take the miss back through the same three signatories.
- **A —** Evidence pack from Aging In-Transit and Replenishment; reframed as promise math (σ ≈ 0.3 d → ~0.49-d window ≈ 95%); rule with on-time floor and kill switch; back-test against receiving history; four regression cells;

released lane by lane on the monthly cadence as an FSD change.

- **R —** Starved lanes restored inside the window; oversell increase = 0; kill switch never fired; re-signed by the same three signatories; lesson institutionalized — the Locus 98% gate (S2) and the 0.5% tolerance (S18) were set from data before the event.

## RESULTS LINE

Sellable units on qualifying in-transit lanes starved → restored inside a back-tested ~0.49-day window (record) · oversell increase = 0; kill switch armed, never fired (record) · four regression cells added (record) · released as a signed FSD change on the normal cadence, no hotfix (record) · ≈$0.5M/yr modeled margin recovered (inputs illustrative).

## THE MATH — STEP-BY-STEP DERIVATION

### Table 1 — before: the problem in numbers.

| STEP | QUANTITY | FORMULA → SUBSTITUTION | INPUTS (BASIS) | VALUE |
|---|---|---|---|---|
| 1 | The v1 rule | `Promisable = 0 for ALL in-transit inventory` | equivalent to claiming σ → ∞ or on-time rate → 0 — infinite variability (record) | everything in transit hidden |
| 2 | One concrete lane | `Promisable = 120 × 0 = 0` | lane with InTransit = 120 units, ETA 0.4 day away (illustrative) | 120 sellable units hidden |
| 3 | Across qualifying lanes | `≈300 fast-mover units/day × 0 promisable` | illustrative volume | ≈300 units/day sat unsellable one day from the building |
| 4 | What starving costs (one SKU) | `100 × 60% × $20 = $1,200/day` | 100 units one day out; 60% would convert; $20 margin (illustrative, not program data) | $1,200/day under-sell |
| 5 | The ignored other side | `100 × (1 − 0.97) = 3 late units; 3 × $15 = $45/day` | on-time rate 97%; $15 expedite (illustrative) | $45/day over-sell exposure — vs $1,200: the “safe” rule priced one side only |

### Table 2 — the fix: step by step, values substituted.

| STEP | QUANTITY | FORMULA → SUBSTITUTION | INPUTS (BASIS) | VALUE |
|---|---|---|---|---|
| 1 | Measure lane variability | `σ = 0.3 day` | trailing receiving history (record) | σ = 0.3 day |
| 2 | Choose coverage | `z at 95% = 1.645` | normal approximation on lane transit (toolkit label) | z = 1.645 |
| 3 | Promise window | `w = z × σ = 1.645 × 0.3 = 0.4935` | rows 1–2 | w ≈ 0.49 day |
| 4 | Coverage check | `Φ(w / σ) = Φ(0.49 / 0.3) = Φ(1.63) ≈ 0.95` | share of arrivals inside the window | ≈95% covered |
| 5 | Corrected rule | `Promisable = InTransit × 1[ETA ≤ w] × 1[OTR ≥ floor]` | OTR = on-time rate over trailing receipts; floor = per-lane admission threshold, 95% here (floor illustrative) | lane by lane, not all-or-nothing |
| 6 | Lane that qualifies | `ETA 0.4 ≤ 0.49 → 1; OTR 97% ≥ 95% → 1; Promisable = 120 × 1 × 1 = 120` | the Table 1 lane, re-evaluated (ETA and OTR illustrative) | its 120 units become promisable |
| 7 | Lane that fails | `ETA 1.2 > 0.49 → 0; Promisable = 80 × 0 = 0` | a slower lane with 80 units in transit (illustrative) | stays at 0 — correctly out |
| 8 | Sample-size gate | `n = z²p(1−p)/e² = 3.84 × 0.95 × 0.05 / 0.05² = 3.84 × 0.0475 / 0.0025 ≈ 73` | OTR estimated within ±5% at 95% CI (illustrative) | a lane qualifies after ~73 receipts |

| STEP | QUANTITY | FORMULA → SUBSTITUTION | INPUTS (BASIS) | VALUE |
|---|---|---|---|---|
| 9 | Kill switch, values in | `trigger = 2 × p90 = 2 × 2.0 = 4.0 days; any unit in transit > 4.0 days → lane out` | lane p90 = 2.0 days (illustrative); armed, reviewed weekly by Inventory Control (record) | never fired (record) |
| 10 | Back-test | `rule replayed against receiving history; 4 regression cells` | before release (record) | no oversell in the back-test |

**Table 3 — after: the metric, before → after.**

| STEP | METRIC | BEFORE | AFTER | BASIS |
|---|---|---|---|---|
| 1 | Sellable units on qualifying in-transit lanes | 0 | ≈300 units/day restored inside the window | record mechanism; volume illustrative |
| 2 | Promise window | none — blanket exclusion | w ≈ 0.49 day at ~95% cover | record (promise math) |
| 3 | Oversell increase from the window | — | 0 | record |
| 4 | Kill switch | not armed | armed, never fired | record |
| 5 | Rule changes after | ad hoc | same 3 signatories, back-test + matrix regression | record — both sides priced from then on |

**Table 4 — the dollars, every multiplication shown.**

| STEP | QUANTITY | FORMULA → SUBSTITUTION | INPUTS (BASIS) | VALUE |
|---|---|---|---|---|
| 1 | Would-be sales | `300 × 60% = 180 sales/day` | units restored (illustrative) × conversion (illustrative) | 180 sales/day |
| 2 | Truly lost without the fix | `180 × 40% = 72 sales/day` | 40% lost-sale share — the rest would have waited (illustrative) | 72 lost sales/day |
| 3 | Margin recovered | `72 × $20 = $1,440/day` | $20 unit margin (illustrative) | ≈$1,440/day |
| 4 | Annualized | `$1,440 × 365 = $525,600` | modeled; the mechanism is record | ≈$0.5M/yr |
| 5 | Cost while v1 stood | `no dollar in the record — not claimed` | record | not claimed |

## TWO PROBES

> **Why not just hotfix it — you knew it was wrong?**
>
> Because the error was in the policy, so the fix had to be too — in the governed artifact, with regression, through the same signatures. A side configuration would have fixed the symptom and left the policy lying about what it allowed.

> **What did the miss cost?**
>
> The record carries no dollar for the conversion lost while v1 stood, and I will not invent one. What I can say is the mechanism: fast movers reliably one day from the building were unsellable on dotcom until the window existed; the ≈$0.5M/yr recovered is my model on illustrative inputs, labeled.

**Canon numbers (basis labels):** Lane σ ≈ 0.3 d; ~0.49-d window ≈ 95% coverage (record promise math) · oversell increase = 0 (record) · 4 regression cells (record) · kill switch never fired (record) · same three signatories re-signed (record) · ≈$0.5M/yr (modeled). Never a lane count; never a dollar for the loss while wrong.

## BACKUP — ONLY IF ASKED S5 — Sephora — the architecture council without owning any system: three decisions and a fence

**LPs this backup can serve:** Have Backbone; Disagree & Commit (third) · Earn Trust (influence without authority) · Think Big (architecture governance). Same program family as S1/S2 — the governance story

**Basis.** I owned none of the systems; the design decision was the deliverable. Zero escalations and the dated gate chain are the record; the dollars are my model of what governance avoided.

### 60-SECOND PITCH — STOP ON THE PRIMARY METRIC

**Situation.** The UDC location and inventory model had to clear architecture council — a challenge gate — with ~80 named participants across ~15 functions, and I owned none of those systems. **Task.** Take the design from discovery through council to sign-off and FSD without a surprise escalation. **Action.** Discovery: both inventory books profiled before any design argument; business sessions closed 4/30. Plan & Design: every impacted row pre-wired with a signed LOE by 7/2; the council asked for three separable decisions plus an out-of-scope fence (reserve math untouched; MCS decommission belongs to Global ERP). Build & Test: the "build" was the pack — architecture, staged timeline, peak-freeze boundary. Cutover: 5/15 challenge → 7/31 sign-off → 8/14 FSD. Hypercare: the fence held in writing for the whole program; a second live warehouse became a monitored shadow copy, not an argument. Lesson: a no that lives in a document survives. **Result.** Primary metric: surprise escalations on a 60+ stakeholder design — the norm on prior programs → zero; the dated gate chain held. ≈$0.5M one-time modeled (avoided rework on ~3 escalations that never happened + six weeks of benefit not deferred + a fenced second reserve engine not built).

### COMPRESSED STAR

- **S —** A dozen core systems (20+ touched), ~80 participants across ~15 functions, a council that re-litigates designs this size — and I owned none of the systems.
- **T —** Discovery (kickoff 3/29, sessions 4/30) → council 5/15 → sign-off 7/31 → FSD 8/14, without a surprise escalation.
- **A —** Profiled both books first — their numbers, not my opinion; systems-impacted table with an owner on every row; one LOE template, one deadline (7/2), one private conversation per owner before the room; three separable decisions plus the fence; the second-truth objection converted into a monitored shadow copy.
- **R —** Surprise escalations → 0 for the program's life; 100% pre-wire coverage by 7/2; gate chain held to the day with zero re-litigation; fence breaches 0; handed clean into the build (S2).

### RESULTS LINE

Surprise escalations → 0 for the program's life (record) · gate chain 3/29 → 4/30 → 5/15 → 7/2 → 7/31 → 8/14 held to the day (record) · pre-wire coverage 100% — every impacted row with a signed LOE by 7/2 (record) · both scope refusals held in writing; fence breaches 0 (record) · ≈$0.5M one-time (modeled — what did not happen).

### THE MATH — STEP-BY-STEP DERIVATION

**Table 1 — before: the problem in numbers.**

| STEP | QUANTITY | FORMULA → SUBSTITUTION | INPUTS (BASIS) | VALUE |
|---|---|---|---|---|
| 1 | Design size | `~80 named participants × ~15 functions; 20+ systems touched` | record | the blast radius |
| 2 | Norm for this size | `~3 surprise escalations per design` | industry benchmark 2–4 | ~3 expected |
| 3 | Price of one escalation | `960 rework hours × $110/h = $105,600` | modeled | ≈$0.1M each |

| STEP | QUANTITY | FORMULA → SUBSTITUTION | INPUTS (BASIS) | VALUE |
|---|---|---|---|---|
| 4 | Bundle decision risk | `P(all pass) = 0.9 × 0.9 × 0.9 = 0.73` | one up/down vote on the whole design; p = 0.9 per decision (illustrative) | 27% chance one objection unwinds everything |

**Table 2 — the fix: step by step, values substituted.**

| STEP | QUANTITY | FORMULA → SUBSTITUTION | INPUTS (BASIS) | VALUE |
|---|---|---|---|---|
| 1 | Three separable decisions | `architecture · timeline · peak boundary → 2³ = 8 outcomes` | any partial approval preserves progress (record design) | a no on one cannot unwind all |
| 2 | Expected approvals | `E = 0.9 + 0.9 + 0.9 = 2.7 of 3` | vs the bundle's 0.73 chance of everything (illustrative p) | 2.7 vs 0.73 |
| 3 | Pre-wire coverage | `rows with signed LOE / impacted rows = 100% by 7/2` | one template, one deadline, one private conversation per owner (record) | 100% before the room |
| 4 | Gate chain held | `3/29 → 4/30 → 5/15 → 7/2 → 7/31 → 8/14 = 77 days council→sign-off` | held to the day (record) | 0 re-litigation |

**Table 3 — after: the metric, before → after.**

| STEP | METRIC | BEFORE | AFTER | BASIS |
|---|---|---|---|---|
| 1 | Surprise escalations | ~3 (the norm) | 0 for the program's life | record; norm is benchmark |
| 2 | Pre-wire coverage before council | partial | 100% by 7/2 | record |
| 3 | Fence breaches in the log | — | 0 | record |
| 4 | Re-litigation after sign-off | — | 0 across 77 days | record |

**Table 4 — the dollars, every multiplication shown.**

| STEP | QUANTITY | FORMULA → SUBSTITUTION | INPUTS (BASIS) | VALUE |
|---|---|---|---|---|
| 1 | Avoided rework | `3 × 960 h = 2,880 h; 2,880 × $110 = $316,800` | ~3 escalations (benchmark) × rework hours × rate (modeled) | ≈$0.3M |
| 2 | Slip weeks avoided | `6 / 52 = 0.115 yr; 0.115 × $900,000 = $103,800` | six weeks of S2's ≈$0.9M/yr labor value not deferred (modeled) | ≈$0.1M |
| 3 | Fenced engine not built | `2 eng × 12 wk × 40 h = 960 h; 960 × $110 = $105,600` | second reserve engine (modeled) | ≈$0.1M |
| 4 | Total one-time | `0.3 + 0.1 + 0.1 = $0.5M` | all modeled — the dollars are what did not happen | ≈$0.5M |

## TWO PROBES

> **How do you get sixty people to sign LOE you don't control?**
>
> A template, a deadline and a private conversation each — so nobody learned their cost in the room. The council was then asked for three separable decisions it could take one at a time; separability was the design.

> **What did you refuse, and did it hold?**
>
> Retuning the reserve math and folding in the MCS decommission — both written, signed and held; later scope requests were answered from the pack in writing. The legitimate concern (a second live warehouse becoming a second truth) was converted into the design as a monitored shadow copy, because converting a concern costs less than winning it.

**Canon numbers (basis labels):** ~80 participants / ~15 functions; a dozen core systems, 20+ touched (record) · 5/15 → 7/31 → 8/14; LOE 7/2 (record dates) · escalations 0; fence breaches 0 (record) · ≈$0.5M one-time (modeled). Never a dollar as record; never pair with S1 or S2 for the same interviewer.

## BACKUP — ONLY IF ASKED S9 — Elementum — 1,000 supplier/carrier routes asked, 597 built first: X12 EDI onboarding behind a ≥95% valid-ASN cohort gate

**LPs this backup can serve:** Have Backbone; Disagree & Commit (primary — 597 vs 1,000, two levels up) · Deliver Results · Insist on the Highest Standards (the gate held twice) · Think Big (the staged 1,000-lane vision) · Broad Responsibility (small carriers kept inside the standard by the portal floor)

**Basis.** I owned the scope analysis and the onboarding standard, influencing an executive two levels above my sponsor; engineers built the maps and harness to the standard; Transportation owned the carrier relationships. Cohort sizes and the number of cohorts are not in the record — do not quote them.

### 60-SECOND PITCH — STOP ON THE PRIMARY METRIC

**Situation.** The Global Head of Supply Chain wanted inbound visibility on 1,000 lanes in two months; shipment-data completeness was 60% and high-value lanes were blind. **Task.** I owned the scope analysis and the onboarding standard, influencing an executive two levels above my sponsor. **Action.** Discovery: a spend Pareto — 597 lanes carried 80% of freight spend — pre-validated with Transportation and Procurement before I took it up. Plan & Design: the onboarding standard (X12 850/855/856/940/945/214/997 with an XML/portal floor — "the standard was the data, not the transport") and a cohort gate of ≥95% valid ASN. Build & Test: a mapping-and-test harness and a scorecard per lane. Cutover: 597 live in month one; one cohort failed the gate twice on a timestamp format and I held it both times. Hypercare: completeness scorecards to 95%; ~200 lanes the next wave behind the same gate. Lesson: automating dirty data manufactures confident wrong answers — the gate is the product. **Result.** Primary metric: inbound shipment-data completeness 60% → 95%. Freight spend −10% ≈ $2M in six months (record); 597 lanes = 80% of spend live in month one; adoption 8×.

### COMPRESSED STAR

- **S —** 1,000 global lanes asked in two months — scope, schedule and quality all fixed; completeness 60%; lanes differed by orders of magnitude in spend and carriers differed in EDI capability.
- **T —** Own the scope analysis and the onboarding standard; take the re-scope two levels up with the evidence attached.
- **A —** Spend Pareto (597 = 80% of spend) pre-validated so no lane owner could contradict it in the room; standard + ≥95% valid-ASN gate published to carriers before cohort one; harness so later cohorts were mostly configuration; held a failing cohort twice; ~400 lanes carrying ~20% of spend deferred — priced in spend share, not lane count.
- **R —** Completeness 60 → 95%; 100% visibility on high-value lanes; 597 lanes live month one; freight −10% ≈ $2M in six months (record); method written up as the sourcing SOP.

### RESULTS LINE

Completeness 60% → 95% (record) · 597 of 1,000 lanes = 80% of freight spend live in month one (record) · freight spend −10% ≈ $2M in six months (record — kept as measured, never annualized) · adoption 8× (record) · ≈$0.4M/yr document-handling labor (modeled add-on).

### THE MATH — STEP-BY-STEP DERIVATION

**Table 1 — before: the problem in numbers.**

| STEP | QUANTITY | FORMULA → SUBSTITUTION | INPUTS (BASIS) | VALUE |
|---|---|---|---|---|
| 1 | Completeness | `480 / 800 = 60%` | expected = 200 shipments × 4 milestones = 800; received 480 (illustrative volumes; 60% record) | 60% — 2 of 5 events missing |

| STEP | QUANTITY | FORMULA → SUBSTITUTION | INPUTS (BASIS) | VALUE |
|---|---|---|---|---|
| 2 | What blindness costs | `off-guide spend unseen · premium modes by habit · chase calls per late shipment` | record problem | no instrument, no lever |
| 3 | Manual documents | `597 lanes × ~50 shipments × 3 docs ≈ 90,000 docs/yr keyed by hand` | illustrative volumes | ~90K docs/yr |

**Table 2 — the fix: step by step, values substituted.**

| STEP | QUANTITY | FORMULA → SUBSTITUTION | INPUTS (BASIS) | VALUE |
|---|---|---|---|---|
| 1 | Spend Pareto | `S(k) = cumulative spend share; S(597) = 80%` | lanes sorted by annual freight spend (record) | 597 of 1,000 lanes carry 80% of spend — do them first |
| 2 | Cohort gate | `validASN rate ≥ 95% in test before activation` | published to carriers before cohort one (record) | one cohort failed twice on a timestamp format — held both times |
| 3 | Long-tail floor | `XML / portal when EDI is absent` | “the standard was the data, not the transport” (record) | small carriers kept inside the standard |
| 4 | Completeness after | `760 / 800 = 95%` | illustrative volumes; 95% record | 95%; high-value lanes 100% |

**Table 3 — after: the metric, before → after.**

| STEP | METRIC | BEFORE | AFTER | BASIS |
|---|---|---|---|---|
| 1 | Shipment-data completeness | 60% | 95%; high-value lanes 100% | record |
| 2 | Lanes live month one | 0 | 597 = 80% of spend | record |
| 3 | Freight spend on onboarded lanes | base | −10% ≈ $2M in 6 months | record |
| 4 | Document handling | ~90K docs/yr keyed | X12/XML validated | modeled |

**Table 4 — the dollars, every multiplication shown.**

| STEP | QUANTITY | FORMULA → SUBSTITUTION | INPUTS (BASIS) | VALUE |
|---|---|---|---|---|
| 1 | Implied addressable base | `$2,000,000 / 10% = $20,000,000/yr` | derived from the record pair, stated as such | ≈$20M/yr |
| 2 | The record result | `$20M × 10% = $2M in 6 months` | record | ≈$2M |
| 3 | Slice — routing-guide compliance | `~$0.8M of the $2M` | off-guide spend caught by 856/214 (modeled split) | ~$0.8M |
| 4 | Slice — mode discipline | `~$0.6M of the $2M` | premium modes → ocean/ground when ETA is trusted (modeled split) | ~$0.6M |
| 5 | Slice — expedites + detention | `~$0.6M of the $2M` | events replace chase calls (modeled split) | ~$0.6M |
| 6 | Split check | `0.8 + 0.6 + 0.6 = $2.0M` | the slices decompose the record — they never add to it | = the record $2M |
| 7 | Document labor | `90,000 × 6 min = 540,000 min; / 60 = 9,000 h; × $45 = $405,000` | illustrative volumes (modeled add-on) | ≈$0.4M/yr |

## TWO PROBES

> **You defied a Global Head of Supply Chain's number. How?**
>
> With the spend curve and the lane owners' validation attached — 597 lanes carried 80% of the spend, so the phased plan was argued in spend share, not lane count. He accepted the phasing off the curve; I then committed to the remainder behind the same gate and delivered it.

> **Is the $2M annualized?**
>
> No — it is the record as measured: minus ten percent on the onboarded lanes' spend in six months. My three-way split (routing-guide compliance ~$0.8M, mode discipline ~$0.6M, expedites and detention ~$0.6M) decomposes the record; it never adds to it.

**Canon numbers (basis labels):** 60% → 95% completeness (record) · 597 / 1,000 lanes = 80% of spend (record) · −10% ≈ $2M in 6 months (record) · ≥95% valid-ASN gate, held twice (record) · 8× adoption (record) · ≈$0.4M/yr labor (modeled). Never: cohort counts or sizes; never double the $2M into a run-rate.

## BACKUP — ONLY IF ASKED Doc bench — Flexport — the FFA Audit Logs PRD: an immutable who/what/when trail across ~100 partner orgs, nine categories at a 100%-captured bar (controls & process integrity)

**LPs this backup can serve:** Insist on the Highest Standards (primary — 100% capture as the written bar) · Ownership (compliance as a launch feature) · Dive Deep (the event taxonomy to the grain) · Earn Trust (sign-off over assumption)

**Basis.** Doc-derived micro-story from the program record (the FFA Audit Logs PRD, 2022) — every figure is RECORD-FROM-DOCS. Same platform as Tripathy's S12: use only for a controls / process-integrity question and keep S12's canon numbers identical to her packet. Program-lead voice: I owned the PRD, the event taxonomy and the sign-off chain; engineering built the pipeline and dashboards. Not in the master packet's S-numbering; supplements canon, never overrides it.

### 60-SECOND PITCH — STOP ON THE PRIMARY METRIC

**Situation.** The Freight Forwarding App serves external partner organizations — roughly a hundred of them, making thousands of changes a day to schedules, routes, documents and permissions. When a partner disputes "who changed this?", the answer has to come from an immutable trail — and the security organization's compliance bar ("important events") was deliberately vague. **Task.** Define the audit-log product: what gets captured, at what standard, phased how — so compliance is a launch feature, not an incident response. **Action.** I wrote the PRD: an immutable who/what/when record; nine event categories — logins/logouts, work-item changes, route updates, messages, document uploads/downloads, schedule updates, exception changes, assignment-rule changes, permission changes — each held to a written 100%-captured bar; priorities P0/P1/P2 per category; delivery phased Alpha (a login/logout dashboard on the log platform, Q4 2022) → Beta → MVP (Q1 2023); and the vague-bar risk mitigated structurally — PM buy-in plus CISO sign-off written into the plan, the "trusted date" for audits recorded in a governed document, and the eight dependent FFA teams mapped before build. **Result.** The capture standard was defined, prioritized and signed before build: 100% of nine categories, a phased path that did not pretend everything could be captured on day one, and zero ambiguity about who approved the scope. RECORD-FROM-DOCS

### COMPRESSED STAR

- **S —** A partner-facing platform with ~100 partner orgs and thousands of changes a day; no immutable trail; a compliance bar stated vaguely enough to move.
- **T —** Own the audit-logs PRD: taxonomy, capture standard, phasing, and the sign-off chain that makes the bar stick.
- **A —** Nine event categories at a written 100%-captured bar with P0/P1/P2 priorities; phased Alpha → Beta → MVP so capture breadth grew honestly; CISO sign-off and PM buy-in as the mitigation for the vague "important events" definition; the "trusted date" for audits recorded in a governed document; eight dependent teams mapped.
- **R —** Capture standard defined and signed before build (record-from-docs). The record carries no post-launch compliance dollar, and none is claimed.

### RESULTS LINE

Nine event categories at a 100%-captured bar (record-from-docs) · ~100 partner orgs · thousands of changes/day (record-from-docs — platform-wide; never merge with canon's 53-partner import migration set) · phased Alpha (Q4 2022) → Beta → MVP (Q1 2023) with P0/P1/P2 (record-from-docs) · CISO sign-off as the written mitigation (record-from-docs) · no post-launch dollar in the record — never invent one.

### THE MATH — STEP-BY-STEP DERIVATION

**Table 1 — before: the problem in numbers.**

| STEP | QUANTITY | FORMULA → SUBSTITUTION | INPUTS (BASIS) | VALUE |
|---|---|---|---|---|
| 1 | Audit coverage | `0 of 9 event categories captured — no dashboard, no tool, no store` | PRD problem statement (record-from-docs) | "who changed this data?" unanswerable |
| 2 | Organizations affected | `≈100 partner organizations on the app` | PRD hypotheses (record-from-docs) | every org's changes invisible |
| 3 | Change volume projected | `thousands of changes/day once all partners and internal operators are live` | PRD hypotheses (record-from-docs) | untracked writes at scale |
| 4 | Explicit non-goals | `reads/views, IP addresses, automated cascades — out of scope this iteration` | PRD scope section (record-from-docs) | scope held to writes by humans |

**Table 2 — the fix: step by step, values substituted.**

| STEP | QUANTITY | FORMULA → SUBSTITUTION | INPUTS (BASIS) | VALUE |
|---|---|---|---|---|
| 1 | Capture bar | `9 of 9 categories at 100% capture, each with a sample query` | PRD metrics section (record-from-docs) | login/logout · work-item edits · route updates · messages · document up/downloads · schedule updates · exception changes · assignment-rule changes · permission changes |
| 2 | Event schema | `8 parameters per event = user FID + date + time + action type + object + field + previous value + current value` | PRD Epic 1 (record-from-docs) | every change fully described |
| 3 | Immutability | `deletion of captured events = 0 — by construction` | PRD acceptance criteria (record-from-docs) | the log cannot be rewritten |
| 4 | Phase arithmetic | `Alpha (Q4 2022): 1 category live — login/logout on a Sumo Logic dashboard → Beta (Q1 2023): 9 of 9 tracked with full parameters → MVP (Q1 2023): every product area publishes links + sample queries` | PRD launch phases (record-from-docs) | proof → coverage → discoverability |
| 5 | Trusted-start mechanism | `each team registers a target date in Confluence; history is trusted only from the latest start date` | PRD risk table (record-from-docs) | honest about partial history |
| 6 | Volume sizing | `2,000 changes/day × 365 = 730,000 events/yr; × 8 parameters = 5,840,000 stored values/yr` | "thousands/day" from Table 1 row 3 (record-from-docs); 2,000 as the low-end reading and the arithmetic are illustrative | ≈0.73M immutable events/yr |

**Table 3 — after: the metric, before → after.**

| STEP | METRIC | BEFORE | AFTER | BASIS |
|---|---|---|---|---|
| 1 | Categories captured at 100% | 0 of 9 | 9 of 9 — the stated success metric | record-from-docs |
| 2 | Alpha exit | — | Sumo Logic dashboard with login/logout metrics | record-from-docs |
| 3 | MVP exit | — | all product areas linked, with sample queries | record-from-docs |
| 4 | Security sign-off | vague CISO requirements | CISO representatives sign the "important events" definition before development | record-from-docs |

**Table 4 — the dollars, every multiplication shown.**

| STEP | QUANTITY | FORMULA → SUBSTITUTION | INPUTS (BASIS) | VALUE |
|---|---|---|---|---|
| 1 | Investigation deflection | `2 h × $60/h = $120 per "who changed this" hunt; × 100 hunts/yr = $12,000/yr` | every input illustrative — no rate or count exists in the record | ≈$12K/yr, illustrative only |
| 2 | Compliance exposure | `no dollar in the record — not claimed` | record-from-docs | not claimed |
| 3 | Post-launch dollar | `none in the record — never invent one` | record-from-docs | not claimed |

## TWO PROBES

### Why 100% capture as the bar — isn't that over-engineering?

Because an audit trail with holes is worse than none — it lends false confidence to a dispute answer. The 100% bar is per category, phased: Alpha proved the pipeline on the highest-volume, simplest events (logins), then breadth grew by priority. The bar was written down precisely so "good enough" could not drift downward under delivery pressure.

### What did the vague "important events" definition actually risk?

Scope that moves after build — the classic compliance trap where the auditor's definition surfaces at the audit, not at the PRD. The mitigation was structural, not heroic: PM buy-in and CISO sign-off on the taxonomy up front, so the definition of "important" was the signed document, not someone's memory. When the bar can move, pin it to a signature.

**Canon numbers (basis labels):** nine categories at 100% capture (record-from-docs) · ~100 partner orgs · thousands of changes/day (record-from-docs) · Alpha Q4 2022 → MVP Q1 2023 (record-from-docs) · CISO sign-off + "trusted date" in a governed document (record-from-docs). Never: merge ~100 partner orgs with S12's 53-partner migration set; never claim a prevented-breach dollar (none in the record).

INTERVIEW-PREP PACKET 5 OF 5 · PER-INTERVIEWER · BUSINESS-PROCESS LENS

# Amazon NACF Supply Chain loop — Packet 5: Venkata Josyula

Amazon — Senior Supply Chain Manager, NACF Supply Chain (L6)

## Venkata Josyula

Principal Product Manager, FBA — treat as BAR RAISER

**Thursday, October 30, 2026 · 12:00 pm PDT · 55-minute video interview**

> **Strategy in one line.** Treat as the Bar Raiser: independent of the hiring team, holding a veto, probing LP depth to the fifth why and consistency across every story and every interviewer's notes. Four stories, one per employer — S16 (Flexport: four systems → one pipeline; the security-review miss owned upward), S15 (Sephora: the gate handed to the skeptic), S10 (Elementum: Austria — the data said measurement), S14 (Manhattan: the $2.5M discrepancy, transparency over a quiet fix). Answer the LP asked, then stop; keep facts identical when the LP switches; every dollar with its basis label.

**LP focus — my guess, not confirmed:** Earn Trust · Ownership or Hire & Develop the Best · Are Right, A Lot or Broad Responsibility. Amazon does not disclose the assignment; the guesses are reasoned from the interviewer's title and the JD in "Read me first" and should be held loosely. No interviewer biography is asserted anywhere in this packet — only title-based inference, labeled as such.

**Bar-raiser lens:** LP depth + cross-story data consistency. This packet carries an LP-coverage grid (which of the four stories can serve which LP) and a consistency-rules box (the numbers a bar raiser cross-checks) — read both before the call.

Canon: amazon/amazon-l6-scm-master-packet.html (§1.2 roster, §4.1 ledger). Every number below is that packet's value; every dollar carries its basis label (record · user anchor · modeled · priced).

## Read me first

### The interviewer's lens (inference from title + JD — not a biography)

**Inference from the title and the slot.** A Principal Product Manager from FBA — outside the hiring team, the last interview of the loop — is the profile of a Bar Raiser: an interviewer trained to be independent of the hiring manager, to hold the bar at "better than half the people at this level," and to hold a veto. Bar raisers are less interested in the supply-chain content than in *how the candidate thinks and whether the stories are consistent*: they drill one story to the fifth "why," they switch LPs mid-story to see whether the candidate re-tells the same facts honestly from a different angle, and they compare numbers across stories and across interviewers' notes. A PM from FBA additionally brings a seller/customer lens — who was the customer, what did they experience, what did you give up for them.

**What that means for the answers.** Four stories, one per employer, so that the bar raiser sees range and the same voice across fifteen years: S16 (Flexport — four systems into one pipeline, the security-review miss owned upward the same day), S15 (Sephora — a 20-person team moved to a four-week cadence by handing the gate to the skeptic), S10 (Elementum — Austria: "the noise said carriers; the data said measurement," one site's fix rolled to 25), S14 (Manhattan — the $2.5M discrepancy, transparency over a quiet fix). Each is told business-first; each ends on a number with a basis label; two of them are failure or near-failure stories, because bar raisers probe for lessons. The LP-coverage grid and the consistency-rules box below exist because this interviewer will test exactly those two things.

### Three guessed LP focus areas, with rationale

> **Labeled guess.** Amazon assigns 2–3 Leadership Principles per interviewer and does not disclose them. These three are inferred from the title and the JD; if the first question lands on a different LP, the twist maps under each story carry the pivot.

| GUESSED LP | WHY THIS INTERVIEWER WOULD CARRY IT | STORY TO OPEN · BACKUP |
|---|---|---|
| **Earn Trust (the flagship)** | Bar raisers almost always carry a trust/integrity LP, and they want the expensive version — the customer told first, the cost priced onto the candidate. S14 is the master map's Earn Trust primary; S16's same-day upward ownership of the miss is the recent echo. | S14 (open) · S16 (owned the miss upward) |
| **Ownership / Hire & Develop the Best (one of the two)** | Bar raisers take LPs the HM did not assign; Ownership (S16 — writing a bus-factor-1 system out of my own hands; owning the scoping miss) and Hire & Develop (S15 — the QA lead who now owns the gate) are both primaries in the master map and both unassigned elsewhere in this loop. | S16 (Ownership) · S15 (Hire & Develop / Earth's Best Employer) |
| **Are Right, A Lot / Success & Scale Bring Broad Responsibility (one of the two)** | Judgment under a confident room, and what you do once you are right: S10 is primary for both in the master map — the data overturning the explanation everyone held, then one site's method becoming 25 sites' instrument with Finance's signature on the method before the number. | S10 (both) · backup: S16 (priced the status quo before build) |

### Answer strategy — business first, systems as enablers

- **Open with the process problem and its cost, not the system.** First sentence names the broken process (Plan, Source, Make, Deliver or Return), who it hurt (customer, Finance, the floor) and the baseline number. The platform appears only as the enabler of the redesign, in one clause.

- **Use the JD's four outcome words as the results frame:** reliability (did the promise hold?), responsiveness (how much earlier did we see it?), cost efficiency ($ with its basis label), asset utilization (capital, labor-hours, capacity released).
- **Action = operating-model redesign in gated phases** — Discovery → Plan & Design → Build & Test → Cutover → Hypercare, each spoken as entry bar · what I delivered · exit bar. Say the gate that fired, not the meeting that happened.
- **Every dollar carries its basis word in the same breath** — record · user anchor · modeled · priced. A record dollar is said flat; a modeled dollar is said with its inputs. Never promote a modeled number to record.
- **Program-lead voice.** I owned definitions, rules, gates and the signature chain; engineers, SIs and vendor teams configured and built to the spec I signed. The only platform personally configured in canon is Manhattan DOM.
- **The pitch ends on the primary metric.** Then stop. Silence after the pitch is the interviewer filling their STAR worksheet; their probes are the second half of the answer.
- **For the bar raiser specifically:** answer the LP asked, then stop — bar raisers fill their worksheet from probes, and over-answering gives them inconsistencies to find. When they switch LPs mid-story, keep the facts identical and change only the emphasis (the twist maps below are built for that). Volunteer the failure beat before they ask for it (S16's two weeks; S14's "what if he had said no").
- **Consistency is the bar:** every number must match what the other four interviewers heard. The consistency-rules box carries the never-say items that would show up as a discrepancy in the debrief: 9.7× not 50×; "six sites" not "six NA sites"; every dollar with its basis label; the $10M won, never saved; the >$100K priced, never claimed.
- **Customer lens (FBA):** name the customer in each story — the VP steering on a wrong number (S16), the QA lead and the release's end users (S15), the site and the carriers who could not dispute the scorecard (S10), the retail customer told first (S14).

**Loop-consistency rule.** One question, one story; never the same story to two interviewers — interviewers compare notes. Held for other packets (do not spend here): S11/S12 (Tripathy) · S6/S8 (Hoerger) · S2/S18 (Goforth) · S1/S3 (Lan). S16 is the same program family as S11 — if the bar raiser has read notes from the HM's interview, say so: "the review the HM heard about is the one this pipeline replaced.". If pushed for "another example" on an LP, use the named backup in the question bank or the backup bench at the end of this packet, never a story allocated to another interviewer.

**How to read the "L6 level evidence" block under each story.** Amazon's L6 functional bar (user-supplied rubric): *Ambiguity* — strategy or network undefined; I select the stakeholders, set the goals, deliver independently. *Scope & influence* — team-level scope; influences one to two levels up. *Execution* — tactical and strategic; simplifies or extends; applies best practices; force-multiplies; mitigates long-term risk; makes the short-versus-long-term trade-off explicit. *Impact* — long-term team goals. *Process improvement* — operational-excellence best practices institutionalized. *Complex* — visible risks, roadblocks and constraints mitigated through expertise, judgment and influence to consensus. Overview themes the block is tuned to: designing and modeling the supply chain rather than working in it; developing new processes for others to own; root-cause mechanisms that resolve trending issues at scale; recommendations to leadership that outline competing metrics. The block sits right after Results so the evidence is on the page when the probing starts.

## Bar-raiser instruments — LP coverage and consistency rules

### LP-coverage grid — which Leadership Principles each of the four stories can serve

**Y** = can open on this LP (the story is primary, backup or third for it in the master map, or the beat is explicit in the record) · **P** = a secondary angle, use only if the Y stories for that LP are spent · blank = do not stretch the story to it. Fifteen of sixteen LPs are covered by at least one Y or P across the four. The one gap is Think Big — its primary (S12) is the hiring manager's and its third (S1) is Lan's; if Think Big lands here, use the unallocated master-packet backup S9 (the 1,000-lane EDI vision, 597 built first behind a ≥95% valid-ASN gate — pointer-level detail only) or offer S16's four-systems-to-one as the nearest honest angle, labeled as such.

| STORY | CO | OWN | I&S | ARL | L&C | H&D | IHS | TB | BFA | FRU | ET | DD | HB | DR | EBE | BR |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **S16 Flexport — pipeline** | | Y | Y | | Y | | P | | | Y | P | P | | P | | |
| **S15 Sephora — cadence** | | | P | | | Y | Y | | | | P | | P | P | Y | |
| **S10 Elementum — Austria** | P | P | | Y | | | P | | | P | Y | Y | P | P | | Y |
| **S14 Manhattan — $2.5M** | Y | Y | | | | | | | Y | | Y | | Y | | | P |

CO Customer Obsession · OWN Ownership · I&S Invent and Simplify · ARL Are Right, A Lot · L&C Learn and Be Curious · H&D Hire and Develop the Best · IHS Insist on the Highest Standards · TB Think Big · BfA Bias for Action · FRU Frugality · ET Earn Trust · DD Dive Deep · HB Have Backbone; Disagree and Commit · DR Deliver Results · EBE Strive to be Earth's Best Employer · BR Success and Scale Bring Broad Responsibility

### Consistency-rules box — the data a bar raiser cross-checks across stories and interviewers

**Say it the same way in every room.**

- **9.7×, never 50× and never 10×.** Flexport customers 7 → 68 = 9.7× (S11, the HM's story). If the bar raiser quotes a multiple back, correct it gently to 9.7×.
- **"Six sites," never "six NA sites."** The Elementum manufacturer is six sites; the CM-ingest program (S7) is six sites, three waves, five months — no region prefix, no "500+ product families."
- **Every dollar carries its basis label** — record · user anchor · modeled · priced. In this packet: $2.5M discrepancy (record) · $10M initiative (record — won, never saved) · >$100K/yr (priced, never claimed) · $500K/yr Austria and ~$8M across 25 sites (record roll-up) · ≈$0.13M/yr (S15, modeled) · ≈$0.1M/yr (S14, modeled) · ≈$0.5M/yr Austria analyst capacity (modeled).
- **Two $2.5M figures exist in my canon and they are different things:** S14's $2.5M is the OMS↔WMS discrepancy at Manhattan (record); S1's ≈$2.5M/yr is Sephora's book-of-record headline (anchor + modeled, Lan's story). Never let them blur.
- **Two ~$1.5M figures exist and neither is in this packet:** S8's ~$1.5M safety-stock release (record, Hoerger) and S11's ~$1.5M pre-production saving (record, Tripathy). S14's ≈$1.5M is *expected corruption avoided* — risk math, not money. Never "$1.5M to $2M."
- **Accuracy 60% → 90% appears twice honestly:** S16 (WBR data accuracy) and S11 (ETA accuracy) are the same program family — the pipeline moved the review's data accuracy; the waves moved ETA accuracy. Say which one.
- **Failure stories are consistent:** S16 — the two-week security-review miss, owned upward the same day; S18 (Goforth) — the RF miss; S4 (Lan's pointer) — the rule I got wrong. The lesson sentences are fixed: "security, privacy and data-access reviews are program dependencies with lead times"; "any single-point dependency in front of an exit criterion gets a what-if plan"; "both sides of a rule get a number before it ships."
- **Voice:** program lead who owned definitions, rules, gates and signatures; engineers, SIs and vendor teams built; the only platform personally configured in canon is Manhattan DOM (S13/S14 era). Never "we" for an action that was mine; name the team's part once.

- **No invented numbers:** no direct-report headcount (S15), no realized dollars on the $10M initiative (S14), no claimed savings on the >$100K (S16), no Austria "$1M" or "99% in two months" (S10). "I don't have a canonical figure and won't invent one; what I can defend is…"

## The stories — one per employer, business-first, canon numbers

Order of preference if the bar raiser lets me choose: S14 (Earn Trust) → S16 (Ownership, with the failure beat) → S15 (Hire & Develop) → S10 (Are Right / Broad Responsibility). Each: 60-second pitch → Situation/Task → Action in gated phases → Results → Probing tuned to a bar raiser (fifth-why, customer, softest number) → the key math → Twist map.

### S16 — Flexport — four systems into one Snowflake pipeline: writing a bus-factor-one operating metric out of my own hands, and the security-review miss I owned upward the day it surfaced

**LPs this story can serve here:** Ownership (primary — bus factor 1 → pipeline; owning the miss) · Invent and Simplify (backup — four systems → one) · Frugality (the >$100K priced baseline) · Learn and Be Curious (learned the stack to write the BRD) · Earn Trust (zero review deliveries skipped while paying for the miss)
**Process family:** Deliver — operating-metric production for a weekly business review; data-integrity and ownership of a decision instrument

**Honesty label.** I wrote the BRD and owned the cutover; the infra team owned the build. The >$100K/yr is a priced baseline in the infra team's own loaded rates — priced, not claimed as savings; my split of it (≈$30K hours + ≈$70K decision risk) is modeled. The two-week security-review miss is mine, owned upward the same day. Same program family as S11 (the HM's story) — never both to one interviewer.

#### 60-SECOND BUSINESS-FIRST PITCH — STOP ON THE PRIMARY METRIC

**Situation.** The Monday business review that steered Flexport's visibility rollout ran on four systems that did not talk, stitched together in spreadsheets by one person — me. Bus factor one, data accuracy 60%, 1.5–2 hours per snapshot and rising, support tickets with no container ID so they could not be joined to anything. A VP was steering a program on a number one person could break and nobody could check. **Task.** Nobody had written requirements for a program's operating metrics; I wrote the business requirements and owned the cutover; the infrastructure team owned the build. **Action.** I documented the as-is and priced the status quo at more than $100K a year in the infra team's own loaded rates — my hours plus the decision risk of a VP steering on a wrong number — so that they prioritized the build themselves. The BRD froze the seven metric definitions, specified the feed, the failure alerts and a cross-reference table for the ticket join while the real key was pushed upstream, and set the acceptance gate: parallel run within 1% per metric for consecutive weeks, every mismatch explained by cause. Mid-build a data-access security review surfaced that I had not planned; it cost two weeks. I owned it upward the same day as a scoping miss, re-sequenced the build, and kept the review alive on the manual pull — zero Mondays skipped. The deck moved onto the pipeline when parity held; the manual pull stayed as rollback a week beyond. **Result.** Primary metric: review data accuracy 60% → 90% once four systems became one pipeline; bus factor one to at least two; the first automated catch was a 4% ticket-sync drop that had been invisible in the manual era. The >$100K a year is a priced baseline, not a claimed saving. Lesson: every kickoff now asks whose approval the data needs and what its lead time is.

#### SITUATION / TASK — THE BUSINESS PROCESS PROBLEM, SCOPE, SCALE, BASELINES

**S —** The Monday review ran on four systems that didn't talk, stitched in Google Sheets by one person — me: bus factor 1, accuracy 60%, 1.5–2 hours per snapshot, tickets carried no container ID. A VP was steering on a number one person could break.

The cost of leaving it: a wrong number steering the rollout, and silent data loss nobody would catch — the 4% ticket-sync drop was invisible in the manual era. The customer in this story is the VP making rollout and funding decisions on the number.

**T —** Nobody had written requirements for a program's operating metrics. I wrote the BRD and owned the cutover; the infra team owned the build. Goal: one pipeline producing the same seven numbers, accurate, and owned by more than one human.

**Baselines on the table:** WBR accuracy 60% · bus factor 1 · snapshot effort 1.5–2 hours per Monday, heading to ~4 hrs/wk · ticket-to-container joinability none · no alarm on silent data loss.

## ACTION — THE PROCESS / OPERATING-MODEL REDESIGN, IN GATED PHASES

- **Discovery** — documented the as-is; named the join gap (tickets without container ID); priced the status quo at >$100K/yr in the infra team's own rates including VP decision risk. *Exit bar:* infra prioritizes it themselves off the price; scope = the seven metrics, nothing new.
- **Plan & Design** — the BRD — seven frozen definitions, daily/hourly feed, failure alerts, cross-reference table then a real key upstream; acceptance = parallel run within 1% for consecutive weeks; the RAID log — missing the data-access security review. *Exit bar:* BRD accepted by infra; acceptance gate published.
- **Build & Test** — extract/load specs per system; parity checks; the parallel-run procedure; when the security review surfaced mid-build, re-sequenced and owned the miss upward the same day. *Exit bar:* review closed (2 weeks); parallel run within 1% for consecutive weeks; mismatches explained (mostly late-arriving tickets on the manual side).
- **Cutover & Go-Live** — the cutover memo; the Monday deck moved onto the pipeline; the manual pull kept as rollback. *Exit bar:* first Monday off the pipeline, same seven numbers; zero review deliveries skipped through the build.
- **Hypercare / Close** — watched the failure alerts; triaged the first automated catch (a 4% ticket-sync drop); definitions into code; institutionalized the kickoff question. *Exit bar:* accuracy 60% → 90%; the pipeline steers the review without me.

**Influence and ownership.** The build was the infra team's — the ETL → S3 → SQS → Snowflake → Looker pipeline and the alerts; mine were the as-is, the pricing, the BRD's frozen definitions, the acceptance gate, the parallel-run procedure and the cutover memo. The lever was pricing the status quo in the owning team's own loaded rates — including decision risk, not just hours — so they prioritized it themselves. The category error was mine: I treated the security review as an engineering concern rather than a program dependency with a lead time.

> **Designed the process; others own it now.** I designed the BRD, the acceptance gate and the cutover memo; the infra team owns the pipeline and its alerts, and the definitions live in code rather than in a person. **Trade-off put to leadership.** The recommendation to the infra lead and the VP outlined the competing metrics — build speed vs the data-access security review and decision risk on the Monday numbers — and re-sequenced the build rather than route around the review.

## RESULTS — $ · # · % · TIME · CUSTOMER · QUALITY (CANON NUMBERS ONLY)

- **$** — >$100K/yr priced baseline in the infra team's own loaded rates (record of the pricing) — priced, *not* claimed as savings; my model splits it ≈$30K hours + ≈$70K decision risk (≈$50K of which the accuracy gain retires), both modeled; bus-factor exposure ≈$15K/yr modeled.
- **#** — four systems → one pipeline; bus factor 1 → ≥2; zero review deliveries skipped through the build (record).
- **%** — review data accuracy 60% → 90%; parity within 1% for consecutive weeks; first automated catch a 4% ticket-sync drop (record).
- **Time** — the security-review miss cost two weeks — owned upward the same day, build re-sequenced, review closed (record).
- **Customer** — the VP steered on the same seven numbers the first Monday off the pipeline; the manual pull stayed as rollback until parity had held a week beyond (record).
- **Quality** — seven frozen definitions moved into code (no drift); every parallel-run mismatch explained by cause (record).

**L6 LEVEL EVIDENCE** — what the interviewer pattern-matches against the Amazon L6 functional rubric; every line is this story's canon, no new numbers

| L6 DIMENSION | EVIDENCE IN THIS STORY |
|---|---|
| **Ambiguity** | A Monday review fed by a manual pull across four systems with a bus factor of one, owned by nobody. I documented the as-is, named the join gap (tickets without container ID) and priced the status quo at >$100K/yr in the infra team's own loaded rates. |
| **Scope & influence** | The infra team — not mine — prioritized the build off the price; the VP steered the review off the pipeline from the first Monday; my seat was the BRD, the acceptance gate and the cutover memo. |
| **Execution trade-off + mechanism others now own** | Build speed vs the data-access security review: when it surfaced mid-build I re-sequenced and owned the miss upward the same day (two weeks). Mechanisms others own now: the pipeline and its failure alerts (infra), the seven frozen definitions in code. |
| **Impact (long-term)** | Four systems → one pipeline; bus factor 1 → ≥2; review data accuracy 60% → 90%; the pipeline steers the review without me. |
| **Process improvement (standard work institutionalized)** | Acceptance = parallel run within 1% for consecutive weeks; the parallel-run procedure and cutover memo; the manual pull kept as rollback until parity held a week beyond; the kickoff question institutionalized (which reviews belong in the RAID log). |
| **Complexity & consensus** | Zero review deliveries skipped through the build; the security review treated as a dependency with a lead time rather than argued around. |
| **Root cause at scale** | The join gap fixed as a cross-reference table then a real key upstream; the first automated catch — a 4% ticket-sync drop — triaged by cause, not patched by hand. |

## PROBING — CHALLENGE Q&A TUNED TO THIS INTERVIEWER

**BR — Why is this Ownership and not just automation?**

Both halves are ownership: taking the bus-factor-one system I had built and writing it out of my own hands, and owning the security-review miss upward the same day it surfaced — with zero review deliveries skipped while I paid for it.

**BR — You are claiming $100K of savings?**

No — pricing, not claiming. The number existed to make the infra team prioritize the build off their own rates, and it worked. The claimed record is the accuracy move and the first automated catch; the split of the price is my model and I will not call any of it realized savings.

**BR — Why didn't you know about the security review? Fifth why.**

Because I treated it as an engineering concern rather than a program dependency with a lead time — a category error, not a scheduling slip. The fix is institutional: every kickoff now asks whose approval the data needs and what its lead time is. I would trade the two weeks again; I would plan them now.

**BR — Who was the customer and what did they experience?**

The VP steering the rollout. What they experienced was continuity — the same seven numbers on the first Monday off the pipeline, zero skipped Mondays — and then a capability they never had: an alert instead of silence when 4% of tickets stopped syncing.

**BR — The decision-risk dollar is soft.**

Yes — one wrong number in ten becoming a wrong steer worth a squad-week; range $12K to $117K. I would firm it by tagging every review action item for a quarter and counting how many were reversed when a number was corrected.

**BR — How does this connect to the review the hiring manager heard about?**

It is the same program: the review in S11 ran by hand for over a year; this pipeline is what retired the manual pull, with the definitions in code. Different story, same facts.

## THE KEY MATH — THE PARITY GATE, THE PRICED BASELINE AND ITS SPLIT (RECORD GATE; ILLUSTRATIVE VALUES LABELED)

| STEP | QUANTITY | FORMULA → SUBSTITUTION | BASIS | VALUE |
|---|---|---|---|---|
| 1 | Parity gate | `\|pipeline − manual\| / manual ≤ 1% per frozen metric, consecutive weeks` | BRD acceptance gate (record) | cutover unarguable |
| 2 | Pass example | `\|70.9 − 71.2\| / 71.2 = 0.42% ≤ 1%` | illustrative coverage values | pass |
| 3 | Investigate example | `\|461 − 456\| / 456 = 1.1% > 1%` | illustrative ticket counts | 5 late tickets — the manual side was wrong |
| 4 | Silence alert | `1 − 0.96 = 4% > 1% threshold → alert` | threshold illustrative; the 4% catch is record | first automated catch |
| 5 | Hours term | `4 h/wk × 52 = 208 h × ≈$145/h ≈ $30K` | modeled, consistent with infra's pricing | ≈$30K/yr |
| 6 | Decision-risk term | `52 × 10% = 5.2 wrong-number weeks × $13.5K ≈ $70K` | illustrative; range $12–117K | ≈$70K/yr |
| 7 | Priced baseline | `30 + 70 = >$100K/yr` | record of the pricing — NOT claimed savings | >$100K/yr priced |

Accuracy 60 → 90% here is the review's data accuracy; S11's 60 → 90% is ETA accuracy — same program family, two different metrics.

**Where this lands at Amazon.** The Ownership pattern: price the status quo in the owning team's own rates so they prioritize it themselves, keep the old path alive until the new one has earned parity, and treat security, privacy and data-access reviews as dependencies with lead times. Reliability (a number that survives its author), responsiveness (alerts instead of silence), cost efficiency (priced, honestly).

## TWIST MAP — LIKELY LP QUESTIONS FROM THIS INTERVIEWER → THE ANGLE TO TAKE WITH S16

| LIKELY QUESTION (THIS INTERVIEWER) | LP | ANGLE — WHAT TO EMPHASIZE | SKIP / NEVER SAY |
|---|---|---|---|
| **Tell me about a time you took ownership of a problem that wasn't yours to fix.** | Ownership | Nobody had written requirements for operating metrics; I wrote the BRD and wrote myself out of the system I had built. | Do not claim the build — infra built; I owned BRD and cutover. |
| **Tell me about a mistake and what you learned.** | Ownership / Earn Trust | The security review: two weeks, owned upward the same day, zero Mondays skipped; the kickoff question that outlived it. | Do not minimize it as "a dependency slipped" — it was my scoping miss. |
| **Tell me about simplifying something.** | Invent and Simplify | Four systems → one pipeline; seven definitions into code; a cross-reference table now, the real key upstream. | Skip the ETL stack names unless asked. |
| **Tell me about a time you made a business case.** | Frugality | The status quo priced at >$100K/yr in the owner's own rates — including decision risk; it got prioritized. | Never "saved $100K." |
| **How do you know when a new system can replace the old one?** | Highest Standards | Parity within 1% per metric for consecutive weeks; mismatches explained by cause; old path alive a week beyond. | — |

| LIKELY QUESTION (THIS INTERVIEWER) | LP | ANGLE — WHAT TO EMPHASIZE | SKIP / NEVER SAY |
|---|---|---|---|
| **Tell me about learning something outside your expertise.** | Learn and Be Curious | Learned enough of the data stack to write a BRD infra could build from; the first requirements ever written for a program's metrics. | Do not overclaim engineering depth. |

## S15 — Sephora — converting a 20-person Product/Engineering/QA team to a four-week cadence in six weeks, by handing the gate to the skeptic (critical defects per release −50%)

**LPs this story can serve here:** Hire and Develop the Best (primary — the QA lead who now owns the gate) · Strive to be Earth's Best Employer (primary — QA elevated, not blamed) · Insist on the Highest Standards (the definition-of-done that can say no) · Have Backbone (naming QA's fear as correct)
**Process family:** Make — product delivery operating model: release cadence, definition-of-done, quality gate ownership

**Honesty label.** VP air cover, no line authority over anyone in the room; the QA lead authored the definition-of-done — I coached the frame, she owned the gate. The −50% and the six weeks are record; the dollars are my model on illustrative defect counts. Never invent a direct-report headcount.

**60-SECOND BUSINESS-FIRST PITCH — STOP ON THE PRIMARY METRIC**

**Situation.** Newly joined at Sephora, the VP of Supply Chain asked me to move a 20-person Product, Engineering and QA team from a six-week hybrid cadence to four-week agile. QA — a contracting firm — and in-house engineering were heading for a standoff: QA feared compressed test windows, and the arithmetic said they were right. The business cost of getting it wrong was a faster cadence shipping more escaped critical defects — the one outcome the VP could not afford. **Task.** Convert the cadence in weeks, not quarters, without line authority and without the standoff becoming the story; success was the four-week cadence with no rise in escaped criticals and the QA window protected by rule. **Action.** One-on-ones with every person first, an influencer named per function, and the problem stated honestly — QA's fear is correct. Then I handed the load-bearing artifact to the most skeptical person: the QA lead authored the definition-of-done with entry criteria that could say no; I coached the frame, she owned the content, engineering shaped it through review rather than veto. Jira got benefit-ranking and review-status fields; late heavy stories got a drop date with a reason code instead of a silent slide into the last two days; QA sat on the product-business calls with pre-approved scenarios so Sev-1s were classified by cause, not blamed. I facilitated the retrospective once and stepped back from the review line. **Result.** Primary metric: critical defects per release — baseline → −50%, while shipping 50% more releases; the 20-person team on a four-week cadence in six weeks; the gate defended by its owner to engineering without me by cycle two, and it survived my leaving. About $130K a year modeled — escaped-critical cost avoided and late-land rework removed — and none of it is a record figure; no headcount was invented to get there.

### SITUATION / TASK — THE BUSINESS PROCESS PROBLEM, SCOPE, SCALE, BASELINES

**S —** Newly joined, the VP of Supply Chain asked me to move a 20-person Product/Engineering/QA team from a six-week hybrid cadence to four-week agile. QA (a contracting firm) and engineering (in-house) were heading for a standoff: QA feared compressed test windows — and they were right.

The risks: the standoff becoming the story, dev consuming the shrunken test window, Sev-1s mis-attributed to QA — and a faster cadence shipping more escapes, the exact failure the VP could not afford. The customers: the release's end users in supply chain operations, and the QA team who would otherwise be blamed for the arithmetic.

**T —** Convert the cadence in weeks, not quarters, without the standoff becoming the story. VP air cover, no line authority over anyone in the room. Success criteria written with the VP: four-week cadence, no rise in escaped criticals, QA window protected by rule.

**Baselines on the table:** release cadence six-week hybrid (≈8.7 releases/yr) · escaped critical defects per release — the baseline the −50% is measured against (count illustrative; the −50% is record) · the QA test window being compressed by arithmetic.

## ACTION — THE PROCESS / OPERATING-MODEL REDESIGN, IN GATED PHASES

- **Discovery (weeks 1–2)** — one-on-ones with every person (objections, motivations); an influencer per function; the problem statement — QA's fear is correct — and the success criteria. *Exit bar:* every function's real objection heard privately; the QA lead agreed to author the gate.
- **Plan & Design (weeks 2–3)** — ceremony calendar for a four-week cycle; the definition-of-done frame (what a gate needs to survive pressure — the QA lead wrote the content); Jira field spec (benefit ranking, review status); user-guide outline. *Exit bar:* DoD with entry criteria written by QA and reviewed with engineering; fields live.
- **Build & Test (weeks 3–5)** — the user guide (grooming → completion); live dashboard; the QA drop date with reason codes; QA seated on product-business calls with pre-approved scenarios. *Exit bar:* cycle 1 completed on the new cadence; drop-date rule in force; Sev-1 cause classification running.
- **Cutover (week 6)** — facilitated the retro once; data read-out to the VP; handed the gate to the QA lead to defend herself; rollback (reverting to six weeks) named as a two-way door at kickoff and never used. *Exit bar:* converted in 6 weeks; critical defects −50% under the drop-date rule.
- **Hypercare / Close** — stepped back from the review line; neighbouring teams copied the ceremony from the user guide. *Exit bar:* the gate survived my leaving; Sev-1 trend from story-accuracy defects reversed.

**Influence and ownership.** Mine: the one-on-ones, the ceremony calendar, the DoD frame, the Jira field spec, the drop-date rule and the retro — facilitated once. Hers: the DoD content and the gate, defended by her. Engineering's objection became co-ownership through review, not veto, with me covering the low-impact stories they had to drop. The assumption I challenged was that QA's fear was resistance — it was correct arithmetic, and saying so is what made the QA lead willing to author the gate.

> **Designed the process; others own it now.** I designed the ceremony calendar, the DoD frame and the drop-date rule; the QA lead owns the gate and defends it, and neighbouring teams copied the ceremony from the user guide. **Trade-off put to leadership.** The read-out to the VP outlined the competing metrics — release frequency (8.7 → 13 per year, modeled) vs escaped critical defects — and showed both moving the right way under a gate that could say no.

## RESULTS — $ · # · % · TIME · CUSTOMER · QUALITY (CANON NUMBERS ONLY)

- **$** — ≈$130K/yr *modeled* (escaped-critical cost avoided ≈$90K + late-land rework removed ≈$40K; range $80–270K); none of it is a record figure, and no headcount was invented to get there.
- **#** — 20-person Product/Eng/QA team; releases per year 8.7 → 13 (*modeled* from the cadence: 52 ÷ cycle weeks).
- **%** — critical defects per release −50% (record) — while shipping 50% more releases.
- **Time** — six-week hybrid → four-week agile, converted in six weeks (record); the QA lead defending the gate by cycle 2.
- **Customer** — QA moved from blamed afterthought to owner of the standard, with the data protecting them; end users got more frequent releases with fewer escaped criticals (record mechanism).
- **Quality** — a QA-authored definition-of-done with entry criteria that can say no; the drop date with reason codes; Sev-1 cause classification — and the gate survived my leaving (record).

**L6 LEVEL EVIDENCE** — what the interviewer pattern-matches against the Amazon L6 functional rubric; every line is this story's canon, no new numbers

| L6 DIMENSION | EVIDENCE IN THIS STORY |
|---|---|
| **Ambiguity** | A VP ask to change a 20-person team's cadence, with QA and engineering heading for a standoff and no line authority over anyone in the room. I chose an influencer per function and set the success criteria. |
| **Scope & influence** | 20-person Product/Engineering/QA team with VP air cover; the QA lead authored and defended the gate; neighbouring teams copied the ceremony from the user guide — influence by frame, not by rank. |
| **Execution trade-off + mechanism others now own** | Release frequency (8.7 → 13/yr modeled) vs test-window quality: a QA-authored definition-of-done with entry criteria that can say no, and a drop date with reason codes. Mechanism others own now: the gate, owned and defended by the QA lead. |
| **Impact (long-term)** | Critical defects per release −50% while shipping 50% more releases; the gate survived my leaving; the Sev-1 trend from story-accuracy defects reversed. |
| **Process improvement (standard work institutionalized)** | A ceremony calendar for the four-week cycle; Jira benefit-ranking and review-status fields; Sev-1 cause classification; a user guide (grooming → completion) that other teams reused. |
| **Complexity & consensus** | QA (a contracting firm) vs engineering (in-house): engineering's objection became co-ownership through review, not a veto; rollback to six weeks named as a two-way door at kickoff and never used. |
| **Root cause at scale** | QA's fear was correct arithmetic about compressed windows — addressed by a mechanism (DoD + drop date), not a heroics week, so the defect reduction held across cycles. |

## PROBING — CHALLENGE Q&A TUNED TO THIS INTERVIEWER

> **BR — Why is this Hire and Develop? You had no reports.**
>
> Because the development mechanism is the story, not the org chart: the most skeptical person authored the load-bearing artifact; I coached the frame, she owned the gate, and she was defending it to engineering without me by cycle two. Development that depends on me staying is not development. I have managed onshore/offshore delivery teams earlier in my career; I have not run a large HR org with comp cycles and won't claim that muscle.

> **BR — Faster cadence usually means more escapes. Why did yours fall?**
>
> Because the gate got stronger as the window got shorter: entry criteria authored by the person with the most to lose, and a drop date that converted silent late lands into visible, reason-coded re-scopes. The −50% is the record; the defect counts behind the dollars are illustrative and labeled.

> **BR — Where does four criticals per release come from?**
>
> It is illustrative, not record — the record gives me the fifty percent drop, not the base count. I would pull the Sev-1 log for the last four six-week releases and use the actual mean.

> **BR — What did engineering give up, and how did you handle it?**
>
> Low-impact stories dropped under the drop-date rule — dev's date protected, QA's window protected, and I covered some of the drops myself. Their objection to the DoD became co-ownership through review, not veto.

> **BR — Fifth why: why hand the artifact to the skeptic?**
>
> A gate the skeptic owns survives pressure; one I write dies when I leave. And naming her fear as correct arithmetic is what made her willing to write it — the standoff was never about resistance.

> **BR — What would you do differently?**
>
> Go to QA first, and put her name on the review line from week one.

## THE KEY MATH — CADENCE TO RELEASES, CRITICALS PER YEAR, AND THE MODELED DOLLARS

### (RECORD −50% AND SIX WEEKS; COUNTS ILLUSTRATIVE)

| STEP | QUANTITY | FORMULA → SUBSTITUTION | BASIS | VALUE |
|---|---|---|---|---|
| 1 | Releases per year | `52 / 6 = 8.7 → 52 / 4 = 13 (+50%)` | record cadence; arithmetic | 8.7 → 13 |
| 2 | Criticals per release | `4 → 2 (−50%)` | record −50%; base count illustrative | −50% |
| 3 | Criticals per year | `4 × 8.7 ≈ 35 → 2 × 13 = 26; 9 fewer` | illustrative counts | 9 fewer/yr while shipping 50% more |
| 4 | Escaped-critical cost avoided | `9 × $10,000 = $90,000` | benchmark $8–15K per critical (modeled) | ≈$90K/yr |
| 5 | Late-land rework removed | `6 person-days × $800 × 8.7 ≈ $41,760 → ~0 under the drop rule` | modeled | ≈$40K/yr |
| 6 | The constraint | `15 eng × 18 d / 3 = 90 stories vs 4 QA × 18 d / 1 = 72 stories` | theory-of-constraints illustration (toolkit label) | QA caps throughput — QA was right |
| 7 | Total | `90 + 40 = $130K` | modeled; range $80–270K; no headcount invented | ≈$130K/yr |

Count per year, not per release — halving per release at 13 releases still nets nine fewer a year; the VP feels escaped criticals per quarter.

**Where this lands at Amazon.** Develop people by giving them the load-bearing artifact and the frame, not by writing it for them — and measure development by whether the gate survives your leaving. The Earth's Best Employer angle: QA moved from blamed afterthought to owner of the standard, with the data protecting them. Reliability (fewer escaped criticals), responsiveness (50% more releases), quality owned at the source.

### TWIST MAP — LIKELY LP QUESTIONS FROM THIS INTERVIEWER → THE ANGLE TO TAKE WITH S15

| LIKELY QUESTION (THIS INTERVIEWER) | LP | ANGLE — WHAT TO EMPHASIZE | SKIP / NEVER SAY |
|---|---|---|---|
| **Tell me about a time you developed someone.** | Hire and Develop the Best | The QA lead: frame coached, content hers, gate defended by her by cycle 2 and after I left. | Never invent a direct-report headcount. |
| **Tell me about making a workplace better for a team.** | Earth's Best Employer | QA from blamed to owner; Sev-1s classified by cause, not blame; the drop date made overload visible and got people help. | Do not claim engagement or attrition numbers — none exist. |
| **Tell me about a time you raised the bar on quality.** | Highest Standards | A definition-of-done that can say no, evaluated at Ready-for-QA; criticals −50% at +50% releases. | Keep the per-release base count labeled illustrative. |
| **Tell me about resolving a conflict between two teams.** | Earn Trust / Backbone | One-on-ones first; QA's fear named as correct arithmetic; engineering's veto turned into review. | Do not make engineering the villain. |
| **How do you change a process without authority?** | Functional — change management | Hear every real objection privately; hand the artifact to the skeptic; name the rollback as a two-way door; step back on evidence. | — |
| **Tell me about a change that stuck after you left.** | Deliver Results / Ownership | Neighbouring teams copied the ceremony; the gate survived my leaving. | No dollars as record — ≈$130K is modeled. |

## S10 — Elementum — Austria: "the noise said carriers; the data said measurement" — routing-guide data quality ~55% → ~99% in three months, $500K/yr on a Finance-signed method, then ~$8M across 25 sites

**LPs this story can serve here:** Are Right, A Lot (primary — the data overturned the room's explanation) · Success & Scale Bring Broad Responsibility (primary — one site's fix rolled to 25 sites) · Dive Deep (backup — measured the measurement) · Earn Trust (the Finance-signed method before any dollar) · Customer Obsession (a scorecard the carriers could not dispute)

**Process family:** Deliver — transportation performance measurement (on-time delivery), routing-guide governance, freight-savings control

**Honesty label.** I owned the analysis and the rules design and pulled the data myself; the data platform team built the engine; Finance signed the savings method before any dollar was claimed. Site-wave counts and dates beyond "Austria first, then 25 sites" are not in the record. Guardrails: $8M, never "$1M"; "~99% in 3 months," never "two months"; 1–2 hrs/wk is the after, never the before.

### 60-SECOND BUSINESS-FIRST PITCH — STOP ON THE PRIMARY METRIC

**Situation.** Transportation at the customer's Austria site could not measure on-time delivery. Reports said data quality was 40%; the team spent 8–10 hours a week reconciling promised against actual dates by hand; the freight base was about $5M a year with no enforceable on-time standard — and everyone, including my own sponsor, believed the carriers were late. **Task.** I owned the analysis and the rules design; the goal was an on-time number the carriers could not dispute, and savings claimed only on a method Finance signed first. **Action.** I pulled three years of shipment history in SQL myself — the room already had an explanation, and a request queue would have been slower and arguable. True data quality was about 55%, not 40%, and the defects were ours: missing rates, wrong-grain lanes, one carrier under five spellings, brochure transit times posing as promises. The problem was measurement, not carriers. I specified the promised-delivery-date as a routing-guide rules engine — lane key by ship-from, ship-to, mode, volume band, Incoterm and SLA, with Incoterm needed on about 8% of lanes and a no-hit paging a human rather than defaulting a carrier; defects worked as classes with reason-code alerts and a weekly burn-down; and the savings method signed with Finance before the number existed — contracted guide rate minus paid rate, per shipment, executed only. The scorecard was released only when automated on-time matched a manual sample per lane cohort; the sponsor was unhappy with the delay for weeks and right to want speed; the carriers never disputed the result. Then scale: a rules file plus one data-quality cycle per site, regional super-users after two shadowed reviews. **Result.** Primary metric: routing-guide data quality — about 55% true, 40% reported → about 99% in three months. $500K a year at Austria, 10% of the freight base on the Finance-signed, executed-only method, then about $8M across 25 EU sites and 400-plus lanes — both record; manual reconciliation 8–10 → 1–2 hours a week; about $0.5M a year of analyst capacity modeled on top.

### SITUATION / TASK — THE BUSINESS PROCESS PROBLEM, SCOPE, SCALE, BASELINES

**S —** Transportation at the customer's Austria site could not measure on-time delivery. Reports said data quality was 40%; the team spent 8–10 hours a week reconciling promised vs actual dates by hand; the freight base was ≈$5M/yr with no enforceable OTD. Everyone — the site, the carriers' account teams, my own sponsor — believed the carriers were late.

The cost of acting on the room's explanation: a scorecard on 55% data the carriers would dispute and win, the manual reconciliation continuing at every site, and any savings claim disputed after the fact. The customers: the site that needed an enforceable standard, and the carriers who deserved to be measured against a real promise.

**T —** Own the analysis and the rules design. Goal: an OTD the carriers could not dispute, and savings claimed only on a method Finance signed first. The real challenge: the data had to overturn the room's explanation.

**Baselines on the table:** routing-guide data quality 40% reported, ~55% true once measured · manual reconciliation 8–10 hours a week per site · enforceable OTD: none, on a ≈$5M/yr freight base.

## ACTION — THE PROCESS / OPERATING-MODEL REDESIGN, IN GATED PHASES

- **Discovery** — pulled three years of shipment history in SQL myself; defect census (missing rates, wrong-grain lanes, five spellings of one carrier, brochure transit times); true DQ ~55% vs 40% reported; the problem reframed as measurement, not carriers. *Exit bar:* sponsor accepts the reframe on the evidence; defect classes named and counted.
- **Plan & Design** — rules-engine spec (lookup by ship-from/to, mode, volume, Incoterm, SLA; a no-hit pages a human, never a default carrier); reason-code alert design; the savings method signed with Finance before any dollar was claimed (contracted guide rate − paid rate, per shipment, executed only); the scorecard DQ gate (automated OTD = manual sample per lane cohort). *Exit bar:* spec signed by transportation; Finance method signed; scorecard gate published.
- **Build & Test** — held engineering to the spec; weekly defect burn-down by reason code; manual-sample comparison per lane cohort (≈97% agreement). *Exit bar:* DQ trending to ~99%; automated OTD matches the manual sample per cohort.
- **Cutover & Go-Live (Austria)** — released the carrier scorecard only after the gate passed; retired the manual reconciliation (8–10 → 1–2 hrs/wk); stood up the savings ledger per shipment. *Exit bar:* a scorecard the carriers could not dispute; $500K/yr on the Finance-signed method.
- **Hypercare / Close → scale** — super-user model per region; one-page runbook (add a lane · read a no-hit · when to escalate); two shadowed rule reviews per site; rolled out site by site. *Exit bar:* 25 EU sites / 400+ lanes ≈ $8M; ~99% DQ; adoption cost per site = a rules file + one DQ cycle, in days.

**Influence and ownership.** I pulled the data myself because the room already had an explanation; the data platform team built the engine; regional super-users ran the rules files after two shadowed reviews each. Mine: the SQL pull, the defect census, the rules-engine spec, the reason-code alert design, the Finance-signed method and the scorecard gate. The sponsor was unhappy with the delay for weeks — answered with the defect classes trending down weekly and the manual-sample gate; he was right to want speed, and the carriers never disputed the result. Refused: a scorecard on 55% data, a silent default carrier on a no-hit, and any savings number without a Finance-signed method.

> **Designed the process; others own it now.** I designed the rules-engine spec, the reason-code alerts and the savings method; regional super-users run the rules files and the DQ cycle at 25 sites without me. **Trade-off put to leadership.** The recommendation to the sponsor and Finance outlined the competing metrics — speed to a carrier conversation vs a baseline the carriers could not dispute — and held the scorecard until DQ reached ~99%, with savings counted executed-only.

## RESULTS — $ · # · % · TIME · CUSTOMER · QUALITY (CANON NUMBERS ONLY)

- **$** — $500K/yr at Austria — 10% of a ~$5M freight base on the Finance-signed, executed-only method (record) — then ~$8M across 25 EU sites / 400+ lanes (record roll-up; ~$320K/site average, derived); plus ≈$0.5M/yr *modeled* analyst capacity (range $0.2–0.7M).
- **#** — 25 EU sites / 400+ lanes on the same rules file and one DQ cycle each; three years of shipment history pulled in SQL for the baseline (record).
- **%** — routing-guide data quality ~55% true (40% reported) → ~99% in 3 months (record); ≈97% agreement automated vs manual sample.
- **Time** — ~99% DQ in 3 months; adoption cost per site = a rules file + one DQ cycle, in days (record).
- **Customer** — a carrier scorecard the carriers could not dispute — automated OTD matched the manual sample per lane cohort before release (record).

- **Quality** — manual reconciliation 8–10 → 1–2 hrs/wk per site; the savings method signed with Finance before any dollar; negotiated-but-not-executed never counted (record).

**L6 LEVEL EVIDENCE** — what the interviewer pattern-matches against the Amazon L6 functional rubric; every line is this story's canon, no new numbers

| L6 DIMENSION | EVIDENCE IN THIS STORY |
|---|---|
| **Ambiguity** | The room already had an explanation — "the carriers." I pulled three years of shipment history in SQL myself, built the defect census and reframed the problem as measurement; the sponsor had not asked for that. |
| **Scope & influence** | The data platform team built the engine; regional super-users ran the rules files after two shadowed reviews each; Finance signed the savings method before any dollar; the sponsor, unhappy with the delay for weeks, was answered with the defect census. |
| **Execution trade-off + mechanism others now own** | Speed (a carrier beating now) vs a baseline that could not be disputed: DQ to ~99% before any scorecard was released. Mechanisms others own now: a rules file + one DQ cycle per site, the super-user model, the one-page runbook and the Finance-signed, executed-only savings method. |
| **Impact (long-term)** | $500K/yr at Austria (record) → ~$8M across 25 EU sites / 400+ lanes (record roll-up); manual reconciliation 8–10 → 1–2 hrs/wk per site. |
| **Process improvement (standard work institutionalized)** | Reason-code alerts; weekly defect burn-down by reason code; a no-hit pages a human, never a default carrier; adoption cost per site measured in days. |
| **Complexity & consensus** | A carrier scorecard the carriers could not dispute — automated OTD matched the manual sample per lane cohort (≈97% agreement) before release. |
| **Root cause at scale** | Five spellings of one carrier, wrong-grain lanes and brochure transit times were the trending cause — fixed in the rules engine and the DQ cycle once, then rolled to 25 sites. |

## PROBING — CHALLENGE Q&A TUNED TO THIS INTERVIEWER

> **BR — Everyone believed the carriers were late. How did you know they were wrong?**
>
> I didn't — I measured. Three years of shipment history in SQL showed true data quality was about 55%, not the 40% proxy, and the defects were ours: missing rates, wrong-grain lanes, five spellings of one carrier, brochure transit times posing as promises. The data overturned the explanation; I did not argue with it.

> **BR — Is the $8M real? Fifth why.**
>
> It is the Finance-signed, executed-only method applied at each site, not a projection: contracted guide rate minus paid rate, per shipment, only on shipments that moved at the guide rate. Negotiated-but-not-executed never counted. The $500K Austria pilot is the method's proof; the $8M is the same ledger across 25 sites; what firms it is the per-shipment ledger Finance read. The assumption underneath is that the ten percent held elsewhere — Austria at $500K sits above the ~$320K/site average because it was a larger site.

> **BR — Your sponsor wanted speed and you delayed weeks. Defend that.**
>
> I traded date for evidence, never scope: a scorecard on 55% data would have been disputed and the carriers would have won. I showed him the defect classes trending down weekly and the gate the scorecard had to pass. He was right to want speed; the carriers never disputed the result.

> **BR — What is Broad Responsibility here?**
>
> One site's fix became 25 sites' instrument — and the responsibility was in the method: the savings rule was signed before the number so no site could claim what it had not executed, and a no-hit paged a human rather than silently defaulting a carrier.

> **BR — Who was the customer?**
>
> The site, which needed an enforceable standard; and the carriers, who deserved to be measured against a real promise — the scorecard's legitimacy with them is why the savings held.

> **BR — What would you do differently?**
> Start with the industry benchmark for lane transit times instead of proving the brochure wrong lane by lane — same answer, faster. And sign the method before the number — which held, and which I would do again.

## THE KEY MATH — TRUE DQ, THE PROMISED-DATE RULE, THE SAVINGS METHOD AND THE ROLL-UP (RECORD FIGURES; AVERAGES DERIVED AND LABELED)

| STEP | QUANTITY | FORMULA → SUBSTITUTION | BASIS | VALUE |
|---|---|---|---|---|
| 1 | Reported vs true DQ | `40% (field-populated proxy) vs valid promised ∧ valid actual ∧ resolvable lane ≈ 55%` | record | ~55% true |
| 2 | Promised-date rule | `promised = ship_date + transit(lane_key); lane_key = (from, to, mode, volume band, Incoterm, SLA)` | record design; Incoterm on ~8% of lanes; no-hit pages a human | a promise per shipment |
| 3 | OTD definition + gate | `OTD = actual ≤ promised / valid promises; automated vs manual sample ≈ 97% agreement per cohort` | record | gate before release |
| 4 | Austria savings | `$5,000,000 × 10% = $500,000/yr` | record; Finance-signed, executed-only | $500K/yr |
| 5 | Network roll-up | `~$8M / 25 sites = ~$320K/site; / 10% ≈ $80M implied base` | record roll-up; averages derived | ~$8M |
| 6 | Analyst capacity | `(9 − 1.5) h/wk × 25 sites × 50 wk × $55 ≈ $515K` | modeled add-on | ≈$0.5M/yr modeled |

Never "$1M routing guide"; never "99% in two months"; 1–2 hrs/wk is the after.

**Where this lands at Amazon.** The cleanest "used data to overturn the room's explanation" story — and the model for how a savings claim should be built: method signed with Finance before the number, executed-only, then scaled site by site with the adoption cost measured in days. Reliability (an enforceable promise), cost efficiency ($500K → ~$8M record), asset utilization (analyst hours returned, modeled).

## TWIST MAP — LIKELY LP QUESTIONS FROM THIS INTERVIEWER → THE ANGLE TO TAKE WITH S10

| LIKELY QUESTION (THIS INTERVIEWER) | LP | ANGLE — WHAT TO EMPHASIZE | SKIP / NEVER SAY |
|---|---|---|---|
| **Tell me about a time you were right and others were wrong.** | Are Right, A Lot | The measurement reframe on three years of data; the fix was a rules engine, not a carrier beating. | Do not gloat — the sponsor was right to want speed. |
| **Tell me about a time your work had impact beyond your team / site.** | Broad Responsibility | Austria's method → 25 sites / 400+ lanes ≈ $8M; adoption in days; super-users after two shadowed reviews. | Never "$1M"; never claim site-wave counts or dates beyond "Austria first." |
| **Tell me about a time you dug into data others had accepted.** | Dive Deep | Pulled the SQL myself; the defect census; true DQ ~55% vs 40% reported. | S8 (measured lanes) is Hoerger's — do not use it here. |
| **How do you make a savings claim credible?** | Earn Trust / Frugality | Finance signs the method before the number; executed-only; negotiated never counted; the ledger per shipment. | — |
| **Tell me about a time you delayed a launch.** | Highest Standards / Backbone | The scorecard held until automated OTD matched the manual sample per cohort; sponsor unhappy for weeks; carriers never disputed it. | Never "99% in two months." |

| LIKELY QUESTION (THIS INTERVIEWER) | LP | ANGLE — WHAT TO EMPHASIZE | SKIP / NEVER SAY |
|---|---|---|---|
| **Who was the customer and what changed for them?** | Customer Obsession | The site got an enforceable standard; the carriers got a fair promise; the team got 8–10 hours a week back. | 1–2 hrs/wk is the after, never the before. |

## S14 — Manhattan Associates — the $2.5M OMS↔WMS discrepancy my manager wanted to fix quietly: transparency as the reconciliation control, two days before an executive review

**LPs this story can serve here:** Earn Trust (primary — the flagship integrity story) · Have Backbone, Disagree & Commit (backup — refusing the quiet fix) · Ownership (the seam between OMS and WMS) · Bias for Action (third — two days to the review) · Customer Obsession (the customer told first and given the priority)
**Process family:** Deliver / Return-to-book — order-to-fulfillment reconciliation control; incident handling under a customer-relationship deadline

**Honesty label.** The census, the risk argument, the customer pack and the war room were mine; the consultants traced the handshake, wrote and verified the per-line patch, and built the permanent fix. The $10M initiative is never counted as savings. Manhattan is the era in which I personally configured Manhattan DOM — the only platform configuration claim in canon.

**60-SECOND BUSINESS-FIRST PITCH — STOP ON THE PRIMARY METRIC**

**Situation.** Two days before an executive review at a Manhattan Associates retail customer, the order-management and warehouse-management systems did not reconcile — about two hundred orders, $2.5 million of order value — and a $10 million initiative rode on that review. My manager and the room favored a quiet back-end correction: a cross-table script, written in under two days, and nobody tells the customer. **Task.** I was the consultant who would run whichever path was chosen; I owned the recommendation and then the war room. **Action.** Facts before path: I built the discrepancy census by cause on a sample of the two hundred orders — which handshake step, which order states — and named the fear of losing the initiative as legitimate. Then I disagreed on risk, not principle: a script across related tables written in two days cannot be verified; an IT audit would turn a data issue into a reputation issue on a $10M relationship; the expected corruption from the silent path was about $1.5M of risk. I offered to run the war room personally so the cost of my position landed on me. He accepted, hesitantly. The customer got root cause, plan and timeline before the review — no apology-first — prioritized their own orders, asked for a per-line, reversible patch on that set with before-and-after evidence per order, and gave time for a clean fix to the handshake. **Result.** Primary metric: the OMS–WMS discrepancy — about 200 orders and $2.5M → zero on the customer's prioritized set before the review, then zero through the permanent fix; a standing reason-coded reconciliation report read before every executive review from then on. The $10M initiative was won — a record fact about the relationship that I never count as savings. Lesson: silent corrections are loans against trust at a terrible interest rate.

### SITUATION / TASK — THE BUSINESS PROCESS PROBLEM, SCOPE, SCALE, BASELINES

**S —** Two days before an executive review, OMS and WMS didn't reconcile — ~200 orders / $2.5M — and a $10M initiative rode on the review. My manager and the room favored a quiet back-end correction.

The risks of the quiet path: a rushed cross-table script corrupting data (the record's own illustration: expected ≈$1.5M of corruption), a near-certain audit finding turning a data issue into a reputation issue on a $10M relationship, and the customer surprised at the review. The customer in this story is the retailer's executives and the IT and finance teams who would otherwise find out from an audit.

**T —** I was the consultant who would run whichever path was chosen; I owned the recommendation and then the war room. Goal: the discrepancy to zero and a review the relationship could survive, without a silent correction.

**Baselines on the table:** OMS↔WMS discrepancy ~200 orders / $2.5M (target zero with before/after evidence per order) · new discrepancies unmonitored (target zero on a standing daily report) · two days to the review.

## ACTION — THE PROCESS / OPERATING-MODEL REDESIGN, IN GATED PHASES

- **Discovery (day 1, morning)** — discrepancy census by cause (handshake step × order state) on a sample of the ~200; exposure statement (review in two days; $10M initiative); both options named honestly. *Exit bar:* the room agrees the facts before the path; the fear of losing the initiative named as legitimate.
- **Plan & Design (day 1, afternoon)** — options memo on risk, not principle: a two-day cross-table script is error-prone; an IT audit turns a data issue into a reputation issue; transparency = root cause + action plan + I run the war room. *Exit bar:* manager accepts, hesitantly; my labor attached to my position.
- **Build & Test (day 2)** — customer communication pack (root cause, plan, timeline — no apology-first); war-room plan (bridge, log, cadence); verification method for any patch (per line, reversible, before/after recorded). *Exit bar:* customer briefed before the review; war room staffed.
- **Cutover (review + patch)** — took the customer's prioritized order list; ran the bulk patch per line with verification; scheduled the clean root-cause fix. *Exit bar:* patch verified; discrepancy to zero on prioritized orders; review held with the facts on the table.
- **Hypercare / Close** — the permanent OMS–WMS handshake fix; the reason-coded reconciliation report as a standing control read before every executive review. *Exit bar:* $10M initiative won; handshake reconciles; new discrepancies = 0 on the daily report.

**Influence and ownership.** Mine: the census, the risk argument, the customer pack and the war room; the consultants traced the handshake, wrote and verified the per-line patch, and built the permanent fix. The argument was about risk, not virtue, and my labor was attached to my position. Had my manager said no, I would have documented the risk assessment in writing and asked for the decision recorded with its owner — disagree and commit, with the tripwire (any audit query on those orders) named; I would not have run the silent script without that record.

> **Designed the process; others own it now.** I designed the customer pack, the war room and the verification method; the standing reason-coded reconciliation is read before every executive review by the account team, not by me.
> **Trade-off put to leadership.** The recommendation to my manager outlined the competing metrics — two days of speed with a cross-table script vs the risk of an error-prone patch and a reputation issue — and attached my own labor to the transparent path.

## RESULTS — $ · # · % · TIME · CUSTOMER · QUALITY (CANON NUMBERS ONLY)

- **$** — ~200 orders / $2.5M discrepancy → zero (record); the $10M initiative won — a record fact about the relationship, never counted as savings; ≈$0.1M/yr *modeled* from the standing reconciliation (range $55–160K); the ≈$1.5M expected corruption from the silent script is *avoided risk*, not a saving.
- **#** — ~200 orders patched per line, reversible, with before/after evidence per order, in the customer's priority order (record).
- **Time** — two days to the executive review; the customer briefed before the review, not in it; the clean root-cause fix scheduled after (record).
- **Customer** — the customer prioritized their own orders, asked for the bulk patch on that list, and gave time for a clean fix — the review held with the facts already on the table (record).
- **Quality** — per-line reversibility — no bulk script across related tables; the standing reason-coded OMS↔WMS reconciliation read before every executive review; new discrepancies = 0 on the daily report (record).
- **%** — no percentage claim exists in the record for this story — say the counts and the control; do not invent a rate.

**L6 LEVEL EVIDENCE** — what the interviewer pattern-matches against the Amazon L6 functional rubric; every line is this story's canon, no new numbers

| L6 DIMENSION | EVIDENCE IN THIS STORY |
|---|---|
| **Ambiguity** | Two days to an executive review, a $10M initiative at stake, and a manager's preferred quiet fix. I ran the discrepancy census by cause (handshake step × order state) and named both options honestly before choosing the path. |
| **Scope & influence** | My manager persuaded on risk, not virtue; the customer's executives briefed before the review; the consultants traced the handshake and built the patch and the permanent fix — influence one level up and outward to the customer. |
| **Execution trade-off + mechanism others now own** | A two-day cross-table script (speed) vs a per-line reversible patch plus transparency (risk) — the recommendation was framed as risk, and my labor was attached to my position. Mechanism others own now: the reason-coded OMS↔WMS reconciliation read before every executive review. |
| **Impact (long-term)** | ~200 orders / $2.5M discrepancy → zero; the $10M initiative won (a record fact about the relationship, never counted as savings); new discrepancies = 0 on the daily report. |
| **Process improvement (standard work institutionalized)** | The verification method for any patch (per line, reversible, before/after recorded); the war-room plan (bridge, log, cadence); the standing reconciliation as a control, not a project. |
| **Complexity & consensus** | The fear of losing the initiative named in the room; the customer prioritized their own orders and gave time for a clean fix — the review held with the facts already on the table. |
| **Root cause at scale** | The census by handshake step × order state led to a permanent OMS–WMS handshake fix, not just the patched orders — the trending cause removed. |

## PROBING — CHALLENGE Q&A TUNED TO THIS INTERVIEWER

**BR — You're claiming the $10M?**

No. The initiative was won in the review the facts survived; I claim the control and the posture, not the contract value. The only recurring dollar I model is ≈$0.1M a year from the standing reconciliation, labeled; the ≈$1.5M is the record's illustration of expected loss from the silent script — avoided risk, not money earned.

**BR — Your manager wanted the quiet fix. What exactly did you say, and what if he had said no?**

Privately first, with the census. I disagreed on risk, not principle: an unverifiable script plus an audit exposure versus a customer told the truth with a plan — and I offered to run the war room so the cost landed on me. He accepted, hesitantly. Had he said no, I would have documented the risk assessment in writing and asked for the decision recorded with its owner, with the tripwire named; I would not have run the silent script without that record.

**BR — Why not just apologize and fix it?**

No apology-first. The customer pack was root cause, action plan and timeline — to their IT and finance before the executive review, not in it. They responded by prioritizing their own orders and giving us time for a clean fix; an apology without a plan would have given them nothing to decide.

**BR — What control outlived the war room?**

The reason-coded OMS↔WMS reconciliation as a standing check, read before every executive review rather than two days before one — status mapping, timing, failed message, manual edit; a row with no class is an investigation, not a bucket. The handshake fix removed the cause; the report made sure the next one surfaced early.

**BR — A hundred thousand a year is small for a flagship story.**

Yes, and I would rather say small and true. The story is about the control and the choice, not the run-rate. The hardest input is how many discrepancies arise per month; I assumed the two hundred built over about sixty days. I would read the daily match report's reason-code counts for a quarter and price exactly what surfaced.

> **BR — Anything more recent where you disagreed upward?**
>
> Yes — re-sequencing my manager's roadmap at Flexport on cost-to-serve evidence (the hiring manager heard that one), and at Sephora the rule I had written myself and had to correct through the same three signatories. Same posture: disagree on evidence, carry the consequence.

## THE KEY MATH — THE DISCREPANCY, THE TWO PATHS PRICED, AND THE STANDING CONTROL (RECORD PAIRS; PROBABILITIES ILLUSTRATIVE)

| STEP | QUANTITY | FORMULA → SUBSTITUTION | BASIS | VALUE |
|---|---|---|---|---|
| 1 | The discrepancy | `$2,500,000 / 200 = $12,500 average per order` | record pair; average derived | ~200 orders, ≈$12.5K each |
| 2 | Quiet path priced | `blast = 2 × $2.5M = $5.0M; E[loss] = 0.3 × $5.0M = $1.5M + a near-certain audit finding` | P(error) 0.3 and 2× blast radius illustrative | ≈$1.5M avoided risk — not a saving |
| 3 | Honest path priced | `a few days' delay on a customer-controlled subset of $2.5M` | record argument | bounded, one-time |
| 4 | Patch | `per line, reversible; evidence rows = set size; customer's priority order first` | record | 0 on the set before the review |
| 5 | Standing control | `match rate = orders with OMS status == map(WMS status) / orders in flight; Σ 4 reason classes = all of Δ` | record design | exposure at detection ~$2.5M/60 d → ~1 day |
| 6 | Recurring value | `manual fixes ≈$30K + stuck cash ≈$30K + war rooms ≈$40K ≈ $0.1M` | modeled; range $55–160K | ≈$0.1M/yr modeled |
| 7 | $10M initiative | `won — excluded from any savings column by rule` | record fact about the relationship | never claimed |

**Where this lands at Amazon.** The purest Earn Trust pattern — tell the customer first, price the honest path's cost onto yourself, and leave a standing control so the courage is rarely needed again. The transferable control is not the courage; it is the reason-coded reconciliation read before every review. Reliability (a seam that reconciles daily), responsiveness (the customer decided the priority), trust that survives an audit.

## TWIST MAP — LIKELY LP QUESTIONS FROM THIS INTERVIEWER → THE ANGLE TO TAKE WITH S14

| LIKELY QUESTION (THIS INTERVIEWER) | LP | ANGLE — WHAT TO EMPHASIZE | SKIP / NEVER SAY |
|---|---|---|---|
| **Tell me about a time you had to deliver bad news to a customer.** | Earn Trust / Customer Obsession | Root cause, plan, timeline before the review; the customer set the priority; no apology-first. | Never count the $10M as impact. |
| **Tell me about a time you disagreed with your manager.** | Have Backbone, Disagree & Commit | On risk, not principle; war room offered; the "what if he had said no" answer ready — document, record the owner, name the tripwire. | Do not make the manager the villain — his fear was legitimate and named as such. |
| **Tell me about acting fast under a deadline.** | Bias for Action | Two days: census by cause in a morning, options memo in an afternoon, customer pack on day two, per-line patch before the review. | Do not imply the permanent fix was done in two days — it was scheduled after. |
| **Tell me about owning a problem across system boundaries.** | Ownership | The seam between OMS and WMS had no owner; the census, the patch method and the standing report gave it one. | Consultants built the patch and the fix — say so. |

| LIKELY QUESTION (THIS INTERVIEWER) | LP | ANGLE — WHAT TO EMPHASIZE | SKIP / NEVER SAY |
|---|---|---|---|
| **How do you design a reconciliation control?** | Functional — control design | Reason-coded classes that sum to the delta; a row with no class is an investigation; read before every review, not two days before one. | No invented percentage — the record has none here. |
| **Why did this matter beyond the one customer?** | Broad Responsibility | A silent fix would have taught the practice that quiet scripts are acceptable; the standing control and the posture became the practice's norm. | Do not claim practice-wide dollars — none exist. |

## Question bank — likely questions for this interviewer, with a one-line answer plan

Bar-raiser questions: LP-depth probes, mid-story LP switches, softest-number challenges and the consistency check. Each plan points at a packet story; stories allocated to other interviewers are pointers only.

| # | LIKELY QUESTION | ANSWER PLAN (ONE LINE) | STORY |
|---|---|---|---|
| 1 | **Tell me about a time you earned trust the hard way.** | S14 pitch — tell the customer first; cost priced onto me; the standing control; $10M won, never claimed. | S14 |
| 2 | **Tell me about a mistake you made and what changed because of it.** | S16 — the security review; two weeks; owned upward the same day; zero Mondays skipped; the kickoff question. | S16 |
| 3 | **Tell me about developing someone.** | S15 — the QA lead who authored and defended the gate; development measured by whether the gate survived my leaving. | S15 |
| 4 | **Tell me about a time the data disagreed with everyone.** | S10 — three years in SQL; true DQ ~55%; measurement, not carriers; Finance-signed method before the number. | S10 |
| 5 | **Switch LP mid-story: "OK — now tell me that same story as Frugality / Customer Obsession / Bias for Action."** | Keep the facts identical; change emphasis per the twist map (e.g., S16 as Frugality = the >$100K priced; S14 as Bias for Action = the two days; S10 as Customer Obsession = the carriers could not dispute). | Twist maps |
| 6 | **Which of your numbers would you least want me to check?** | Name the softest input in the story just told and how to firm it (S16 decision risk; S15 criticals base count; S10 the $0.5M analyst capacity; S14 discrepancies per month) — never defend a soft number as record. | All four |
| 7 | **What did you give up for the customer?** | S14 — my time in a war room and the risk of being wrong in front of them; S15 — the stories I covered so dev could drop them; S10 — weeks of schedule for evidence. | S14 / S15 / S10 |
| 8 | **What is your biggest weakness as a leader?** | I convert verbal agreements into documents people must sign — slower in the week; and I under-planned a dependency with a lead time once (S16) and now over-ask that question at every kickoff. | S16 |
| 9 | **Tell me about a time you had to scale something.** | S10 — Austria → 25 sites at a rules file plus one DQ cycle each; the super-user model; the method signed before the number so scale could not inflate the claim. | S10 |
| 10 | **How do you decide what not to build?** | S16 — ship with the cross-reference table, push the real key upstream; S15 — the drop-date rule; S14 — no bulk script across related tables. | S16 / S15 / S14 |
| 11 | **Why should we hire you over someone who has run an Amazon network region?** | Fifteen years designing the supply chain — definitions, lead times, buffers, one truth — and keeping it honest in a weekly review; I have not run an Amazon region and won't pretend otherwise; the pattern transfers, the scale is the reason I am here. | Master TMAY close |
| 12 | **Is there anything you've told the other interviewers that you'd state differently to me?** | No — and name the two numbers most likely to be cross-checked: 7 → 68 is 9.7×, and the two $2.5M figures (Manhattan discrepancy; Sephora headline) are different things. | Consistency box |

## Cheat-numbers box — canon values with basis labels

Canon numbers for S16, S15, S10 and S14 with their basis labels, plus the cross-loop figures a bar raiser is most likely to cross-check — from the master packet's §4.1 ledger.

| METRIC | VALUE TO SAY | BASIS | NEVER |
|---|---|---|---|
| **S16 accuracy / bus factor** | WBR data accuracy 60% → 90%; bus factor 1 → ≥2; parity within 1% for consecutive weeks; 4% ticket-sync drop caught | record | conflating with S11's ETA accuracy 60 → 90% without saying which |
| **S16 dollars** | >$100K/yr priced baseline in infra's own loaded rates; ≈$30K hours + ≈$70K decision risk (≈$50K retired) + ≈$15K bus factor | priced (record of the pricing) / modeled | "saved $100K" |
| **S16 miss** | data-access security review — 2 weeks lost; owned upward the same day; zero WBR deliveries skipped | record | "a dependency slipped" |
| **S15 team / cadence** | 20-person Product/Eng/QA team; six-week hybrid → four-week agile in six weeks; releases 8.7 → 13 | record / modeled (releases) | a direct-report headcount |
| **S15 quality / dollars** | critical defects per release −50%; ≈$0.13M/yr (≈$90K + ≈$40K; range $80–270K) | record / modeled | the four-per-release base count as record |
| **S10 data quality** | ~55% true (40% reported) → ~99% in 3 months; ≈97% agreement auto vs manual sample; manual 8–10 → 1–2 hrs/wk | record | "99% in two months"; 1–2 hrs as the before |
| **S10 dollars** | $500K/yr Austria (10% of ~$5M, Finance-signed, executed-only) → ~$8M across 25 EU sites / 400+ lanes; ~$320K/site derived; ≈$0.5M/yr analyst capacity | record / record roll-up / modeled | "$1M routing guide"; counting negotiated-not-executed |
| **S10 design** | lane key (from, to, mode, volume band, Incoterm, SLA); Incoterm on ~8% of lanes; no-hit pages a human | record | site-wave counts or dates beyond "Austria first" |
| **S14 discrepancy** | ~200 orders / $2.5M → 0 on the prioritized set before the review; per-line reversible patch with before/after evidence | record | a percentage — none exists |
| **S14 initiative** | $10M initiative won | record — never counted as savings | any realized dollar on it |
| **S14 modeled** | ≈$0.1M/yr standing reconciliation (range $55–160K); ≈$1.5M expected corruption from the silent script | modeled / avoided risk | calling the $1.5M a saving |
| **Cross-loop multiples** | 7 → 68 = 9.7× (S11, HM's story) | record | "10×" / "50×" |
| **Cross-loop site count** | "six sites" (Elementum) | record | "six NA sites" |
| **Platform configuration** | Manhattan DOM — the only platform personally configured | record | "I configured S/4" / BlueYonder / Locus / Snowflake |

**Never-say, one glance before the call.**

- Never count the $10M initiative as savings; never claim the >$100K as savings; never call S14's ≈$1.5M a saving.
- Never "10×" or "50×" (9.7×); never "six NA sites"; never "$1.5M to $2M"; never "$1M routing guide"; never "99% in two months."
- Never invent a direct-report headcount, a days-to-tie figure, a percentage for S14, or site-wave counts for S10.
- Never let the two $2.5M figures blur (Manhattan discrepancy vs Sephora headline) — they belong to different interviewers' stories.
- Never "we" for an action that was mine; name the team's part once — consultants (S14), infra (S16), data platform team (S10), the QA lead (S15).
- S11/S12, S6/S8, S2/S18, S1/S3 are the other four interviewers' — pointers only; if the bar raiser has read their notes, connect (S16 ↔ S11; S14 ↔ S12 "more recent upward disagreement") without re-telling.

## Backup bench — use only if a primary doesn't fit

Six fallbacks for the bar raiser, none of them another interviewer's primary, chosen to cover the LPs the four primaries leave thin: S19 (Elementum · Walmart-client — the promise buffer as one governed dial; Invent & Simplify / Frugality / Dive Deep, all model), S20 (Elementum · Walmart-client — the pre-purchase WBR and the Andon; Highest Standards / Earn Trust on mechanism design — keep separate from S11's production WBR), S13 (Manhattan — Cyber Monday at 4 AM; Bias for Action / Customer Obsession) and S9 (Elementum — 597 lanes two levels up; Backbone / Think Big — the Think Big gap in the LP grid); plus two doc-derived Flexport benches from the program record — the RAPID Crux customer transition (Earn Trust / Customer Obsession: 79 customers, transition-or-cancel decision rules) and the FFA Asia latency investigation (Dive Deep; facts identical to Hoerger's bench) — both record-from-docs, both from the S11/S12 program family, so keep the shared numbers identical to Tripathy's packet. Tell S16, S15, S10 and S14 first; a bar raiser compares notes, so a bench story must carry exactly the master's numbers.

**Bench discipline.** The two (or four) primaries above are the stories to tell first. Reach for a bench story only when the question genuinely does not fit a primary, or when the interviewer asks for "another example" on an LP already spent. Each entry is a 60-second pitch, a compressed STAR, one results line and two probes — enough to answer and stop; no full probing bank or derivation. Where a bench story is another interviewer's primary, the red note says so: use it only if the question demands it, and keep the facts identical to that packet.

### BACKUP — ONLY IF ASKED S19 — Elementum (client: Walmart Marketplace) — the delivery-promise buffer rebuilt as priced risk: one governed dial, +32 points of fast-promise share at the same 95% floor (all model, no record dollar)

**LPs this backup can serve:** Dive Deep (alternate when S8/S10 are spent) · Invent and Simplify (alternate) · Frugality (speed bought with zero new spend) — design-flavored probes

**Basis.** Engagement lead for the promise-policy workstream; the client's promise-engineering and data-science teams owned and built the engine and the transit models — I never configured or deployed any platform. Every quantity is from the engagement's illustrative twelve-cohort model — say "the model" in the same breath; there is no production dollar and none is ever promoted. Labels (metro two-day, mountain rural) are the model's illustrative cohort names.

**60-SECOND PITCH — STOP ON THE PRIMARY METRIC**

**Situation.** A marketplace client printing delivery dates on inventory it never touches: promised date = order time + seller handling + predicted transit + a safety buffer — and the buffer was a flat 1.10 days on every lane regardless of variability. In the engagement model — twelve lane cohorts, about a million orders a week, transit σ from 0.22 to 1.60 days — that flat buffer printed 95.8% accuracy but put only 7.6% of orders on a fast promise: dense metro lanes over-insured and fat-tailed rural lanes under-insured at the same time. **Task.** Own the decomposition, the alternatives, the validation gates and the governance recommendation; the client's data science and engineering did not report to me. **Action.** Discovery: the buffer-waste audit — the flat buffer bought $\Phi(5.0)$, near-certainty, on the tightest cohort and only $\Phi(0.69) \approx 75\%$ on the widest. Plan & Design: four policies on one comparison table — uniform, σ-proportional with one governed dial, minimum-days at the floor, profit-optimal on priced breach economics. Build & Test: $b = 1.645\sigma$ holds 95% on every cohort by construction; the equimarginal check proved the optimum; the Q4 drill showed a frozen summer dial breaking the floor. Cutover: four shadow weeks against logged live orders before any displayed promise. Hypercare: the dial published in the weekly review header, re-solved before peak. **Result.** Primary metric: fast-promise share 7.6% → 39.6%, +32 points at the held 95% floor; average promised day 3.09 → 2.75; +$255K a week, about $13.3M a year — all on the illustrative book, and I say "the model" in the same breath.

## COMPRESSED STAR

- **S —** A flat 1.10-day buffer on every lane: 95.8% headline accuracy hiding over-insurance on tight lanes and under-insurance on loose ones; fast-promise share 7.6% (model).
- **T —** Lead the promise-policy workstream: decompose the waste, put priced alternatives on one table, design the gates, leave a governance instrument — not a tuning habit.
- **A —** Promise identity and accuracy factorization; buffer-waste audit; four policies on one table; proofs (Φ(1.645) = 0.95 per cohort, equimarginal check, Q4 drill at ~1.45× σ); shadow mode for four weeks; the dial k owned by a business owner and printed every Monday.
- **R —** Fast-promise share 7.6 → 39.6% (+32 pts) at exactly the 95% floor; average promised day 3.09 → 2.75; +$255K/wk ≈ $13.3M/yr — model, labeled; the widest cohorts got honestly slower promises (1.10 → 2.63 days) on purpose.

## RESULTS LINE

Fast-promise share 7.6% → 39.6% (+32.0 points) at a held 95.0% floor (model) · average promised day 3.09 → 2.75 (model) · +$255K/wk ≈ $13.3M/yr vs uniform (modeled on the illustrative ~1M-order book) · the profit-optimal frontier (97.8% accuracy, ≈ +$20M/yr) handed to Finance as a priced decision — shown, never claimed, never merged with the $13.3M.

## THE MATH — STEP-BY-STEP DERIVATION

### Table 1 — before: the problem in numbers.

| STEP | QUANTITY | FORMULA → SUBSTITUTION | INPUTS (BASIS) | VALUE |
|---|---|---|---|---|
| 1 | The promise identity | `EDD = t_order + handling + transit_pred + b; kept ⇔ (handling_err + transit_err) ≤ b` | per order; b = buffer in days (record-from-docs identity) | the buffer is the only policy term |
| 2 | Kept-promise probability per lane | `P(kept) = Φ(b / σ)` | ε ~ N(0, σ²) per cohort — whiteboard version; production uses the empirical quantile (record-from-docs) | accuracy is a function of b and σ only |
| 3 | Incumbent buffer | `b = 1.10 days on every lane` | uniform policy A (illustrative model) | 1.10 d everywhere |
| 4 | Tightest cohort, z-score | `z = b / σ = 1.10 / 0.22 = 5.0; Φ(5.0) ≈ 0.9999997` | metro cohort σ = 0.22 d (illustrative model) | near-certainty bought — promise needlessly slow |
| 5 | Widest cohort, z-score | `z = 1.10 / 1.60 = 0.6875; Φ(0.69) ≈ 0.754` | mountain-rural cohort σ = 1.60 d (illustrative model) | ≈75.4% kept — a known-loss book |
| 6 | Portfolio print | `accuracy 95.8% · fast share 7.6% · avg promised day 3.09 · profit $5.69M/wk` | order-weighted across 12 cohorts, ~1M orders/wk (illustrative model) | a green headline hiding both failures |
| 7 | The asymmetry | `padding fails silently (lost conversion files no ticket); cutting fails loudly (breaches do)` | record-from-docs — why the system drifted slow | bias toward slow, unowned |

### Table 2 — the fix: step by step, values substituted.

| STEP | QUANTITY | FORMULA → SUBSTITUTION | INPUTS (BASIS) | VALUE |
|---|---|---|---|---|
| 1 | One governed dial | `b_lane = k × σ_lane; Φ(k) = floor ⇒ k = 1.645 for 95%` | Φ(1.645) = 0.95 under normal errors (record-from-docs design) | k = 1.645, one number for the book |
| 2 | Tightest cohort re-buffered | `b = 1.645 × 0.22 = 0.36 days` | σ = 0.22 d (illustrative) | buffer 1.10 → 0.36 d; 0.74 d of speed harvested |

| STEP | QUANTITY | FORMULA → SUBSTITUTION | INPUTS (BASIS) | VALUE |
|---|---|---|---|---|
| 3 | Widest cohort re-buffered | `b = 1.645 × 1.60 = 2.63 days` | σ = 1.60 d (illustrative) | buffer 1.10 → 2.63 d — slower on purpose, kept honest |
| 4 | Every cohort at the floor | `P(kept) = Φ(1.645 × σ / σ) = Φ(1.645) = 0.95 on each cohort` | by construction (design) | 95.0% everywhere; no cohort subsidizes another |
| 5 | Portfolio after | `accuracy 95.0% · fast share 39.6% · avg day 2.75 · profit $5.94M/wk` | illustrative model output, policy B | +32 pts fast share at the floor |
| 6 | Equimarginal proof (policy C) | `φ(b/σ)/σ equal across all 12 cohorts (spread < 1e-15)` | solver optimality condition (record-from-docs) | the last buffer-day buys the same accuracy everywhere |
| 7 | Peak regime rule | `σ_Q4 ≈ 1.45 × σ; frozen k ⇒ z = 1.645/1.45 = 1.13; Φ(1.13) ≈ 0.87 per fully-inflated lane; portfolio prints 91.9% < 95%` | Q4 drill (illustrative model) | k re-solved and published in October, never frozen |
| 8 | Shadow gate | `4 consecutive weeks logged ≥ floor, contacts flat → promote; else hold` | promotion bar (design) | rehearse against reality before any displayed date |

**Table 3 — after: the metric, before → after.**

| STEP | METRIC | BEFORE | AFTER | BASIS |
|---|---|---|---|---|
| 1 | Fast-promise share (≤2-day) | 7.6% | 39.6% (+32 pts) | illustrative model |
| 2 | Portfolio promise accuracy | 95.8% (split: ~100% / 75.4%) | 95.0% — the floor, held on every cohort | illustrative model |
| 3 | Average promised day | 3.09 | 2.75 (−0.34) | illustrative model |
| 4 | Weekly modeled profit | $5.69M | $5.94M (+$255K) | illustrative model |
| 5 | Owner of the speed-accuracy trade | implicit, in the last model tune | one governed dial, published weekly, re-solved per regime | engagement design record |

**Table 4 — the dollars, every multiplication shown.**

| STEP | QUANTITY | FORMULA → SUBSTITUTION | INPUTS (BASIS) | VALUE |
|---|---|---|---|---|
| 1 | Weekly profit, policy A | `$5,690,000 / wk, taken from the model` | illustrative model output | $5.69M/wk |
| 2 | Weekly profit, policy B | `$5,940,000 / wk, taken from the model` | illustrative model output | $5.94M/wk |
| 3 | Weekly delta | `$5,940,000 − $5,690,000 = $250,000 (the model's precise print: +$255K)` | rows 1, 2 (illustrative) | ≈ +$255K/wk |
| 4 | Annualized | `$255,000 × 52 = $13,260,000` | row 3 × 52 weeks (modeled) | ≈ $13.3M/yr — "the model," said in the same breath |
| 5 | Frontier: policy D delta | `$6,070,000 − $5,690,000 = $380,000/wk; $380,000 × 52 = $19,760,000` | illustrative model output; ×52 modeled | ≈ $20M/yr — handed to Finance, never claimed |
| 6 | Why D holds 97.8% | `breach EV $12 > margin-per-promise-day trade at $6.50 ⇒ optimum above the 95% floor` | $12 and $6.50 are the model's stated assumptions | the floor and the economics disagree — a governance finding |
| 7 | Headline | `≈ $13.3M/yr modeled; record dollars: none` | rows 3–4 | ≈ $13.3M/yr, always labeled |

## TWO PROBES

### The whole story rests on an illustrative cohort set.

It does, deliberately — the model exists so the argument happens in numbers instead of adjectives. If the real σ looks nothing like the model's, k changes and the lane ranking changes; what survives any parameter set is the structure: buffers risk-proportional, the last buffer-day buying the same accuracy everywhere, and a floor that disagrees with the breach economics treated as a governance finding.

### Why one dial instead of per-lane learned buffers?

Data science wanted per-lane learned buffers — more model, more value, invisible governance. I priced the difference and recommended the one-dial policy first: it captures most of the value, it is auditable, and σ improvements flow into speed automatically and visibly. The no went into the comparison table as a priced row, not a meeting minute.

**Canon numbers (basis labels):** 7.6% → 39.6% fast-promise share (+32 pts, model) · 95.8% → 95.0% accuracy held at the floor (model) · 3.09 → 2.75 promised days (model) · +$255K/wk ≈ $13.3M/yr (modeled) · 12 cohorts, ~1M orders/wk, σ 0.22–1.60 days, k = 1.645 (illustrative model) · breach ≈$12 vs margin ≈$6.50 (model assumptions). Never a production dollar.

## BACKUP — ONLY IF ASKED S20 — Elementum (client: Walmart Marketplace) — the pre-purchase WBR: input levers under output metrics, the five-minute red-week decomposition, and the Andon that replaced a 30-day lag

**LPs this backup can serve:** Insist on the Highest Standards (alternate — hard reds on policy floors) · Earn Trust (alternate — mix-adjusted twins, honest denominators) · Dive Deep (alternate — the factor math) — cadence/mechanism probes

**Basis.** I designed the weekly business review — metric architecture, decomposition math, thresholds and control loop — and led the program that stood it up; the client's engineering built the pipelines and dashboards and their teams own the lines. I never configured or deployed any platform and never claim a production output metric moved. The production-record WBR story is S11 (Flexport) — keep the two separate.

### 60-SECOND PITCH — STOP ON THE PRIMARY METRIC

**Situation.** A marketplace client whose delivery promise was owned by a committee: data science owned the model, engineering the engine, seller ops the handling, carrier programs the lanes — and promise accuracy, the one number the customer feels, was owned by nobody between Tuesdays. Scorecards ran on 30-day rolling windows, so a degrading seller or carrier did weeks of customer damage before any floor bit, and a red week was a blame negotiation. **Task.** Design and stand up the weekly business review — outputs a customer feels, input levers a named owner can turn, and the math that connects them — plus a control loop faster than the 30-day clock. **Action.** Discovery: one glossary, one event source, audited denominators. Plan & Design: seven output metrics over ten input levers, each input with a roll-up and a named owner; the factorization — accuracy ≈ seller-ships-in-window × carrier-meets-transit × Φ(buffer/σ) — as the decomposition; red/green at ±5% relative week-over-week with hard reds on policy floors. Build & Test: the sample-week proof — 0.97 × 0.985 × 0.996 ≈ 95.2% — and the drill: a regional carrier miss (97% nationally, 82% in the Southeast) decomposed to its factor in minutes. Cutover: Monday data freeze, Tuesday review, reds pre-wired with cause, owner, date. Hypercare: mix-adjusted twins next to every raw output, and the Andon — a 7-day control chart per seller-lane cell, act within 24 hours. **Result.** Primary metric: red-week attribution — a blame negotiation → minutes, by factor math; detection-to-action 30 days → under 24 hours by design; one joint headline, Promise Yield = fast share × accuracy, 41.5% baseline on the docs' sample. The only dollar is modeled: about $580K per caught degradation episode, and I say "modeled" in the same breath.

### COMPRESSED STAR

- **S —** Output metric owned by a committee; 30-day rolling windows smearing step changes into drift; three teams keeping three clocks; a red week discovered live and negotiated.
- **T —** Design the WBR for the pre-purchase delivery experience and lead the program that stood it up — then hand it to owners and leave.
- **A —** One glossary and denominator audit; 7 outputs over 10 input levers with roll-ups and named owners; the factor identity as the decomposition; ±5% WoW red/green with hard reds on floors; Monday freeze → Tuesday review, reds pre-wired; mix-adjusted twins; a cell-grain Andon (7-day EWMA/CUSUM, 2× baseline breach, n ≥ 30, act in 24 h, false-trip bar < 5%).
- **R —** Attribution days → minutes; detection-to-action ~30 days → < 24 h (designed); Promise Yield 43.6% × 95.2% ≈ 41.5% baseline on the docs' sample, four-quarter design target 50%; ≈$0.58M per caught episode (modeled).

### RESULTS LINE

Red-week attribution: a cross-team negotiation → minutes by factor math (proven on the sample-data drill) · detection-to-action ~30 days → under 24 hours by design (the Andon) · 7 outputs / 10 inputs / 1 joint headline / 6 sample weeks (engagement design record) · ≈$0.58M per caught degradation episode (modeled; no annual figure — episode frequency is the client's data and I won't invent it).

## THE MATH — STEP-BY-STEP DERIVATION

### Table 1 — before: the problem in numbers.

| STEP | QUANTITY | FORMULA → SUBSTITUTION | INPUTS (BASIS) | VALUE |
|---|---|---|---|---|
| 1 | The accuracy identity | `O1 ≈ I3 × I4 × Φ(b/σ) — seller ships in window × carrier meets transit × buffer absorbs the residual` | record-from-docs factorization; three factors, three owners | every breach belongs to exactly one factor |
| 2 | Healthy sample week | `O1 = 0.97 × 0.985 × 0.996 = 0.9516 ≈ 95.2%` | I3 = 0.97, I4 = 0.985, Φ = 0.996 (docs' illustrative sample) | 95.2% — matches the sample print |
| 3 | The aggregate trap | `carrier prints 97% nationally while one region runs 82%` | Southeast after a hub consolidation (record-from-docs example) | the average hides the cell |
| 4 | The rolling-window lag | `a step change entering a 30-day window moves the average ≈1/30 per day` | 30-day scorecard mechanics (record-from-docs issue) | weeks before a floor bites — enforcement lag |
| 5 | The gameable pair | `fast share 43.6% and accuracy 95.2% on separate lines, separate owners` | docs' sample; either can be juiced by bleeding the other | no joint headline |
| 6 | Baseline joint number | `Promise Yield = 0.436 × 0.952 = 0.415` | rows 2, 5 (docs' sample) | ≈ 41.5% |

### Table 2 — the fix: step by step, values substituted (the red-week drill).

| STEP | QUANTITY | FORMULA → SUBSTITUTION | INPUTS (BASIS) | VALUE |
|---|---|---|---|---|
| 1 | The episode | `Southeast cell: carrier adherence 0.985 → 0.82 on w = 10% of volume` | 82% is the docs' example; the 10% weight is illustrative | one regional carrier miss |
| 2 | New carrier factor | `I4' = 0.90 × 0.985 + 0.10 × 0.82 = 0.8865 + 0.082 = 0.9685` | volume-weighted blend (illustrative) | I4: 0.985 → 0.9685 (−1.7%) |
| 3 | New accuracy | `O1' = 0.97 × 0.9685 × 0.996 = 0.9357 ≈ 93.6%` | rows 1–2; other factors unchanged | 95.2% → 93.6% — a red week |
| 4 | The attribution read | `WoW deltas: I3 0.0% · I4 −1.7% · Φ 0.0%` | computed per factor (the decomposition) | the red names I4 and the carrier PM — in minutes |
| 5 | Segmentation | `factor red isolated to one region ⇒ C-family cause, not model or seller` | lane/cohort cut of I4 (design) | cause class identified before the meeting |
| 6 | The Andon trigger | `cell 7-day breach rate sustained ≥ 2 × baseline, n ≥ 30 ⇒ act < 24 h` | EWMA/CUSUM parameters (record-from-docs design) | buffer widened or badge pulled on the cell, same day |
| 7 | Auto-restore | `control chart recovers ⇒ badge/buffer restored, no meeting needed` | design | surgical, reversible, seller-fair |

### Table 3 — after: the metric, before → after.

| STEP | METRIC | BEFORE | AFTER | BASIS |
|---|---|---|---|---|
| 1 | Red-week attribution time | days (blame negotiation) | minutes (factor math) | engagement design record, proven on the drill |
| 2 | Detection-to-action, degrading cell | up to ~30 days | < 24 h from sustained signal | designed (Andon) |
| 3 | Output ownership | committee | one named owner of the output block + log | engagement design record |

| STEP | METRIC | BEFORE | AFTER | BASIS |
|---|---|---|---|---|
| 4 | Joint headline | none | Promise Yield, 41.5% sample baseline → 50% 4-qtr target | docs' sample; design target |
| 5 | Mix honesty | raw outputs only | mix-adjusted twin + mix line per output | engagement design record |

**Table 4 — the dollars, every multiplication shown.**

| STEP | QUANTITY | FORMULA → SUBSTITUTION | INPUTS (BASIS) | VALUE |
|---|---|---|---|---|
| 1 | Episode accuracy loss | `95.2% − 93.6% = 1.6 points` | Table 2 rows 2–3 (illustrative episode) | 1.6 pts of excess breaches |
| 2 | Excess breaches per week | `1,000,000 × 1.6% = 16,000` | ~1M orders/wk illustrative book | ≈16,000 breaches/wk |
| 3 | Cost per breach | `$12, taken as stated` | the model's assumption: contact + refund risk + trust decay (record-from-docs) | $12 each |
| 4 | Excess cost per week | `16,000 × $12 = $192,000` | rows 2 × 3 (modeled) | ≈$192K/wk |
| 5 | Detection advantage | `~4 weeks (rolling-window era) − 1 week (Andon era) = 3 weeks` | row 4 of Table 1; Andon design | 3 weeks earlier action |
| 6 | Avoided per episode | `$192,000 × 3 = $576,000` | rows 4 × 5 (modeled) | ≈$0.58M per caught episode |
| 7 | Annual total | `not computed — $576,000 × (episodes/yr) requires a frequency I don't have` | deliberately uninvented | per-episode only, labeled modeled |

## TWO PROBES

> **Your baseline Promise Yield is sample data.**
>
> Correct, and I say so: 43.6% × 95.2% ≈ 41.5% is the engagement docs' own illustrative six-week set, used to prove the mechanics before live lines existed. The 50% target is a target on the real book, and the first live job is re-baselining it.

> **How is this different from the Flexport review (S11)?**
>
> S11 is the production record — a review I ran by hand every Monday for over a year with record numbers (automation 40 → 90%, accuracy 60 → 90%). This engagement is where the review's mathematics — the factor decomposition and the control loop — was carried furthest as a design for a client. I keep the two separate and never borrow a number across them.

**Canon numbers (basis labels):** 7 outputs / 10 input levers / 1 joint headline (design record) · 0.97 × 0.985 × 0.996 ≈ 95.2% (sample-week proof) · 97% national vs 82% Southeast (engagement's worked example) · 30 days → < 24 h (designed) · Promise Yield 41.5% baseline, 50% target (docs' sample) · ≈$0.58M per caught episode (modeled: ≈$192K/wk × 3 weeks). Never a production output claim.

## BACKUP — ONLY IF ASKED S13 — Manhattan — Cyber Monday, 4 AM: allocation across 3 DCs and 100 stores when the scheduler died

**LPs this backup can serve:** Bias for Action (primary) · Customer Obsession (primary backup — the promise ranked first) · Deliver Results

**Basis.** OMS consultant on allocation, promising and the OMS–WMS handshake; engineering fixed the scheduler once I isolated it; the post-mortem mechanisms were mine to drive. $8M / $6M / 99% are record; the margin figures are my model at benchmark rates. Older story — if asked for recent, pair with the UDC robotics throttle (S2): same muscle, 2024–25.

### 60-SECOND PITCH — STOP ON THE PRIMARY METRIC

**Situation.** Manhattan OMS consultant on allocation and promising across 3 DCs and 100 stores; 4 AM Cyber Monday the VP called — backorders piling with inventory present, ~$6M of sales at risk against a 95% fill-rate goal. **Task.** Own the response before the root cause was known. **Action.** Discovery: the contradiction (ATP > 0, backorders rising) was the entry bar to declare an incident. Plan & Design: I ranked by promise-at-risk, not FIFO; node choice = the promise — near 0.85 days vs far 4.60. Build & Test: one bridge, every rebalance timestamped; reason-coded alerts I had already built. Cutover: rebalanced so no DC drowned; status every two hours until 9 PM. Hypercare: root cause — the allocation scheduler died silently at the daylight-saving change; the post-mortem shipped a heartbeat, a DST checklist, a 3σ page rule and a rehearsed rebalance playbook. Lesson: decide reversibly, log it, find the cause later. **Result.** Primary metric: peak fill rate — 95% goal → 99% delivered. ~$8M of peak sales protected against ~$6M at risk (record); my model puts that at ≈$3.2M of gross margin at a 40% benchmark margin; structural stockouts −20% over the engagement.

### COMPRESSED STAR

- **S —** 4 AM Cyber Monday: backorder lines piling with inventory present; promises aging; ~$6M at risk against a 95% fill goal; nobody had written a clearing rule for failure.
- **T —** Own the response before the cause was known, without making it worse.
- **A —** Declared the incident on the contradiction; clearing rule — promise-at-risk, not FIFO — written on the bridge log in the first hour so nobody re-argued it at noon; rebalanced across three DCs; two-hourly status; isolated the DST scheduler death through my own reason-coded alerts; blameless post-mortem → heartbeat, DST checklist, 3σ page rule, rehearsed playbook.
- **R —** Fill 99% vs the 95% goal; ~$8M protected vs ~$6M at risk (record pair); structural stockouts −20% over the engagement; the next peak ran boring.

### RESULTS LINE

Peak fill 99% vs the 95% goal (record) · ~$8M of peak sales protected vs ~$6M at risk (record pair — one-time sales that didn't die, never a saving) · ≈$3.2M gross margin at a 40% benchmark (modeled) · structural stockouts −20% over the engagement (record) · detection moved from a VP phone call to one heartbeat interval.

### THE MATH — STEP-BY-STEP DERIVATION

**Table 1 — before: the problem in numbers.**

| STEP | QUANTITY | FORMULA → SUBSTITUTION | INPUTS (BASIS) | VALUE |
|---|---|---|---|---|
| 1 | 4 AM state | `scheduler dead; backlog = 30,000 orders; ~$6M of sales at risk vs the 95% fill goal` | record risk pair; backlog illustrative | the bridge call |
| 2 | FIFO failure | `FIFO clears by arrival — cheap, distant-promise orders first while at-risk promises age` | record rule problem | value-blind clearing |

| STEP | QUANTITY | FORMULA → SUBSTITUTION | INPUTS (BASIS) | VALUE |
|---|---|---|---|---|
| 3 | Naive node loads | `DC1: 22,000 / 10,000 = 2.2 days; DC3: 2,000 / 6,000 = 0.33 days` | capacities 10K / 8K / 6K orders/day (illustrative) | worst node 2.2 days behind while another idles |
| 4 | No liveness detection | `a dead scheduler was found by a VP at 4 AM, not by a page` | record problem | detection by seniority |

**Table 2 — the fix: step by step, values substituted.**

| STEP | QUANTITY | FORMULA → SUBSTITUTION | INPUTS (BASIS) | VALUE |
|---|---|---|---|---|
| 1 | Promise-at-risk ranking | `risk = value × urgency(promise − now); clear descending` | on the bridge log in hour 1 (record rule) | at-risk promises first, not first-in |
| 2 | Node choice = the promise | `T = handling + transit: near 0.85 d vs far 4.60 d; 4.60 − 0.85 = 3.75 d` | record pair | −3.75 days per re-pointed order |
| 3 | Load levelling | `rebalance 30,000 → 12,500 / 10,000 / 7,500: 12,500/10,000 = 1.25; 10,000/8,000 = 1.25; 7,500/6,000 = 1.25 days` | load = queue / capacity (illustrative) | worst slip 2.2 → 1.25 days |
| 4 | Liveness page (post-mortem) | `page if x < μ − 3σ = 1,200 − 3 × 150 = 1,200 − 450 = 750/hr` | μ = 1,200 allocations/hr, σ = 150 (illustrative); a dead scheduler reads 0/hr | caught in one interval, not by a VP |

**Table 3 — after: the metric, before → after.**

| STEP | METRIC | BEFORE | AFTER | BASIS |
|---|---|---|---|---|
| 1 | Peak fill rate | 95% goal | 99% delivered | record |
| 2 | Peak sales | ~$6M at risk | ~$8M protected | record pair — one-time event |
| 3 | Worst-case promise slip | 2.2 days | 1.25 days | illustrative walk |
| 4 | Detection | a VP at 4 AM | one interval via the 3σ page | post-mortem design |
| 5 | Structural stockouts | — | −20% over the engagement | record |

**Table 4 — the dollars, every multiplication shown.**

| STEP | QUANTITY | FORMULA → SUBSTITUTION | INPUTS (BASIS) | VALUE |
|---|---|---|---|---|
| 1 | Margin protected | `$8,000,000 × 40% = $3,200,000` | benchmark margin 35–45% (modeled) | ≈$3.2M one-time |
| 2 | Orders at risk | `$6,000,000 / $150 = 40,000 orders` | AOV benchmark | 40,000 orders |
| 3 | Cancels / CS avoided | `40,000 × $6.50 = $260,000` | CS contact $5–8 (benchmark; modeled) | ≈$0.26M |
| 4 | Expedite premium avoided | `40,000 × 40% = 16,000 re-pointed; 16,000 × $8 = $128,000` | re-pointed share and premium midpoint of the $5–12 benchmark (both illustrative) | ≈$0.13M |
| 5 | Structural-stockout run-rate | `$400M × 4% = $16M lost sales; × 20% = $3.2M recovered; × 40% = $1,280,000` | illustrative retailer × lost-sales benchmark (2–6%) × −20% record × margin | ≈$1.3M/yr modeled |

## TWO PROBES

### Why act before you knew the cause?

Because waiting had an hourly price — the queue was aging while inventory sat idle — and a reversible, logged intervention did not. I checked with engineering first that manual clears could not corrupt allocation, timestamped every rebalance, and hunted the cause in parallel with alerts I had already built.

### Isn't this a hero story?

The event is; the post-mortem is the point. Carrying the first stretch solo is one of my do-differentlys. What I would present as the result is the mechanism set — heartbeat, DST checklist, 3σ page rule, rehearsed playbook — and that the next peak ran boring.

**Canon numbers (basis labels):** 99% fill vs 95% goal (record) · ~$8M protected vs ~$6M at risk (record) · 3 DCs / 100 stores (record) · near 0.85 d vs far 4.60 d (record) · −20% structural stockouts (record) · ≈$3.2M margin at 40% (modeled). Never "$2MM baseline"; never present $8M as a saving.

## BACKUP — ONLY IF ASKED S9 — Elementum — 1,000 supplier/carrier routes asked, 597 built first: X12 EDI onboarding behind a ≥95% valid-ASN cohort gate

**LPs this backup can serve:** Have Backbone; Disagree & Commit (primary — 597 vs 1,000, two levels up) · Deliver Results · Insist on the Highest Standards (the gate held twice) · Think Big (the staged 1,000-lane vision) · Broad Responsibility (small carriers kept inside the standard by the portal floor)

**Basis.** I owned the scope analysis and the onboarding standard, influencing an executive two levels above my sponsor; engineers built the maps and harness to the standard; Transportation owned the carrier relationships. Cohort sizes and the number of cohorts are not in the record — do not quote them.

### 60-SECOND PITCH — STOP ON THE PRIMARY METRIC

**Situation.** The Global Head of Supply Chain wanted inbound visibility on 1,000 lanes in two months; shipment-data completeness was 60% and high-value lanes were blind. **Task.** I owned the scope analysis and the onboarding standard, influencing an executive two levels above my sponsor. **Action.** Discovery: a spend Pareto — 597 lanes carried 80% of freight spend — pre-validated with Transportation and Procurement before I took it up. Plan & Design: the onboarding standard (X12 850/855/856/940/945/214/997 with an XML/portal floor — "the standard was the data, not the transport") and a cohort gate of ≥95% valid ASN. Build & Test: a mapping-and-test harness and a scorecard per lane. Cutover: 597 live in month one; one cohort failed the gate twice on a timestamp format and I held it both times. Hypercare: completeness scorecards to 95%; ~200 lanes the next wave behind the same gate. Lesson: automating dirty data manufactures confident wrong answers — the gate is the product. **Result.** Primary metric: inbound shipment-data completeness 60% → 95%. Freight spend −10% ≈ $2M in six months (record); 597 lanes = 80% of spend live in month one; adoption 8×.

### COMPRESSED STAR

- **S —** 1,000 global lanes asked in two months — scope, schedule and quality all fixed; completeness 60%; lanes differed by orders of magnitude in spend and carriers differed in EDI capability.
- **T —** Own the scope analysis and the onboarding standard; take the re-scope two levels up with the evidence attached.
- **A —** Spend Pareto (597 = 80% of spend) pre-validated so no lane owner could contradict it in the room; standard + ≥95% valid-ASN gate published to carriers before cohort one; harness so later cohorts were mostly configuration; held a failing cohort twice; ~400 lanes carrying ~20% of spend deferred — priced in spend share, not lane count.
- **R —** Completeness 60 → 95%; 100% visibility on high-value lanes; 597 lanes live month one; freight −10% ≈ $2M in six months (record); method written up as the sourcing SOP.

### RESULTS LINE

Completeness 60% → 95% (record) · 597 of 1,000 lanes = 80% of freight spend live in month one (record) · freight spend −10% ≈ $2M in six months (record — kept as measured, never annualized) · adoption 8× (record) · ≈$0.4M/yr document-handling labor (modeled add-on).

### THE MATH — STEP-BY-STEP DERIVATION

**Table 1 — before: the problem in numbers.**

| STEP | QUANTITY | FORMULA → SUBSTITUTION | INPUTS (BASIS) | VALUE |
|---|---|---|---|---|
| 1 | Completeness | `480 / 800 = 60%` | expected = 200 shipments × 4 milestones = 800; received 480 (illustrative volumes; 60% record) | 60% — 2 of 5 events missing |

| STEP | QUANTITY | FORMULA → SUBSTITUTION | INPUTS (BASIS) | VALUE |
|---|---|---|---|---|
| 2 | What blindness costs | `off-guide spend unseen · premium modes by habit · chase calls per late shipment` | record problem | no instrument, no lever |
| 3 | Manual documents | `597 lanes × ~50 shipments × 3 docs ≈ 90,000 docs/yr keyed by hand` | illustrative volumes | ~90K docs/yr |

**Table 2 — the fix: step by step, values substituted.**

| STEP | QUANTITY | FORMULA → SUBSTITUTION | INPUTS (BASIS) | VALUE |
|---|---|---|---|---|
| 1 | Spend Pareto | `S(k) = cumulative spend share; S(597) = 80%` | lanes sorted by annual freight spend (record) | 597 of 1,000 lanes carry 80% of spend — do them first |
| 2 | Cohort gate | `validASN rate ≥ 95% in test before activation` | published to carriers before cohort one (record) | one cohort failed twice on a timestamp format — held both times |
| 3 | Long-tail floor | `XML / portal when EDI is absent` | “the standard was the data, not the transport” (record) | small carriers kept inside the standard |
| 4 | Completeness after | `760 / 800 = 95%` | illustrative volumes; 95% record | 95%; high-value lanes 100% |

**Table 3 — after: the metric, before → after.**

| STEP | METRIC | BEFORE | AFTER | BASIS |
|---|---|---|---|---|
| 1 | Shipment-data completeness | 60% | 95%; high-value lanes 100% | record |
| 2 | Lanes live month one | 0 | 597 = 80% of spend | record |
| 3 | Freight spend on onboarded lanes | base | −10% ≈ $2M in 6 months | record |
| 4 | Document handling | ~90K docs/yr keyed | X12/XML validated | modeled |

**Table 4 — the dollars, every multiplication shown.**

| STEP | QUANTITY | FORMULA → SUBSTITUTION | INPUTS (BASIS) | VALUE |
|---|---|---|---|---|
| 1 | Implied addressable base | `$2,000,000 / 10% = $20,000,000/yr` | derived from the record pair, stated as such | ≈$20M/yr |
| 2 | The record result | `$20M × 10% = $2M in 6 months` | record | ≈$2M |
| 3 | Slice — routing-guide compliance | `~$0.8M of the $2M` | off-guide spend caught by 856/214 (modeled split) | ~$0.8M |
| 4 | Slice — mode discipline | `~$0.6M of the $2M` | premium modes → ocean/ground when ETA is trusted (modeled split) | ~$0.6M |
| 5 | Slice — expedites + detention | `~$0.6M of the $2M` | events replace chase calls (modeled split) | ~$0.6M |
| 6 | Split check | `0.8 + 0.6 + 0.6 = $2.0M` | the slices decompose the record — they never add to it | = the record $2M |
| 7 | Document labor | `90,000 × 6 min = 540,000 min; / 60 = 9,000 h; × $45 = $405,000` | illustrative volumes (modeled add-on) | ≈$0.4M/yr |

## TWO PROBES

> **You defied a Global Head of Supply Chain's number. How?**
>
> With the spend curve and the lane owners' validation attached — 597 lanes carried 80% of the spend, so the phased plan was argued in spend share, not lane count. He accepted the phasing off the curve; I then committed to the remainder behind the same gate and delivered it.

> **Is the $2M annualized?**
>
> No — it is the record as measured: minus ten percent on the onboarded lanes' spend in six months. My three-way split (routing-guide compliance ~$0.8M, mode discipline ~$0.6M, expedites and detention ~$0.6M) decomposes the record; it never adds to it.

**Canon numbers (basis labels):** 60% → 95% completeness (record) · 597 / 1,000 lanes = 80% of spend (record) · −10% ≈ $2M in 6 months (record) · ≥95% valid-ASN gate, held twice (record) · 8× adoption (record) · ≈$0.4M/yr labor (modeled). Never: cohort counts or sizes; never double the $2M into a run-rate.

**BACKUP — ONLY IF ASKED** **Doc bench — Flexport — the RAPID Crux customer transition: 79 customers, $185,892 ARR, 294,581 containers — transition where it fits, cancel honestly where it doesn't (Earn Trust / Customer Obsession)**

**LPs this backup can serve:** Earn Trust (primary — honest cancels) · Customer Obsession (per-customer calls for the largest accounts) · Have Backbone, Disagree & Commit (internal pushback priced, not overridden) · Deliver Results (every customer decided by end 2022)

**Basis.** Doc-derived micro-story from the program record (RAPID — the Crux customer transition, 2022) — every figure is RECORD-FROM-DOCS. Crux was the acquired visibility product; its customers moved to Shipment Visibility (Tripathy's S11 program family) — if the room connects them, keep her packet's canon numbers identical. Program-lead voice: I owned the transition design and the decision rules; account owners ran the per-customer calls. Not in the master packet's S-numbering; supplements canon, never overrides it.

### 60-SECOND PITCH — STOP ON THE PRIMARY METRIC

**Situation.** The acquired visibility product, Crux, still carried 79 customers — $185,892 ARR and 294,581 containers a year — and the decision was taken to move them onto Shipment Visibility by end 2022. The catch: SV did not fit every segment; some customers were better served by an honest cancel than a force-fit. **Task.** Decide every customer — transition or cancel — without burning the revenue that fit or the trust of those that didn't. **Action.** Segment-level decision rules instead of 79 one-offs: truck brokers to a white-label partner product or cancel; forwarders, customs brokers and 3PLs to the SV API or cancel; the two competitor SCM-SaaS accounts canceled on published dates (August 1 and October 24, 2022). The two largest accounts by volume got per-customer calls — one moving a national home-improvement retailer's freight, the other a 600-truck carrier relationship. Internal pushback — migration keep-the-lights-on cost, feature parity, the timing of consignee acquisition — was answered with priced options in the decision document, not overridden by title. **Result.** Every one of the 79 customers had a decided, dated path — transition where SV fit, cancel honestly where it did not — with the segment table and the decision rules as the record. RECORD-FROM-DOCS

### COMPRESSED STAR

- **S —** 79 acquired-product customers ($185,892 ARR; 294,581 containers) across eight segments, on a product being retired by end 2022; the replacement did not fit every segment.
- **T —** Own the transition design: decision rules per segment, dates per account, and the handling of internal dissent.
- **A —** A segment table with a transition-or-cancel rule per segment; a white-label partner product for truck brokers; the SV API for forwarders/brokers/3PLs; competitor SaaS accounts canceled on published dates; per-customer calls for the two largest-volume accounts; pushback on KTLO cost, feature parity and consignee timing priced into the decision document.
- **R —** 79 of 79 customers decided and dated (record-from-docs). The record carries no preserved-ARR figure, and none is claimed.

### RESULTS LINE

79 customers · $185,892 ARR · 294,581 containers (record-from-docs) · eight segments with a rule each (record-from-docs) · two competitor SaaS accounts canceled on published dates (record-from-docs) · two largest accounts handled per-customer (record-from-docs) · no preserved-ARR or churn dollar in the record — never invent one.

### THE MATH — STEP-BY-STEP DERIVATION

**Table 1 — before: the problem in numbers.**

| STEP | QUANTITY | FORMULA → SUBSTITUTION | INPUTS (BASIS) | VALUE |
|---|---|---|---|---|
| 1 | Customers on the retiring product | `79 across the printed segment table` | RAPID doc, customer-by-type table (record-from-docs) | 79 accounts to decide |
| 2 | Revenue carried | `$15,491 MRR × 12 = $185,892 ARR` | printed totals (record-from-docs) | $185,892 ARR |
| 3 | Volume carried | `294,581 containers tracked (2021)` | printed total (record-from-docs) | 294,581 containers/yr |
| 4 | The fit problem | `SV fits shippers and consignees; carriers, brokers, forwarders, 3PLs and competitor SaaS are not consignees` | RAPID opportunity section (record-from-docs) | one product cannot absorb all 79 |

**Table 2 — the fix: step by step, values substituted.**

| STEP | QUANTITY | FORMULA → SUBSTITUTION | INPUTS (BASIS) | VALUE |
|---|---|---|---|---|
| 1 | Segment count check | `2 + 14 + 39 + 9 + 3 + 5 + 1 + 4 + 2 = 79` | GKA · Enterprise · SMB · Motor Carrier · Truck Broker · Freight Forwarder · Customs Broker · 3PL · SCM SaaS rows (record-from-docs) | rows reconcile to the printed total |
| 2 | Container check | `17,061 + 17,983 + 4,153 + 12,359 + 221,044 + 11,871 + 432 + 4,218 + 5,460 = 294,581` | segment rows (record-from-docs) | reconciles exactly |
| 3 | Concentration | `221,044 ÷ 294,581 = 0.7504 ≈ 75% of volume on 3 truck-broker accounts` | segment rows (record-from-docs) | the decision's center of gravity |
| 4 | Rule — green segments | `2 + 14 + 39 = 55 customers → transition to Shipment Visibility` | RAPID opportunity section (record-from-docs) | 55 transitioned |
| 5 | Rule — carriers & brokers | `9 + 3 = 12 → white-label through Transmission, else cancel at wind-down` | RAPID decision section (record-from-docs) | 12 decided |
| 6 | Rule — forwarders, customs brokers, 3PLs | `5 + 1 + 4 = 10 → SV API, else cancel at wind-down` | RAPID decision section (record-from-docs) | 10 decided |
| 7 | Rule — competitor SaaS | `2 → cancel: Shippabo 2022/08/01; Shipwell 2022/10/24` | RAPID appendix (record-from-docs) | 2 canceled on published dates |
| 8 | Closure check | `55 + 12 + 10 + 2 = 79` | rows 4–7 (record-from-docs) | every customer under a rule — eight segments carry a designed forward rule; the ninth printed row (SCM SaaS) was immediate cancel, already executed on the dates in row 7 |
| 9 | Two largest accounts | `per-customer handling, not a segment rule — AV Logistics and ITS Logistics, the volume leaders` | RAPID sections 3 and 7 (record-from-docs) | Transmission partnership modeled on ITS's existing usage |

**Table 3 — after: the metric, before → after.**

| STEP | METRIC | BEFORE | AFTER | BASIS |
|---|---|---|---|---|
| 1 | Customers decided | 0 of 79 | 79 of 79, each with a dated path | record-from-docs |
| 2 | Paths | one undifferentiated book | 55 → SV · 12 → Transmission-or-cancel · 10 → SV-API-or-cancel · 2 → canceled | record-from-docs |
| 3 | Largest accounts | segment averages | per-customer calls for the two biggest-volume accounts | record-from-docs |
| 4 | Internal dissent | — | KTLO cost, feature parity and consignee-timing objections priced inside the decision document — answered, not overridden | record-from-docs |

**Table 4 — the dollars, every multiplication shown.**

| STEP | QUANTITY | FORMULA → SUBSTITUTION | INPUTS (BASIS) | VALUE |
|---|---|---|---|---|
| 1 | ARR identity check | `$15,491 × 12 = $185,892` | printed MRR and ARR totals (record-from-docs) | totals internally consistent |
| 2 | ARR on the committed SV path | `$16,500 + $22,560 + $30,000 = $69,060` | GKA + Enterprise + SMB rows (record-from-docs) | $69,060 ARR transitioning to SV |
| 3 | ARR under transition-or-cancel decision | `$17,760 + $139,512 + $14,820 + $600 + $2,400 + $10,800 = $185,892` | the six non-green rows (record-from-docs) | equals the printed ARR total exactly |
| 4 | Concentration of the decision | `$139,512 ÷ $185,892 = 0.7505 ≈ 75% of decided ARR in 3 truck-broker accounts` | segment rows (record-from-docs) | why the two largest got per-customer calls |
| 5 | Average account | `$185,892 ÷ 79 = $2,353 ARR; 294,581 ÷ 79 = 3,729 containers` | printed totals (record-from-docs) | the typical account was small — rules, not 79 one-offs |
| 6 | Preserved-ARR claim | `no preserved-ARR or churn dollar in the record — not claimed` | record-from-docs | never invent one |

## TWO PROBES

### How do you justify canceling a paying customer?

By pricing the alternative honestly: force-fitting a segment the product doesn't serve converts this year's revenue into next year's churn plus support cost, and it burns the reference. A clean cancel with notice and a path — a white-label partner product, or the API if they fit it — kept the relationship honest; two of the 79 were competitor SaaS accounts where canceling was the only truthful option.

### How did you handle the internal pushback?

As inputs to price, not objections to defeat: migration KTLO cost, feature parity and consignee-acquisition timing each got an option with a cost in the decision document, and the decider chose with the prices on the table. Pushback that is priced becomes design input; pushback that is overridden becomes a later surprise.

**Canon numbers (basis labels):** 79 customers · $185,892 ARR · 294,581 containers (record-from-docs) · segment table incl. Enterprise 14 · SMB 39 · Motor Carrier 9 (record-from-docs) · two SaaS cancels dated Aug 1 / Oct 24, 2022 (record-from-docs) · transition deadline end 2022 (record-from-docs). Never: a saved-ARR percentage; never merge with S11's 7 → 68 customer canon.

## BACKUP — ONLY IF ASKED Doc bench — Flexport FFA — the Asia latency investigation: a 21-second P95 traced to routing, decomposed to the layer, fixed in sequence (Dive Deep)

**LPs this backup can serve:** Dive Deep (primary — decomposition to the layer) · Deliver Results · Ownership (the standing rule imported into later launches) · Insist on the Highest Standards (a smoke-test protocol per change)

**Basis.** Doc-derived micro-story from the program record (the FFA Asia latency investigation, post-MVP 2022) — every figure is RECORD-FROM-DOCS: the investigation document, written when the FF App MVP had launched to 11 partners in 10 countries and 48 of 53 import partners (90.5%) had migrated. Program-lead voice: I owned the investigation, the decomposition and the fix sequence; infrastructure teams executed the WAF and CDN changes. Not in the master packet's S-numbering; supplements canon, never overrides it. The same bench appears in Hoerger's packet — facts identical.

### 60-SECOND PITCH — STOP ON THE PRIMARY METRIC

**Situation.** Weeks after the FF App MVP launched to 11 partners in 10 countries — with 48 of 53 import partners already migrated — the monthly business review showed Asia page loads at a P95 of 21.19 seconds against 3.3 in the US. **Task.** Find the layer, not the anecdote, and fix it without a re-architecture in the middle of a live partner-facing launch. **Action.** I decomposed the problem before prescribing: static assets ran ~5s P95 in China against ~500ms in the US, and the app's domain had no geo-based DNS or reverse proxy — Asia traffic was riding the public internet across the Pacific; a double CDN sat on the static path. The fix was a sequence, not a project: enable the WAF, remove the double CDN (direct to S3), then a regional CDN proposal with a smoke-test protocol per change, then China tuning, then Asia tuning — each with a date and an owner. **Result.** Root cause isolated to routing — not the app, not the partners; two fixes shipped inside the investigation window and the regional plan scheduled with dates; and a standing rule imported into every later launch: the pilot's geography is a performance requirement, not a rollout detail.

RECORD-FROM-DOCS

### COMPRESSED STAR

- **S —** Post-MVP, the MBR surfaced Asia P95 page load at 21.19s vs 3.3s in the US (17-day measurement window); 48 of 53 import partners (90.5%) already migrated — the app had to hold where the partners were.
- **T —** Isolate the layer causing Asia latency and sequence the fix set without destabilizing a live partner-facing launch.
- **A —** Split static from dynamic: static JS ~5s P95 in China vs ~500ms US; found no geo-based DNS / reverse proxy on the domain (public-internet routing) and a double CDN on static assets; sequenced WAF enablement → double-CDN removal (direct to S3) → regional CDN proposal → China tuning → Asia tuning, each gated by a smoke-test protocol with a named date and owner.
- **R —** Root cause isolated to routing; WAF and double-CDN fixes shipped inside the window; CDN / China / Asia work scheduled with owners; the investigation became the template for launch-geography checks.

### RESULTS LINE

P95 page load 21.19s Asia vs 3.3s US (record-from-docs) · static ~5s China vs ~500ms US (record-from-docs) · 48 of 53 partners migrated = 90.5% (record-from-docs; canon's "48 of 53 within a month" — consistent) · two fixes shipped, three sequenced with dates (record-from-docs) · no post-fix latency figure exists in the record — never invent one.

### THE MATH — STEP-BY-STEP DERIVATION

**Table 1 — before: the problem in numbers.**

| STEP | QUANTITY | FORMULA → SUBSTITUTION | INPUTS (BASIS) | VALUE |
|---|---|---|---|---|
| 1 | Page-load P95 by region | `Asia = 21.19s; US = 3.3s; Others = 4.45s` | Datadog @view.loading_time, 04/24–05/10 window (record-from-docs) | Asia is the outlier |
| 2 | The gap factor | `21.19 ÷ 3.3 = 6.42` | row 1 (record-from-docs) | 6.42× slower than the US |
| 3 | Traffic behind the number | `6.31 ÷ 2.19 = 2.88` | Asia 6.31k vs US 2.19k page views, same window (record-from-docs) | ≈2.9× the US traffic riding the worst latency |
| 4 | Static-asset leg | `JS P95 = 5s China vs 500ms US; 5 ÷ 0.5 = 10` | static.flexport.com CDN behavior (record-from-docs) | 10× gap on the static leg |
| 5 | Dynamic leg | `no geo-DNS / reverse proxy for trade.flexport.com → Asia GraphQL hairpins over the public internet` | infra review (record-from-docs) | routing penalty on every data call |
| 6 | Known defect on the static path | `double CDN in China — Cloudflare stacked in front of Tencent` | infra review (record-from-docs) | one avoidable hop |

**Table 2 — the fix: step by step, values substituted.**

| STEP | QUANTITY | FORMULA → SUBSTITUTION | INPUTS (BASIS) | VALUE |
|---|---|---|---|---|
| 1 | Decompose before touching | `page load = static resources + GraphQL requests + JS rendering` | the metric's own definition (record-from-docs) | three legs, measured and owned separately |
| 2 | Step 1 — security gate first | `enable WAF for trade.flexport.com: China, then global` | incident-207 follow-up; done 2022/06/30 (record-from-docs) | shipped — fix 1 of 2 |
| 3 | Step 2 — remove the double CDN | `Tencent CDN → direct to the S3 bucket; Cloudflare hop removed for China` | done 2022/06/24 (record-from-docs) | shipped — fix 2 of 2 |
| 4 | Step 3 — static proposal, CN/Asia | `alternatives written, reviewed, signed off` | ETA 2022/07/15 (record-from-docs) | sequenced with a date |
| 5 | Step 4 — China tuning | `e.g. replicate static assets in-region` | ETA 2022/07/30; depends on step 4 (record-from-docs) | sequenced with a date |
| 6 | Step 5 — Asia tuning | `the same playbook beyond China` | ETA 2022/08/15 (record-from-docs) | sequenced with a date |
| 7 | Held in reserve | `global CDN for trade.flexport.com — built only if the metrics still demand it` | 2022 Q3 or later, metrics-gated (record-from-docs) | no speculative spend |
| 8 | Verification per change | `2 user types × 2 instances = 4 smoke passes before + 4 after = 8` | login → workview → logout → re-login; internal SSO + partner users; US + China hosts (record-from-docs; the multiplication counts the checklist) | 8 workflow passes around every change |

**Table 3 — after: the metric, before → after.**

| STEP | METRIC | BEFORE | AFTER | BASIS |
|---|---|---|---|---|
| 1 | Root cause | one number — "Asia is slow" | static leg (10×) and dynamic routing isolated, each with an owner and a sequence | record-from-docs |
| 2 | Fixes | none | 2 shipped (WAF; double-CDN removal); 3 sequenced with dates (07/15, 07/30, 08/15) | record-from-docs |

| STEP | METRIC | BEFORE | AFTER | BASIS |
|---|---|---|---|---|
| 3 | Post-fix P95 | 21.19s | no post-fix figure exists in the record — never claimed | record-from-docs |
| 4 | Spend discipline | — | global CDN gated on metrics, not built on speculation | record-from-docs |

**Table 4 — the dollars, every multiplication shown.**

| STEP | QUANTITY | FORMULA → SUBSTITUTION | INPUTS (BASIS) | VALUE |
|---|---|---|---|---|
| 1 | Partner base exposed | `48 ÷ 53 = 0.9057 ≈ 90.5% of import partners already migrated when the gap surfaced` | migration tracker (record-from-docs) | 90.5% of partners on the affected app |
| 2 | Excess wait served per window | `gap vs US = 21.19 − 3.3 = 17.89s; 6,310 × 17.89 = 112,886 partner-seconds; ÷ 3,600 ≈ 31.4 hours` | 6.31k Asia page views in the 17-day window (record-from-docs); gap arithmetic illustrative | ≈31.4 partner-hours of excess wait every 17 days |
| 3 | Dollar value of the wait | `no dollar in the record — not claimed` | record-from-docs | not claimed |
| 4 | Post-fix saving | `no post-fix latency figure exists → no savings chain exists` | record-from-docs | never invent one |

## TWO PROBES

### How did you know it was routing and not the application?

By decomposition before remediation: the static/dynamic split put ~5s of the China time on assets the app server never touched, and the same assets served in ~500ms from the US — the code was constant, the path was not. The domain's missing geo-DNS / reverse proxy and the double CDN explained both halves of the gap without a single application change.

### Why not just stand up a China region?

A region is a re-architecture with a compliance tail; the evidence said the bottleneck was routing and CDN topology, which fixes in weeks. The sequenced plan — WAF, double-CDN removal, regional CDN, tuning — bought most of the win at a fraction of the risk, and the smoke-test protocol meant each step was reversible.

**Canon numbers (basis labels):** 21.19s vs 3.3s P95 (record-from-docs) · ~5s vs ~500ms static (record-from-docs) · 48/53 = 90.5% migrated (record-from-docs) · 11 partners / 10 countries MVP (record — matches S12 canon) · fix sequence WAF → double-CDN → regional CDN → China → Asia with smoke tests (record-from-docs). Never: a post-fix latency number (none in the record); never present scheduled items as delivered.

INTERVIEW-PREP BOOKLET · NACF INBOUND FLUENCY · BUSINESS-FIRST, SYSTEMS AS ENABLERS

# NACF Inbound — how capacity and inventory are managed

Amazon — Senior Supply Chain Manager, NACF Supply Chain (L6) · interview loop October 29–30, 2026

## The panel this booklet serves

1 · Snigdha Tripathy — Sr Manager, Supply Chain — NACF · **HIRING MANAGER** · ex-EU IXD / placement leader ($80MM savings 2021; built the IXD team ground-up, +$50MM 2020; 50+ FC launches Amazon.in; Kaizen/Lean trainer; 11+ yrs Amazon)

2 · Ryan Hoerger — Manager, Placement / Transfer Analytics · ex-Sr BIE · Chicago Booth MBA, Siebel Scholar · Nike analytics · statistical rigor

3 · Rocky Goforth — Sr Manager, Infrastructure

4 · Zhaowen Lan — Sr Manager, Supply Chain — peer

5 · Venkata Josyula — Principal PM, FBA · bar raiser

> **Strategy in one line.** Sound like someone who has run inbound — not someone who memorized a glossary. Every mechanism below is spoken as a business process with a constraint, a metric and a lever; every number is worked end-to-end so the arithmetic can be done out loud in the room. Inbound in one breath: *the right inventory, reserved into the right node, received and stowed inside capacity, made available where the demand is — with the plans traveling ahead of the freight and the variance closing the loop.*

**Honesty labels, on every mechanism:** **[well-known public Amazon practice]** — published, talkable-in-public Amazon mechanisms (AWD, IXD, appointment scheduling, random stow, IPI components, WBR) · **[industry standard]** — textbook supply chain (UPH labor planning, z·σ·√LT safety stock, S&OP, queueing approximations) · **[illustrative]** — numbers invented to make the math legible. No confidential Amazon internals are asserted anywhere; no number is claimed as Amazon's own.

**How to use.** §1–§4 are the fluency: read the diagrams until you can redraw them, then recompute every green math block until the substitutions feel like your own. §5 maps your master-packet stories onto inbound concepts. §6 is per-interviewer: ten questions each of the five will likely ask, with a one-to-two sentence answer pointer, then three questions to ask them. §7 is the math appendix — every formula symbolic → substituted → computed, for the night before. §8 is one week at one FC — the same 12,000 units walked from the forecast through the close, for the whiteboard.

Canon: amazon/amazon-l6-scm-master-packet.html (stories S2, S6, S8, S9, S11, S12, S13, S18, S19 with their basis labels) · panel packets 1–5 (per-interviewer lenses). Sibling booklet file; the five panel packets are untouched by this document.

## The 60-second answer — "what is NACF inbound?"

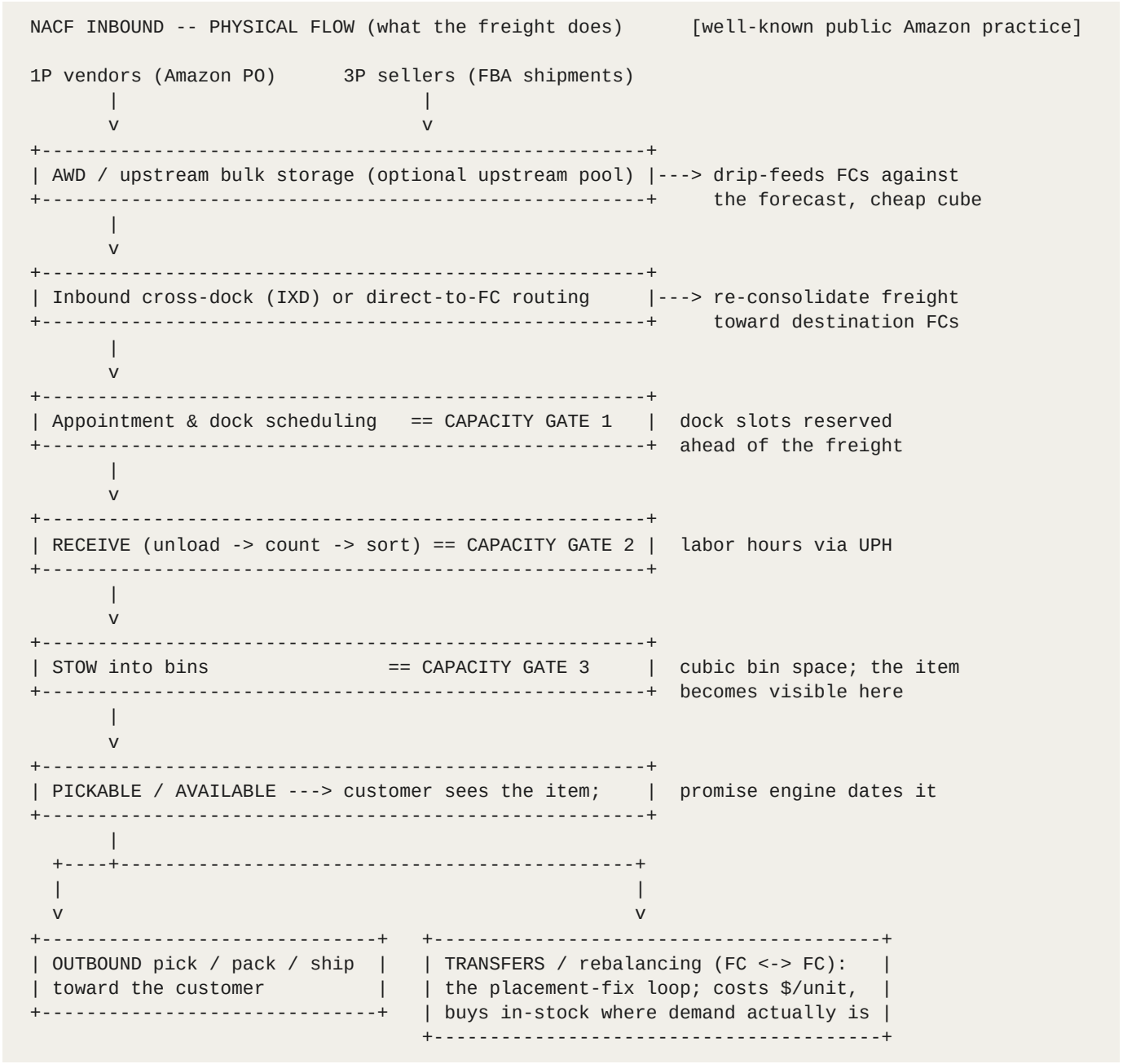
> **SAY THIS VERSION OUT LOUD, THEN STOP**
>
> "Inbound is everything from the vendor or seller's dock to the moment an item is pickable in front of a customer promise — and it is managed as two coupled problems. **Capacity**: three physical currencies — dock-appointment slots, labor hours, and cubic storage — planned from the demand forecast through a receive plan and a UPH-based labor plan, with peak buffers sized off the forecast distribution, not the average. **Inventory**: where the units land and how long they last — placement by distance-to-customer against cost, weeks of cover by SKU, safety stock positioned where the variability actually is, and a transfer loop that rebalances what placement got wrong. The information flow travels ahead of the freight: forecast → inbound plan → placement decision → capacity reservation, and the actuals come back through the WBR as variance. My job in a seat like this is to keep that loop honest — measured lead times, not brochure ones; buffers released on evidence; capacity added against the constraint that actually binds."

Everything after this page unpacks that paragraph. The discipline that makes it credible: every mechanism carries its label, and every quantity in the worked examples can be recomputed from the row above it — do the arithmetic out loud in the interview and the fluency reads as experience.

> **The one mental model to keep on top.** Inbound is a *reservation system* feeding a *positioning system*: you reserve capacity (slots, hours, cube) against a plan, and you position inventory (placement, cover, buffers) against demand. Most inbound failures are one system lying to the other — freight arrives that capacity never reserved, or inventory sits where demand isn't. The metrics in §2–§3 exist to catch exactly those two lies early.

## 1 · The inbound machine, end-to-end

The physical flow first — what the freight does. Each node is labeled with the constraint that binds it; the three capacity gates are the whole of §2. **[well-known public Amazon practice]** for the node names (AWD, IXD, appointment scheduling, random-stow-style bin visibility); the diagram itself is **[illustrative]**.




| NODE | WHAT IT IS (BUSINESS LANGUAGE) | WHAT CONSTRAINS IT | THE METRIC THAT WATCHES IT | LABEL |
|---|---|---|---|---|
| **Vendor / seller ship** | 1P purchase orders and 3P FBA shipments; the plan's first promise is the ship date and the ASN (advance shipment notice). | Vendor lead time and OTIF (on-time-in-full); ASN data quality. | OTIF %, fill rate, ASN accuracy. | **[industry standard]** |

| NODE | WHAT IT IS (BUSINESS LANGUAGE) | WHAT CONSTRAINS IT | THE METRIC THAT WATCHES IT | LABEL |
|---|---|---|---|---|
| **AWD / upstream** | Low-cost bulk storage upstream of the FC network; holds inventory cheap and replenishes the network as the forecast firms. | Upstream cube; replenishment lead time into FCs. | Upstream weeks of cover; replenishment dwell. | **[well-known public Amazon practice]** |
| **IXD / direct FC** | Inbound cross-dock sorts and re-consolidates inbound freight toward destination FCs; some freight books direct. | Cross-dock throughput; linehaul schedule. | IXD dwell time; trailer utilization. | **[well-known public Amazon practice]** |
| **Appointment / dock** | Carriers book delivery appointments; the dock door is the first scarce, schedulable resource in the building. | Dock doors × receiving hours × unload rate. | Dock utilization %, trailer dwell, missed-appointment count. | **[well-known public Amazon practice]** |
| **Receive** | Unload, count, inspect, sort to stowable state; where physical freight becomes system inventory. | Labor hours (units per hour, UPH) and conveyor/sort rates. | Receive UPH, backlog in trailers/cartons, receive accuracy. | **[industry standard]** |
| **Stow** | Items into bins; the item becomes findable and sellable — availability starts here, not at the dock. | Cubic bin space and stow labor. | Stow UPH, bin utilization %, time-to-available. | **[well-known public Amazon practice]** |
| **Pickable / available** | Inventory the promise engine can sell against; the boundary between inbound and outbound. | Nothing inbound — this is the finish line. | In-stock rate; time from dock to available. | **[industry standard]** |
| **Transfers / rebalancing** | FC-to-FC moves that fix placement error after the fact; the network's insurance policy, priced per unit. | Transfer cost vs stockout cost (the breakeven in §3). | Transfer units/wk, transfer cost/unit, avoided stockouts. | **[well-known public Amazon practice]** |

## The information flow — plans travel ahead of the freight

Every physical move above is preceded by an information move; when the two drift apart, the WBR is where it gets caught. **[industry standard]** structure; Amazon vocabulary (WBR, promise engine) **[well-known public Amazon practice]**.

```
INFORMATION FLOW -- plans travel ahead of the freight     [industry standard]

demand forecast (SKU x region x week)
       |
       v
inbound plan: what to buy / book, and when it must land  <---> vendor & seller
       |                                                       lead times, OTIF
       v
placement decision: WHICH node receives it --
  distance-to-customer vs cost vs node capacity      <--- the hiring manager's
       |                                                  home turf (EU IXD placement)
       v
capacity reservation: dock slots + labor plan (UPH -> hours) + cube
       |
       v
receive schedule -> stow plan -> availability signal
       |
       v
promise engine / search in-stock ---> the customer sees a delivery date
       |
       v
actuals vs plan --> variance --> next WBR input (the loop that keeps it honest)
```

> **How to sound like you've run it.** Name the seam, not the node. "The dock" is trivia; "the seam where the inbound plan meets the appointment calendar — and what happens when a vendor ships three days early into a full dock" is experience. Each worked example in §2–§3 is one of those seams, with the arithmetic attached.

## 2 · Capacity management — the three currencies and the planning loop

Inbound capacity is three currencies that convert into each other only through the plan. **[industry standard]** framework; all numbers **[illustrative]**.

| CAPACITY TYPE | UNIT OF ACCOUNT | WHAT DRIVES CONSUMPTION | HOW YOU PLAN IT | WHEN IT BINDS |
|---|---|---|---|---|
| **Dock / appointment slots** | Trailer-unloads per day = doors × hours × unload rate | Truckload arrivals; vendor/carrier appointment discipline; trailer mix (floor-loaded vs palletized). | Receive plan → trailers/day → doors at an unload rate, capped at ~85% utilization (worked in CAP-A, CAP-D). | First, at peak — it is the least elastic currency (you cannot stack trailers at a door). |
| **Labor hours** | Hours = units ÷ UPH, grossed up for indirect work and learning curves | Units received/stowed; associate proficiency; process (conveyable vs hand-stack). | UPH standards → direct hours → paid hours → headcount by shift (CAP-B). | Second — elastic but slow: hiring and the learning curve lag the plan by weeks. |
| **Cubic storage** | Cubic feet = on-hand units × cube per unit | Inventory target (days of cover), unit cube, bin geometry / honeycomb loss. | Inventory plan → cube needed vs effective bin capacity (CAP-C). | Last, then suddenly — it looks fine at 49% and is a wall at 90%; the least flexible currency mid-peak. |

### The planning loop — forecast to floor

```
CAPACITY PLANNING LOOP                                          [industry standard]

demand forecast (units/day, with a distribution -- p50 ... p90)
       |
       v
receive plan: units/day -> cartons -> pallets -> trailers/day (CAP-A)
       |
       +---------------> dock reservation: trailers/hour vs doors at ~85% ceiling
       |
       v
labor plan: units / UPH = direct hours -> +indirect % -> /shift-hours = heads (CAP-B)
       |
       v
cube check: on-hand units x cube/unit vs effective bin capacity (CAP-C)
       |
       v
peak buffer: fixed capacity to ~p50-p70, flex (temp labor, OT, appointment
smoothing) for the tail (CAP-E) -- then WBR reads actuals vs all of it
```

### The metrics that watch capacity

| METRIC | DEFINITION / FORMULA | RED FLAG | LABEL |
|---|---|---|---|
| **Dock utilization %** | `utilization = scheduled unloads ÷ (doors × hours × unload rate)` | Sustained > ~85% — dwell goes nonlinear (CAP-D). | **[industry standard]** |
| **Dwell time** | `dwell = trailer arrival → door departure (hours)` | Rising dwell with flat volume = variability, not volume, is the problem. | **[industry standard]** |
| **Receive backlog** | `backlog_days = unreceived units ÷ receive capacity (units/day)` | > 0.5 day carried into the next shift; it compounds. | **[industry standard]** |

| METRIC | DEFINITION / FORMULA | RED FLAG | LABEL |
|---|---|---|---|
| **Forecast variance** | `variance = (actual − plan) ÷ plan, by week and by node` | Same-sign variance three weeks running = a bias, not noise; fix the plan, not the floor. | **[industry standard]** |

### CAP-A · REQUIRED RECEIVE CAPACITY — UNITS/DAY → TRAILERS → DOCK DOORS

One illustrative FC receiving 120,000 units/day. Every row recomputes from the rows above; say the chain out loud and you sound like the floor.

| STEP | QUANTITY | FORMULA → SUBSTITUTION | INPUTS (BASIS) | VALUE |
|---|---|---|---|---|
| 1 | Inbound volume | `V = 120,000 units/day` | illustrative receive plan | 120,000 units/day |
| 2 | Cartons per day | `120,000 ÷ 6 = 20,000` | 6 units/carton (illustrative) | 20,000 cartons/day |
| 3 | Pallets per day | `20,000 ÷ 40 = 500` | 40 cartons/pallet (illustrative) | 500 pallets/day |
| 4 | Truckloads per day | `500 ÷ 26 = 19.2` | 26 pallets per 53-ft trailer [industry standard] | 19.2 trailers/day |
| 5 | Trailer arrival rate | `19.2 ÷ 10 = 1.92` | 10 receiving hours/day (illustrative) | 1.92 trailers/hour |
| 6 | Raw doors required | `1.92 ÷ 0.5 = 3.84` | unload rate 0.5 trailer/door-hour = 2 h/trailer (illustrative) | 3.84 doors |
| 7 | Doors at the 85% ceiling | `3.84 ÷ 0.85 = 4.5 → round up` | utilization ceiling ~85% [industry standard] | **5 dock doors planned** |

**The sentence to say:** "Twelve trailers an hour isn't the plan — the plan is five doors, because at 85% utilization the doors stop being fungible and dwell starts compounding."

### CAP-B · LABOR HOURS FROM UPH — WITH THE LEARNING CURVE PRICED IN

| STEP | QUANTITY | FORMULA → SUBSTITUTION | INPUTS (BASIS) | VALUE |
|---|---|---|---|---|
| 1 | Direct receive hours | `120,000 ÷ 300 = 400` | mature-associate 300 UPH (illustrative) | 400 direct hours/day |
| 2 | Paid hours (indirect gross-up) | `400 × 1.15 = 460` | 15% indirect allowance (illustrative) | 460 paid hours/day |
| 3 | Headcount | `460 ÷ 10 = 46` | 10-hour shifts (illustrative) | 46 associates/day |
| 4 | Learning curve, week 1 | `300 × 0.60 = 180 UPH` | new associate at 60% of standard (illustrative) | 180 UPH effective |
| 5 | Blended UPH at 30% new | `0.7×300 + 0.3×180 = 210 + 54 = 264` | Q4 ramp: 30% of staff in week 1–2 (illustrative) | 264 UPH blended |
| 6 | Direct hours at blended UPH | `120,000 ÷ 264 = 454.5; 454.5 − 400 = 54.5; 54.5 ÷ 400 = 13.6%` | same volume as step 1 | 454.5 h = **+54.5 h (+13.6%)** |

**The sentence to say:** "The hiring plan isn't the headcount math — it's the learning-curve math. Thirty percent new hires isn't a 30% problem, it's a 13.6% tax on every received unit for two weeks, and the peak plan has to pay it before the first trailer arrives."

## CAP-C · CUBIC CAPACITY — UNITS → CUBE → % OF THE BUILDING

| STEP | QUANTITY | FORMULA → SUBSTITUTION | INPUTS (BASIS) | VALUE |
|---|---|---|---|---|
| 1 | On-hand target | `I = 2,000,000 units` | illustrative network-position target for this FC | 2.0M units |
| 2 | Cube required | `2,000,000 × 0.25 = 500,000` | 0.25 ft³/unit average (illustrative) | 500,000 ft³ |
| 3 | Nominal utilization | `500,000 ÷ 1,200,000 = 41.7%` | 1.2M ft³ usable bin capacity (illustrative) | 41.7% of nominal |
| 4 | Effective capacity (honeycomb) | `1,200,000 × 0.85 = 1,020,000` | 15% partial-bin loss [industry standard] | 1,020,000 ft³ effective |
| 5 | True utilization | `500,000 ÷ 1,020,000 = 49.0%` | against effective, not nominal | **49.0% — plan against this** |
| 6 | Days of cover check | `2,000,000 ÷ 120,000 = 16.7 days` | same 120K units/day flow as CAP-A | 16.7 days |
| 7 | Surge consumption | `3 × 120,000 × 0.25 = 90,000 ft³; 90,000 ÷ 1,020,000 = 8.8%` | a 3-day inbound surge ahead of forecast | 8.8% of effective cube in 3 days |

**The sentence to say:** "Cube is the currency that lies — 42% nominal is 49% effective, and a three-day vendor early-ship surge eats nine points of the building before the WBR sees it."

## CAP-D · WHY DWELL EXPLODES ABOVE ~85% DOCK UTILIZATION

Queueing approximation: with variability in arrivals and unload times, expected wait scales with ρ/(1−ρ) and total dwell with service ÷ (1−ρ), where ρ = utilization **[industry standard — Kingman-style approximation, qualitative use]**.

| STEP | UTILIZATION P | QUEUE FACTOR P/(1−P) | DWELL AT 2.0 H SERVICE: 2 ÷ (1−P) | READING |
|---|---|---|---|---|
| 1 | 70% | `0.70 ÷ 0.30 = 2.33` | `2.0 ÷ 0.30 = 6.7 h` | comfortable |
| 2 | 85% | `0.85 ÷ 0.15 = 5.67` | `2.0 ÷ 0.15 = 13.3 h` | the knee — the ceiling in CAP-A |
| 3 | 95% | `0.95 ÷ 0.05 = 19.0` | `2.0 ÷ 0.05 = 40.0 h` | a 2-hour unload becomes a 40-hour trailer stay |
| 4 | 85% → 95% | `19.0 ÷ 5.67 = 3.4×` | `40.0 ÷ 13.3 = 3.0×` | **+10 points of utilization ≈ 3× the dwell** |

**The sentence to say:** "Above about 85% the building stops failing linearly — the last ten points of utilization triple your dwell. So the plan holds dock slack the way inventory holds safety stock: you don't apologize for either until the week you needed it."

### CAP-E · PEAK BUFFER SIZING — P50 VS P90

| STEP | QUANTITY | FORMULA → SUBSTITUTION | INPUTS (BASIS) | VALUE |
|---|---|---|---|---|
| 1 | Peak-week distribution | `p50 = 120,000; p90 = 156,000; 156,000 − 120,000 = 36,000` | illustrative; p90/p50 = 1.30 | +36,000 units/day at p90 |
| 2 | Build to p50 — one p90 day | `36,000 ÷ 120,000 = 0.30 day` | shortfall carried as backlog | 0.30 day of backlog |
| 3 | Build to p50 — three p90 days | `3 × 36,000 = 108,000; ÷ 120,000 = 0.9` | backlog compounds across the shift handoffs | ~1 full day of backlog |
| 4 | Build to p90 — a p50 day | `36,000 ÷ 156,000 = 23%` | idle capacity on the median day | 23% idle |
| 5 | Labor to flex the tail | `156,000 ÷ 264 = 591 h; 591 − 454.5 = 136 h; 136 ÷ 10 = 13.6 ≈ 14` | blended 264 UPH from CAP-B; 10-h shifts | **≈14 flex associates** |

**The sentence to say:** "Fixed capacity to the median, flex the tail. Building the building to p90 means paying 23% idle on the average day; building to p50 means a three-day p90 run is a full day of backlog — so the answer is temp labor, overtime and appointment smoothing, sized at about fourteen heads on this volume."

## 3 · Inventory management, inbound lens — where the units land and how long they last

Capacity decides how fast freight moves; inventory decisions decide whether the right units are in the right building. This is the hiring manager's home turf — placement was her EU IXD charter — so the placement math (INV-C) gets said slowly, with the constraint and the shadow price named.

### Placement strategy — distance-to-customer vs cost

**The mechanism.** Amazon positions inventory close to forecasted demand so that the outbound leg — the expensive, promise-bound leg — is short **[well-known public Amazon practice]**. Inbound placement is the upstream half of that decision: which node receives the units, given regional demand, node capacity and inbound cost. The trade is never "cheapest inbound" — it is distance-weighted delivered cost under a capacity constraint, with the customer promise as the boundary condition. Speed and cost are one table, not two arguments.

**Network safety-stock positioning.** Because pooling variability upstream beats duplicating buffers everywhere **[industry standard]**, the cheap place to hold uncertainty is upstream (AWD-style bulk) and the fast place is forward (FC pickface); the worked question is how much buffer each layer carries (INV-B) and when a transfer beats a stockout (INV-D).

### Weeks of cover, health, and the FBA seller view

| CONCEPT | DEFINITION | OPERATOR'S ACTION RULE | LABEL |
|---|---|---|---|
| **Weeks of cover (WOC)** | WOC = on-hand ÷ weekly forecast | < 2 → replenish / expedite · 2–4 → healthy · 4–6 → watch · > 6 → excess action (worked in INV-A). | **[industry standard]** |
| **Excess / aged** | Units above the cover ceiling, and units aging against storage-fee cliffs. | Price, transfer, or remove — in that order of preference. | **[industry standard]** |
| **Stranded** | Physically present, not sellable (listing, hold, or data break) — inbound's silent inventory. | Fix the listing/hold at the source; a stow without a sellable record is not availability. | **[well-known public Amazon practice]** |
| **In-stock rate** | Share of demand-facing selection available to promise. | The north-star inventory output; every inbound metric is an input to it. | **[industry standard]** |
| **IPI (FBA seller view)** | Inventory Performance Index components: excess inventory %, sell-through, stranded inventory, in-stock rate. | Sellers manage to the same four levers; restock limits make network capacity legible to sellers **[well-known public Amazon practice]**. | **[well-known public Amazon practice]** |
| **Transfers / rebalancing** | FC↔FC moves that correct placement error. | Run them against the breakeven (INV-D), never as a reflex. | **[well-known public Amazon practice]** |

### INV-A · WEEKS OF COVER — THREE SKUS, THREE ACTIONS

| STEP | SKU | FORMULA → SUBSTITUTION | WOC | ACTION (RULE FROM THE TABLE ABOVE) |
|---|---|---|---|---|
| 1 | A | `4,200 ÷ 600/wk = 7.0` | 7.0 | > 6 → excess action: to reach 4.0 WOC move 4,200 − 4.0×600 = 4,200 − 2,400 = **1,800 units** (transfer feeds INV-D) |
| 2 | B | `1,350 ÷ 450/wk = 3.0` | 3.0 | 2–4 → healthy; no action, watch variance |
| 3 | C | `500 ÷ 500/wk = 1.0` | 1.0 | < 2 → expedite: stockout in 500 ÷ 7 = 71.43/day; 500 ÷ 71.43 = **7.0 days** absent receipt; pull the inbound appointment forward or re-source |

**The sentence to say:** "WOC turns a warehouse full of units into three sentences: A is seven weeks of stock and I'm moving 1,800 units; B is fine; C is a stockout next Friday unless the appointment moves."

### INV-B · SAFETY STOCK — Z·Σ·√LT, SUBSTITUTED, AND THE TWO LEVERS

| STEP | QUANTITY | FORMULA → SUBSTITUTION | INPUTS (BASIS) | VALUE |
|---|---|---|---|---|
| 1 | Demand and variability | `d = 500/day; σ_d = 120/day` | illustrative SKU at one FC | — |
| 2 | Demand over lead time | `500 × 7 = 3,500` | lead time LT = 7 days (illustrative) | 3,500 units cycle exposure |
| 3 | σ over lead time | `σ_LT = 120 × √7 = 120 × 2.6458 = 317.5` | independent daily demand [industry standard] | 317.5 units |
| 4 | Safety stock at 95% | `SS = 1.645 × 317.5 = 522; 522 ÷ 500 = 1.044 ≈ 1.04 days` | z(95%) = 1.645 [industry standard] | **522 units = 522 ÷ 500 = 1.04 days** |
| 5 | Service lever: 99% | `SS = 2.326 × 317.5 = 739; 739 − 522 = +217; 217 ÷ 522 = 41.6%` | z(99%) = 2.326 | +217 units = **+41.6% buffer for +4 pts** |
| 6 | Lead-time lever: 7 → 5 days | `SS = 1.645 × 120 × √5 = 1.645 × 268.3 = 441; 522 − 441 = −81; 81 ÷ 522 = 15.5%` | √5 = 2.2361; 120 × 2.2361 = 268.3 | −81 units = **−15.5% buffer, free** |

**The sentence to say:** "Service points are bought at an accelerating price — 95 to 99 costs forty percent more buffer — but lead time is the free lever: two days off the lane releases fifteen percent of the buffer with the service level untouched. That's why measured lead times, not brochure ones, are an inventory program" (this is exactly the S8 story — §5).

### INV-C · PLACEMENT MINI-MODEL — 2 FCS × 3 DEMAND REGIONS, CAPACITY-CONSTRAINED

Distance-weighted delivered cost per unit **[illustrative]**. Demand: R1 = 2,000/wk, R2 = 3,000/wk, R3 = 1,000/wk (6,000 total). FC-West capacity 2,500 units/wk; FC-East 4,000.

| COST $/UNIT | R1 (2,000) | R2 (3,000) | R3 (1,000) |
|---|---|---|---|
| **FC-East** | $1.20 | $2.40 | $3.60 |
| **FC-West** | $3.40 | $2.10 | $1.30 |

| STEP | DECISION | FORMULA → SUBSTITUTION | WEEKLY COST |
|---|---|---|---|
| 1 | Unconstrained optimum: R1→East, R2→West, R3→West | 2,000×1.20 + 3,000×2.10 + 1,000×1.30 = 2,400 + 6,300 + 1,300 | $10,000 |
| 2 | West capacity binds at 2,500: give West its cheapest regions first — R3 (1,000), then R2 partial (1,500) | 1,000×1.30 + 1,500×2.10 = 1,300 + 3,150 | West serves 2,500 |
| 3 | East takes R1 + residual R2 (1,500); load check 2,000 + 1,500 = 3,500 ≤ 4,000 | 2,000×1.20 + 1,500×2.40 = 2,400 + 3,600 | East serves 3,500 ✓ |
| 4 | Constrained total | 2,400 + 1,300 = 3,700; 3,700 + 3,150 = 6,850; 6,850 + 3,600 = 10,450 | **$10,450** |
| 5 | Capacity premium | 10,450 − 10,000 = 450; 450 ÷ 10,000 = 4.5% | **$450/wk (+4.5%)** |
| 6 | Shadow price check | (2.40 − 2.10) × 1,500 = 0.30 × 1,500 = 450 | each added West slot is worth $0.30 on R2 — the model agrees with itself |

**The sentence to say:** "Placement is a constrained assignment, not a map. Here the constraint costs four and a half points, and the shadow price — thirty cents a unit — is the number I'd take to the capacity team to argue for one more West slot, and to the finance review to explain the premium."

### INV-D · TRANSFER BREAKEVEN — WHEN REBALANCING BEATS A STOCKOUT

| STEP | QUANTITY | FORMULA → SUBSTITUTION | INPUTS (BASIS) | VALUE |
|---|---|---|---|---|
| 1 | Transfer cost per unit | $1.10 handling + $1.90 freight = $3.00 | illustrative FC↔FC economics | $3.00/unit |
| 2 | Expected stockout cost per unit | $12 × 0.40 = $4.80 | lost margin $12/unit; P(stockout before inbound lands) = 40% (illustrative) | $4.80/unit |
| 3 | Decision rule | 3.00 < 4.80 → transfer | transfer when cost < expected loss | **transfer wins by $1.80/unit** |
| 4 | Breakeven probability | p* = 3.00 ÷ 12 = 0.25 | transfer when stockout probability > 25% | p* = 25% |
| 5 | At scale (SKU A's 1,800 units) | spend 1,800 × 3.00 = 5,400; expected loss 1,800 × 4.80 = 8,640; net 8,640 − 5,400 = 3,240 | the 1,800 units from INV-A | **+$3,240 net — and the premium is priced, not argued** |

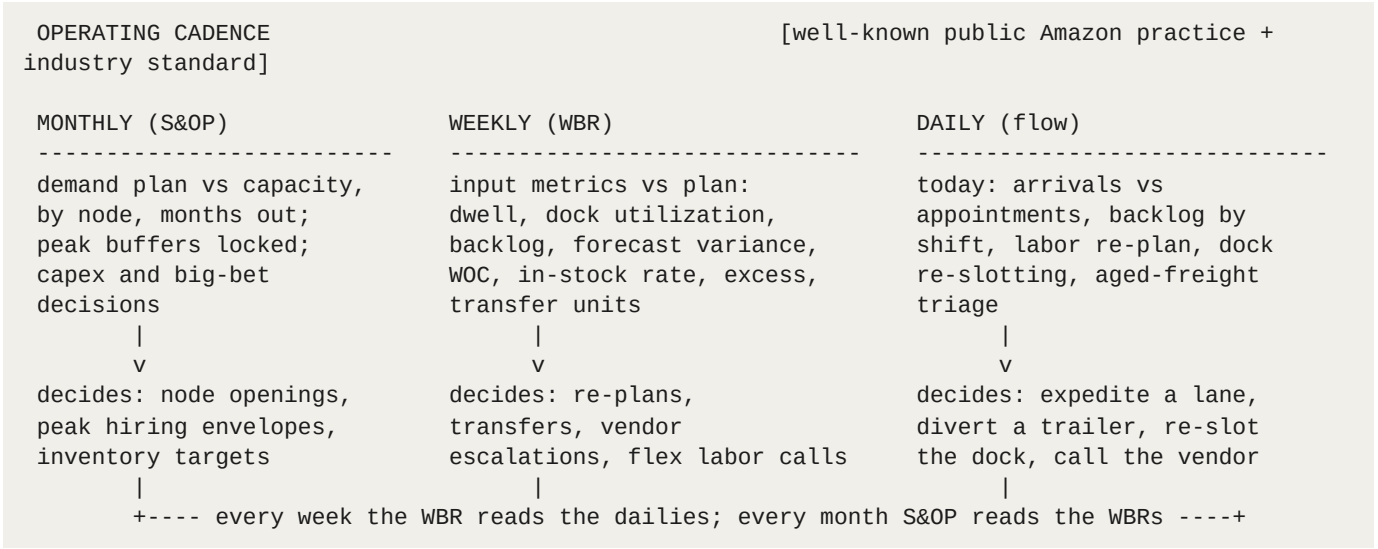
**The sentence to say:** "Transfers are insurance with a deductible. At a twelve-dollar margin the breakeven is a twenty-five percent stockout probability — below that I let it ride, above it I move the freight, and either way the decision is arithmetic, not adrenaline."

### INV-E · VENDOR OTIF / FILL-RATE → THE IN-STOCK CHAIN

| STEP | LINK | FORMULA → SUBSTITUTION | VALUE |
|---|---|---|---|
| 1 | Planned PO volume | `10,000 ÷ 7 = 1,428.6 ≈ 1,429/day` | 10,000 units (illustrative) |
| 2 | × fill rate | `10,000 × 0.92 = 9,200` | 9,200 shipped (800 short) |
| 3 | × on-time | `9,200 × 0.88 = 8,096` | 8,096 OTIF units = **81.0% of plan** |
| 4 | Weekly deficit | `10,000 − 8,096 = 1,904` | 1,904 units (19.0% of plan) |
| 5 | Safety-stock absorption | `522 ÷ 1,429 = 0.37 days; residual 1,904 − 522 = 1,382` | the INV-B buffer covers about a third of a day — it was sized for demand noise, not vendor failure |
| 6 | In-stock proxy for the week | `10,000 − 1,382 = 8,618; 8,618 ÷ 10,000 = 86.2%; 100 − 86.2 = 13.8 ≈ 14 pts` | **~86% in-stock absent recovery — a 14-point hit from one vendor's 92/88 week** |

**The sentence to say:** "OTIF isn't a vendor scorecard, it's the first link of the in-stock chain: fill times on-time is eighty-one percent of plan, and a cycle buffer sized for demand noise absorbs a third of a day of it. So vendor management is inventory management — measured lead times, escalation thresholds, and a recovery path that starts at the ASN, not at the stockout."

## 4 · Operating cadence — one drumbeat, three tempos




### Peak (Q4) planning rhythm [industry standard]

| WHEN | WHAT GETS DECIDED | THE INBOUND-SPECIFIC WORK |
|---|---|---|
| **Q2–early Q3** | Network peak plan: capacity envelopes by node, hiring envelopes, inventory targets. | p50/p90 forecast distribution locked (CAP-E); cube plan vs effective capacity (CAP-C); vendor appointment and freight-plan commitments. |
| **Late Q3** | Readiness: flex labor pipelines, overflow/storage contingency, freeze windows. | Learning-curve staffing math paid early (CAP-B step 5–6); appointment smoothing rules published to vendors/carriers. |
| **Q4 in-peak** | Execution: daily flow cadence dominates; WBR shifts to peak inputs. | Dwell and backlog watched hourly (CAP-D); transfers only against the breakeven (INV-D); escalations on same-sign forecast variance. |
| **January** | Debrief: what the plan got wrong, written down while it's fresh. | Variance autopsy by node; buffers released on evidence, not relief (the S6/S8 release rule, §5). |

**Escalation norms.** Escalate with the arithmetic attached: the metric, its frozen definition, the variance, the root cause so far, and the decision you need [industry standard]. Same-sign variance three periods running is a plan problem and goes up as a re-plan request, not a floor problem and goes down as pressure. A miss you report with a recovery path is a WBR input; a miss discovered by someone else is a trust problem — the S11 discipline (§5).

## 5 · Tie-to-your-stories — the master packet, spoken in inbound

Each story maps to one inbound concept; the third column is the sentence that makes the mapping out loud. Canon numbers carry their basis labels; stories stay allocated to their panel owners — here they are *vocabulary*, not re-told STARs.

| STORY (CANON) | INBOUND CONCEPT | THE SENTENCE THAT TIES THEM |
|---|---|---|
| **S8 — Elementum measured lanes** (200 lanes; promise at p99; 99% lead-time accuracy; 580K units / ~$1.5M buffer released, record, one-time) | Inbound lead-time reliability | "Every inbound plan and every safety-stock formula is only as honest as its lead times — I replaced brochure lead times with measured distributions and released buffer on evidence; INV-B's free lever is that story." |
| **S6 — Elementum supply/demand view** (forecast accuracy 70% → 85%; $2M safety stock released under the both-errors rule, record, one-time) | Network safety-stock positioning | "Buffer gets released where forecast error AND lead-time error both improved — a signed release rule, not a victory lap; that's how I'd run NACF's inventory health reviews." |
| **S2 — Sephora UDC comingle** (two warehouses → one pool; +15% efficiency, signed; robotics behind a published 98% gate) | Capacity pooling | "Pooling variance beats adding capacity: one pool removed duplicate handling before anyone asked for a bigger building — the same logic as holding cube slack upstream instead of over-building the FC." |
| **S13 — Manhattan Cyber Monday** (3 DCs / 100 stores; 99% vs 95% fill; ~$8M protected vs ~$6M at risk, record pair) | Peak capacity under failure | "Peak is when the CAP-D nonlinearity stops being theory — I rebalanced by promise-at-risk at 4 AM, and the post-mortem shipped the heartbeat and the playbook so the next peak ran boring." |
| **S18 — Sephora dual-DC WMS migration** (0.5% tolerance published with Finance; mocks 7% → 1.8% → 0.3%; rehearsed fallback for ~100 of ~400 RF devices) | Launch / cutover readiness | "New capacity or a new process goes live behind a published tolerance and a rehearsed fallback — exactly how I'd bring a new IXD lane or a peak process into the network." |
| **S19 — Elementum promise dial** (k = 1.645σ one governed dial; fast-promise share 7.6% → 39.6% at a held 95% floor — all model, labeled) | Promise from inventory position | "The date the customer sees is priced from inventory position and lane variability — placement (INV-C) sets the position, the dial prices the promise." |

| STORY (CANON) | INBOUND CONCEPT | THE SENTENCE THAT TIES THEM |
|---|---|---|
| **S11 / S12 — Flexport visibility & roadmap** (boarding behind published gates; CTS $0.66 → $0.03 across 7 → 68 customers; month-18 plan → value in month 6) | Upstream visibility into inbound | "Inbound starts before the truck: ASN-grade visibility and graduation gates upstream are what let a dock plan survive contact with real vendors — and re-sequencing on cost-to-serve is how I'd order the roadmap to get there." |
| **S9 — Elementum lane onboarding** (597 lanes = 80% of spend live in month one; ≥95% valid-ASN gate held twice; completeness 60% → 95%) | ASN / inbound data quality | "Bad ASNs manufacture surprise at the dock — a gate on data quality upstream is capacity protection downstream." |

> **Discipline in the room.** These are one-sentence bridges, not re-tellings — the full STARs belong to their allocated interviewers per the panel packets. Never mix the loop packet's conflicting numbers into these bridges; never present any modeled dollar as record.

## 6 · Per-interviewer questions — what they'll ask, and what to ask them

For each interviewer: ten likely NACF-inbound questions with a one-to-two sentence answer pointer (the booklet section carries the depth), then three questions to ask them. The first six in each list are the questions to expect from that seat's lens; the last four press the part of the job that person actually owns — placement, the distribution, the building, the peer handoff, or the seller.

### Snigdha Tripathy — Sr Manager, Supply Chain · HIRING MANAGER

> **Lens.** Placement and IXD are her home turf (EU IXD leader; built the team ground-up; 50+ FC launches). She judges fit for the actual work: network process design, value made legible, and judgment under unit economics. Talk placement slowly and precisely.

| # | LIKELY QUESTION THEY WILL ASK | HOW TO ANSWER |
|---|---|---|
| 1 | **Walk me through how you'd decide where inbound inventory lands across our network.** | Walk INV-C slowly: a cost matrix, the capacity that binds, and a shadow price that reproduces the premium — $450 either way in that example. §8 stage 6 is the same move on one week, where splitting 2,350 units saves $5,640 and the shadow-price check prints that same $5,640; then bridge to S8, because those lane costs are only honest if the lead times were measured. |
| 2 | **How do you know the network has enough inbound capacity for Q4?** | Say CAP-A, CAP-B, and CAP-E as one chain: units to trailers to doors held at the 85% ceiling, UPH to heads, fixed capacity near the median and flex on the tail. §8 runs it on 12,000 units — the dense equivalent is 1.92 trailers, the tender is 6, and a 7th arrival on a 6-slot block is what actually breaks the day. |
| 3 | **Tell me about a time you built a process or capability from nothing.** | Use S11: a boarding process stood up behind gates that were published before wave 1, with a weekly input/output review. That is the same shape as a ground-up IXD build, mechanism first and volume second, and S18 is the launch version of it if she pulls on the fifty-plus FC openings. |
| 4 | **A strategic vendor misses OTIF three weeks running into peak. What do you do?** | Price the miss before the conversation. INV-E turns 92% fill times 88% on-time into 8,096 units, 81% of plan, and about a 14-point in-stock hit a 522-unit cycle buffer was never sized to absorb — §8's 120-unit shortage is the small version of that pass-through — and the escalation carries the recovery path and a re-plan, the way S8 replaced brochure lead times with measured ones. |
| 5 | **How do you run your weekly business review — inputs, outputs, decisions?** | Use §4's three tempos and S11's mechanics: a short list of frozen inputs, each signed by its owner and tagged to the output it moves, reds already decomposed with a date. Three weeks of same-sign variance comes up as a re-plan request, which is how §8's 1,650-unit slow-mover over-order would be handled the week it repeats. |
| 6 | **How do you trade speed against cost in placement without losing the customer promise?** | Put the promise in as the boundary condition, then show one priced table. INV-C's constrained premium was 4.5% with a $0.30 shadow price on the displaced unit, INV-D transfers only when cost sits below margin times probability (breakeven 25% on the $3 and $12 example), and S19 is the promise-side version: one governed dial, with the option you decline kept in the table as a priced row. |

| # | LIKELY QUESTION THEY WILL ASK | HOW TO ANSWER |
|---|---|---|
| 7 | **Where would you run a Kaizen on the dock-to-available path, and what stays frozen through peak?** | Start on the wait that adds no units — §8 stage 2, where 6 doors × 2.0 h × 0.5 rate = 6 slots and the 7th trailer dwells 4.0 h. Through peak, CAP-E keeps the flex on labor, overtime, and appointment smoothing, and the placement objective and the safety-stock z stay frozen until the January debrief writes down what the plan got wrong. |
| 8 | **IXD dwell and an FC dock are both red the same week. Which do you pull, and how do you make the cost of waiting visible?** | Pull the pool whose units are already inside a customer promise, and price the hours on the other one. §8's marginal trailer is 4.0 h of dwell, and under a FIFO yard that delay lands on an expected 1,714 units — that slide is the escalation, the same habit as making an IXD saving legible in dollars rather than as a red cell. |
| 9 | **How do you ramp inbound at a new FC — doors, the labor curve, and the first units you place there?** | Doors from CAP-A off the launch forecast, labor from CAP-B with the learning curve paid before trailer one (30% of the crew at 60% UPH blends to 264 and costs 13.6% more hours), and the first inventory as an INV-C assignment into a building CAP-C still shows as empty. S18 is the gate around the ramp: a published tolerance, mocks, and a fallback already rehearsed. |
| 10 | **Infrastructure wants the dock held near 70% and Finance wants the cube earning. How do you settle it?** | Put both constraints on one page with a price. CAP-D is why dock slack is insurance — from 85% to 95%, dwell goes from 13.3 h to 40 h, about 3× — CAP-C is why cube is read on effective capacity, 49.0% in that example against 41.7% nominal, and the settlement is INV-C's shadow price, what the next slot is worth per unit, taken to both teams as the same number. |

### Questions to ask them

- "Which inbound outcome does this seat own end-to-end in the first year — placement, the capacity envelope, or inventory health — and which two or three WBR inputs would tell you, six months in, that the hire was right?"
- "Where does NACF inbound hurt most going into this peak — dock slots, the labor curve, cube, or placement accuracy — and what would you want moved first?"
- "You built EU IXD and placement from the ground up. What did that teach you about the order of people, process, and mechanisms that I should copy walking in?"

## Ryan Hoerger — Manager, Placement / Transfer Analytics

> **Lens.** Ex-Sr BIE, Booth MBA, Siebel Scholar, Nike analytics — statistical rigor is the love language. Show the distribution behind the number, name your assumptions, and let the math be audited.

| # | LIKELY QUESTION THEY WILL ASK | HOW TO ANSWER |
|---|---|---|
| 1 | **How would you model optimal inventory placement across FCs?** | Start from the cost matrix, let capacity bind, then check that the shadow price reproduces the premium. INV-C does it in six steps and the $450 appears twice, which is the audit; §8 stage 6 is the two-node case, where the $5,640 of savings also equals transfer cost avoided minus the $0.60 of extra inbound, so the model agrees with itself before anyone is asked to believe it. |
| 2 | **When is a transfer worth doing?** | Transfer when its cost is below margin times the probability of stocking out before the next inbound lands. On INV-D's economics that is $3.00 against $12 × 0.40 = $4.80, and the breakeven probability is 3.00 ÷ 12 = 25%; §8 applies the same rule once the trailers are already at ONT9-style, and the fast mover fails it while the medium and slow movers pass. |
| 3 | **How do you size and position safety stock across a network?** | Size it with INV-B, SS = z × σ × √LT, and name the lever you pulled. Moving z from 1.645 to 2.326 added 217 units (+41.6%), while cutting lead time from 7 days to 5 released 81 units (−15.5%) with the service level held; pool the rest upstream and release it only under S6's both-errors rule, because §8's fast mover is a buy with the z term left out — cover lands on the 7-day lead time and a 120-unit shortage passes straight through. |
| 4 | **How do you measure forecast quality, and what do you do with the variance?** | Use the §2 definition, (actual − plan) ÷ plan, by week and by node. Three weeks of the same sign is bias, so the action is a re-plan; S6 is the program behind that sentence, forecast accuracy from 70% to 85%, with buffer released only where both error terms had actually moved. |
| 5 | **Do you size buffers to p50 or p90?** | Fix the building near the center of the distribution and flex the tail. CAP-E is the arithmetic: a p50 plan carries 0.30 day of backlog on one p90 day and about a full day across three, a p90 building sits 23% idle on the median day, and the flex on these numbers is 36,000 ÷ 264 ÷ 10 ≈ 14 heads. |
| 6 | **How do you keep a model from going stale or getting gamed?** | Put a drift flag and a refresh cadence on the distributions, which is what S8 did for lanes, and freeze the metric definitions under the owner's signature the way S11 did. A truncated week stays a data miss: show the hole in the denominator before you show a ratio that has been cleaned up. |
| 7 | **The placement residual looks biased on slow movers. What do you inspect before you change the model?** | Separate bias from noise on the velocity slice before you touch the objective. §8's slow mover is the picture: sell-through 350 ÷ 3,400 = 10.3%, 8.71 weeks of cover against a 4.0 target, an over-order of 1,650 units that a WOC residual would have flagged — and only then do you decide whether the feature, the capacity constraint, or the loss is the thing to change. |

| # | LIKELY QUESTION THEY WILL ASK | HOW TO ANSWER |
|---|---|---|
| 8 | **How would you test, out of sample, that a transfer policy beats its breakeven?** | Freeze the rule on a prior window — here p* = 3.00 ÷ 12 = 25% — and score the stockouts you actually saw against the transfers the rule would have fired. INV-D step 5 is the in-sample version, +$3,240 on 1,800 units at an assumed 40% probability; the out-of-sample question is whether that net survives when the probability is estimated from history rather than typed in. |
| 9 | **Two planners hand you two different sigmas for the same SKU. What has to be true before safety stock moves?** | Same series, same lead-time window, and a stated assumption about independent days, because INV-B's √LT is exactly that assumption. §8's sigmas — 80, 40, and 15 units a day — stay labeled inputs until the window and the autocorrelation are in the room; resizing safety stock off an unstated σ is how the fast mover's 348-unit buffer quietly disappears from the buy. |
| 10 | **The capacity team wants average dwell as the WBR headline. What do you put next to it, and when do you refuse the formula?** | Put the marginal trailer next to the mean, because CAP-D's queue factor moves from 5.67 at 85% to 19 at 95% while an average can still look calm. §8 reports both from the same yard — expected dwell (6 × 2.0 + 4.0) ÷ 7 = 2.29 h, and 4.0 h on the trailer that waits — and once ρ = 7/6 the Kingman form is undefined, so the residual count replaces it. |

## Questions to ask them

- "When operators override the placement or transfer recommendation, do you score the override against the model's counterfactual — and who is usually right?"
- "What is the refresh cadence and the drift test on the demand and lead-time distributions the placement model actually sees?"
- "Which input do you trust least today — σ, the stockout probability, or the transfer cost per unit — and what observation would change it?"

## Rocky Goforth — Sr Manager, Infrastructure

> **Lens.** Infrastructure owns the physical envelope — buildings, docks, automation, peak readiness. Speak capacity as physics: constraints, nonlinearities, launch discipline.

| # | LIKELY QUESTION THEY WILL ASK | HOW TO ANSWER |
|---|---|---|
| 1 | **What actually constrains an FC's inbound throughput — and which one binds first at peak?** | Three currencies, in a fixed order: dock slots first, because a door takes one trailer; labor second, because hiring lags the plan by weeks; cube last, and then as a wall. CAP-A and CAP-C are the two ends of that, and §8 is a week where the Wednesday block fails at 7 trailers on 6 slots while the mod is still at 25.7% effective before the receipt even lands. |
| 2 | **Why does dwell blow up when docks run hot?** | The wait scales like ρ ÷ (1 − ρ), and CAP-D runs it: at 85% the factor is 5.67 and a 2.0 h unload dwells about 13.3 h; at 95% the factor is 19 and dwell is 40 h, roughly 3× for ten more points of utilization. §8 is the region past that formula — ρ = 7/6 makes (1 − ρ) negative — so the residual trailer replaces the approximation. |
| 3 | **How should we be planning infrastructure for Q4?** | Follow §4's peak rhythm with CAP-E underneath: lock the p50/p90 distribution and the effective-cube plan in Q2 and early Q3, pay the learning-curve hours in late Q3, and in peak watch dwell and backlog on the daily tempo. S13 is the in-peak posture I want Infrastructure beside me on — rebalance by promise at risk, then leave the heartbeat and the playbook so the next peak is dull. |
| 4 | **Tell me about bringing automation into a live building.** | S2. Robotics came in behind a published 98% gate and a throttle that existed before go-live; week two printed 94%, the fleet was held near 30% of volume, the cause was found, and full volume was back in week three. The gate is what made the install an infrastructure change with a stop condition. |
| 5 | **How do you launch new capacity or process without wobbling the network?** | S18: a 0.5% tolerance published with Finance, three mocks that moved discrepancies from 7% to 1.8% to 0.3%, and a fallback already rehearsed when about 100 of about 400 RF devices failed. A new door bank or a new inbound lane gets those same three things — a published tolerance, mocks against it, and a fallback someone has already run. |
| 6 | **When capacity and cost fight, how do you decide?** | Price the constraint and take that price to both sides. INV-C turned one more West slot into $0.30 a unit and $450 a week, because (2.40 − 2.10) × 1,500 = 450 and that matched the premium; §8 turns "send 2,350 units direct to the sister node" into $5,640 versus transferring them after all six trailers have arrived here, and CAP-E is the building-sized version of the same choice, idle percent versus backlog days, already computed. |
| 7 | **Six doors, and a carrier puts seven trailers on one block. What do you change — the building, the appointment grid, or the carrier?** | Change the grid and the tender. §8's full receiving day is 6 × 10 × 0.5 = 30 trailer-slots and this vendor's six trailers are 6 ÷ 150 = 4% of the week, so the 2.0 h block is what failed — 6 × 2.0 × 0.5 = 6 slots and the seventh trailer dwells 4.0 h — and the operating rule is that ASN trailers take those six doors while the unscheduled yard-drop holds the 4.0 h. |

| # | LIKELY QUESTION THEY WILL ASK | HOW TO ANSWER |
|---|---|---|
| 8 | **Cube reads fine, then one bulky receipt puts a mod near 80%. How early do you want that signal, and in what unit?** | On the ASN, in cubic feet against effective capacity, before the door is reserved. CAP-C is the warning that nominal utilization understates the building — 41.7% nominal was 49.0% effective — and §8's slow mover is 16.8% of the units and 43.5% of the receipt cube, taking the mod from 25.7% to 79.9% effective in one drop, with another 860 units of that SKU (688 ft³) touching the 90% line. |
| 9 | **Where should Supply Chain stop asking Infrastructure for peak space and change the flow instead?** | When the cube is a placement error the flow can still decline. §8's 1,650 excess slow-mover units are 1,650 × 0.80 = 1,320 ft³, 19.4% of that mod's effective capacity, so moving them is a flow decision, while a new rack is the ask when CAP-A says the doors are the currency that binds — bring the shadow price of the space and the WOC of the units in the same sentence as the square footage. |
| 10 | **We staffed mature UPH, the crew is full of new hires, and the unload rate misses. Who owns that, and why was it a surprise?** | The rate standard belongs to the building, the share of new heads belongs to the plan, and the miss belongs on the readiness page either way. CAP-B prices it ahead of the trailer: 0.7 × 300 + 0.3 × 180 = 264 blended UPH, and 120,000 ÷ 264 = 454.5 h against 400 h, a 13.6% tax for the first two weeks — that line sits next to the door plan at the late-Q3 review. |

## Questions to ask them

- "Which physical constraint is actually binding NACF inbound right now — a door bank, a sort or conveyor rate, or a mod that is fuller on effective cube than the nominal number says — and how do you want that called at the WBR?"
- "How far ahead are door counts and appointment blocks locked, and what is the path on a morning when a carrier's tender overruns the block the way seven trailers on six slots does?"
- "What should Supply Chain put on the ASN or the capacity reservation — cube by SKU, trailer count, or the learning-curve headcount — that Infrastructure still receives too late?"

## Zhaowen Lan — Sr Manager, Supply Chain · peer

> **Lens.** A peer who shares the cadence. He is listening for how you run mechanisms other people own — the WBR, the handoffs, the disagreement that stays professional.

| # | LIKELY QUESTION THEY WILL ASK | HOW TO ANSWER |
|---|---|---|
| 1 | **How do you run a WBR that people actually trust?** | Keep the list short and the definitions frozen. S11 ran five to seven input metrics, each signed by the person who owns it, each tagged to the output it is supposed to move, and a red arrived already decomposed with a date on the fix — the trust is in that pre-read. |
| 2 | **An input metric owned by another team is red every week. What do you do?** | Decompose it with the owner before the meeting and bring the dated fix. When the definition itself is wrong, correct it in the room: S11 discarded a 99.4% week rather than defend a denominator that had been cut short, and the number stays the owner's while the escalation carries the variance and the date. |
| 3 | **How do you manage the S&OP-to-WBR-to-daily handoff?** | §4 assigns each tempo a decision: S&OP locks the envelope, the WBR reads variance against that envelope, and the day re-slots the dock and calls the vendor. §8's Wednesday seven-on-six is a daily decision; the 1,650-unit over-order on the slow mover, once it is the same sign across weeks, is the re-plan the two of us would take back to S&OP. |
| 4 | **Tell me about a disagreement with a partner team and how it resolved.** | S12. The numbers went to the partner privately first, the reversal trigger was written before the plan moved, and the plan's owner ratified the change in the room. The disagreement stayed on the sequence and the cost, and both were still visible after the meeting. |
| 5 | **How do you keep inventory healthy across the network — excess vs in-stock?** | Apply INV-A per SKU and keep in-stock as the output. Under two weeks you expedite, two to four you watch, over six you move units; §8's close is that policy on one receiving week — fast mover 0.966 WOC, medium 3.00, slow mover 8.71 — and a stranded unit is not available just because it is inside the building. |
| 6 | **What would your first 90 days look like in this seat?** | Book of record and the primary metric first, then a stocks-and-flows map of inbound, then the invariant that is already broken, and one input/output review actually running by week six. I would rather walk into your WBR at week six with §4's short list alive than with a ninety-day plan that has not yet met a red metric. |
| 7 | **I call a SKU excess and you call it strategic cover. How do we leave the room with one number?** | Compute the band and the safety stock, and let the adjective catch up to the arithmetic. On §8's slow mover, safety stock is 65 units and the close is 3,050, so the word "strategic" has to explain 2,985 units above the safety stock, 1,650 of them past a 4.0-week target; if there is a real spike, it goes into the forecast and the WOC moves with it. |
| 8 | **Walk me through the handoff you don't want reopened: placement's recommendation versus the trailers the FC actually receives.** | That handoff gets signed at reservation, while the information is still ahead of the freight. §8 stage 6 values the split at $5,640 before the six trailers are tendered entirely to ONT9-style; once they are in the yard the inbound cost is sunk and the only open decision is the INV-D test on what to transfer, so Wednesday is an unload of a node that was already chosen. |

| # | LIKELY QUESTION THEY WILL ASK | HOW TO ANSWER |
|---|---|---|
| 9 | **A metric we both touch goes red. Who speaks first, and what would you hear as me stepping on your team?** | The owner of the input speaks first, with the decomposition already written — that is the S11 rule I want between peers. A new number you have not seen, or a redefinition of dwell or WOC in the meeting, is me stepping on your team; three weeks of same-sign variance is the escalation §4 already gives both of us. |
| 10 | **What do you want me to just fix, and what do you want me to bring to you?** | Fix what sits inside the envelope and is reversible today: a door re-slot, a vendor call, a 120-unit discrepancy claim. Bring the plan-level breaks — three weeks of same-sign forecast variance, a SKU that will cross its lead time with no safety stock the way §8's fast mover does at 6.76 days against 7.00, or a transfer that spends past the breakeven — and bring them before the WBR does. |

### Questions to ask them

- "Where are the seams between your team and this seat — especially who decides where a unit is placed, versus what this building actually receives?"
- "Which shared input is the last to turn green at your WBR, and what do you want a peer to do about it before they put it on your agenda?"
- "When you and a peer have split on excess versus in-stock, what made the call feel fair once the meeting was over?"

## Venkata Josyula — Principal PM, FBA · bar raiser

**Lens.** Principal PM for FBA plus bar-raiser duties: the seller is a customer, LP answers must be clean STARs with honest numbers, and every mechanism should end at a customer outcome.

| # | LIKELY QUESTION THEY WILL ASK | HOW TO ANSWER |
|---|---|---|
| 1 | **How does inbound look from the seller's side of FBA?** | From the seller's dock it is still a promise: ship, appointment, check-in, available. IPI's four components — excess, sell-through, stranded, in-stock — are the §3 health math made visible to the seller, and restock limits are network capacity translated into a constraint they can see; §8's three SKUs are that scoreboard with the arithmetic showing, sell-through 50.9%, 25.0%, and 10.3%, one SKU you replenish and one you stop sending. |
| 2 | **Why Amazon — and why this role?** | The three reasons in the master packet land on the same seam. Elementum was the forecast made honest, Flexport was the weekly review with frozen inputs, Sephora was one version of the truth across buildings — and this seat is where the forecast becomes a reservation, the review closes the variance, and two nodes have to share one set of numbers. |
| 3 | **Tell me about a time you failed.** | S18's RF miss: I had scored the risk by how small the task looked, and the right scale was impact — about 100 of about 400 devices failed at cutover. The rehearsed fallback fired, so the customer outcome held, and the lesson I kept is that a single point of failure gets a what-if plan on the page before the weekend, whatever its size on the punch list. |
| 4 | **Seller experience and network cost pull apart — how do you trade them?** | Price the trade, publish the price, and keep the rejected option in the table. INV-D is that page for a unit already in the network — $3.00 against a $12 margin, breakeven probability 25% — S19 is the same discipline on the promise, one dial, with the decline written as a priced row, and a seller's in-stock and our transfer bill belong on that page together. |
| 5 | **How would you improve FBA inbound?** | Start upstream of the door, where a surprise is still cheap. S9 and S11 are ASN quality and visibility before the truck, the appointment and the check-in-to-available clock are the seller-visible half of the §1 flow, AWD is the cheap cube that keeps a slow bulky unit out of an FC pick mod, and §8's slow mover — 0.80 ft³ and 10.3% sell-through — is the unit that pool is for. |
| 6 | **Tell me about a time you chose long-term over short-term pressure.** | S12. A committed 18-month roadmap was re-sequenced so the value landed in month 6, at about 66% less effort. The reversal trigger was written down before the plan swung, and the cost of delaying the operators was priced and published, so the long-term choice was a document the room could audit. |
| 7 | **A restock limit just cut a seller whose selection was in stock. Was that capacity working, or a defect?** | Check which currency was scarce, and on which SKU. If the FC was already at a CAP-D dwell and the limit stopped a trailer the block could not take, the limit was doing the job of the 85% ceiling; if the cut SKU looks like §8's fast mover — 50.9% sell-through, cover inside one lead time — while a 10.3% sell-through unit still flowed, the limit used an average and constrained the healthy unit, so I would want velocity and cube in the rule, the split stage 5 already shows inside one receipt. |

| # | LIKELY QUESTION THEY WILL ASK | HOW TO ANSWER |
|---|---|---|
| 8 | **Tell me about a time the data changed your mind.** | S6. I was ready to release safety stock on a better forecast alone, and the rule we wrote required forecast error and lead-time error both to have moved, so the buffer stayed where only one error had improved. INV-C's shadow-price check is the same habit on a placement number: the premium and the shadow price have to print the same dollar, $450 in that example and $5,640 in §8, before I will defend it. |
| 9 | **When is holding a seller's inventory upstream, AWD-style, the wrong answer for that seller?** | When the unit is fast, the demand is local, and the replenishment lead time into the FC consumes the promise. INV-B puts lead time under the square root, so days there are the expensive term, and §8's fast mover burned a 1,000-unit opening in 2.00 days and then sat at 6.76 days against a 7-day bridge with no safety stock under it — the upstream pool is the right home for the slow bulky unit (0.80 $ft^3$, 8.71 weeks of cover) and the wrong home for the unit a customer will miss this week. |
| 10 | **What does raising the bar look like on an inbound mechanism — a higher target, or something else?** | A frozen definition, a figure the room can recompute from the row above it, and a recommendation you will walk back when the arithmetic flips. S11's discarded 99.4% week, INV-D's 25% breakeven, and §8's choice to keep the fast mover in place once a transfer would only relocate a $6,000 stockout and add $1,500 of cost — that is the bar I would want a candidate to clear in the room. |

### Questions to ask them

- "Where does FBA inbound lose seller trust most sharply right now — the appointment, the check-in-to-available clock, or restock limits — and which of those can this Supply Chain seat actually move?"
- "How has AWD changed the conversation with sellers about who holds the slow, bulky units, and where does that conversation still stall?"
- "In the first month, what does a Supply Chain Manager do that makes a Principal PM's mechanism easier to run — and what do they do that you end up routing around?"

## 7 · Math appendix — every formula, symbolic → substituted → computed

The night-before page. Each row: the formula in symbols, the substitution from the booklet's worked example, the computed result, and where it lives. All inputs **[illustrative]** unless labeled otherwise; recompute each row from the row above it.

| # | FORMULA (SYMBOLIC) | SUBSTITUTION | COMPUTED | SECTION |
|---|---|---|---|---|
| A1 | `trailers/day = V ÷ u_c ÷ c_p ÷ p_t` | `120,000 ÷ 6 ÷ 40 ÷ 26` | `= 19.2 trailers/day` | CAP-A |
| A2 | `doors = (trailers/day ÷ hours) ÷ unload_rate ÷ ρ_max` | `(19.2 ÷ 10) ÷ 0.5 ÷ 0.85` | `= 4.5 → 5 doors` | CAP-A |
| A3 | `direct hours = V ÷ UPH; paid = ×(1+indirect)` | `120,000 ÷ 300 = 400; ×1.15` | `= 460 paid h; ÷10 = 46 heads` | CAP-B |
| A4 | `blended UPH = (1−n)·UPH + n·UPH·LC` | `0.7×300 + 0.3×300×0.6 = 264` | `120,000 ÷ 264 = 454.5 h (+13.6%)` | CAP-B |
| A5 | `util_true = I·cube ÷ (C × 0.85)` | `2.0M×0.25 ÷ (1.2M×0.85)` | `= 500,000 ÷ 1,020,000 = 49.0%` | CAP-C |
| A6 | `queue factor = ρ ÷ (1−ρ); dwell ≈ service ÷ (1−ρ)` | `ρ=0.95: 19.0; 2.0 ÷ 0.05` | `= 40 h dwell (vs 13.3 h at 85%)` | CAP-D |
| A7 | `flex heads = (p90 − p50) ÷ UPH_blend ÷ shift` | `36,000 ÷ 264 ÷ 10` | `= 13.6 ≈ 14 flex associates` | CAP-E |
| B1 | `WOC = on-hand ÷ weekly forecast` | `A: 4,200÷600; B: 1,350÷450; C: 500÷500` | `= 7.0 / 3.0 / 1.0 weeks` | INV-A |
| B2 | `excess units = on-hand − WOC_target × forecast` | `4,200 − 4.0×600` | `= 1,800 units to move` | INV-A |
| B3 | `SS = z × σ_d × √LT` | `1.645 × 120 × √7 = 1.645 × 317.5` | `= 522 units = 1.04 days` | INV-B |
| B4 | `SS(99) − SS(95) = (z99 − z95)·σ_LT` | `(2.326 − 1.645) × 317.5` | `= +217 units (+41.6%)` | INV-B |
| B5 | `SS(LT=5) = z × σ_d × √5` | `1.645 × 120 × 2.2361` | `= 441 units (−81; −15.5%)` | INV-B |
| B6 | `cost = Σ d_r × c_{r,f} s.t. node capacity` | `2,000×1.20 + 1,000×1.30 + 1,500×2.10 + 1,500×2.40` | `= $10,450/wk (vs $10,000 unconstrained)` | INV-C |
| B7 | `shadow price = (c_far − c_near) × units_displaced` | `(2.40 − 2.10) × 1,500` | `= $450/wk = the capacity premium` | INV-C |
| B8 | `transfer if cost_t < margin × P(stockout); p* = cost_t ÷ margin` | `3.00 < 12 × 0.40 = 4.80; p* = 3.00 ÷ 12` | `= transfer; p* = 25%; net +$3,240 on 1,800 units` | INV-D |
| B9 | `OTIF_units = PO × fill × on_time` | `10,000 × 0.92 × 0.88` | `= 8,096 units = 81.0% of plan` | INV-E |
| B10 | `in-stock proxy = 1 − (deficit − SS) ÷ PO` | `1 − (1,904 − 522) ÷ 10,000` | `= 86.2% weekly in-stock proxy` | INV-E |

Recompute discipline: A1→A2 share 19.2; A3→A4 share the 120,000 and 300 UPH; B3→B4→B5 share σ_d = 120 and z values; B2's 1,800 feeds B8's scale row; B9's deficit feeds B10. If any row doesn't reproduce from its own substitution, the row is wrong — say the formula, not the memory.

## §8 · One week at one FC — the complete worked example

One building, one week, one vendor, three SKUs, and the same figures carried from the forecast through the close. Every stage below is symbolic formula, then the substitution, then the result — the same shape as the CAP and INV blocks. If a later row quotes a number you cannot find above it, the row is wrong. All inputs are **[illustrative]**; the rates, the z, and the transfer economics are the ones already used in §2–§3 so the arithmetic is familiar out loud. ONT9-style and PHX7-style are labels for the whiteboard, not claims about any real building.

> **Assumptions to say before the first number.** Demand is flat across seven days. The Wednesday receipt is stowed the same day, so a 2.0 h unload slips the promise and does not lose Wednesday's sales. PHX7-style opening on-hand for this vendor is zero. In the stress block, service is deterministic and all seven trailers are in the yard at block open. √7 = 2.6458 and z = 1.645, as in INV-B. Transfer cost is $3.00/unit and lost margin is $12/unit, as in INV-D. Placement is decided on the planned PO; the 2% shortage is discovered at receive, so health at the close uses actuals.

### The scenario card — week 44, reused everywhere below

| CONSTANT | VALUE | WHERE IT IS REUSED |
|---|---|---|
| Building, week | ONT9-style FC, week 44 | Every stage. Illustrative; not a real site's data. |
| The shipment | One vendor, 3 SKUs, 12,000 units, 6 trailers | Stages 1–7. 12,000 ÷ 6 = 2,000 units/trailer. |
| Doors, day, block | 6 doors; 10 h receiving day; stress block is 2.0 h | Stage 2. The block is one service deep. |
| Unload rate | 0.5 trailer/door-hour, so service = 1 ÷ 0.5 = 2.0 h | Stage 2. Same rate as CAP-A. |
| Pack factors | 6 units/carton, 40 cartons/pallet, 26 pallets per dense trailer | Stage 1. Same factors as CAP-A. |
| Labor standards | Receive 300 UPH, stow 240 UPH, indirect 15%, shift 10 h | Stages 3 and 5. Receive UPH matches CAP-B. |
| The mod | 4,000 bins × 2.0 ft³ = 8,000 ft³ nominal; × 0.85 = 6,800 ft³ effective | Stage 5. The 15% honeycomb matches CAP-C. |
| Service and lead time | z = 1.645; LT = 7 days; √7 = 2.6458 | Stage 7. Same constants as INV-B. |
| Money | Inbound ONT9-style $0.80/unit; PHX7-style $1.40/unit; transfer $3.00; margin $12 | Stage 6. Transfer and margin match INV-D. |
| Sister node | PHX7-style demand: F 500, M 700, S 1,050 per week; opening 0 | Stage 6 only. This vendor's first units there. |

| SKU | ROLE | UNITS ON THE PO | WEEKLY FORECAST | PER DAY | OPENING ON-HAND | Σ / DAY | FT³ / UNIT |
|---|---|---|---|---|---|---|---|
| **F** | Fast | 6,000 | 3,500 | 500 | 1,000 | 80 | 0.15 |
| **M** | Medium | 4,000 | 1,400 | 200 | 1,600 | 40 | 0.30 |
| **S** | Slow | 2,000 | 350 | 50 | 1,400 | 15 | 0.80 |
| **Total** | | 12,000 | 5,250 | | 4,000 | | |

Buy targets that produced the PO, proved in stage 1: F back to 1.0 week of cover, M to 3.0 weeks, S would have needed 350 units to stay at 4.0 weeks and was ordered to a 2,000-unit multiple instead. The 1,650-unit gap on S is the number stage 6 and stage 7 both spend.

## Stage 1 · Inbound forecast → units per week → trailer equivalents

### 1A · THE BUY THAT EQUALS 12,000 — RECEIPT = TARGET ON-HAND − OPENING + FORECAST SALES

| STEP | QUANTITY | FORMULA → SUBSTITUTION | INPUTS (BASIS) | VALUE |
|---|---|---|---|---|
| 1 | Opening cover, F | `1,000 ÷ 500 = 2.00 days; 1,000 ÷ 3,500 = 0.286 wk` | scenario card | 2.00 days — gone by Tuesday close |
| 2 | Opening cover, M | `1,600 ÷ 1,400 = 1.143 wk` | scenario card | 1.14 weeks |
| 3 | Opening cover, S | `1,400 ÷ 350 = 4.00 wk` | scenario card | 4.00 weeks — already the healthy center |
| 4 | F receipt | `target OH = 1.0 × 3,500 = 3,500; 3,500 − 1,000 + 3,500 = 6,000` | target 1.0 WOC; sales = forecast | **6,000 units** |
| 5 | M receipt | `target OH = 3.0 × 1,400 = 4,200; 4,200 − 1,600 + 1,400 = 4,000` | target 3.0 WOC | **4,000 units** |
| 6 | S receipt the band wants | `target OH = 4.0 × 350 = 1,400; 1,400 − 1,400 + 350 = 350` | target 4.0 WOC, the INV-A center | 350 units |
| 7 | S over-order | `2,000 − 350 = 1,650` | vendor order multiple is 2,000 | **1,650 units — remembered at stages 6 and 7** |
| 8 | Week inbound | `6,000 + 4,000 + 2,000 = 12,000` | F + M + the multiple on S | **12,000 units** |

F's target close equals one lead time of demand and contains no safety stock. That choice is quiet here and loud in stage 4.

### 1B · UNITS → CARTONS → PALLETS → TRAILERS — THE TENDER IS NOT THE DENSE EQUIVALENT

| STEP | QUANTITY | FORMULA → SUBSTITUTION | INPUTS (BASIS) | VALUE |
|---|---|---|---|---|
| 1 | Cartons | `12,000 ÷ 6 = 2,000` | 6 units/carton, as CAP-A | 2,000 cartons |
| 2 | Pallets | `2,000 ÷ 40 = 50` | 40 cartons/pallet, as CAP-A | 50 pallets |
| 3 | Dense trailer-equivalents | `50 ÷ 26 = 1.923` | 26 pallets per 53-ft trailer | **1.92 trailers** |
| 4 | What the vendor tendered | `6 trailers; 12,000 ÷ 6 = 2,000 units each` | the ASN | 6 trailers |
| 5 | How full each trailer is | `50 ÷ 6 = 8.333 pallets; 8.333 ÷ 26 = 0.321` | 50 pallets across 6 trailers | **32.1% of a dense trailer** |
| 6 | Door currency vs cube currency | `6 ÷ 1.923 = 3.12` | tendered ÷ dense | **3.12× — plan doors off 6, cube off 50 pallets** |

**The sentence to say:** "Twelve thousand is a cover equation plus an order multiple — six thousand, four thousand, two thousand — and the two thousand includes sixteen hundred fifty the band did not ask for. Dense, that is one-point-nine trailers; the tender is six, a third full, so the dock feels six arrivals and the building feels fifty pallets."

## Stage 2 · Dock-door capacity — 7 trailers against 6 slots, and the dwell that forms

The week was reserved as six trailers spread across five days. Wednesday morning the yard has all six ASN trailers (the 12,000 units) plus one unscheduled yard-drop that is not on this PO. The block they have fallen into is sized at six door-slots. Carried in from stage 1: 6 trailers, 2,000 units each, service rate 0.5.

### 2A · DOES THE BLOCK HAVE THE SLOTS? — DOORS × HOURS × UNLOAD RATE

| STEP | QUANTITY | FORMULA → SUBSTITUTION | INPUTS (BASIS) | VALUE |
|---|---|---|---|---|
| 1 | Slots in the block | `S = 6 × 2.0 × 0.5 = 6.0` | doors × hours × unload rate | **6 door-slots** |
| 2 | Arrivals | `6 ASN + 1 unscheduled = 7` | the week's 6 trailers, bunched, plus a yard-drop with none of the 12,000 | **7 trailers** |
| 3 | Utilization | `ρ = 7 ÷ 6 = 1.1667` | arrivals ÷ slots | **116.7%** |
| 4 | The 85% ceiling | `0.85 × 6 = 5.1` | same ceiling as CAP-A | even 6 scheduled trailers were over the ceiling |
| 5 | Why Kingman stops here | `1 − ρ = 1 − 7/6 = −0.167` | dwell ≈ service ÷ (1 − ρ) needs ρ below 1 | the form is undefined — count the residual instead |
| 6 | The building's week, for scale | `6 × 10 × 0.5 = 30/day; 30 × 5 = 150; 6 ÷ 150 = 0.040` | full receiving day, five days | 30 slots/day, 150/week; this vendor is **4.0% of the week** |

The week had the doors. The two-hour block did not. That is a bunching failure, and it is the one to say out loud before anyone proposes a seventh door.

### 2B · EXPECTED DWELL ONCE P IS ABOVE 1 — DETERMINISTIC, ALL SEVEN PRESENT AT BLOCK OPEN

| STEP | QUANTITY | FORMULA → SUBSTITUTION | INPUTS (BASIS) | VALUE |
|---|---|---|---|---|
| 1 | Service time | `1 ÷ 0.5 = 2.0 h` | the unload rate on the card | 2.0 h on a free door |
| 2 | Who starts at once | `min(7, 6) = 6 trailers start; 7 − 6 = 1 waits` | 6 doors, simultaneous arrival | 1 trailer in queue |
| 3 | Dwell of a trailer that starts | `wait 0 + service 2.0 = 2.0` | door free at t = 0 | 2.0 h |
| 4 | Dwell of the marginal trailer | `wait = 2.0; dwell = 2.0 + 2.0 = 4.0` | first door frees at the end of one service | **4.0 h** |
| 5 | Expected dwell | `(6 × 2.0 + 1 × 4.0) ÷ 7 = 16 ÷ 7 = 2.286` | FIFO: each of the 7 equally likely to be last | **2.29 h** |
| 6 | Expected wait | `(6 × 0 + 1 × 2.0) ÷ 7 = 2 ÷ 7 = 0.286` | same weights | 0.29 h |
| 7 | Spill into the next block | `7 × 2.0 − 6 × 2.0 = 14 − 12 = 2.0 door-hours` | work demanded minus work the block can do | **1 trailer, and 1 of the next block's 6 slots** |
| 8 | FIFO hit on this PO | `6 × (1/7) × 2,000 = 12,000 ÷ 7 = 1,714 units` | P(a given ASN trailer waits) = 1/7; 2,000 units on it | **1,714 units delayed 2.0 h** |
| 9 | ASN-priority alternative | `delayed PO units = 0; unscheduled dwell = 4.0 h` | the 6 ASN trailers take the 6 doors | the 12,000 stay on a 2.0 h clock; the yard-drop holds the 4.0 h |

**The sentence to say:** "Seven over six is past the formula — one minus rho is negative — so the number is the residual trailer. Six start and dwell two hours; the seventh dwells four; the average is sixteen over seven, two-point-two-nine hours. Give the ASN the doors and the twelve thousand stay on a two-hour clock. Leave it FIFO and about seventeen hundred units slip two hours, and the next block opens already down one slot."

## Stage 3 · Labor plan — UPH to hours to heads on the shift

Labor is reserved on the plan, 12,000 units, before the shortage in stage 4 is known. The plan spread those units across five days; the yard in stage 2 put them on one shift. Both rosters are below, because the average is what gets staffed and the bunch is what shows up. Carried in: 12,000 units, 300 UPH, 15% indirect, 10 h shifts.

### 3 · RECEIVE LABOR — THE BUNCH IS THE SHIFT THAT HAS TO BE ROSTERED

| STEP | QUANTITY | FORMULA → SUBSTITUTION | INPUTS (BASIS) | VALUE |
|---|---|---|---|---|
| 1 | Direct hours, the plan | `12,000 ÷ 300 = 40.0` | mature receive UPH, as CAP-B | 40.0 direct hours |
| 2 | Paid hours | `40.0 × 1.15 = 46.0` | 15% indirect | 46.0 paid hours |
| 3 | Heads if the week arrives on one shift | `46.0 ÷ 10 = 4.6 → round up to 5` | 10 h shift; a partial head is not a person you can roster | **5 receivers** |
| 4 | Slack on that crew | `5 × 10 − 46.0 = 4.0; 4.0 ÷ 50 = 0.080` | paid hours the roster actually contains | 4.0 h slack, 8.0% idle |
| 5 | The spread the average suggests | `12,000 ÷ 5 = 2,400/day; 2,400 ÷ 300 = 8.0; 8.0 × 1.15 = 9.2; 9.2 ÷ 10 = 0.92` | five even days | 0.92 → **1 receiver looks like enough** |
| 6 | What 1 receiver does with the bunch | `40.0 ÷ 1 = 40.0 direct hours of work on a 10 h shift` | stage 2 put the week on one morning | about 4 shifts of backlog from one rostering choice |

**The sentence to say:** "Forty direct hours, times one-fifteen, is forty-six paid hours, divided by a ten-hour shift is four-point-six, so five people. The daily average says one person and point-nine of a head. Staff the bunch. The one hundred twenty units we find short in the next stage free four-tenths of an hour — you do not re-roster a shift for that."

## Stage 4 · Receive accuracy — a 2% shortage on the fast mover, and what it does to AFS

AFS here means units the promise can see: opening on-hand, which was already sellable, plus units that clear receive and will clear stow. The shortage never reaches stow, so it never becomes AFS. Carried in: PO of 6,000 / 4,000 / 2,000, opening 1,000 / 1,600 / 1,400, daily demand 500 / 200 / 50.

### 4A · THE COUNT — 2% OF F, AND WHY 99% OVERALL IS THE WRONG HEADLINE

| STEP | QUANTITY | FORMULA → SUBSTITUTION | INPUTS (BASIS) | VALUE |
|---|---|---|---|---|
| 1 | Shortage | `0.02 × 6,000 = 120` | 2% short on F only; M and S count clean | **120 units, all SKU-F** |
| 2 | Received, F | `6,000 − 120 = 5,880` | step 1 | 5,880 units |
| 3 | Received, the PO | `5,880 + 4,000 + 2,000 = 11,880` | M and S at plan | 11,880 units |
| 4 | Accuracy | `11,880 ÷ 12,000 = 0.990; 5,880 ÷ 6,000 = 0.980` | actual ÷ PO | **99.0% overall, 98.0% on F, 100% on M and S** |
| 5 | F AFS at the moment of receipt | `planned 1,000 + 6,000 = 7,000; actual 1,000 + 5,880 = 6,880; 7,000 − 6,880 = 120` | opening was already AFS | **AFS lower by 120** |
| 6 | Those 120 as demand | `120 ÷ 500 = 0.24 days; 120 ÷ 3,500 = 0.0343` | F demand from the card | 0.24 days, 3.43% of the week |

#### 4B · DOES THE SHORTAGE MISS THIS WEEK'S PROMISE, OR THE BRIDGE TO THE NEXT RECEIPT?

| STEP | QUANTITY | FORMULA → SUBSTITUTION | INPUTS (BASIS) | VALUE |
|---|---|---|---|---|
| 1 | Mon–Tue sales, F | `2 × 500 = 1,000` | opening cover was 2.00 days (stage 1) | opening is consumed exactly; AFS hits zero at Tuesday close |
| 2 | Wed–Sun sales, F | `5 × 500 = 2,500; 5,880 − 2,500 = 3,380 left` | assumption: stow finishes Wednesday, so the day is slipped by 2.0 h and not lost | 2,500 sold; the receipt covers them |
| 3 | Week sales, all three | `F 3,500; M 1,400; S 350` | each week's forecast, fully served under the assumption | this week's in-stock holds |
| 4 | Close, F, if the count had been clean | `1,000 + 6,000 − 3,500 = 3,500; 3,500 ÷ 500 = 7.00 days` | the stage-1 buy, which targeted 1.0 WOC and no safety stock | 7.00 days — exactly the lead time |
| 5 | Close, F, actual | `1,000 + 5,880 − 3,500 = 3,380; 3,380 ÷ 500 = 6.76 days` | the 120 removed | **3,380 units, 6.76 days** |
| 6 | The pass-through | `7.00 − 6.76 = 0.24 days; 0.24 × 500 = 120` | steps 4 and 5 | **the shortage equals the entire lead-time miss** |
| 7 | Close, M | `1,600 + 4,000 − 1,400 = 4,200` | M counted clean | 4,200 units — used again in stages 5–7 |
| 8 | Close, S | `1,400 + 2,000 − 350 = 3,050` | S counted clean | 3,050 units — used again in stages 5–7 |

**The sentence to say:** "Two percent of the fast mover is one hundred twenty units. The building's accuracy reads ninety-nine percent, and that average is hiding the only SKU that was bought to land exactly on its lead time. This week's sales still clear. The one hundred twenty comes off the bridge: seven-point-zero-zero days becomes six-point-seven-six, and there was no safety stock underneath it."

### Stage 5 · Stow and cubic fill — the mod before, at the peak, and at the close

This vendor's three SKUs live in one illustrative pick mod, so the fill move is visible. A whole FC would swallow 12,000 units; the mod is the grain a stow lead actually manages. Carried in: cube per unit from the card, received units from stage 4 (5,880 / 4,000 / 2,000), closes 3,380 / 4,200 / 3,050.

### 5A · BIN CAPACITY, THEN FILL BEFORE AND AFTER THE RECEIPT

| STEP | QUANTITY | FORMULA → SUBSTITUTION | INPUTS (BASIS) | VALUE |
|---|---|---|---|---|
| 1 | Nominal capacity | `4,000 × 2.0 = 8,000 ft³` | bins × cube per bin | 8,000 ft³ |
| 2 | Effective capacity | `8,000 × 0.85 = 6,800; per bin 2.0 × 0.85 = 1.70` | 15% honeycomb, as CAP-C | **6,800 ft³ effective** |
| 3 | Opening cube | `1,000×0.15 + 1,600×0.30 + 1,400×0.80 = 150 + 480 + 1,120 = 1,750` | opening units × cube | 1,750 ft³ |
| 4 | Opening fill | `1,750 ÷ 8,000 = 0.21875 = 21.9% nominal; 1,750 ÷ 6,800 = 0.2574 = 25.7% effective` | against both denominators | **25.7% effective — plan against this** |
| 5 | Receipt cube, actual | `5,880×0.15 + 4,000×0.30 + 2,000×0.80 = 882 + 1,200 + 1,600 = 3,682` | stage-4 received units | 3,682 ft³ |
| 6 | Who ate the cube | `units: 2,000 ÷ 11,880 = 0.168 = 16.8%; cube: 1,600 ÷ 3,682 = 0.435 = 43.5%` | S versus the receipt | **S is 16.8% of units and 43.5% of cube** |
| 7 | Peak, before the week's sales | `1,750 + 3,682 = 5,432; 5,432 ÷ 6,800 = 0.7988 = 79.9%; 5,432 ÷ 8,000 = 0.679 = 67.9% nominal` | opening + receipt | **79.9% effective (67.9% if you trust nominal)** |
| 8 | Headroom to the 90% wall | `0.90 × 6,800 = 6,120; 6,120 − 5,432 = 688; 688 ÷ 0.80 = 860` | CAP-C's "wall at 90%" | **688 ft³, or 860 more units of S** |
| 9 | Bins the receipt touches | `3,682 ÷ 1.70 = 2,165.9 → 2,166; 2,166 ÷ 4,000 = 0.5415 = 54.2%` | receipt cube ÷ effective cube per bin | 2,166 bins, 54.2% of the mod |

### 5B · STOW LABOR, AND THE CUBE THAT IS STILL THERE AT THE CLOSE

| STEP | QUANTITY | FORMULA → SUBSTITUTION | INPUTS (BASIS) | VALUE |
|---|---|---|---|---|
| 1 | Stow hours on the plan | `12,000 ÷ 240 = 50.0 direct; 50.0 × 1.15 = 57.5 paid` | stow 240 UPH; same 15% indirect | 57.5 paid hours |
| 2 | Stow heads on the bunched shift | `57.5 ÷ 10 = 5.75 → round up to 6; slack = 6 × 10 − 57.5 = 2.5 h` | same rounding as the receivers | **6 stowers, 2.5 h slack** |
| 3 | What the shortage saves | `120 ÷ 240 = 0.5 direct hour; 11,880 ÷ 240 = 49.5` | the 120 units from stage 4 | half an hour — hold the roster of 6 |
| 4 | Close cube | `3,380×0.15 + 4,200×0.30 + 3,050×0.80 = 507 + 1,260 + 2,440 = 4,207` | closes from stage 4 | 4,207 ft³ |
| 5 | Close fill | `4,207 ÷ 6,800 = 0.6187 = 61.9%; S share 2,440 ÷ 4,207 = 0.580 = 58.0%` | effective denominator | **61.9% effective; S is 58% of what is left** |

**The sentence to say:** "The mod opens at twenty-five-point-seven percent effective, not the twenty-two percent nominal. The receipt takes it to seventy-nine-point-nine, and the slow mover is sixteen-point-eight percent of the units and forty-three-point-five percent of the cube. Eight hundred sixty more of those units would put the mod on the ninety-percent wall. Stow is six people on the shift that the freight actually arrived."

## Stage 6 · Placement — two nodes, put the units where the demand is, pick the cheaper plan

This decision belongs with the information flow, before the tender. It is worked here, after the freight is in the building, so both numbers can be said: what the split was worth on Tuesday, and what is still worth doing once the six trailers are in the ONT9-style yard. Planned units, from stage 1: F 6,000, M 4,000, S 2,000. PHX7-style weekly demand, from the card: F 500, M 700, S 1,050, opening zero.

| COST TO HAVE THE UNIT THERE, $/UNIT | SERVE ONT9-STYLE DEMAND | SERVE PHX7-STYLE DEMAND |
|---|---|---|
| **Inbound to ONT9-style, then transfer if needed** | `$0.80` | `$0.80 + $3.00 = $3.80` |

| COST TO HAVE THE UNIT THERE, $/UNIT | SERVE ONT9-STYLE DEMAND | SERVE PHX7-STYLE DEMAND |
|---|---|---|
| **Inbound to PHX7-style, then transfer if needed** | `$1.40 + $3.00 = $4.40` | `$1.40` |

### 6A · WHERE EACH SKU GOES — DEMAND FIRST, LEFTOVER ONLY WHERE COVER CAN TAKE IT

| STEP | DECISION | FORMULA → SUBSTITUTION | WEEKLY RESULT |
|---|---|---|---|
| 1 | F stays entirely at ONT9-style. Even the whole 6,000 leaves that node at one week. | `close = 1,000 + 6,000 − 3,500 = 3,500; WOC = 3,500 ÷ 3,500 = 1.00` | ONT 6,000 / PHX 0. PHX's 500 is a separate buy, not a slice of this PO. |
| 2 | M: give PHX its week, keep the rest at ONT. ONT's band survives the cut. | `ONT keeps 4,000 − 700 = 3,300` | ONT 3,300 / PHX 700 |
| 3 | S: ONT keeps the 350 the 4.0 WOC band asked for in stage 1. The over-order goes where the demand is. | `PHX gets 2,000 − 350 = 1,650` | ONT 350 / PHX 1,650 — the same 1,650 |
| 4 | Totals | `ONT 6,000 + 3,300 + 350 = 9,650; PHX 0 + 700 + 1,650 = 2,350; 9,650 + 2,350 = 12,000` | **9,650 / 2,350** |
| 5 | Trailer-equivalents at this vendor's 2,000 units per trailer | `9,650 ÷ 2,000 = 4.825; 2,350 ÷ 2,000 = 1.175` | about 5 trailers and about 1, not 6 and 0 |

### 6B · COST OF THE SPLIT VERSUS COST OF SIX TRAILERS HERE, THEN A TRANSFER TO REPAIR IT

| STEP | PLAN | FORMULA → SUBSTITUTION | COST |
|---|---|---|---|
| 1 | Split, direct inbound, no transfer | `9,650 × 0.80 = 7,720; 2,350 × 1.40 = 3,290; 7,720 + 3,290 = 11,010` | **$11,010** |
| 2 | All 12,000 inbound to ONT9-style | `12,000 × 0.80 = 9,600` | $9,600 sunk once the trailers are here |
| 3 | Then transfer the 2,350 the split would have sent direct | `2,350 × 3.00 = 7,050; 9,600 + 7,050 = 16,650` | $16,650 |
| 4 | Premium for deciding late | `16,650 − 11,010 = 5,640` | **$5,640** |
| 5 | Shadow-price check — the model has to agree with itself | `extra inbound = 2,350 × (1.40 − 0.80) = 2,350 × 0.60 = 1,410; transfer avoided = 7,050; 7,050 − 1,410 = 5,640` | $5,640 again |
| 6 | The same check by SKU | `M: 700 × (3.00 − 0.60) = 700 × 2.40 = 1,680; S: 1,650 × 2.40 = 3,960; 1,680 + 3,960 = 5,640` | F contributes $0 — it was not split |

Optimum on Tuesday: **split, 9,650 here and 2,350 direct to PHX7-style.** The booking that was actually made — six trailers, all to ONT9-style — leaves $5,640 on the table. After the yard, the $9,600 inbound is sunk, and the open question is the INV-D test on each SKU, not a wish to rerun Tuesday.

### 6C · AFTER THE TRAILERS ARE HERE — TRANSFER ONLY WHAT PASSES THE BREAKEVEN, ON ACTUAL CLOSES

| STEP | SKU | FORMULA → SUBSTITUTION | DECISION |
|---|---|---|---|
| 1 | F, the test | `transfer 500 × $3.00 = $1,500; PHX loss avoided = 500 × $12 = $6,000; ONT then shorts the same 500 × $12 = $6,000` | net +$1,500 worse. **Keep F. Expedite, do not transfer.** |
| 2 | M, the test | `cost 700 × 3.00 = 2,100; PHX loss if you do not = 700 × 12 = 8,400; 2,100 < 8,400` | **Transfer 700. Wins by $6,300.** |
| 3 | M, ONT after the move | `(4,200 − 700) ÷ 1,400 = 3,500 ÷ 1,400 = 2.50 WOC` | still inside the 2–4 band (actual close from stage 4) |
| 4 | S, three quantities you might move | `to the 6.0 ceiling: 3,050 − 6×350 = 3,050 − 2,100 = 950; to cover PHX's week: 1,050; to the 4.0 center: 3,050 − 4×350 = 3,050 − 1,400 = 1,650` | 950 leaves PHX short 100; 1,050 leaves ONT at 5.71 WOC, still in the watch band |
| 5 | S, the quantity chosen | `1,650 × 3.00 = 4,950; PHX week loss avoided ≥ 1,050 × 12 = 12,600; ONT left at 1,400 ÷ 350 = 4.00 WOC` | **Transfer 1,650. It is the stage-1 over-order, and it is the INV-A move down to 4.0 WOC.** |
| 6 | Cash still to spend, forward-looking | `2,100 + 4,950 = 7,050 on M and S; $0 on F` | $7,050. The $5,640 from Tuesday is no longer available to save; this $7,050 is the repair. |

**The sentence to say:** "Tuesday's optimum was ninety-six fifty here and twenty-three fifty direct to the sister node, eleven thousand ten dollars against sixteen thousand six fifty if I haul them later — a fifty-six forty gap, and the shadow price prints the same fifty-six forty, so I trust it. The trailers came here anyway. Forward from the yard I keep the fast mover, because moving it spends fifteen hundred dollars to relocate a stockout, and I transfer seven hundred of the medium and the sixteen fifty of the slow. That sixteen fifty is the same over-order the buy created in stage one."

## Stage 7 · Inventory health at the close — who is thin, who is fine, who gets moved

Closes are the stage-4 actuals: F 3,380, M 4,200, S 3,050. Sales this week equaled forecast, so sell-through uses units sold ÷ (opening + received). Safety stock uses the card: z = 1.645, √7 = 2.6458, σ of 80 / 40 / 15. The action rule is INV-A's: under 2 weeks expedite, 2–4 watch, over 6 move.

### 7A · WOC, SAFETY STOCK, AND SELL-THROUGH — THREE SKUS, THREE ANSWERS

| STEP | QUANTITY | FORMULA → SUBSTITUTION | INPUTS (BASIS) | VALUE |
|---|---|---|---|---|
| 1 | WOC, F | `3,380 ÷ 3,500 = 0.9657 ≈ 0.966` | close ÷ weekly forecast | **0.966 weeks — under 2, expedite** |
| 2 | SS, F | `σ_LT = 80 × 2.6458 = 211.664; SS = 1.645 × 211.664 = 348.19 ≈ 348; 348 ÷ 500 = 0.696 days` | z × σ × √LT | 348 units, 0.70 days |
| 3 | F against the bridge | `LT units = 7 × 500 = 3,500; LT+SS = 3,500 + 348 = 3,848; gap = 3,848 − 3,380 = 468; and 120 + 348 = 468` | the shortage plus the safety stock the buy never held | **short 120 vs the lead time, short 468 vs lead time plus safety stock** |
| 4 | Sell-through, F | `3,500 ÷ (1,000 + 5,880) = 3,500 ÷ 6,880 = 0.5087 = 50.9%` | sold ÷ (open + received) | 50.9% |
| 5 | WOC, M | `4,200 ÷ 1,400 = 3.00` | stage-4 close | **3.00 weeks — healthy** |
| 6 | SS, M | `σ_LT = 40 × 2.6458 = 105.832; SS = 1.645 × 105.832 = 174.09 ≈ 174; cover days = 4,200 ÷ 200 = 21.0; LT+SS days = 7 + 174 ÷ 200 = 7.87; 21.0 ÷ 7.87 = 2.67×` | same z and √LT | 174 units of safety stock; position is 2.67× the floor |
| 7 | Sell-through, M | `1,400 ÷ (1,600 + 4,000) = 1,400 ÷ 5,600 = 0.250 = 25.0%` | sold ÷ (open + received) | 25.0% |
| 8 | WOC, S | `3,050 ÷ 350 = 8.714 ≈ 8.71; days = 3,050 ÷ 50 = 61.0` | stage-4 close | **8.71 weeks, 61 days — over 6, move it** |
| 9 | SS, S | `σ_LT = 15 × 2.6458 = 39.687; SS = 1.645 × 39.687 = 65.29 ≈ 65; 3,050 − 65 = 2,985 above the safety-stock floor` | same z and √LT | 65 units. "Strategic cover" would have to explain the other 2,985. |
| 10 | Sell-through, S | `350 ÷ (1,400 + 2,000) = 350 ÷ 3,400 = 0.1029 = 10.3%` | sold ÷ (open + received) | 10.3% — the excess showing up as a rate |

### 7B · THE CORRECTIVE ACTION — ONE PER SKU, EACH ONE A NUMBER ALREADY ON THIS PAGE

| STEP | SKU | FORMULA → SUBSTITUTION | ACTION |
|---|---|---|---|
| 1 | F — at risk of stockout | `expedite 468 = 120 shortage + 348 safety stock; 468 ÷ 500 = 0.94 days of demand; 3,380 + 468 = 3,848 = the LT+SS target` | **Pull the next ASN forward by 468 units. Do not fund it by transferring F out — stage 6c already priced that as a loss.** |
| 2 | M — healthy | `WOC 3.00 is the center of the 2–4 band. After the stage-6 transfer of 700: 2.50 WOC, still inside the band.` | **No inventory action. The 700-unit move is a placement repair, and the band still holds.** |
| 3 | S — overstocked | `move 1,650; 3,050 − 1,650 = 1,400; 1,400 ÷ 350 = 4.00 WOC; cost 1,650 × 3.00 = 4,950` | **Transfer 1,650 to PHX7-style. Same 1,650 as the stage-1 over-order.** |
| 4 | Cube the S move gives back | `1,650 × 0.80 = 1,320 ft³; 1,320 ÷ 6,800 = 0.194 = 19.4% of the mod` | 19.4 points of effective fill, freed without a rack |

**The sentence to say:** "Three SKUs, three sentences. Fast mover, zero-point-nine-seven weeks, six-point-seven-six days against a seven-day lead time — expedite four hundred sixty-eight, which is the one hundred twenty we never received plus the three hundred forty-eight of safety stock we never bought, and keep every unit of it here. Medium, three-point-zero-zero weeks, healthy, and it can still spare seven hundred for the sister node and land at two-and-a-half. Slow mover, eight-point-seven weeks and ten percent sell-through — transfer the sixteen fifty, four thousand nine hundred fifty dollars, back to four weeks, and that sixteen fifty is the order multiple from the first page of the week."

## The one-page close — stage, metric, value, what breaks

| STAGE | METRIC | VALUE | WHAT BREAKS IF IT IS MISMANAGED |
|---|---|---|---|
| **1 · Forecast** | Units and trailer-equivalents | 12,000 = 6,000 + 4,000 + 2,000. 50 pallets = 1.92 dense trailers. Tendered 6, at 32.1% full. | Ordering S to the 2,000-unit multiple overstocks by 1,650. Buying F to 1.0 WOC with no safety stock leaves a 7-day bridge and nothing under it. |
| **2 · Dock** | Slots, ρ, dwell | 6 × 2.0 × 0.5 = 6 slots against 7 arrivals. ρ = 116.7%. Marginal dwell 4.0 h. E[dwell] = 2.29 h. The week was 4.0% full. | FIFO delays an expected 1,714 units by 2.0 h and spills one slot into the next block. Quoting Kingman past ρ = 1 hides the residual trailer. A seventh door was never the constraint. |
| **3 · Labor** | Hours and heads | 12,000 ÷ 300 = 40 h × 1.15 = 46 h ÷ 10 = 4.6 → 5 receivers on the bunched shift. The spread averages 0.92 → 1. | Rostering the average leaves ~40 direct hours with one person, and the backlog carries into Thursday. The 120-unit shortage frees 0.4 h — hold the crew. |
| **4 · Accuracy** | Shortage and AFS | 0.02 × 6,000 = 120 of F. Received 11,880 (99.0% overall, 98.0% on F). F AFS 6,880 vs 7,000. Cover 6.76 days vs 7.00. | A 99% headline hides the only SKU sitting on its lead time. This week's sales still clear; the 120 is the entire miss versus the next receipt. |
| **5 · Stow / cube** | Fill and stow heads | Effective mod 6,800 ft³. Fill 25.7% → 79.9% peak → 61.9% at close. Nominal at the peak would have said 67.9%. S is 43.5% of receipt cube. 6 stowers. | Another 860 units of S, 688 ft³, hits the 90% wall. Reading nominal fill understates the peak by 12 points. The slow mover is the cube problem. |
| **6 · Placement** | Split vs all six trailers here | Optimum 9,650 / 2,350. Cost $11,010 vs $16,650. Savings $5,640, matched by the shadow-price check. After arrival: keep F, transfer M 700 and S 1,650, spend $7,050. | Six trailers into ONT9-style leave PHX's medium and slow demand unfilled and leave 8.71 weeks of S in this mod. Deciding after the yard turns a $1,410 inbound premium into a $7,050 transfer. |
| **7 · Health** | WOC and the action | F 0.966 WOC, expedite 468. M 3.00 WOC, hold (2.50 after the 700). S 8.71 WOC, transfer 1,650 for $4,950, back to 4.00, freeing 1,320 ft³. | F stocks out inside the next lead time. S sits until the storage cost and the wasted cube. M is the control: inside the band, and still able to fund the sister node. |

**SAY THIS VERSION OUT LOUD — ABOUT SIXTY SECONDS, THEN STOP**

"Week 44, an ONT9-style building: one vendor, three SKUs, twelve thousand units on six trailers. Cover targets rebuild that — fast mover to one week, medium to three — except the slow mover arrives as a two-thousand-unit multiple against a three-fifty need, sixteen fifty too many. Dense, one-point-nine trailers; tendered, six, so the dock feels six arrivals. Wednesday, seven trailers hit a six-slot block: the queueing formula does not apply, the seventh dwells four hours, the average is two-point-two-nine. Staff the bunch, five receivers and six stowers. Receive is short one hundred twenty of the fast mover — the whole gap from a seven-day lead time to six-point-seven-six days, because safety stock was never bought. The mod goes from twenty-five-point-seven percent full to seventy-nine-point-nine. Splitting twenty-three fifty units to the sister node was worth five thousand six hundred forty dollars. At the close: expedite four sixty-eight of the fast mover and keep it, leave the medium at three weeks, transfer the same sixteen fifty of the slow mover. Same twelve thousand, three actions."